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(54) **CONTEXTUAL-BASED RENDERING OF
VIRTUAL AVATARS**

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H04N 7/15 (2006.01)

(52) **U.S. Cl.**

CPC **G06T 19/006** (2013.01); **G06F 3/011** (2013.01); **G06F 3/012** (2013.01); **G06F 3/013** (2013.01); **G06F 3/017** (2013.01); **G06F 3/0304** (2013.01); **G06F 17/18** (2013.01); **G06V 40/20** (2022.01); **H04N 7/157** (2013.01); **G06T 13/40** (2013.01)

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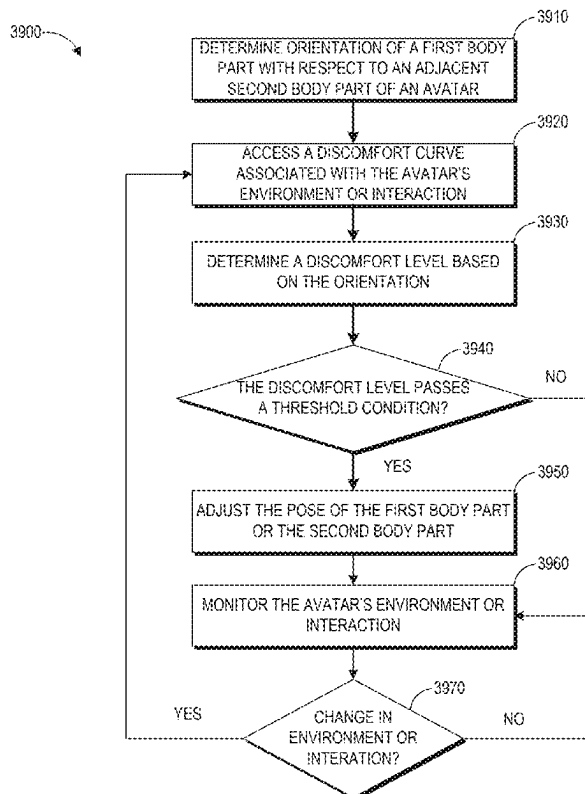
(62) Division of application No. 18/320,904, filed on May 19, 2023, now Pat. No. 12,340,477, which is a division of application No. 17/833,508, filed on Jun. 6, 2022, now Pat. No. 11,699,255.

(57)

ABSTRACT

Examples of systems and methods for rendering an avatar in a mixed reality environment are disclosed. The systems and methods may be configured to automatically scale an avatar or to render an avatar based on a determined intention of a user, an interesting impulse, environmental stimuli, or user saccade points. The disclosed systems and methods may apply discomfort curves when rendering an avatar. The disclosed systems and methods may provide a more realistic interaction between a human user and an avatar.

EXAMPLE PROCESS OF ANIMATING A VIRTUAL
AVATAR WITH A DISCOMFORT CURVE



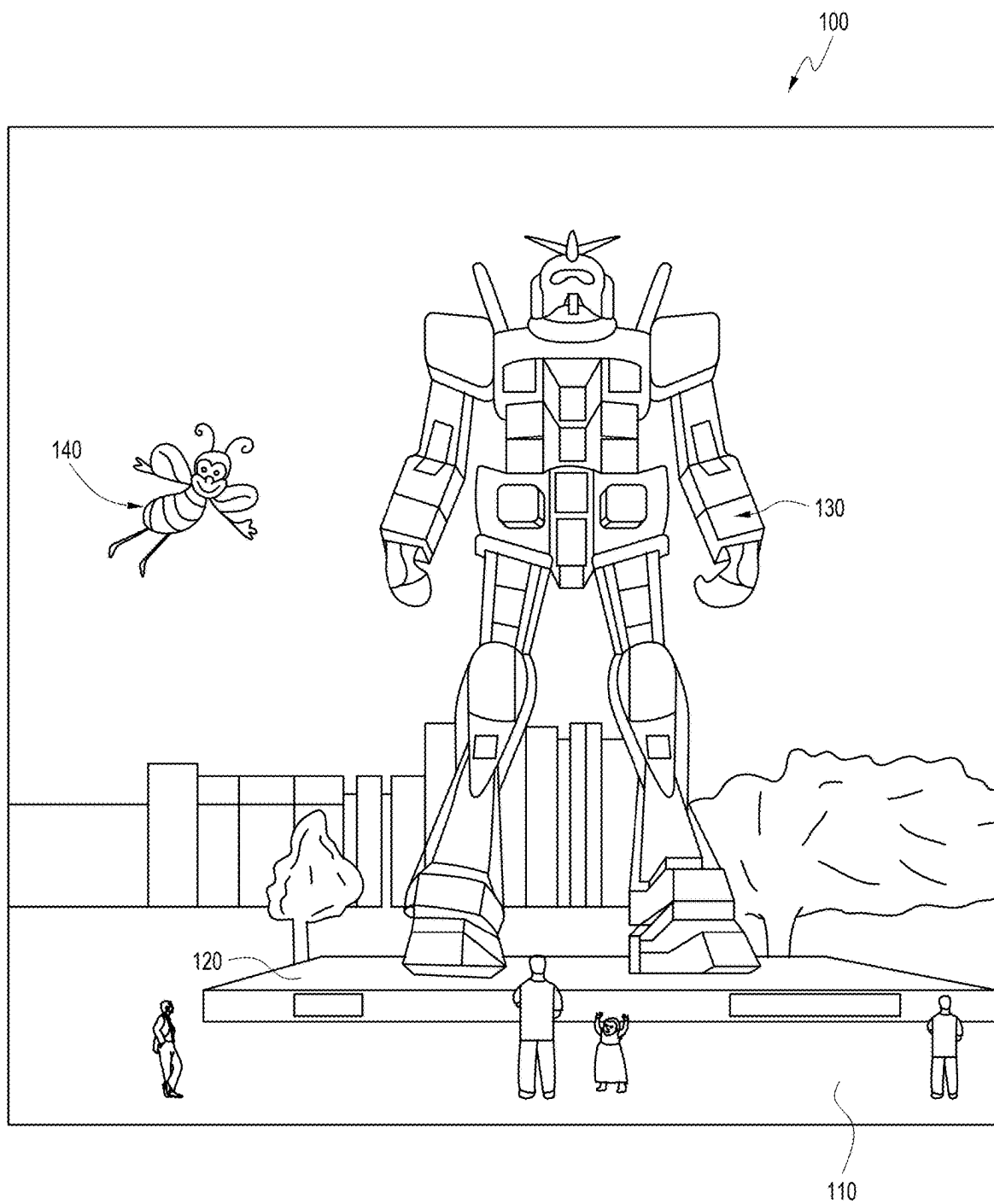


FIG. 1

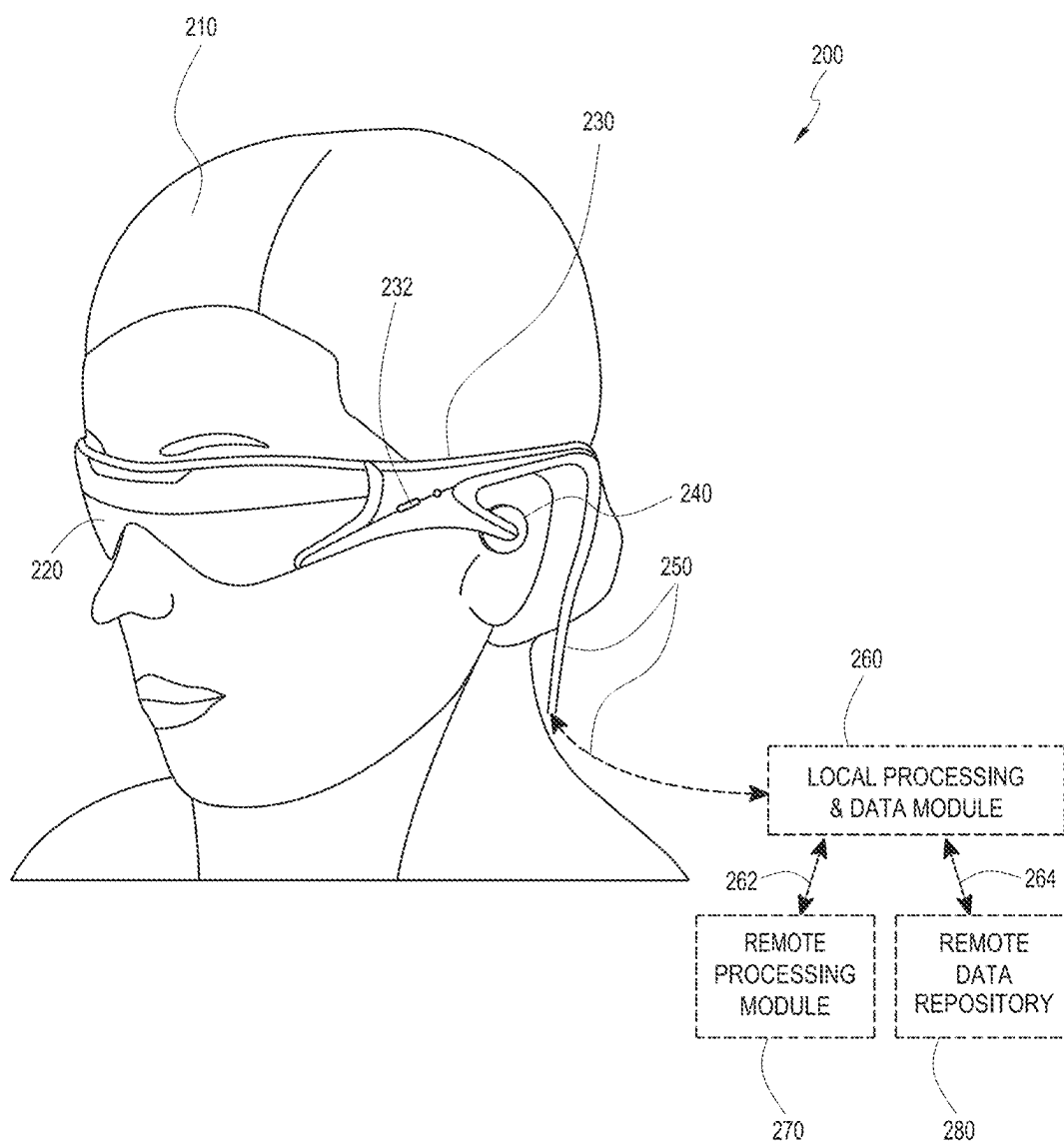
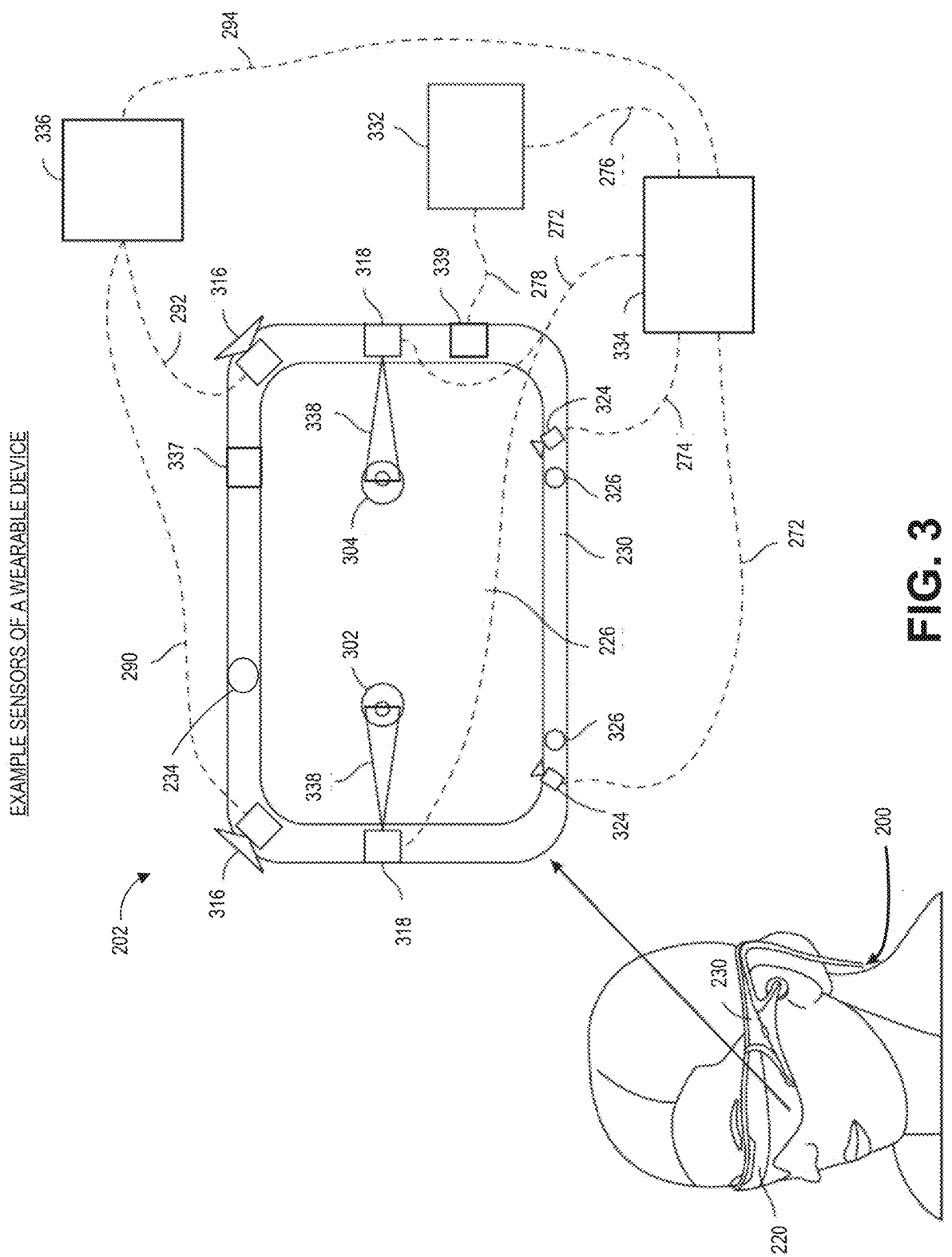
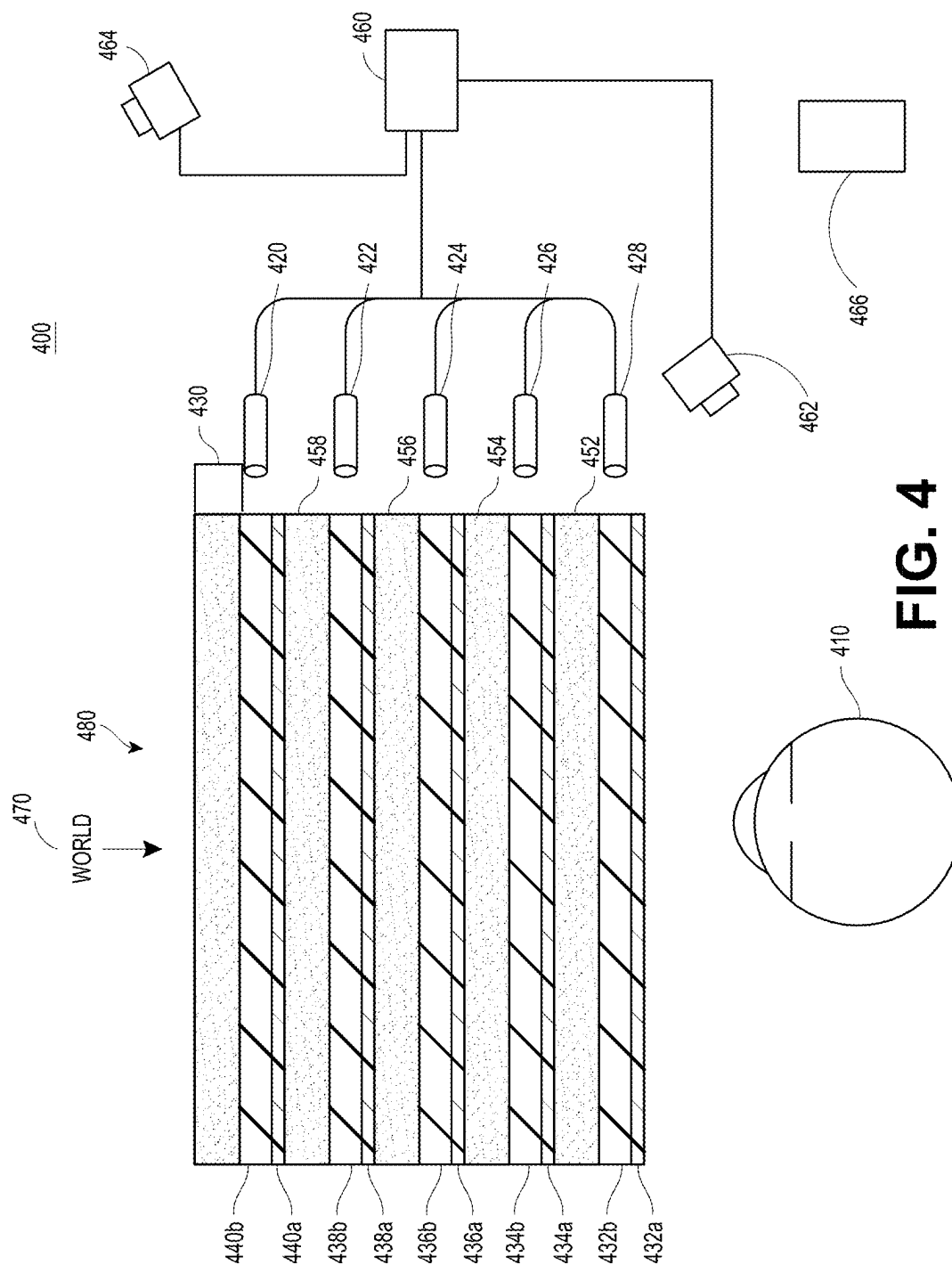


FIG. 2





EXAMPLE USER INTERACTIONS WITH A WEARABLE DEVICE

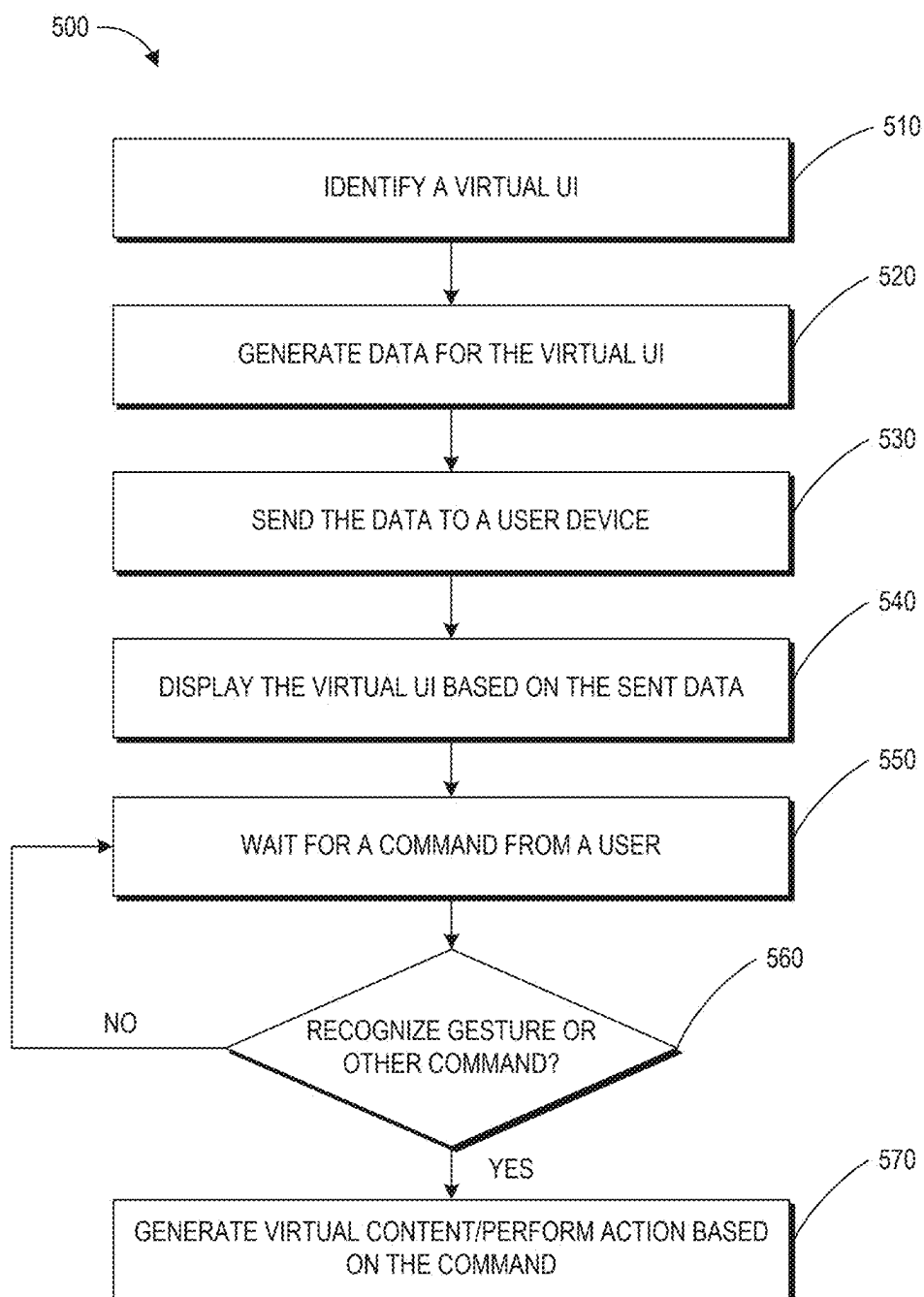


FIG. 5

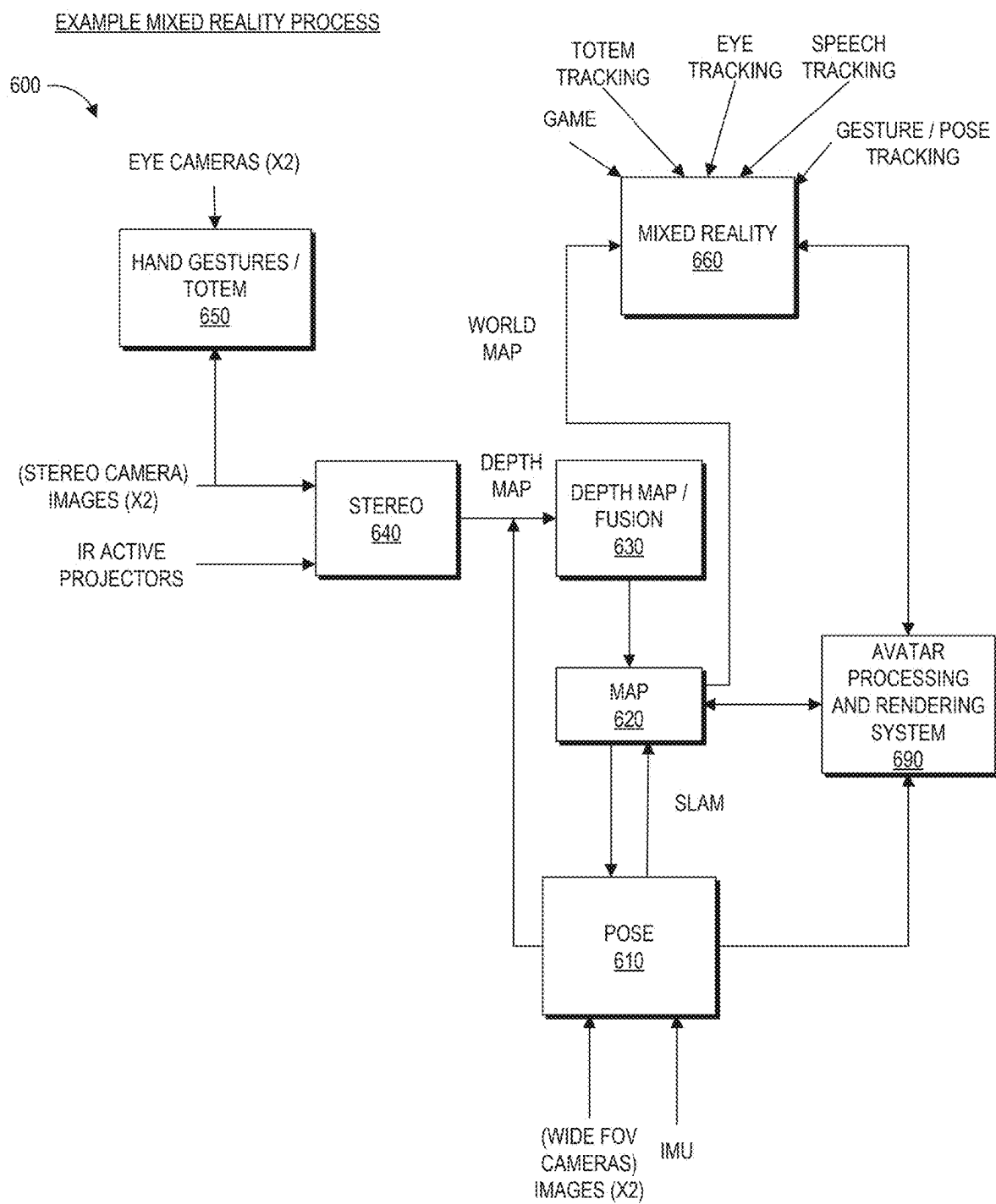


FIG. 6A

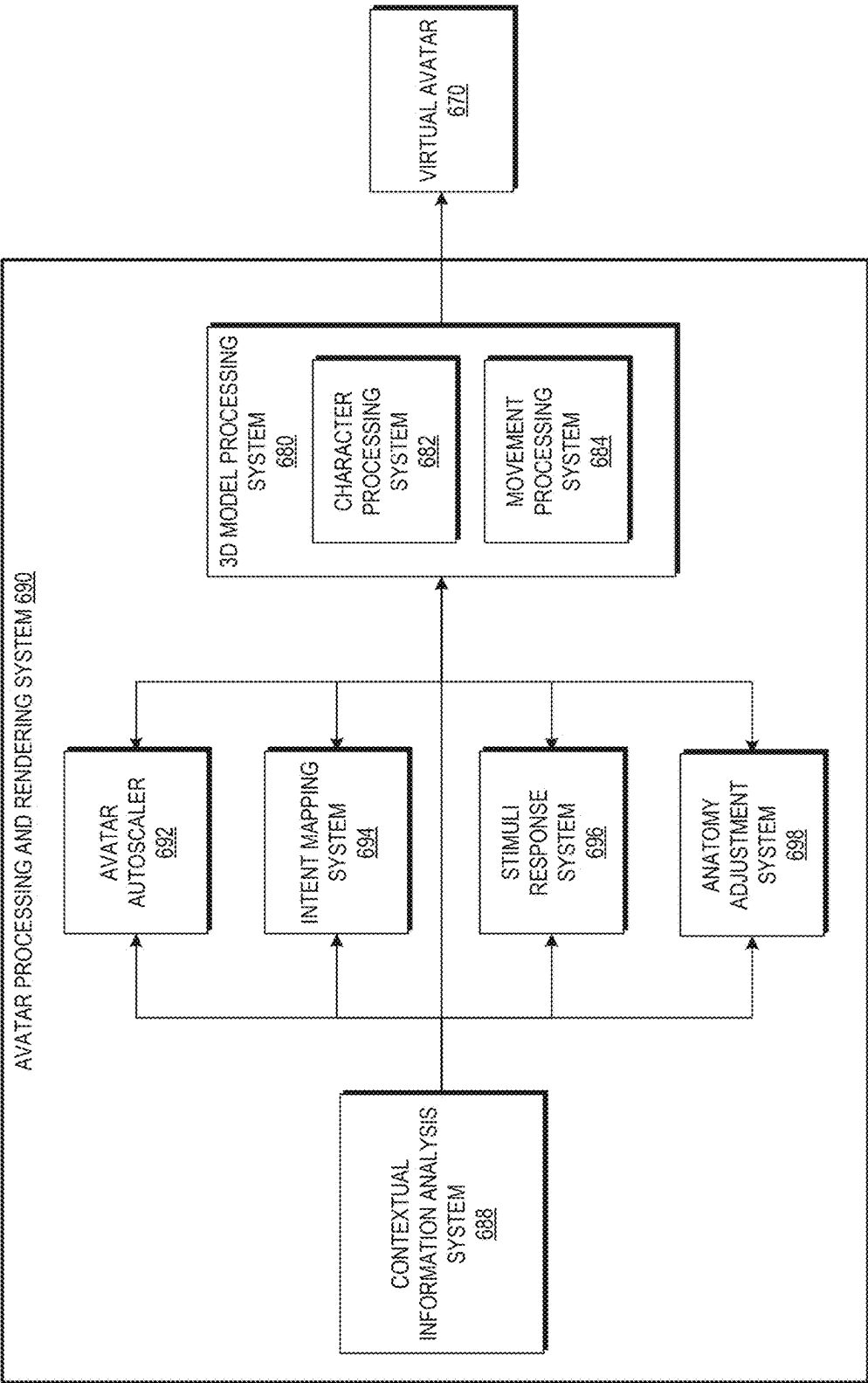


FIG. 6B

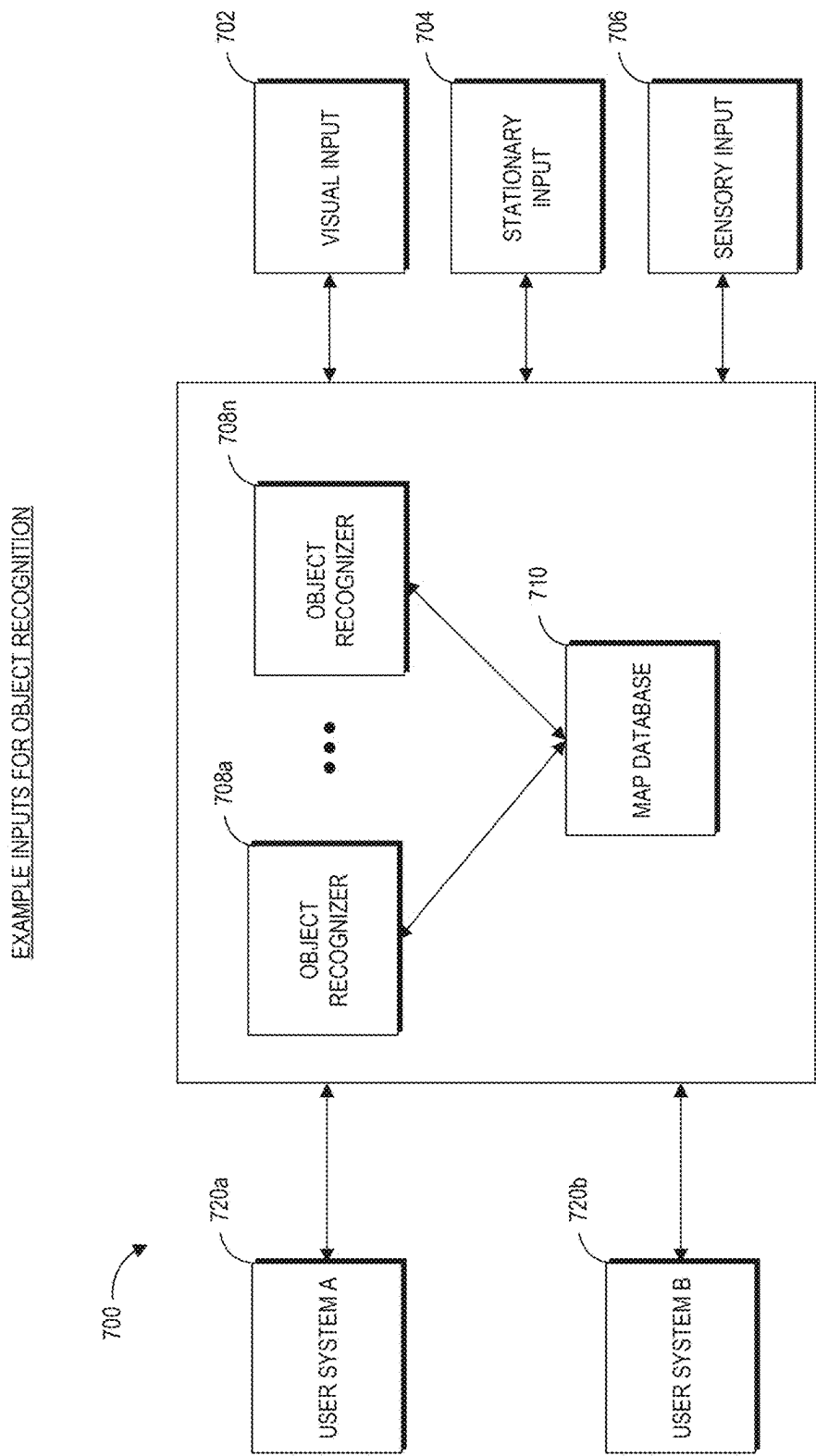
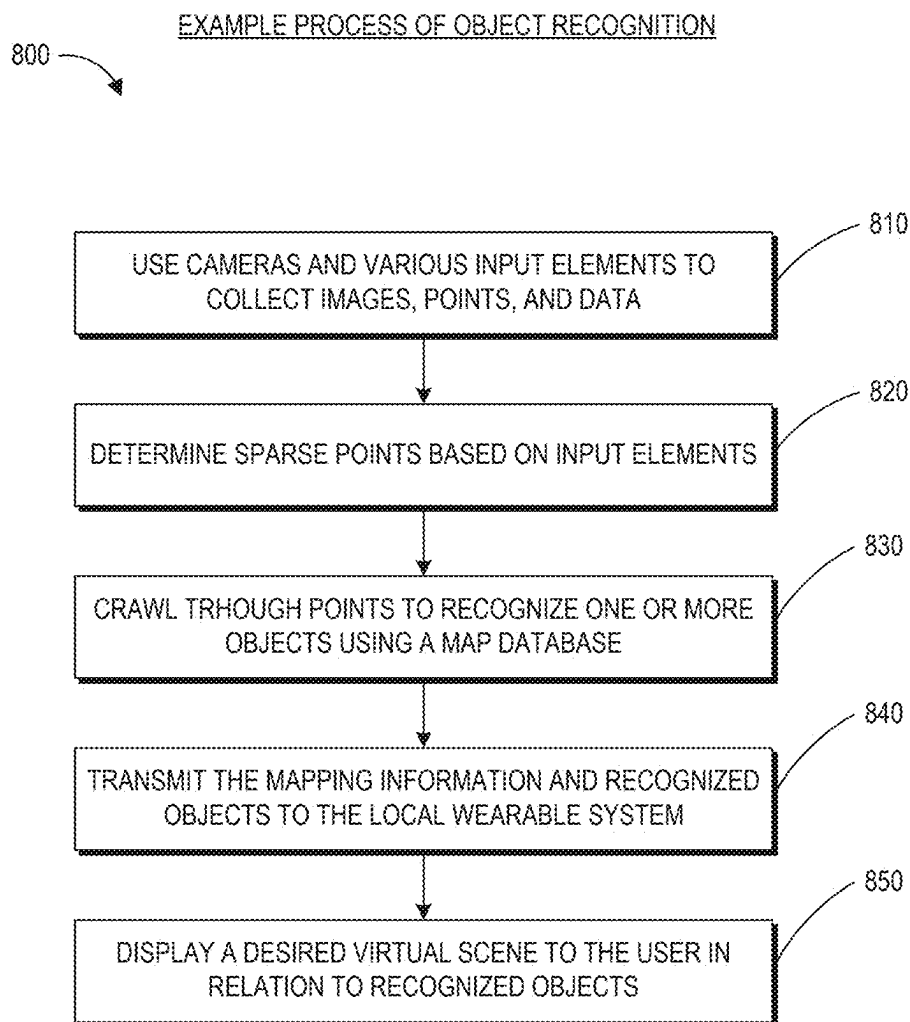


FIG. 7

**FIG. 8**

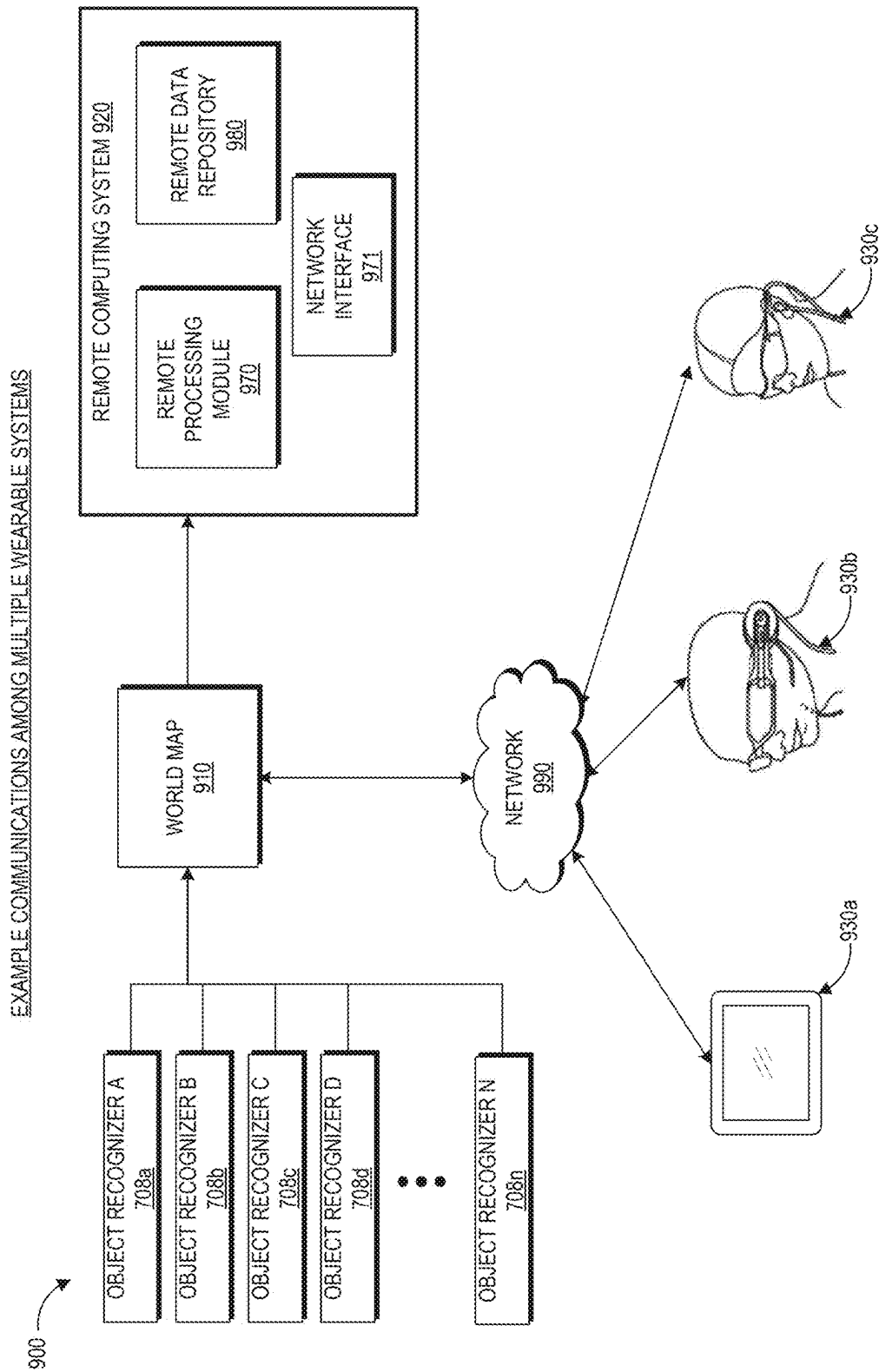


FIG. 9A

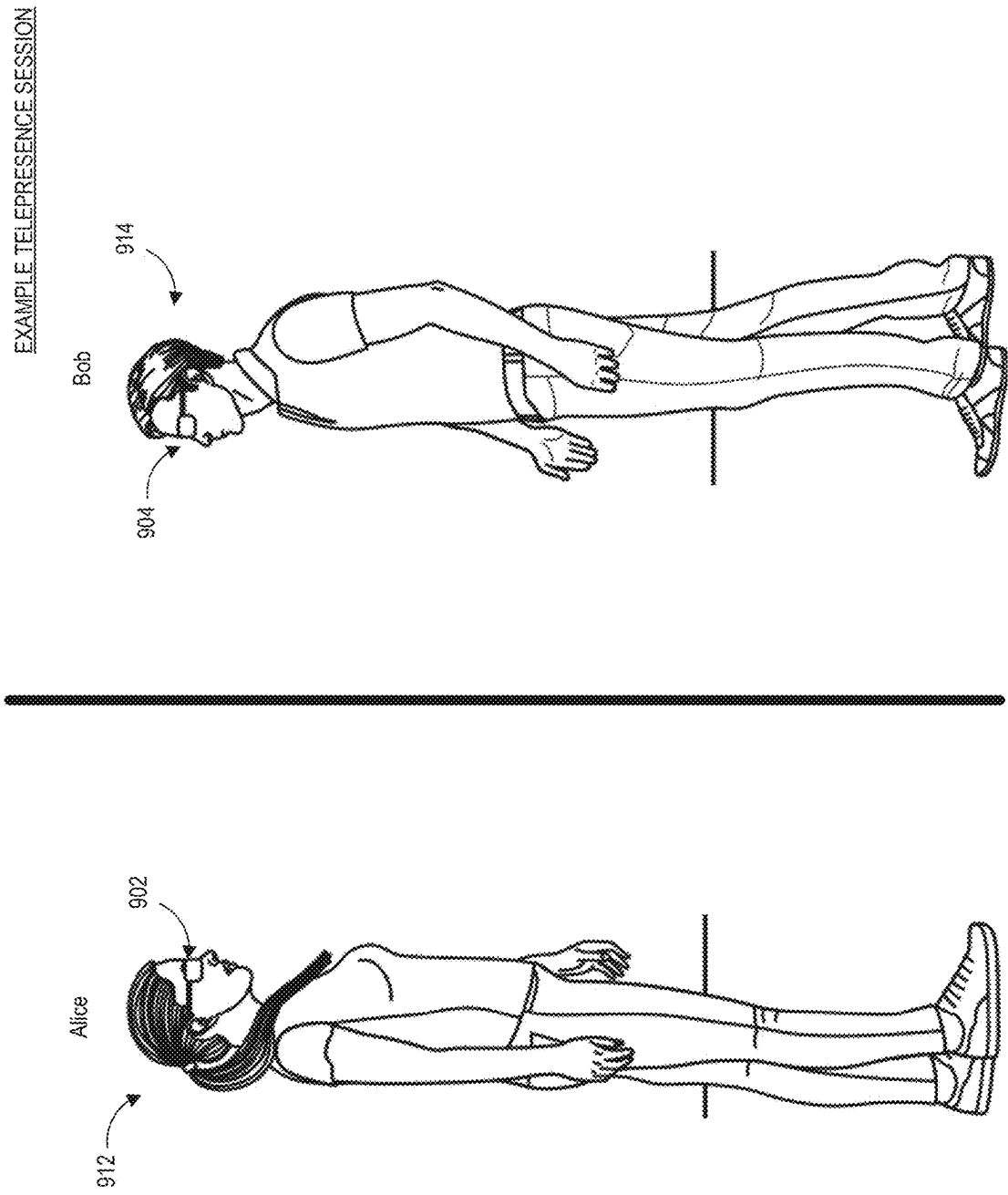


FIG. 9B

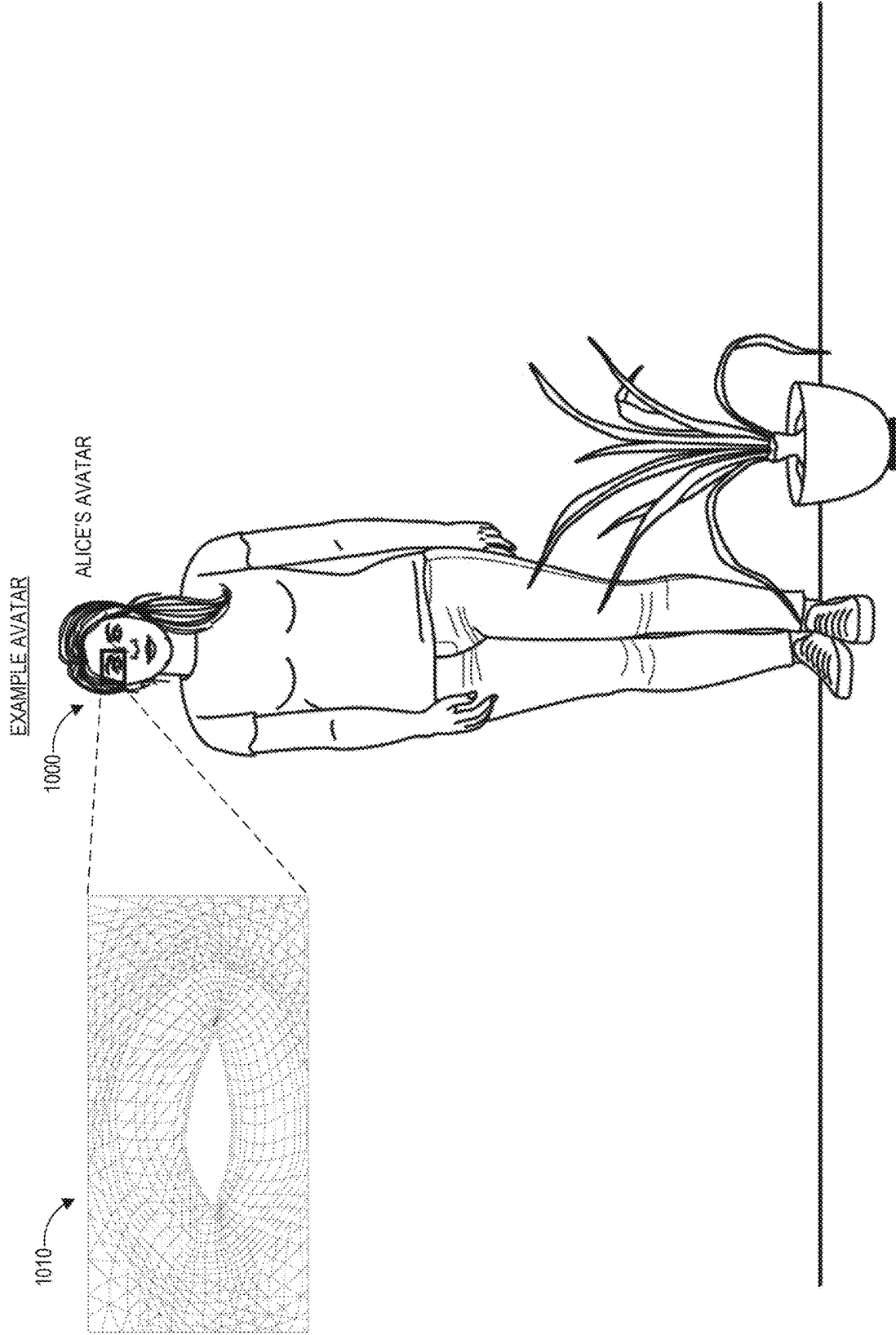


FIG. 10

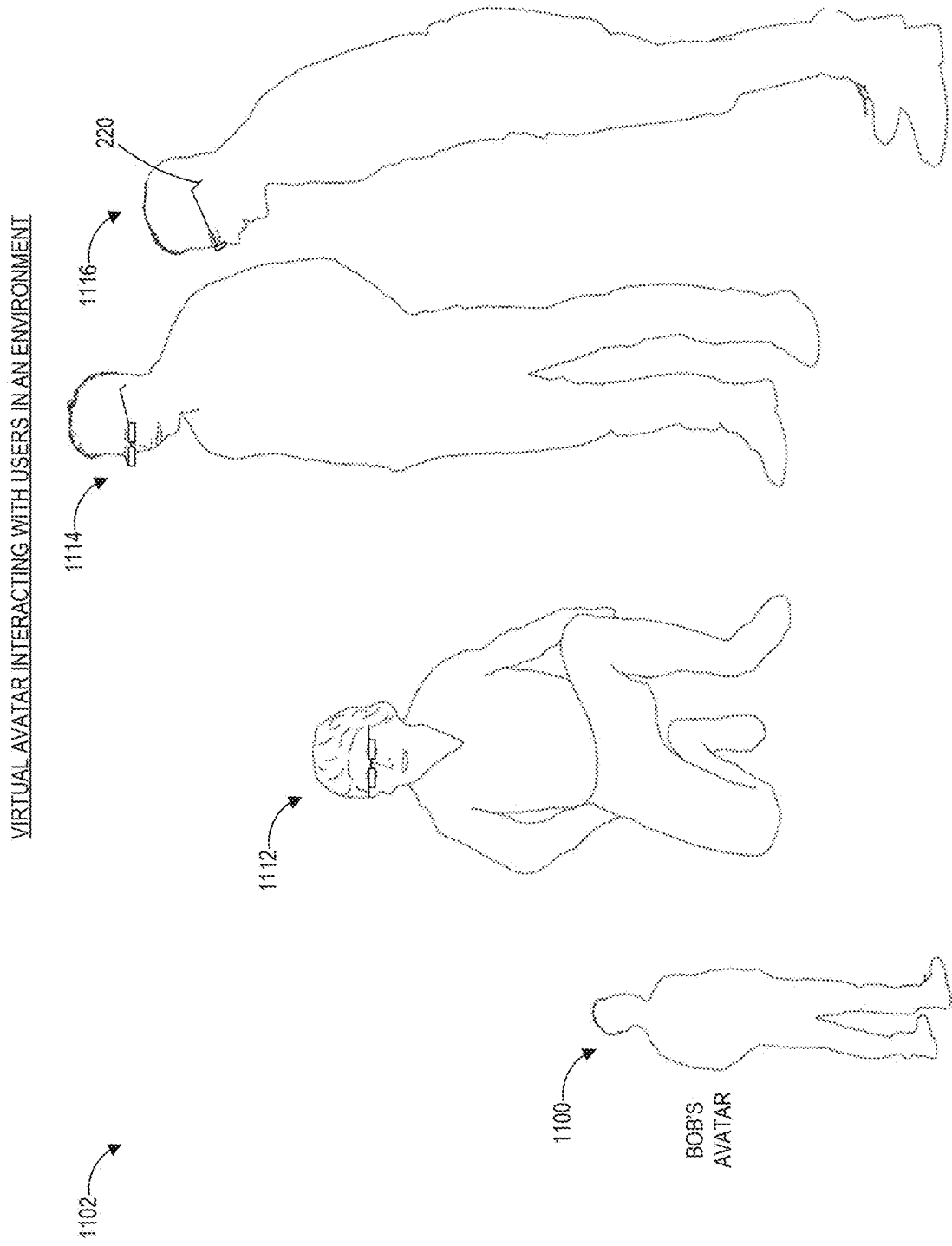


FIG. 11A

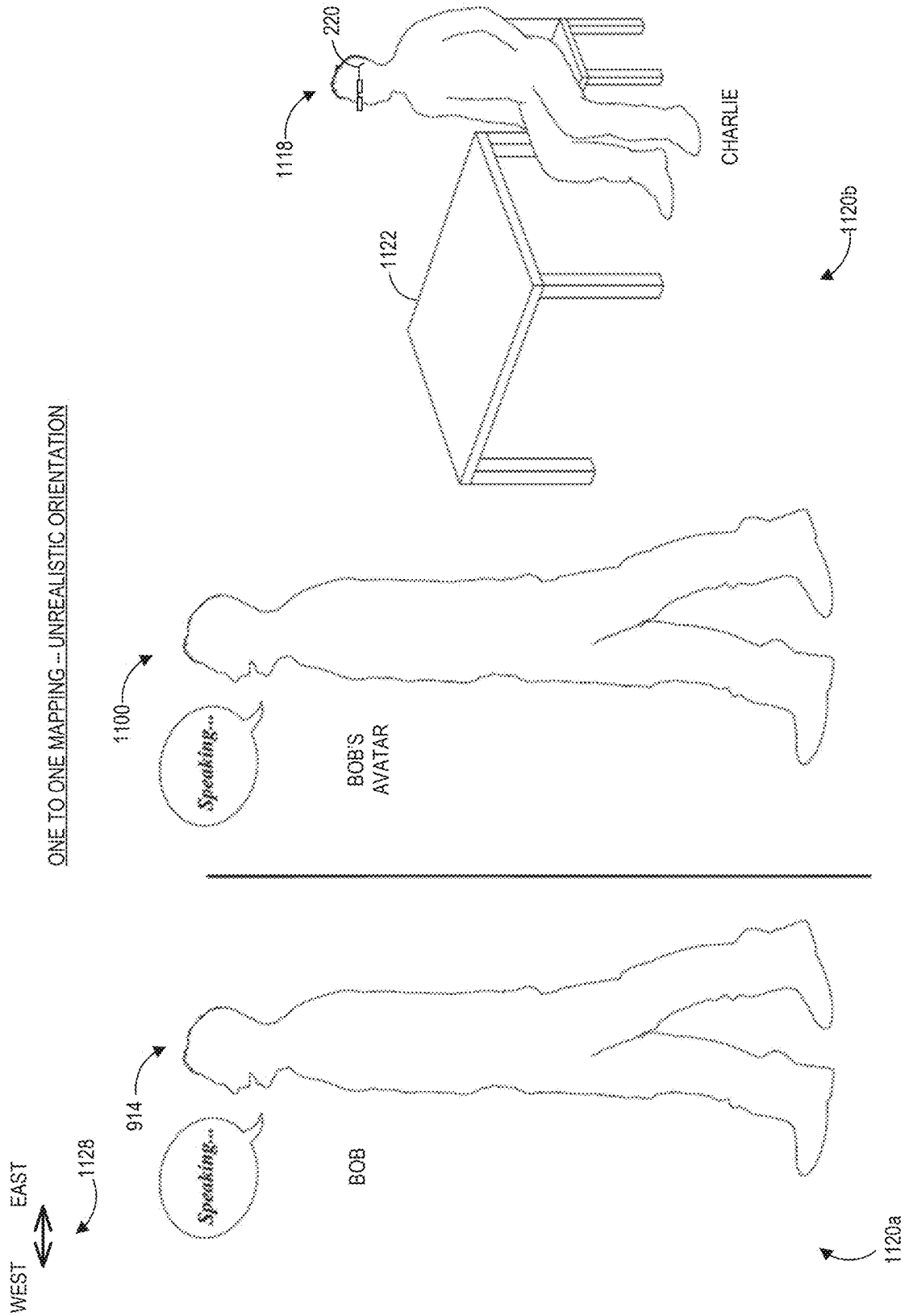


FIG. 11B

ONE TO ONE MAPPING – UNREALISTIC POSITION

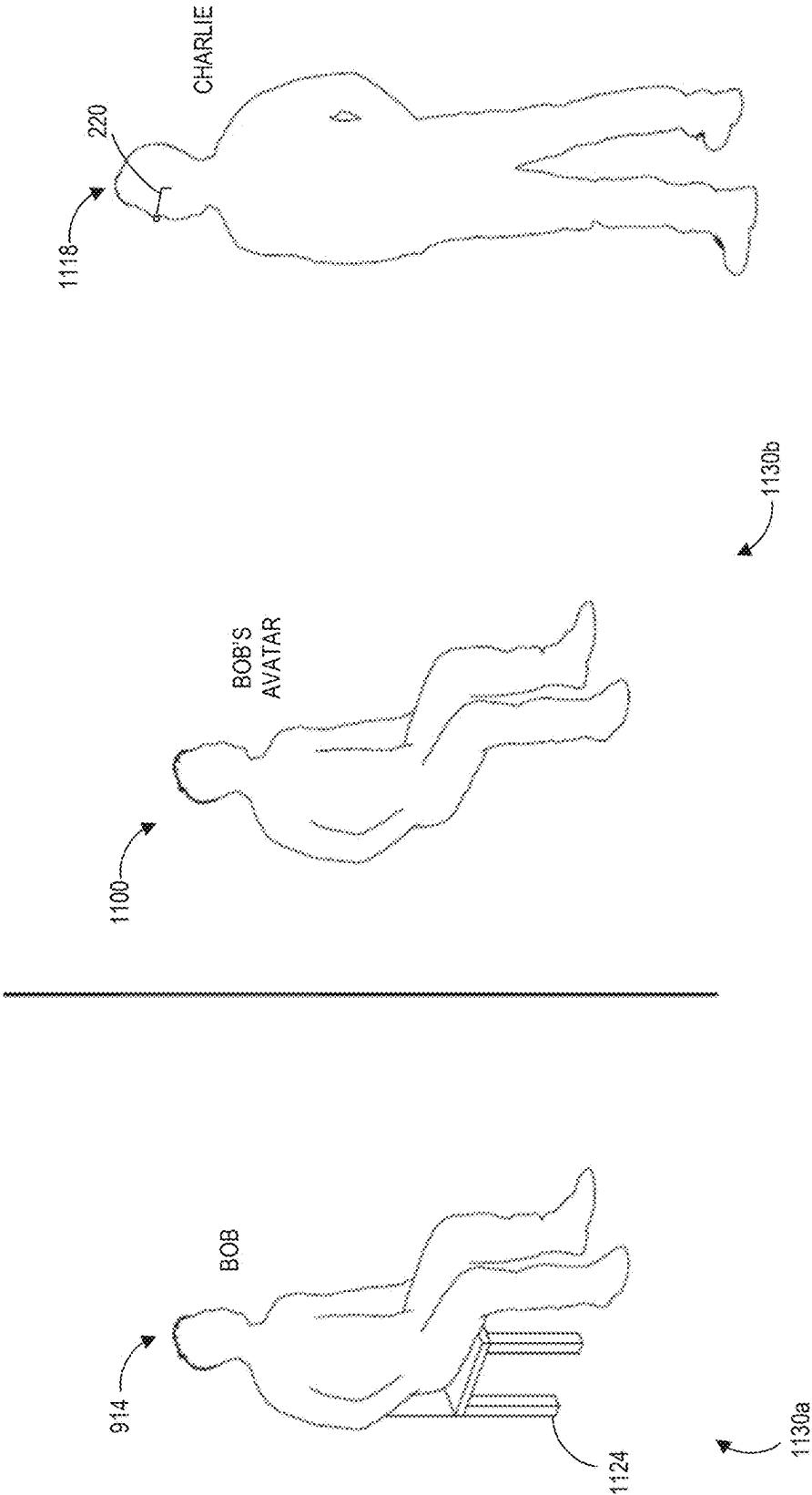


FIG. 11C

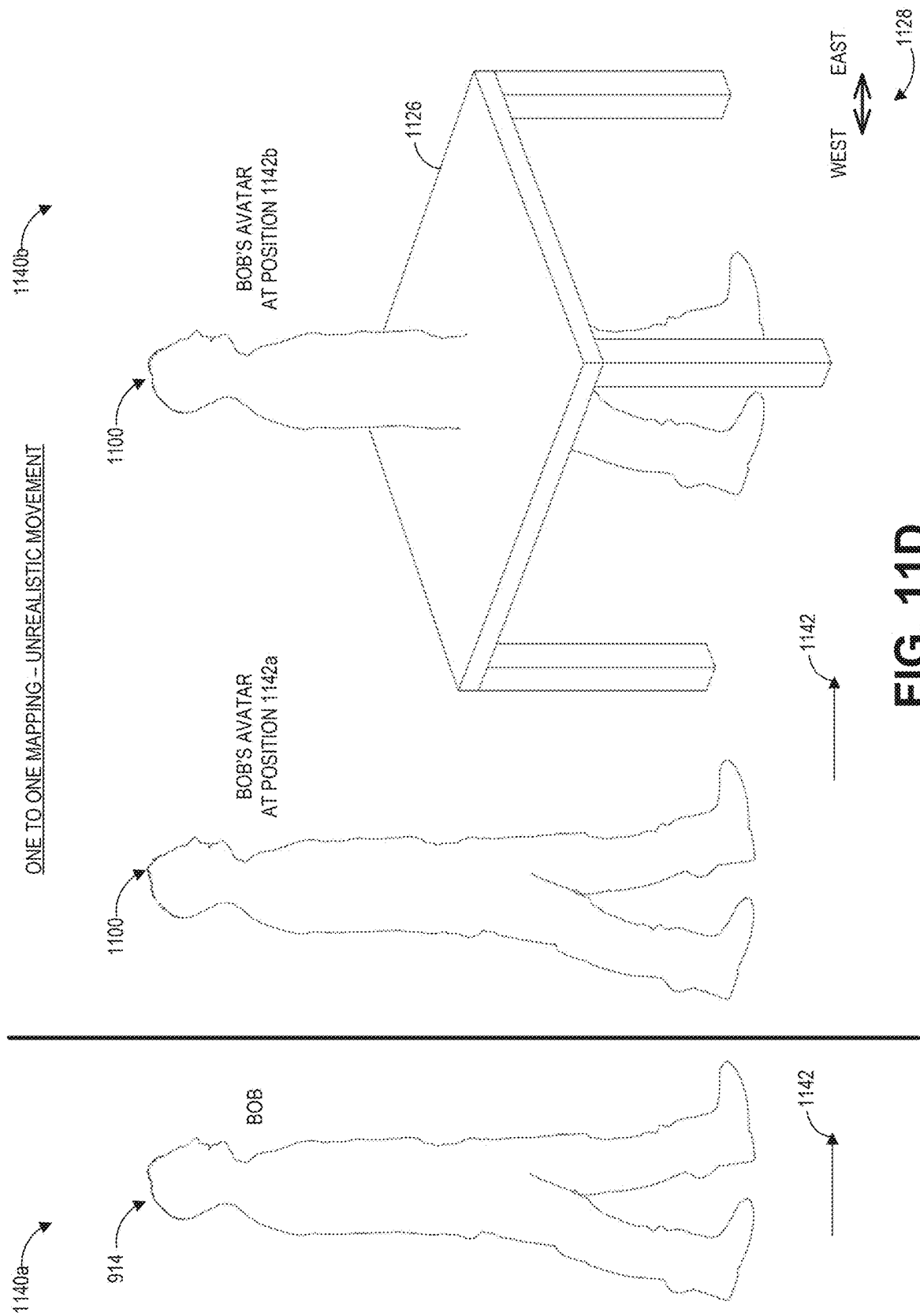


FIG. 11D

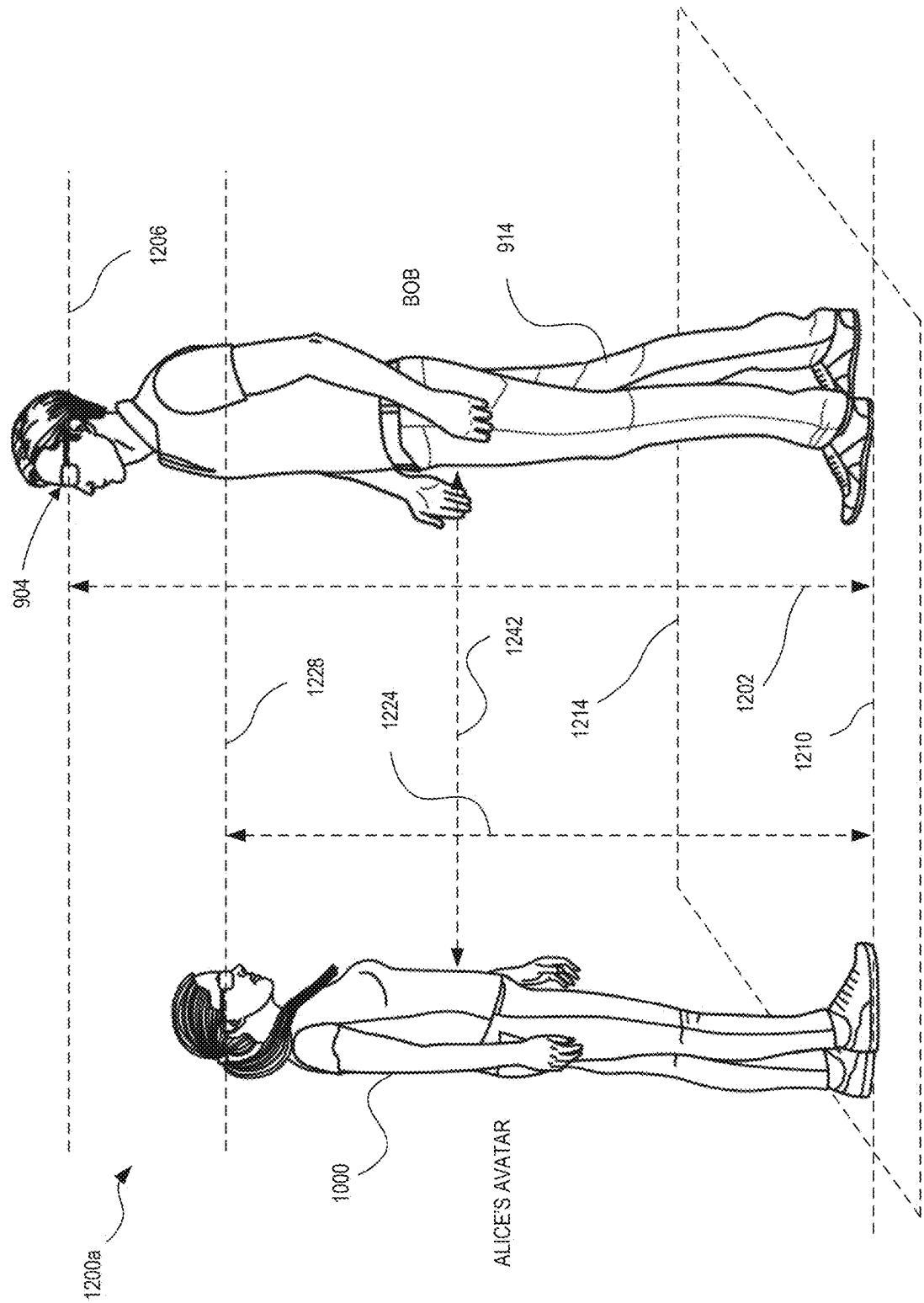


FIG. 12A

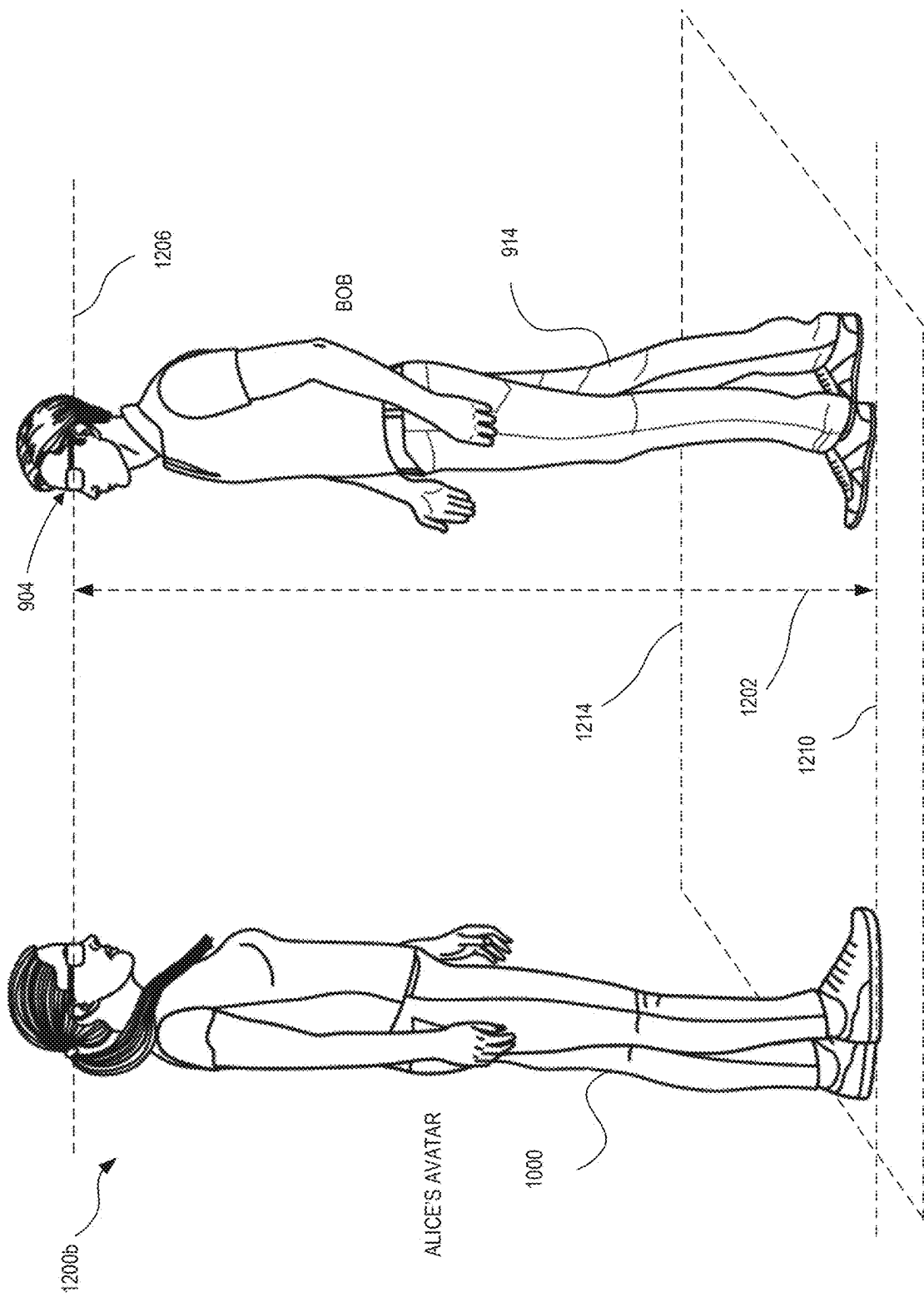


FIG. 12B

1300a

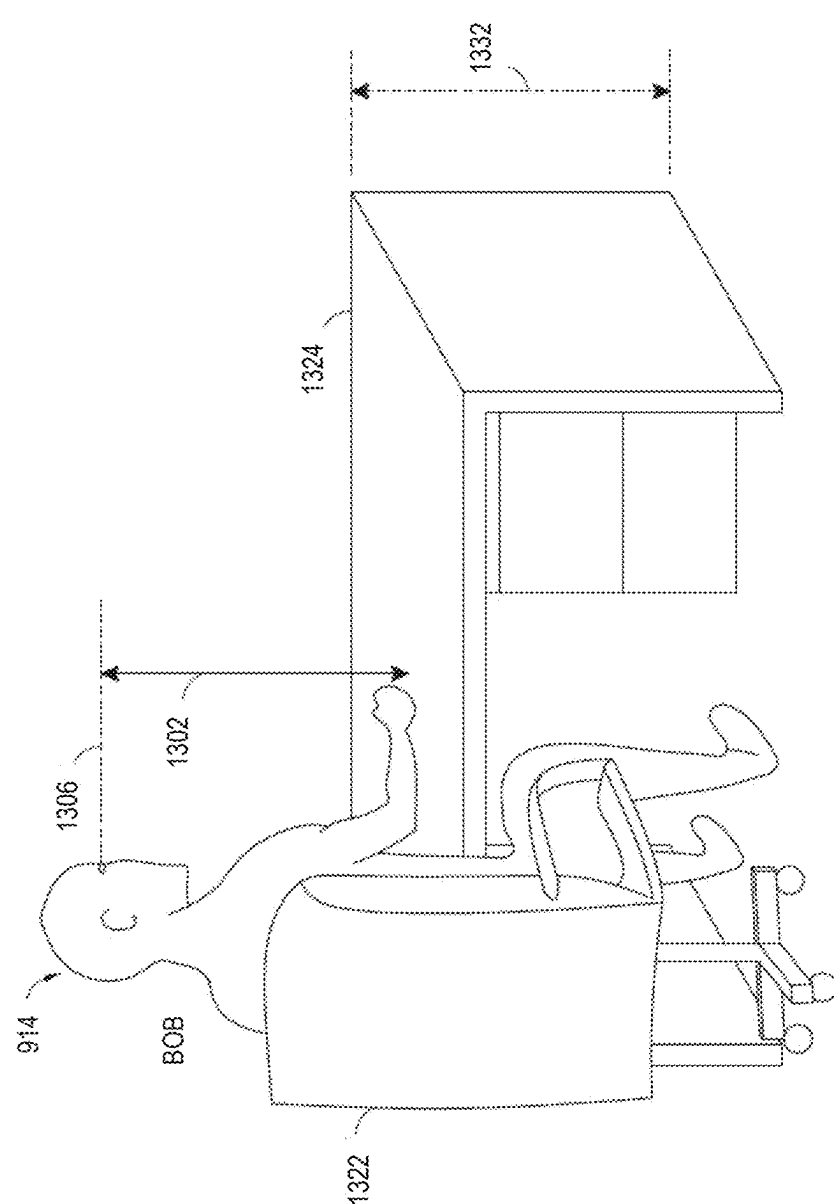


FIG. 13A

1300b

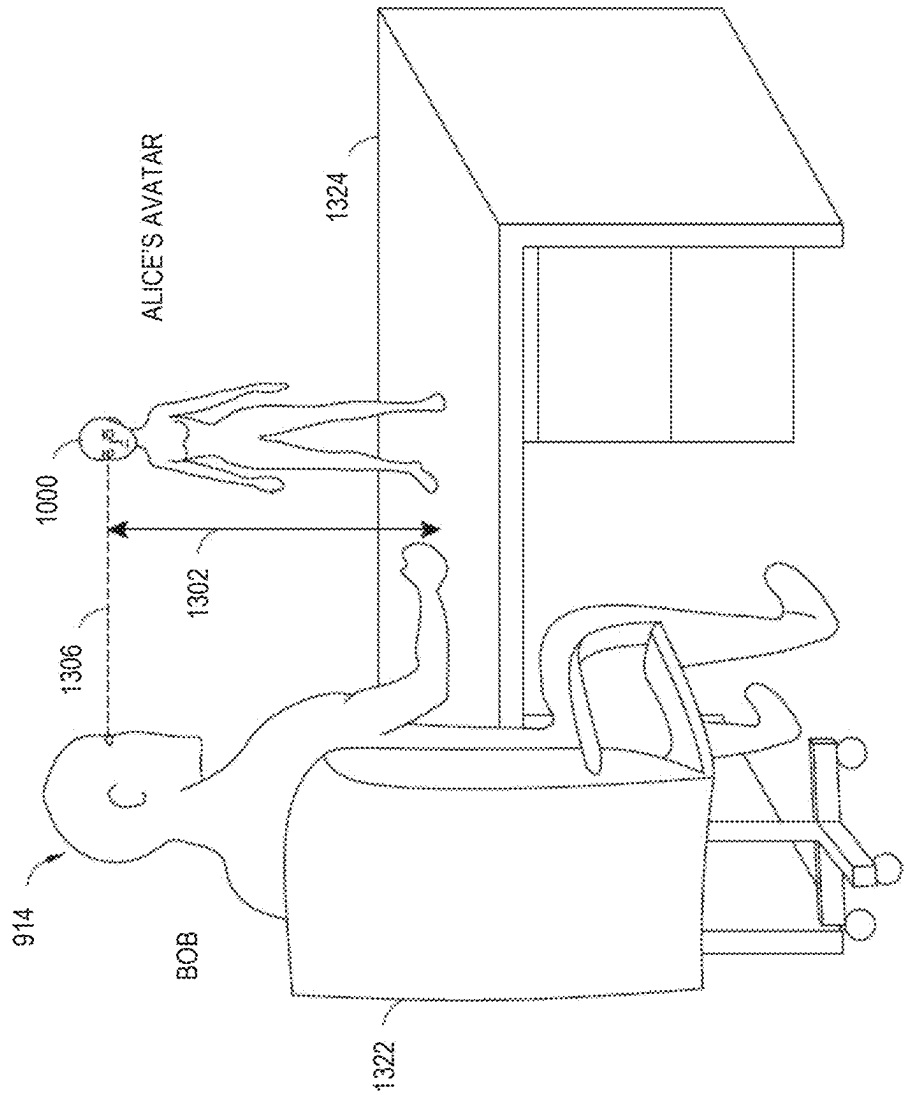


FIG. 13B

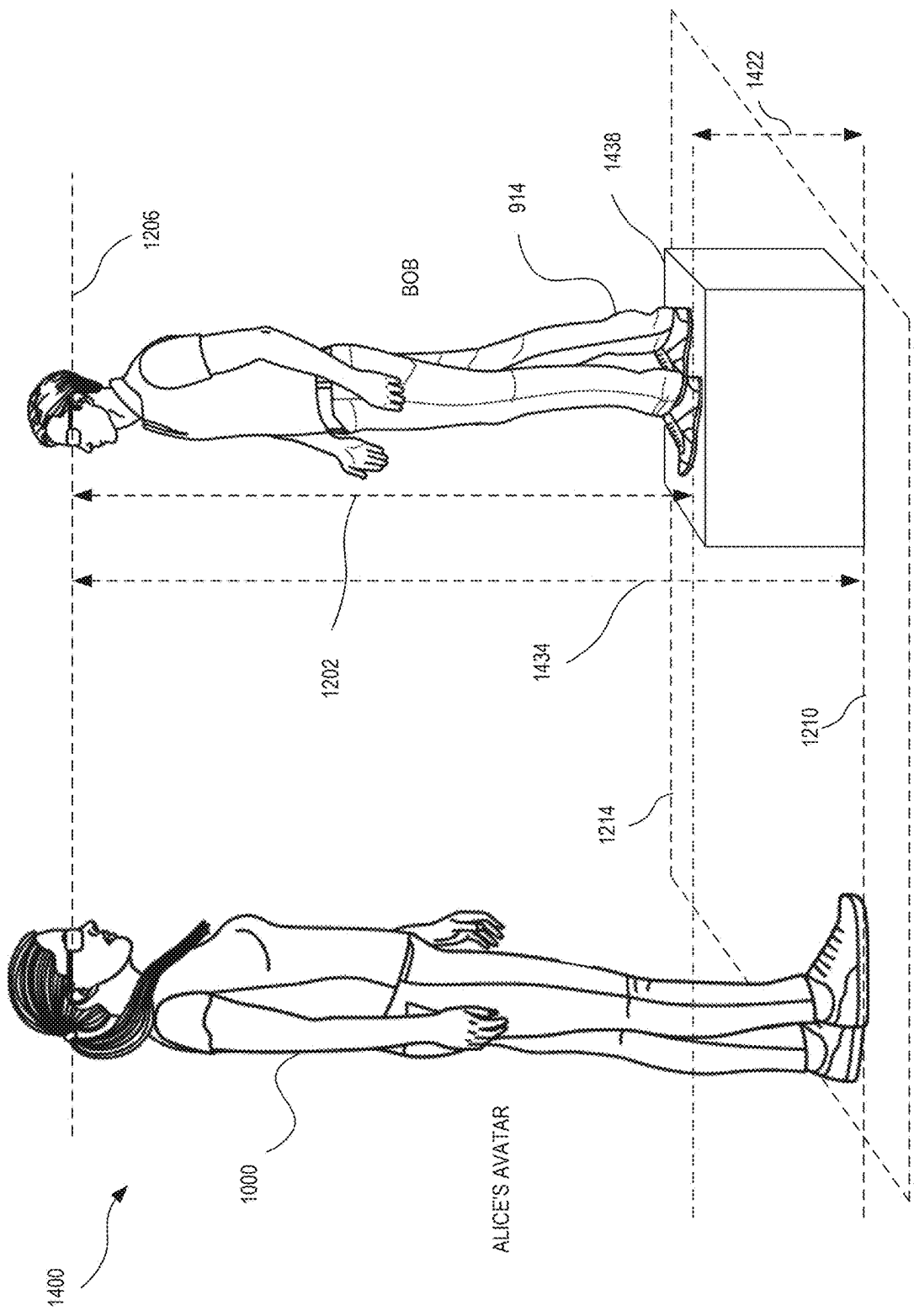


FIG. 14

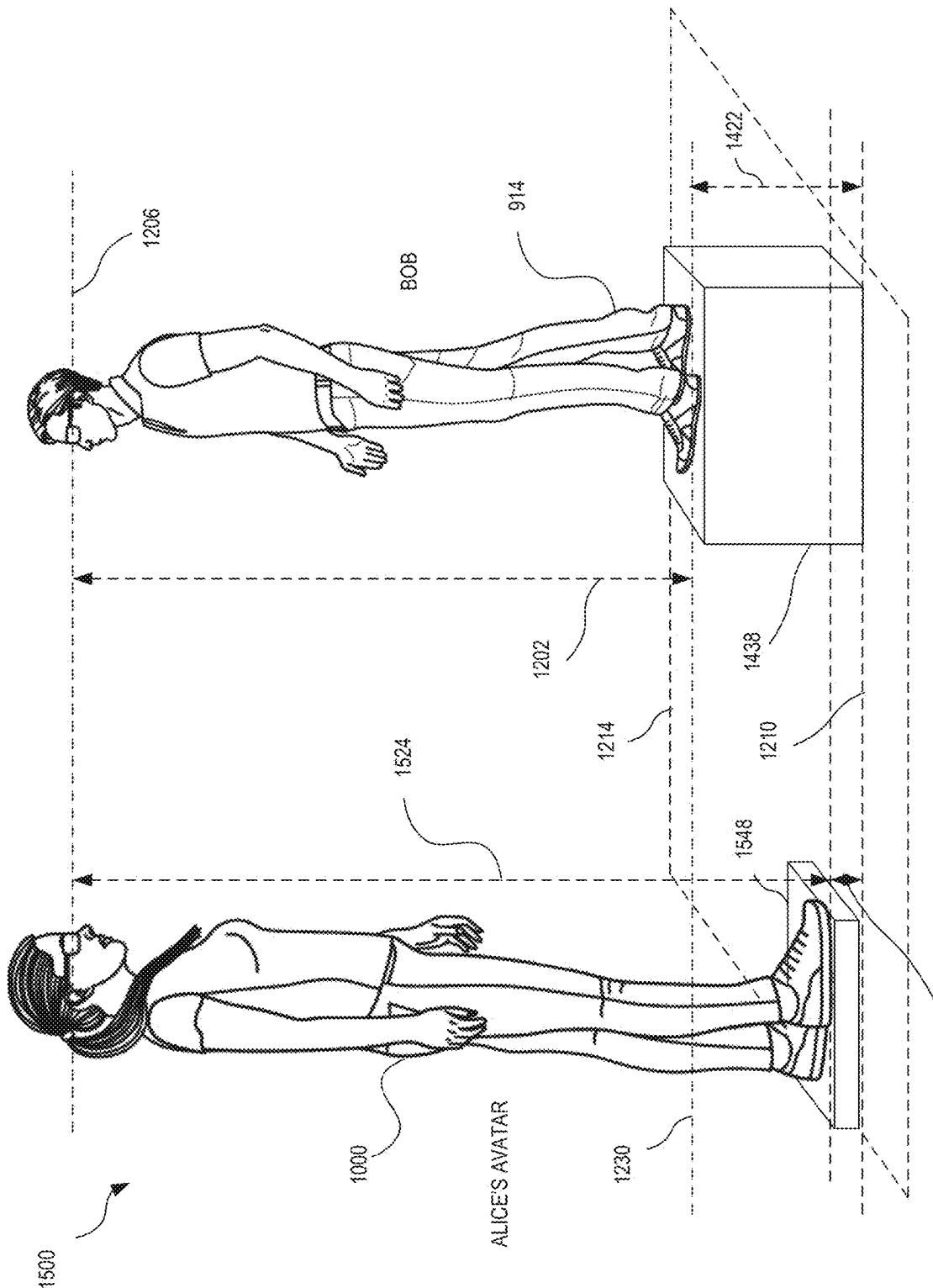


FIG. 15

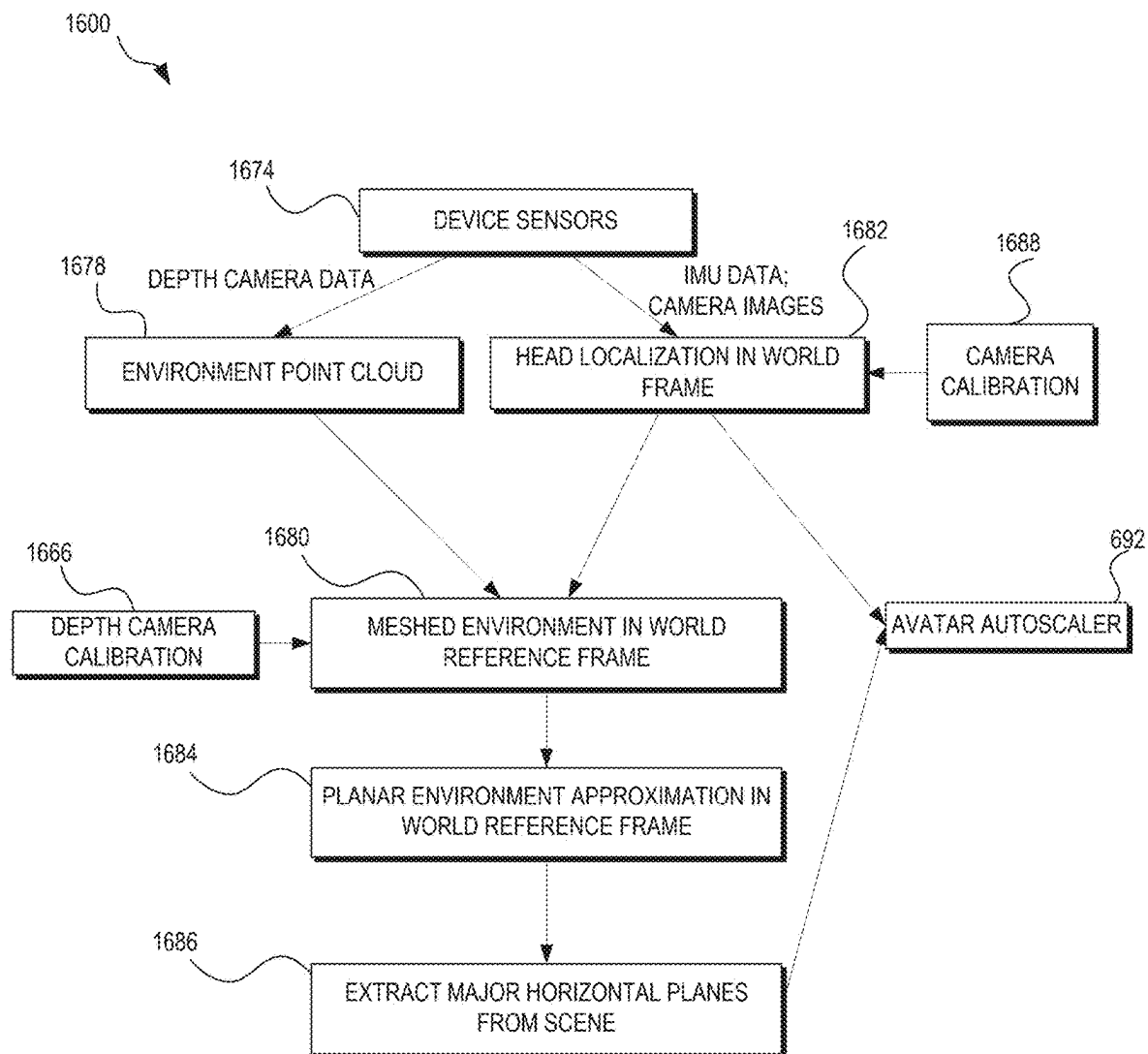


FIG. 16A-1

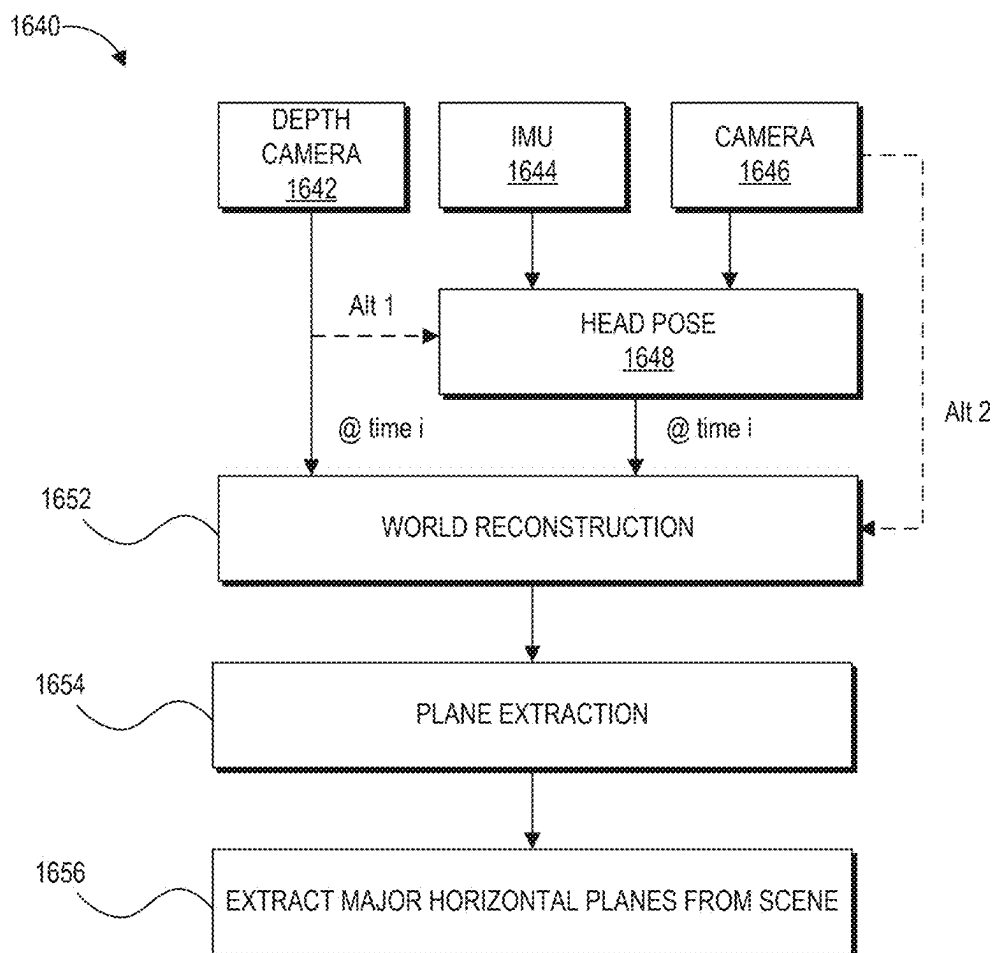


FIG. 16A-2

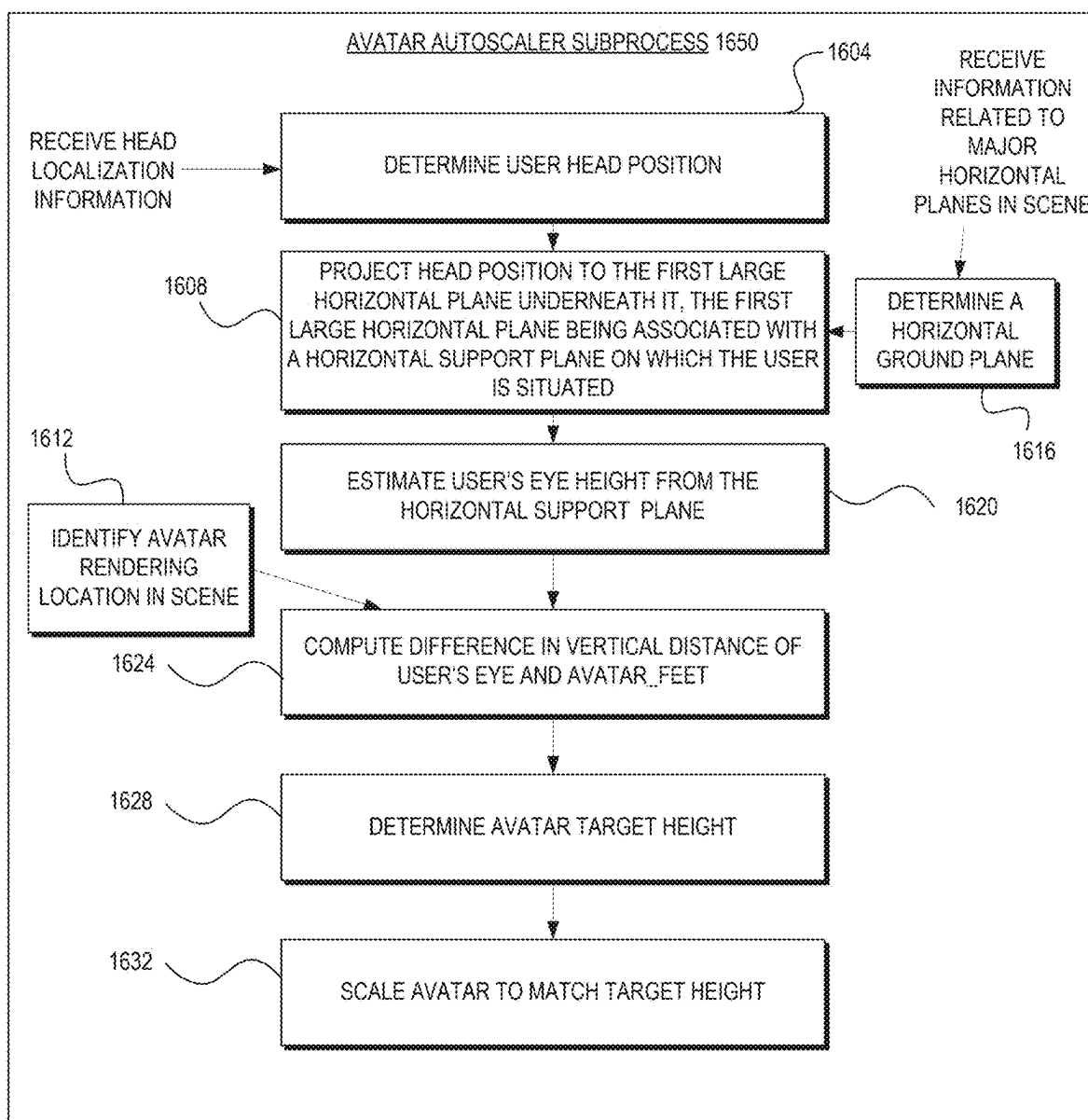


FIG. 16B

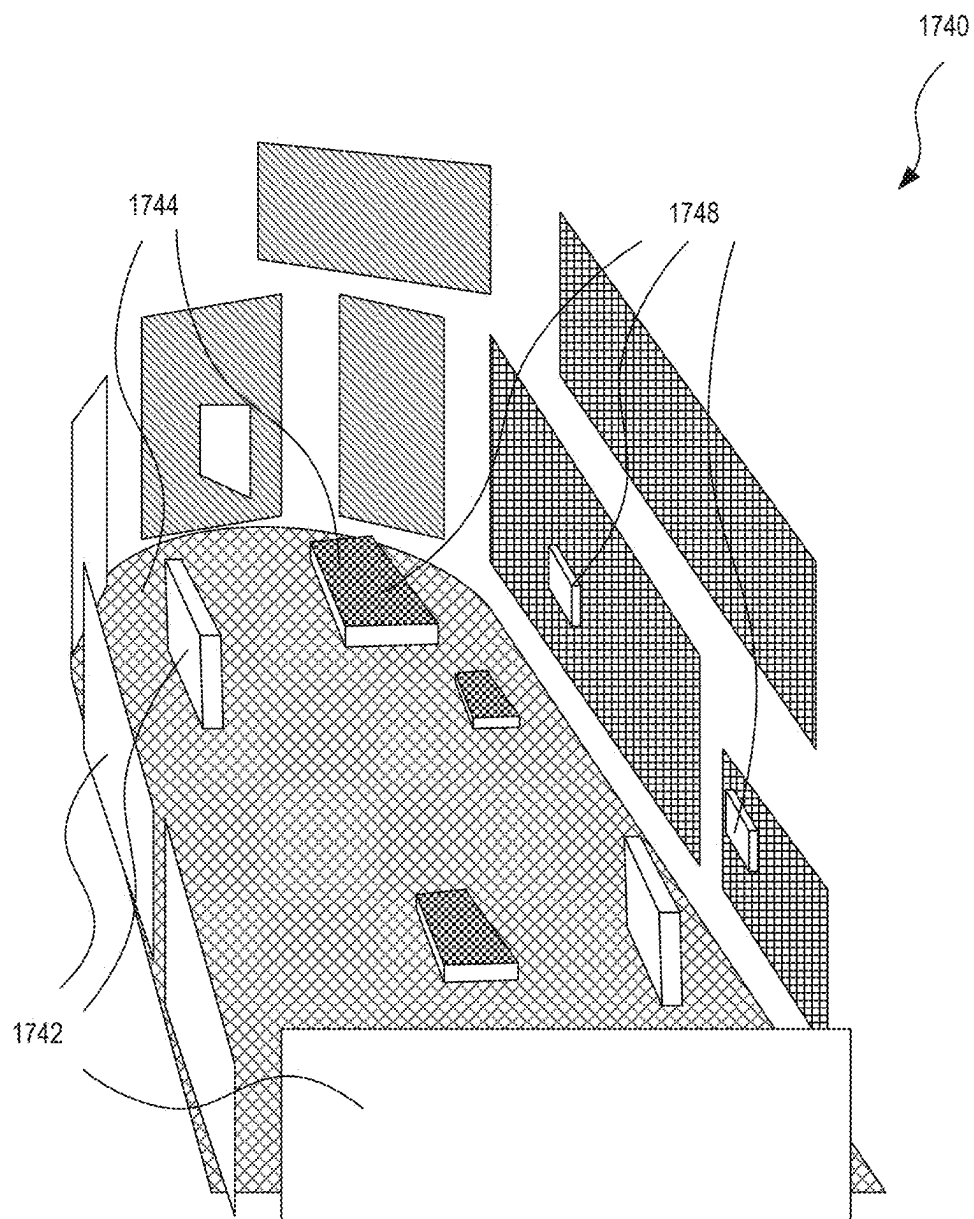


FIG. 17A

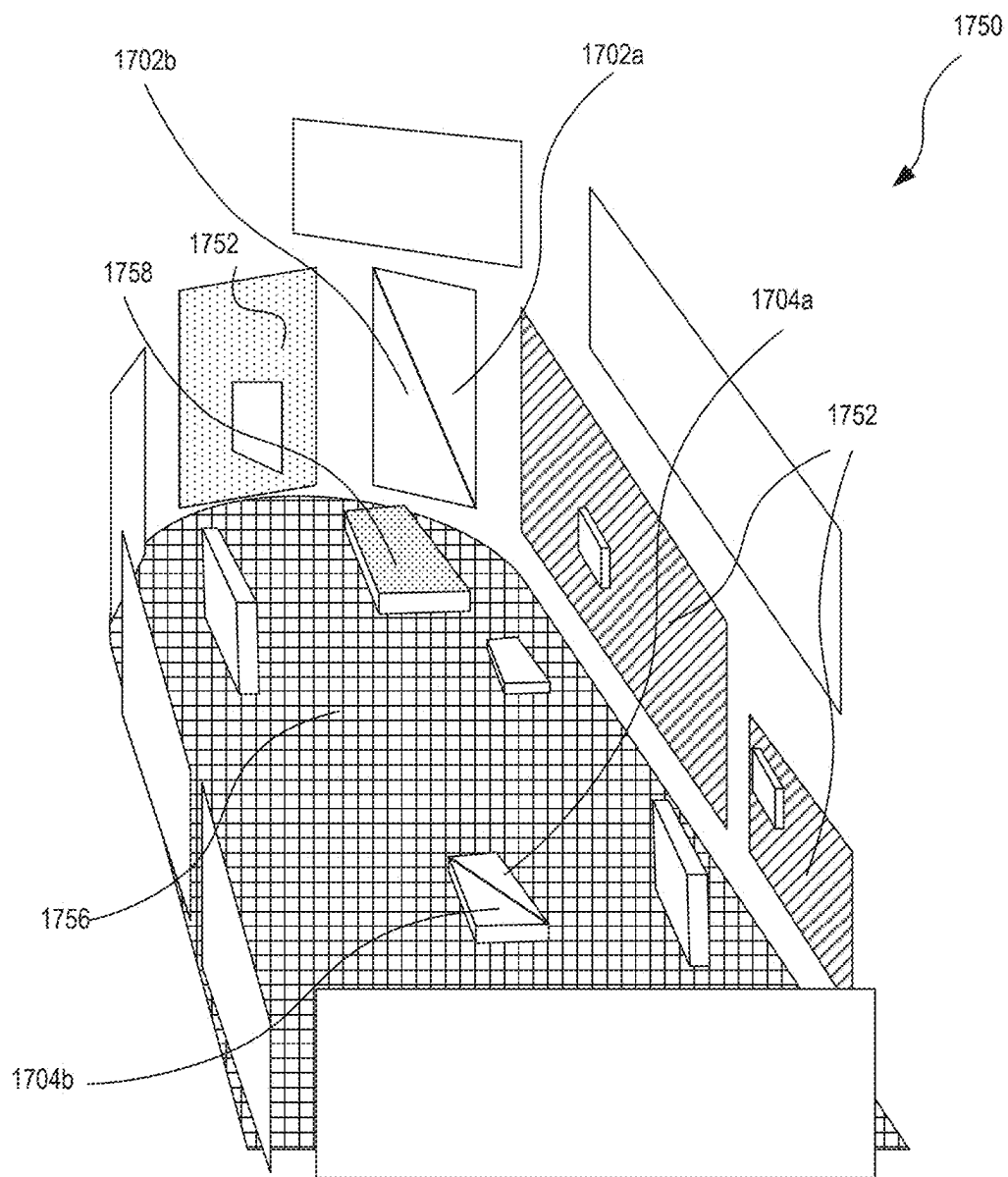


FIG. 17B

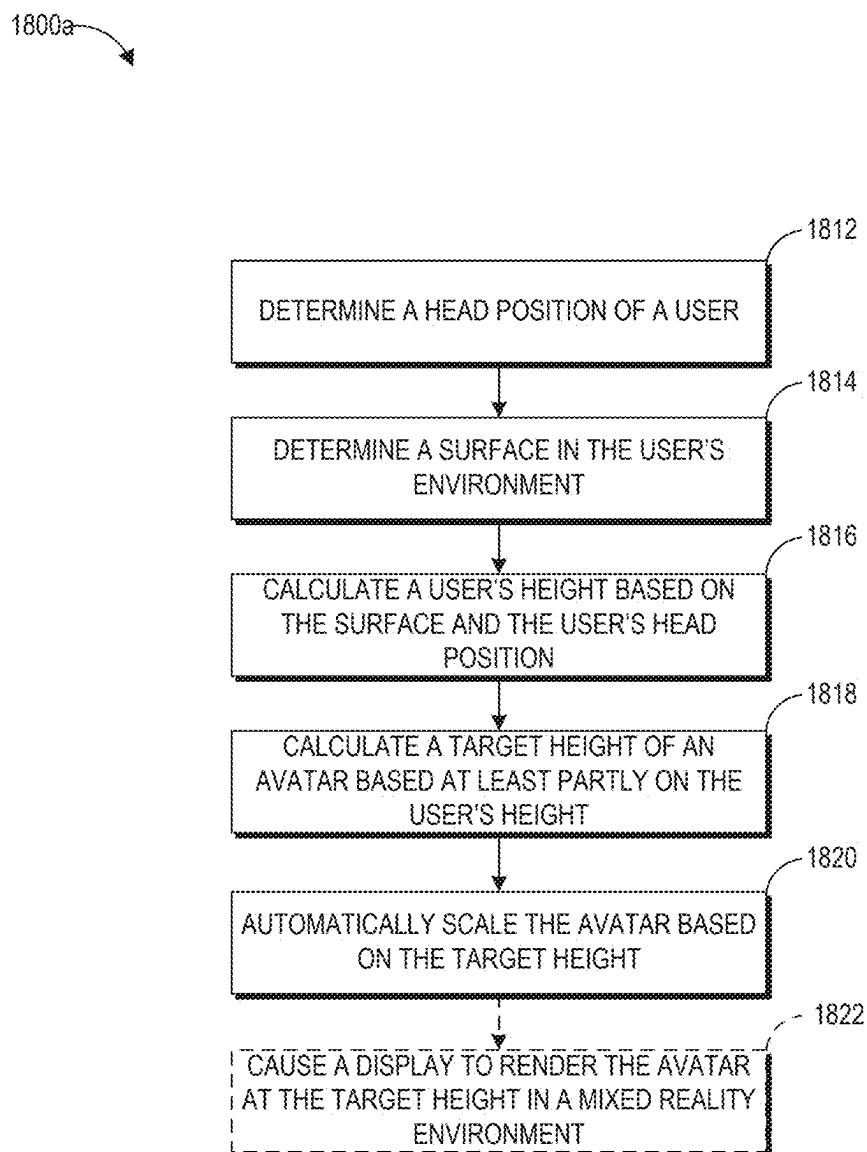


FIG. 18A

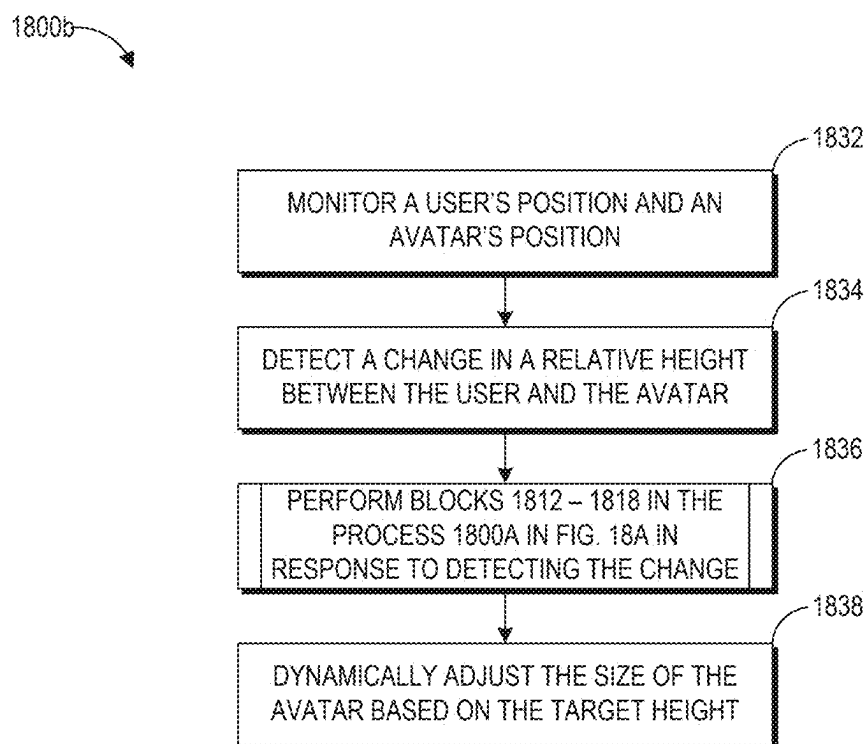


FIG. 18B

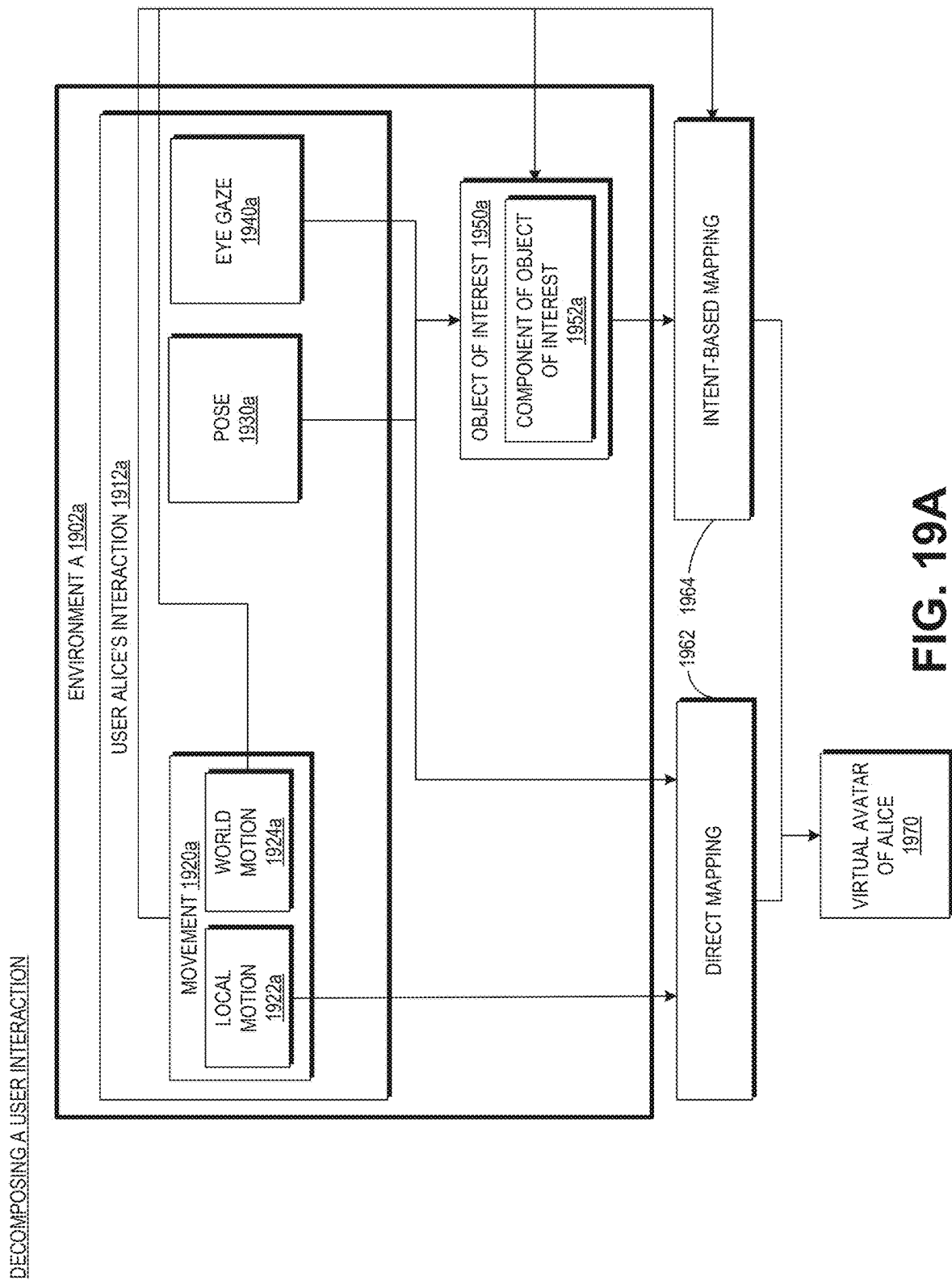


FIG. 19A

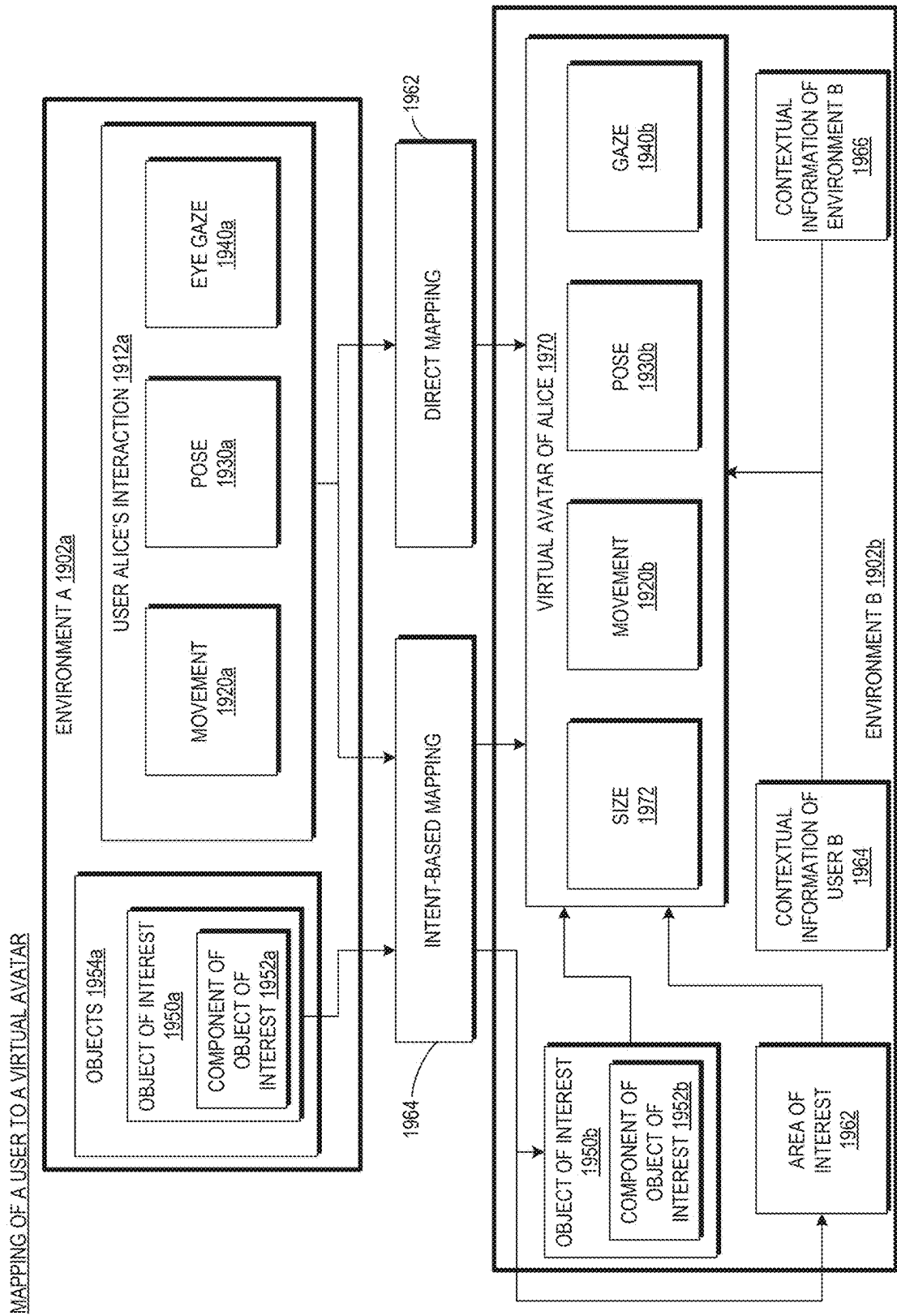


FIG. 19B

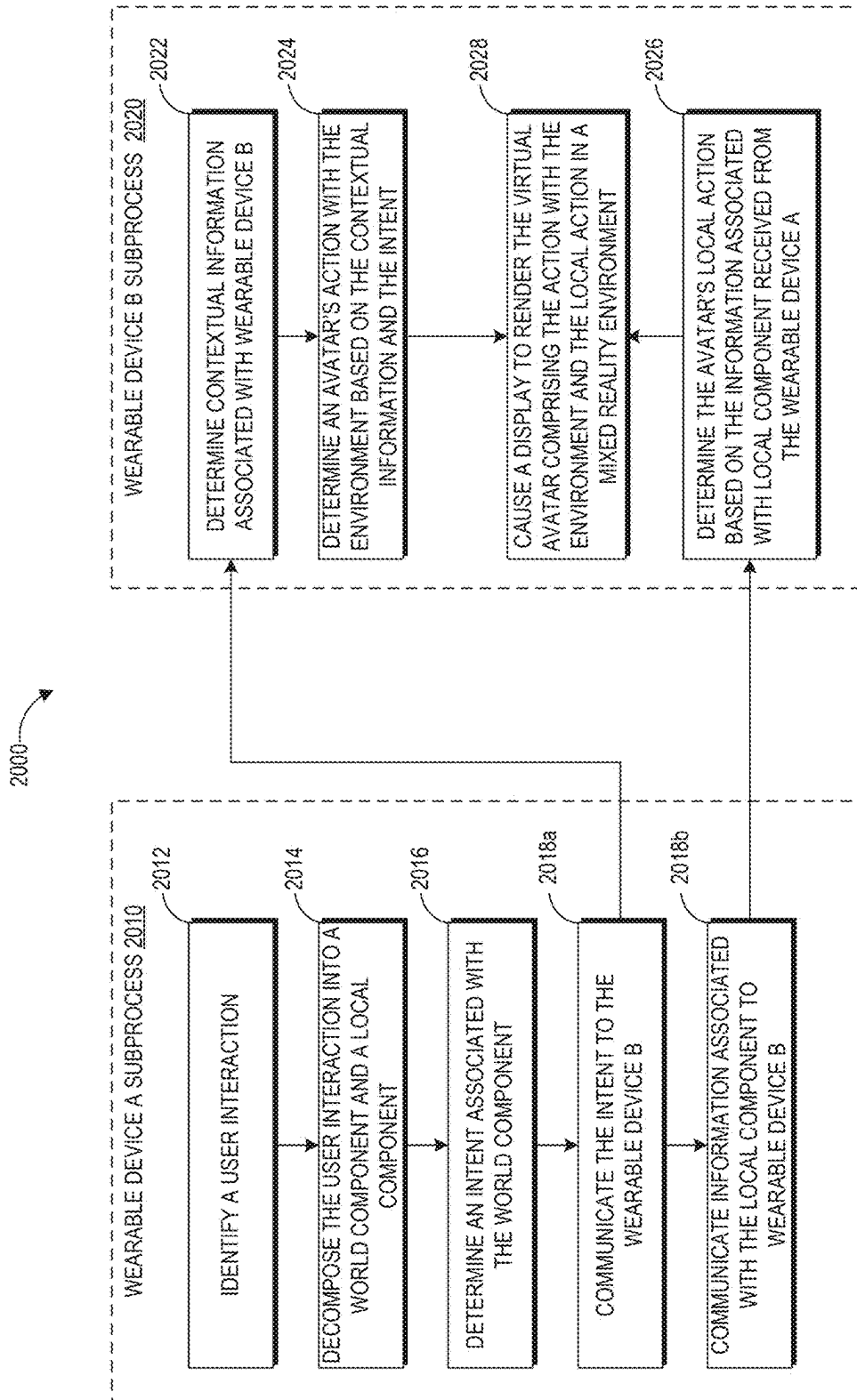


FIG. 20

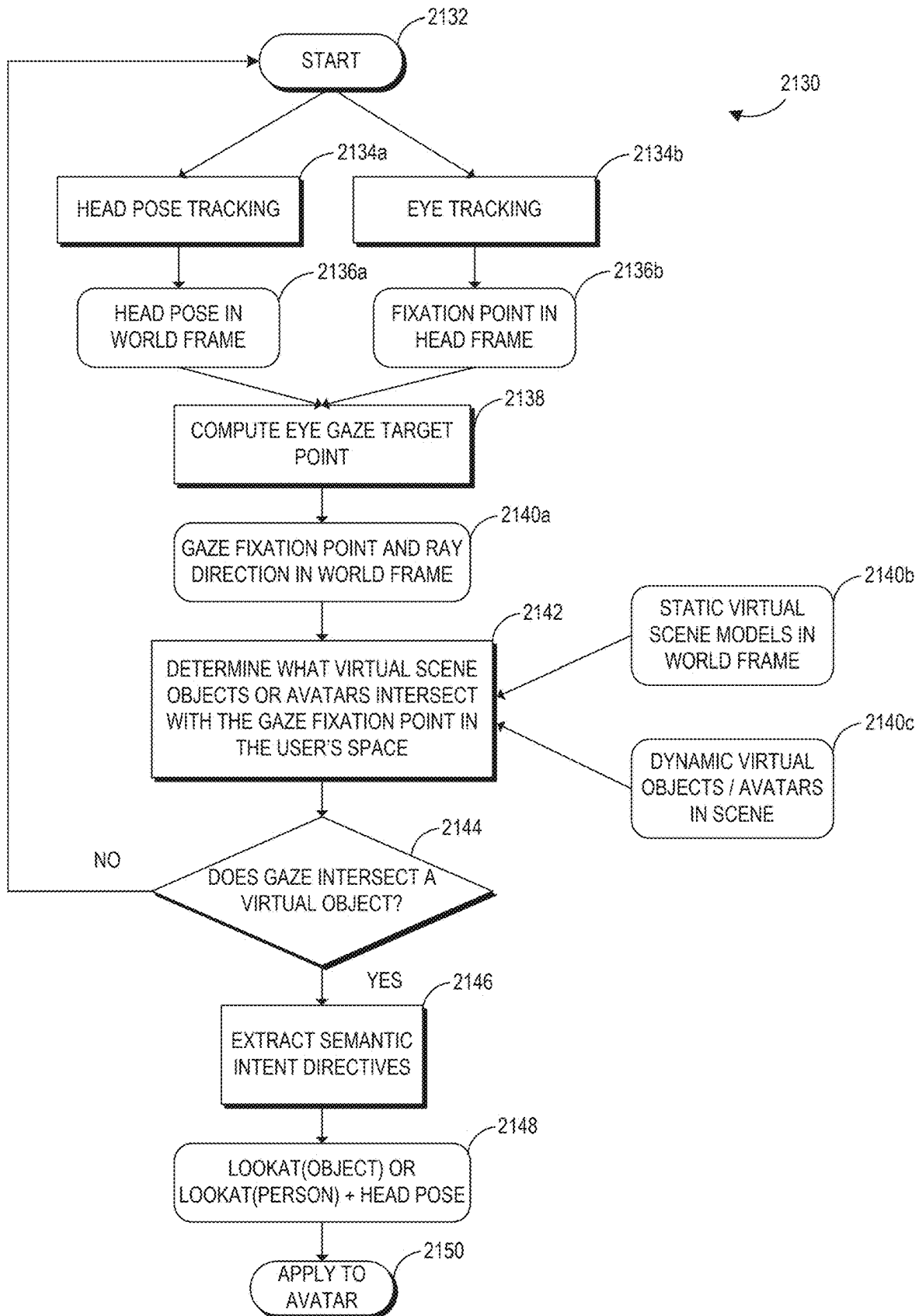


FIG. 21A

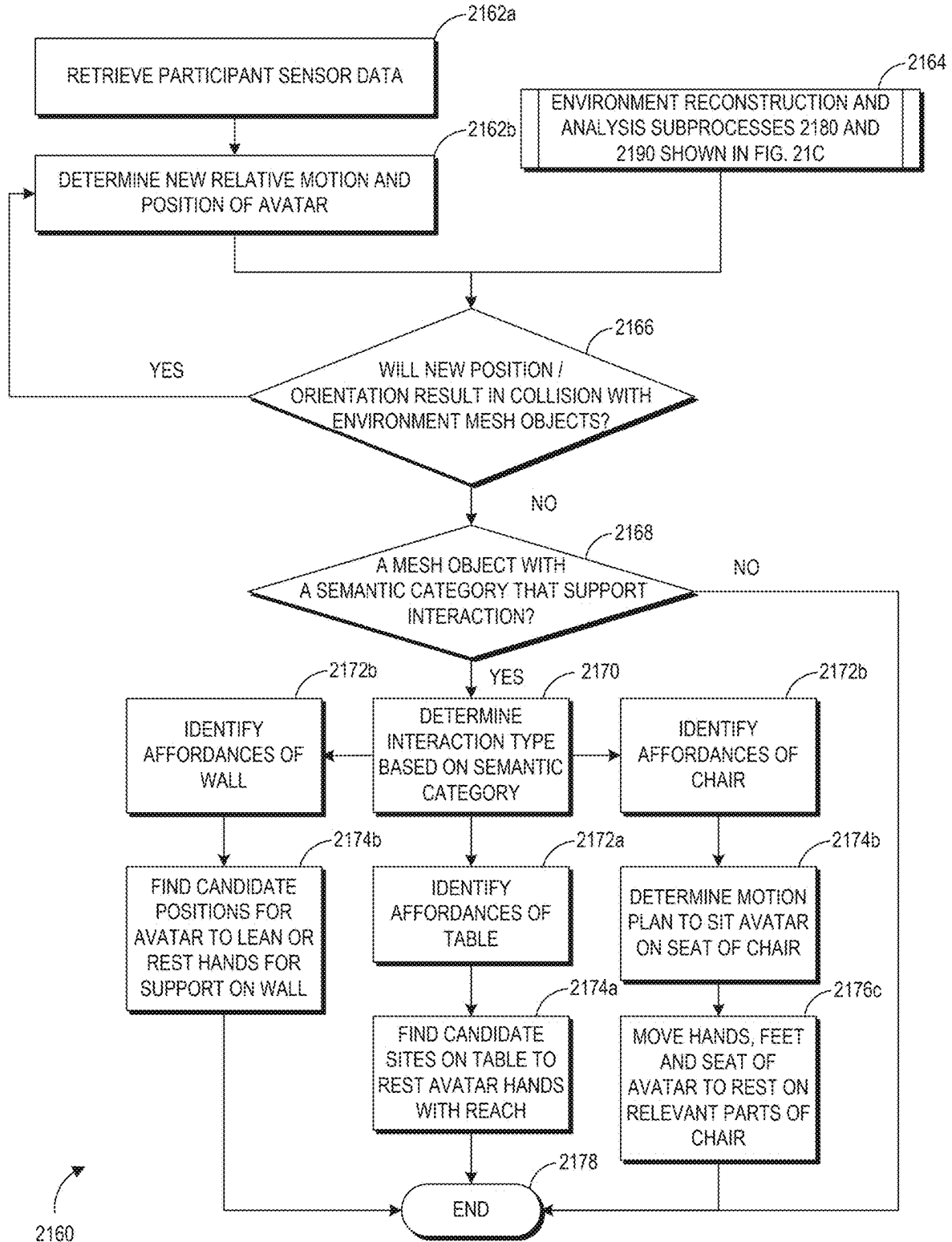


FIG. 21B

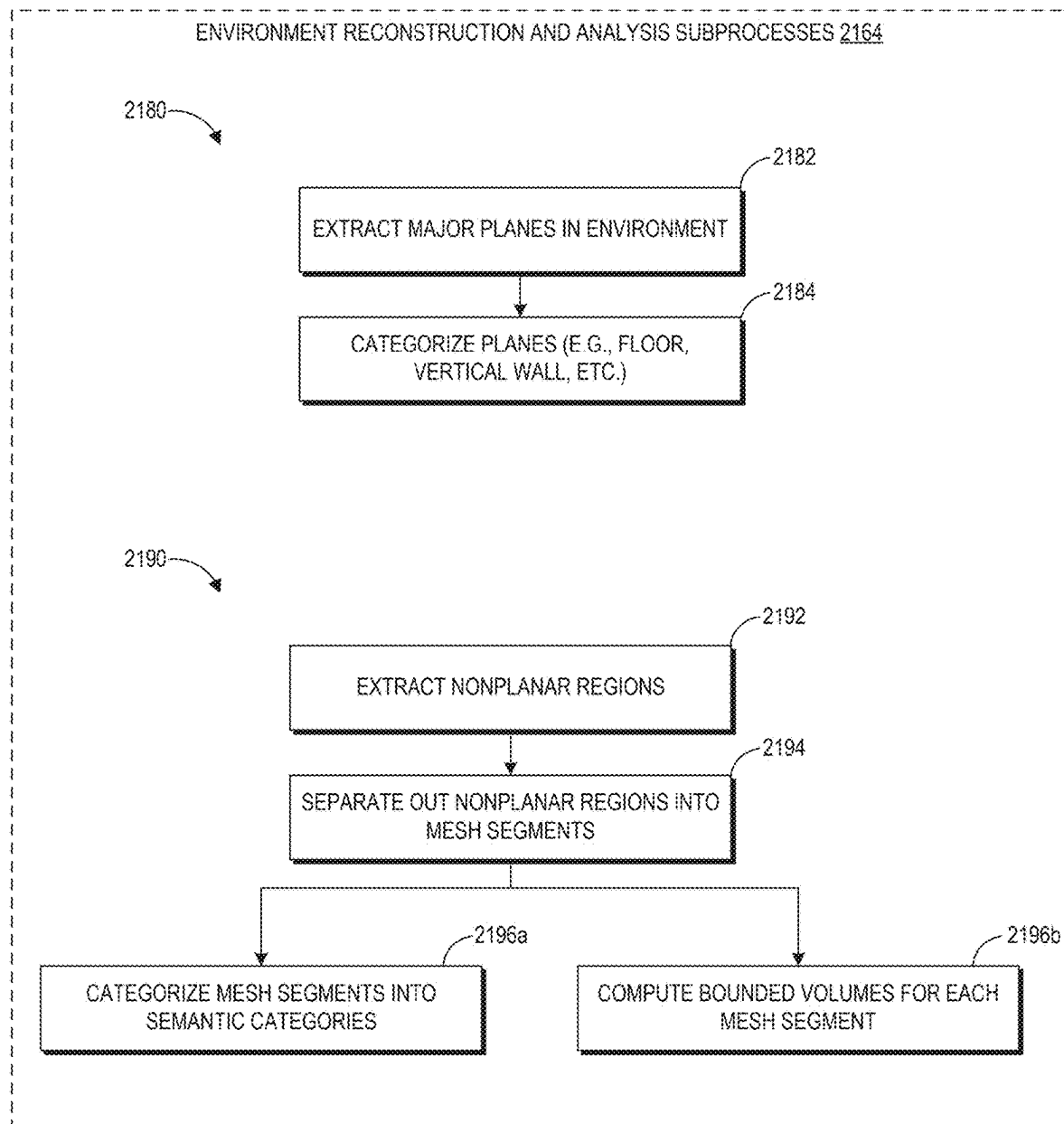


FIG. 21C

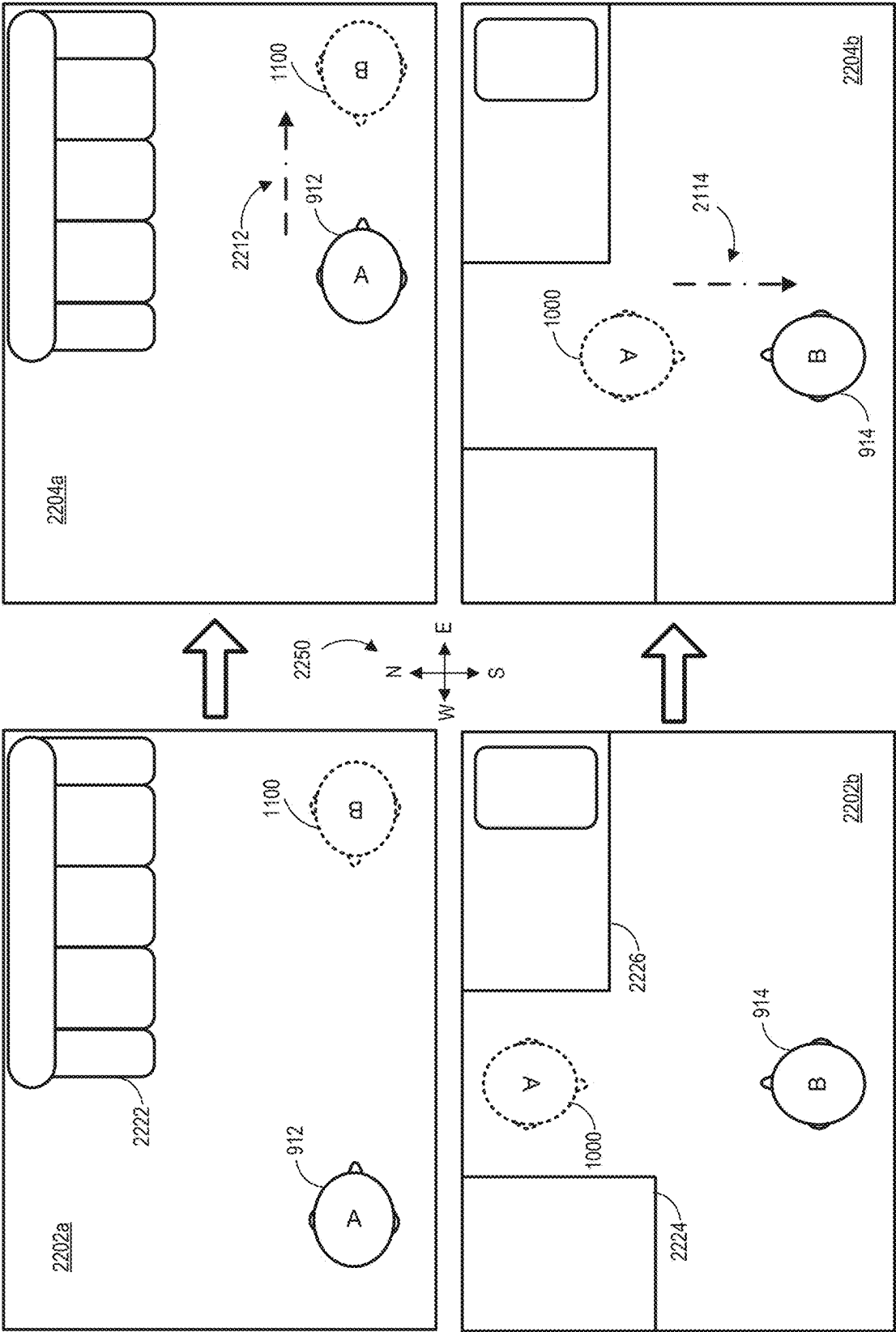


FIG. 22

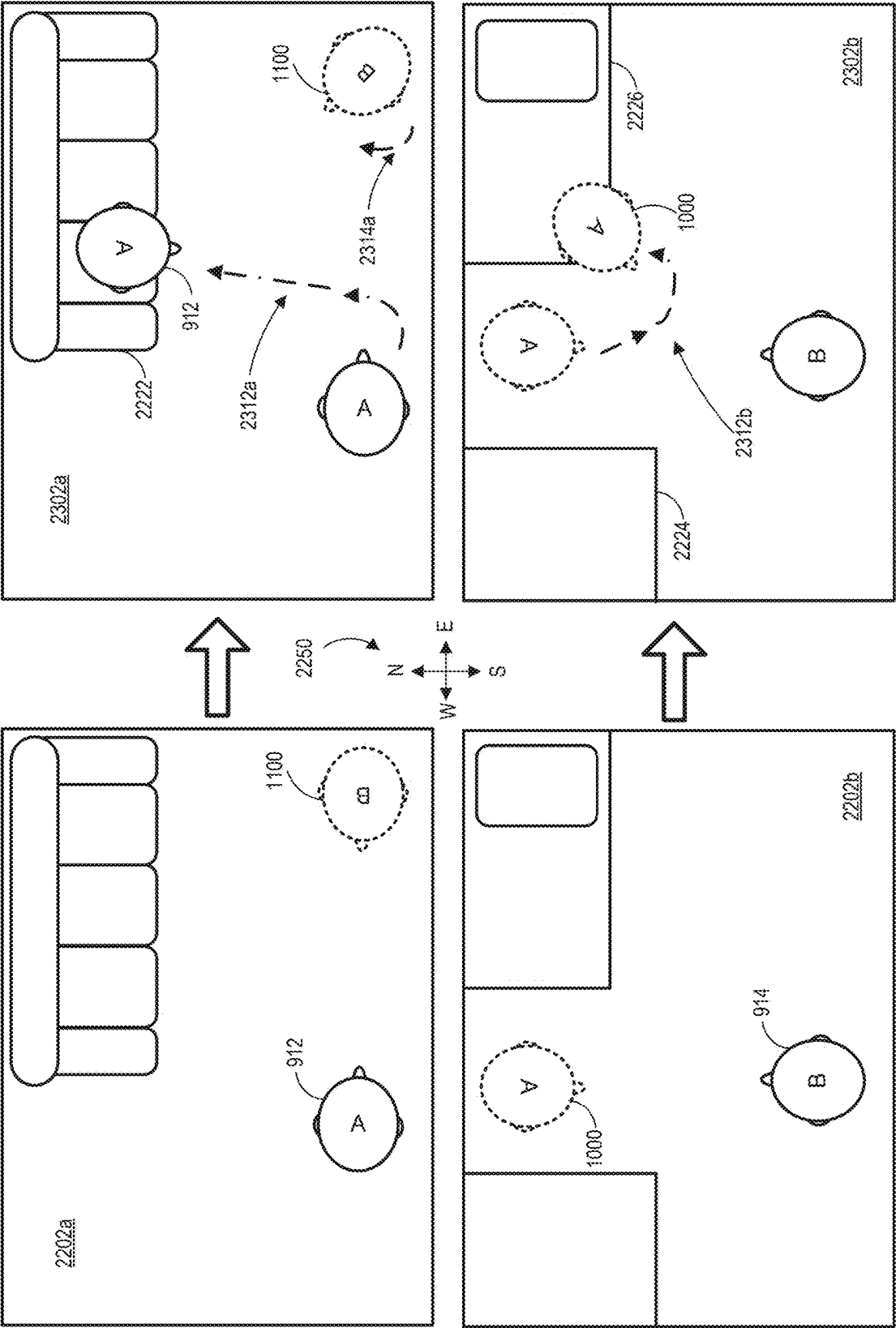


FIG. 23

OVERVIEW OF AVATAR RENDERING BASED ON AN OBJECT OF INTEREST

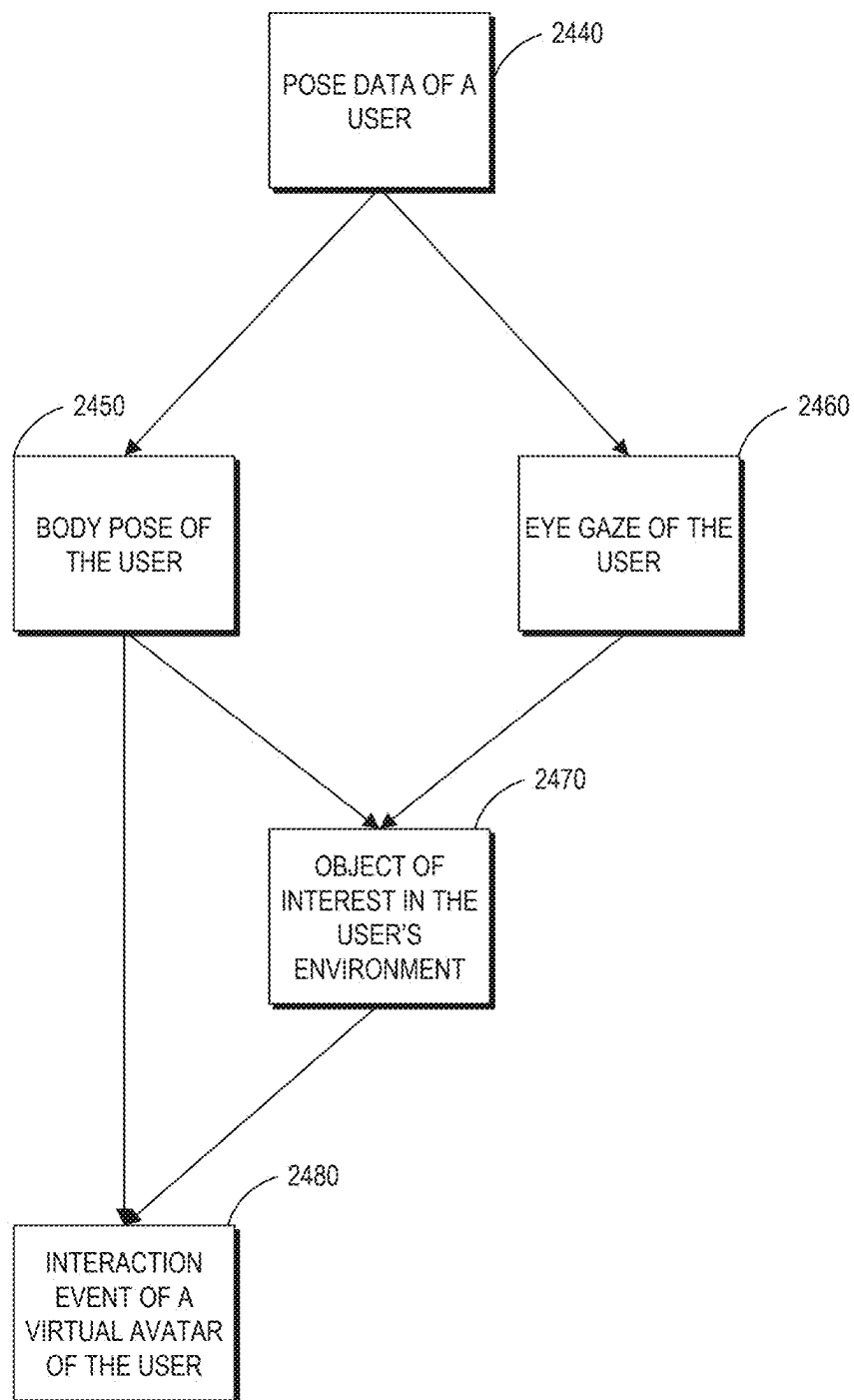


FIG. 24

IDENTIFY AN OBJECT OF INTEREST IN A USER'S ENVIRONMENT

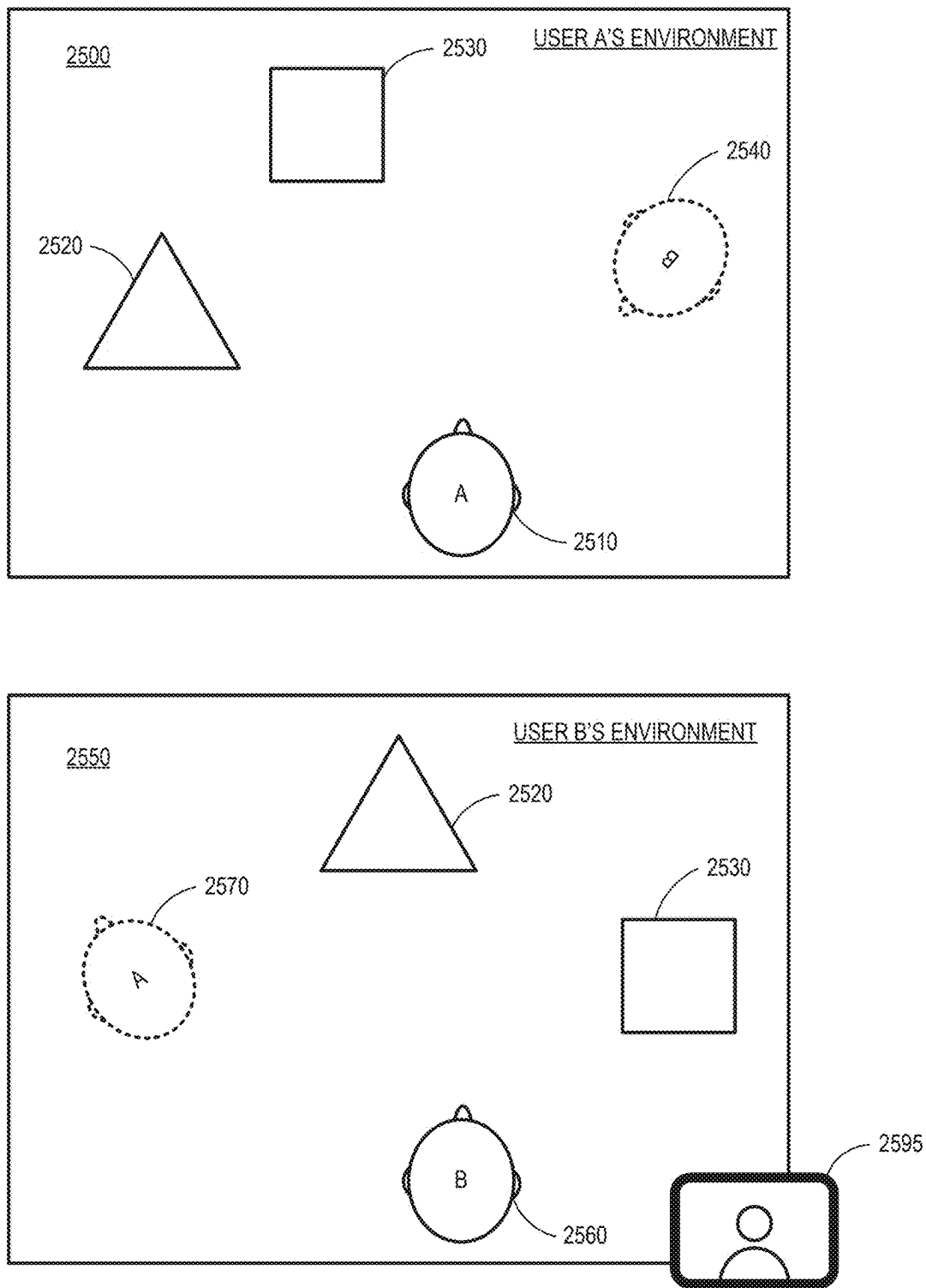


FIG. 25A

IDENTIFY AN OBJECT OF INTEREST BASED ON HEAD POSE AND EYE GAZE

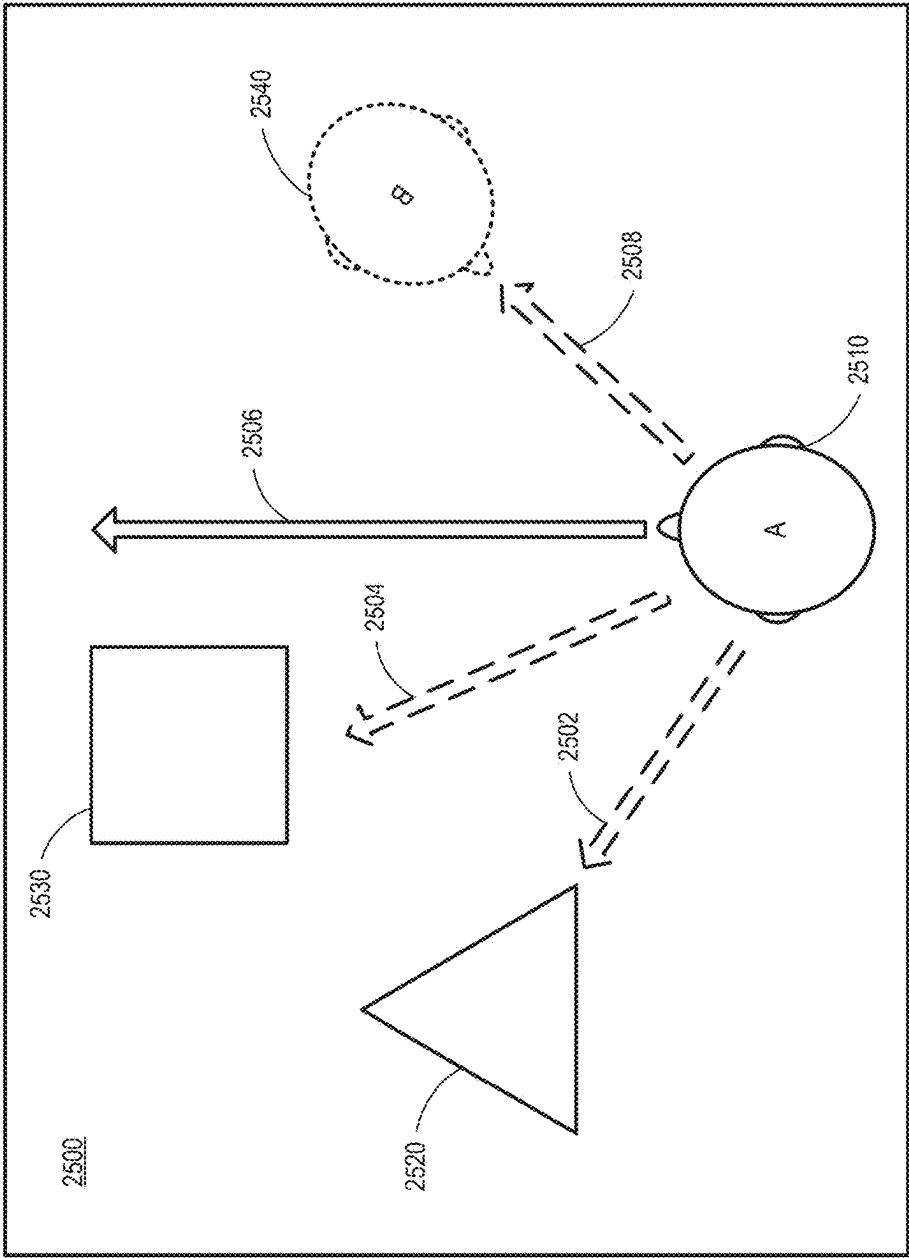


FIG. 25B

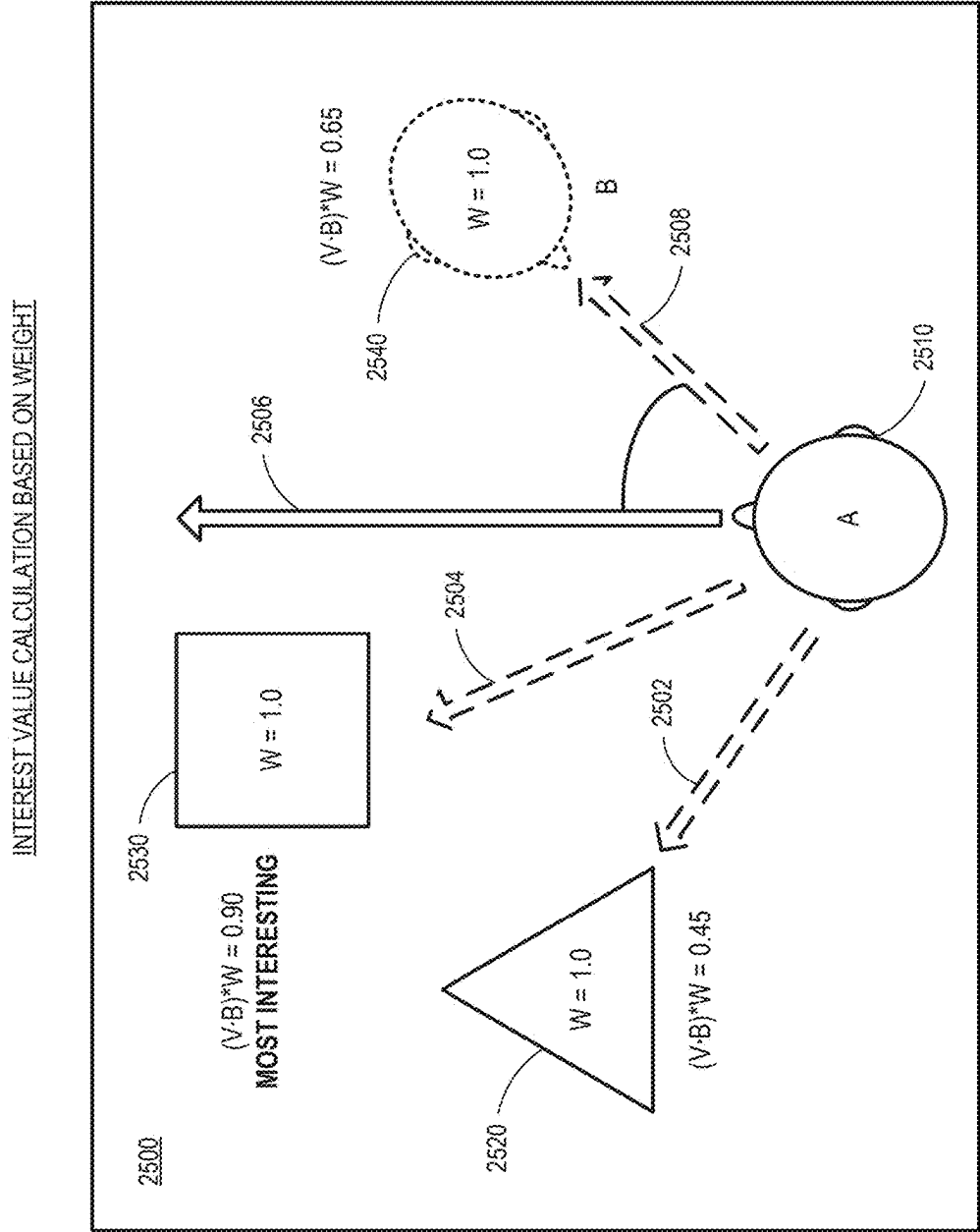


FIG. 25C

INTEREST VALUE CALCULATION BASED ON WEIGHT - CONT'D

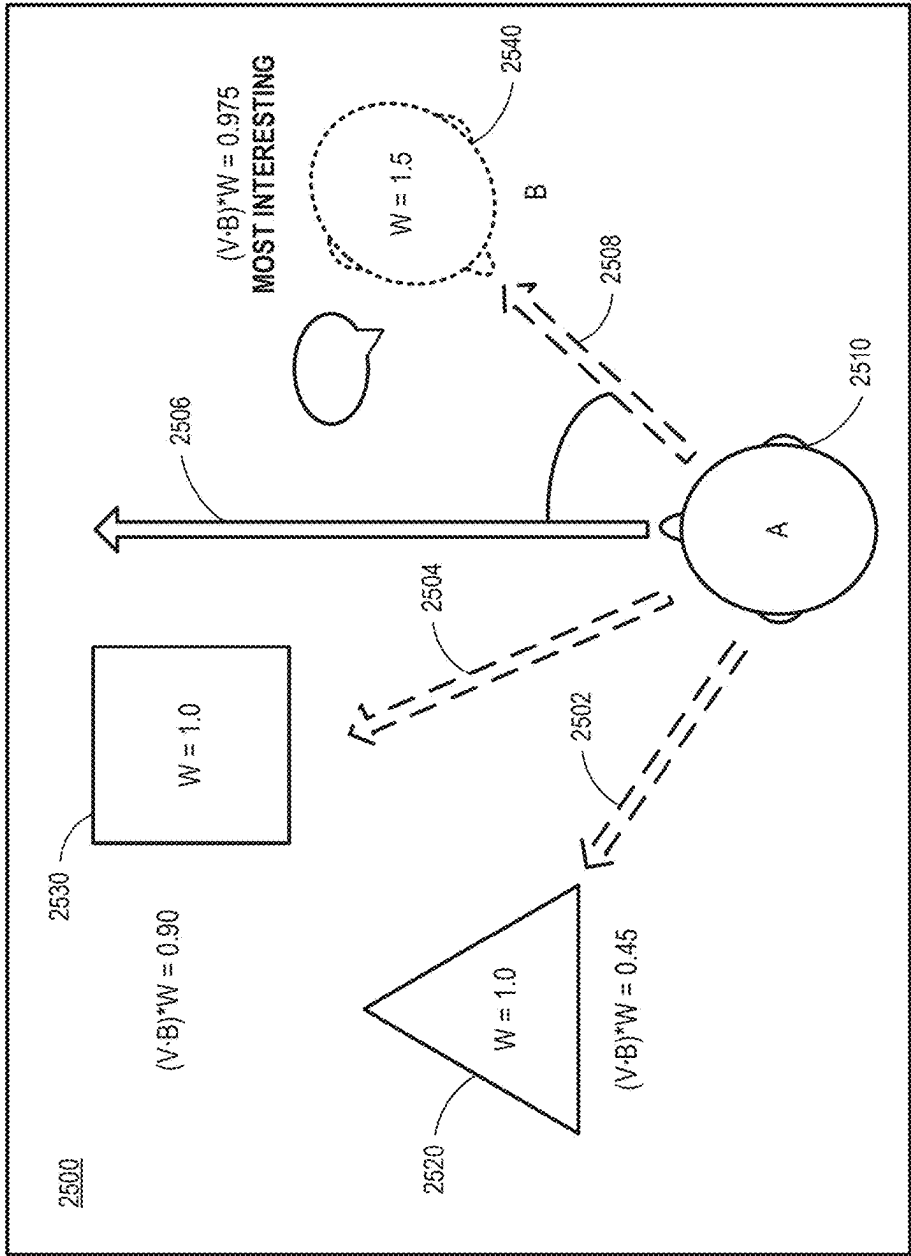


FIG. 25D

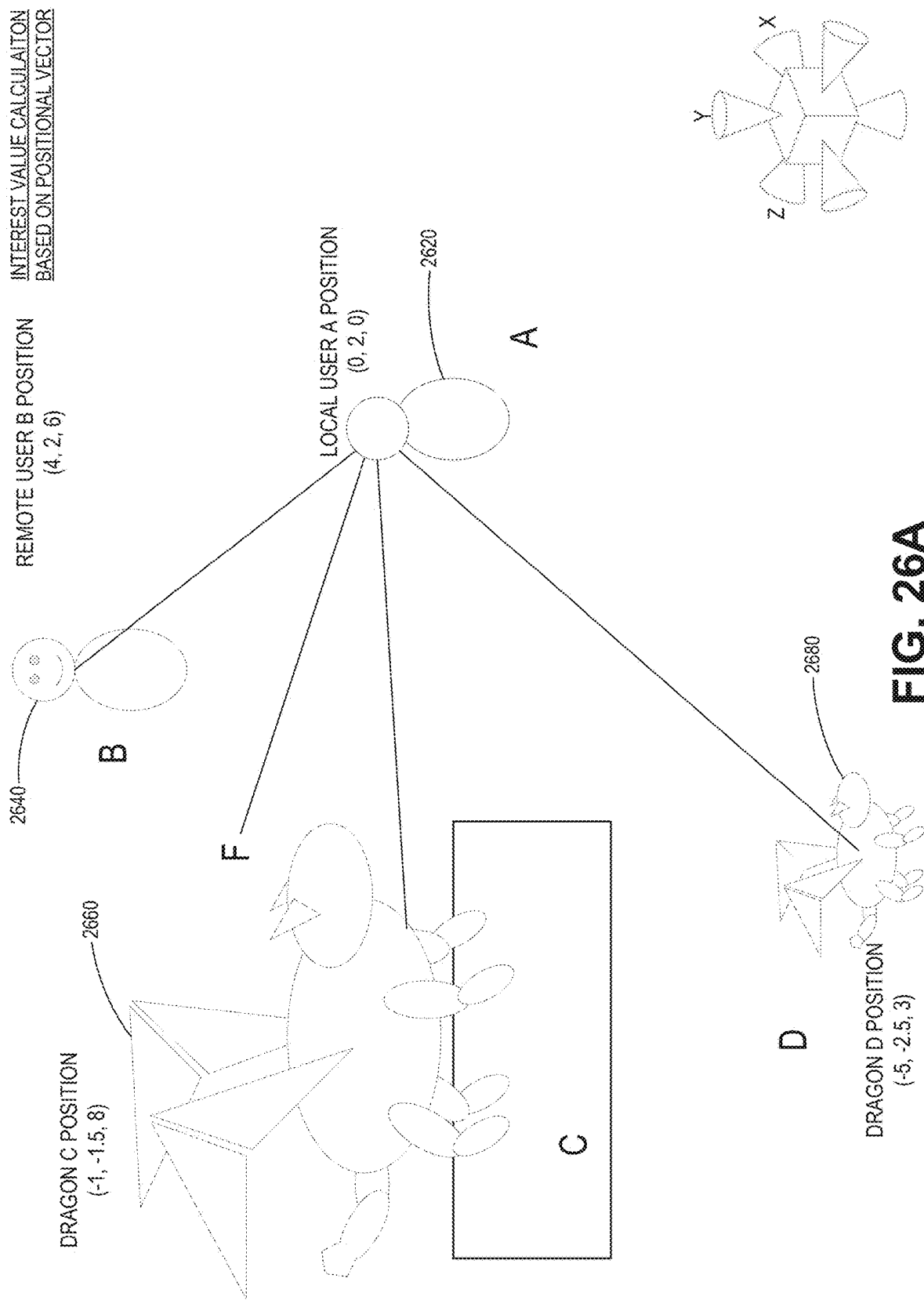
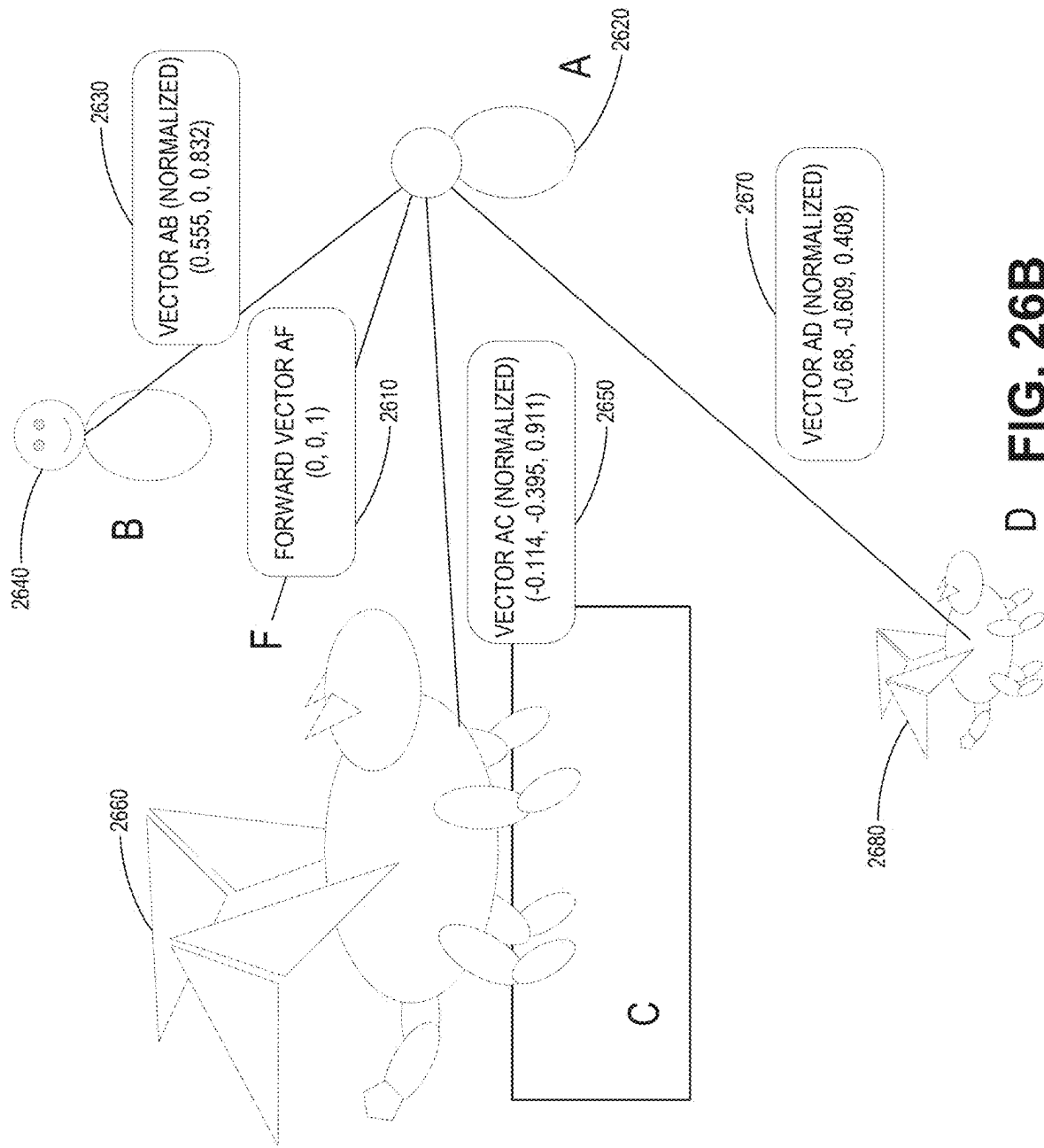
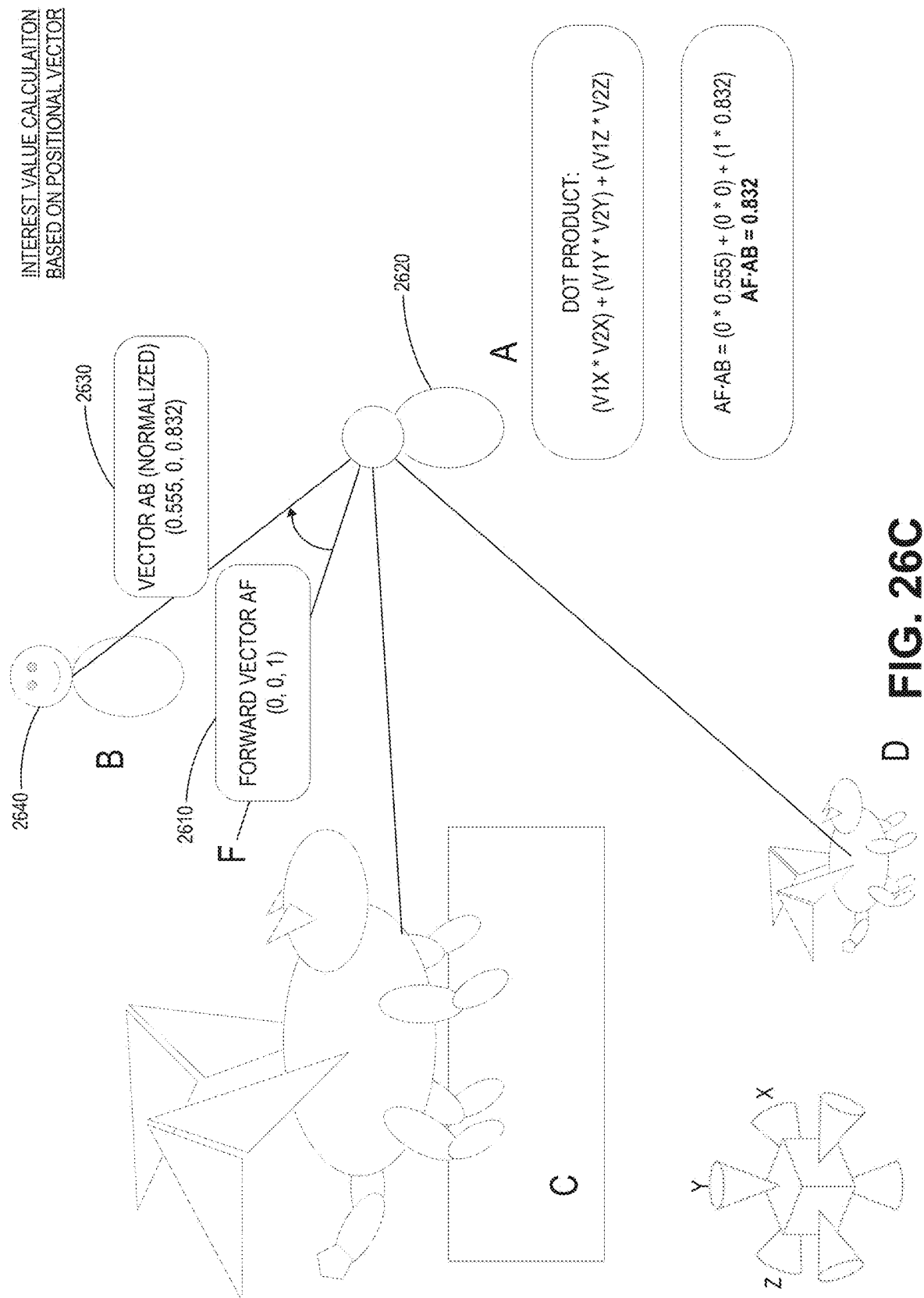


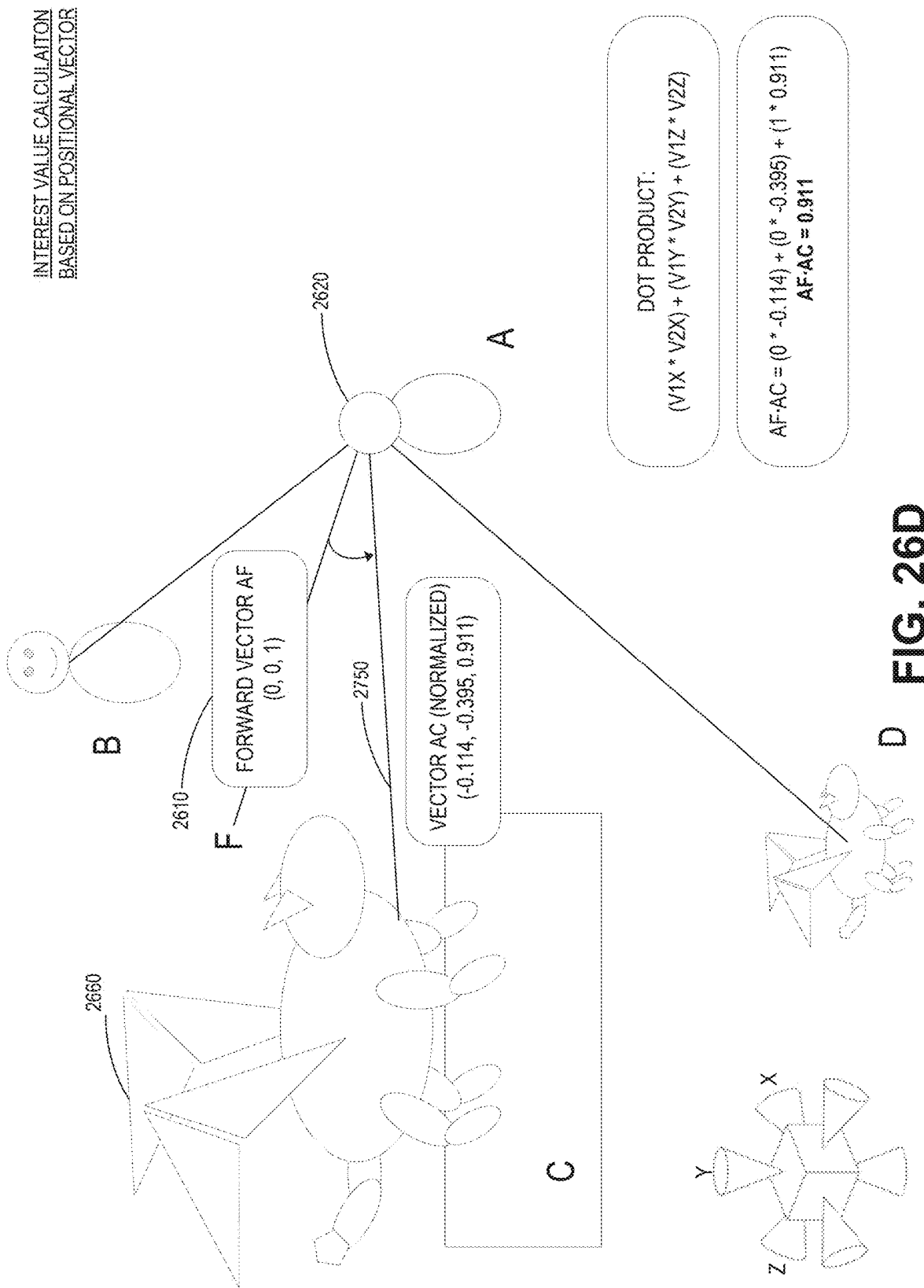
FIG. 26A

INTEREST VALUE CALCULATION
BASED ON POSITIONAL VECTOR



D **FIG. 26B**





INTEREST VALUE CALCULATION
BASED ON POSITIONAL VECTOR

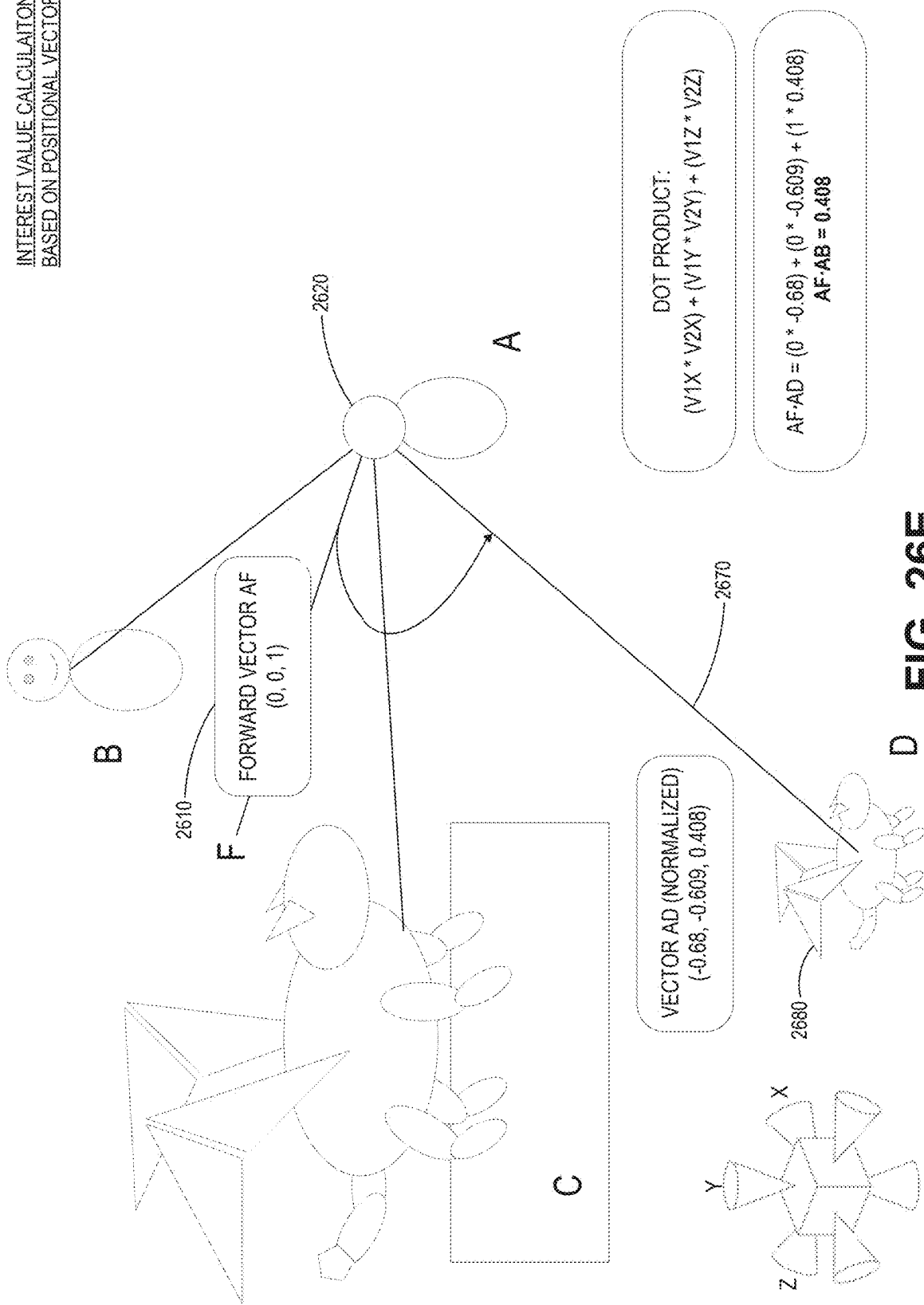
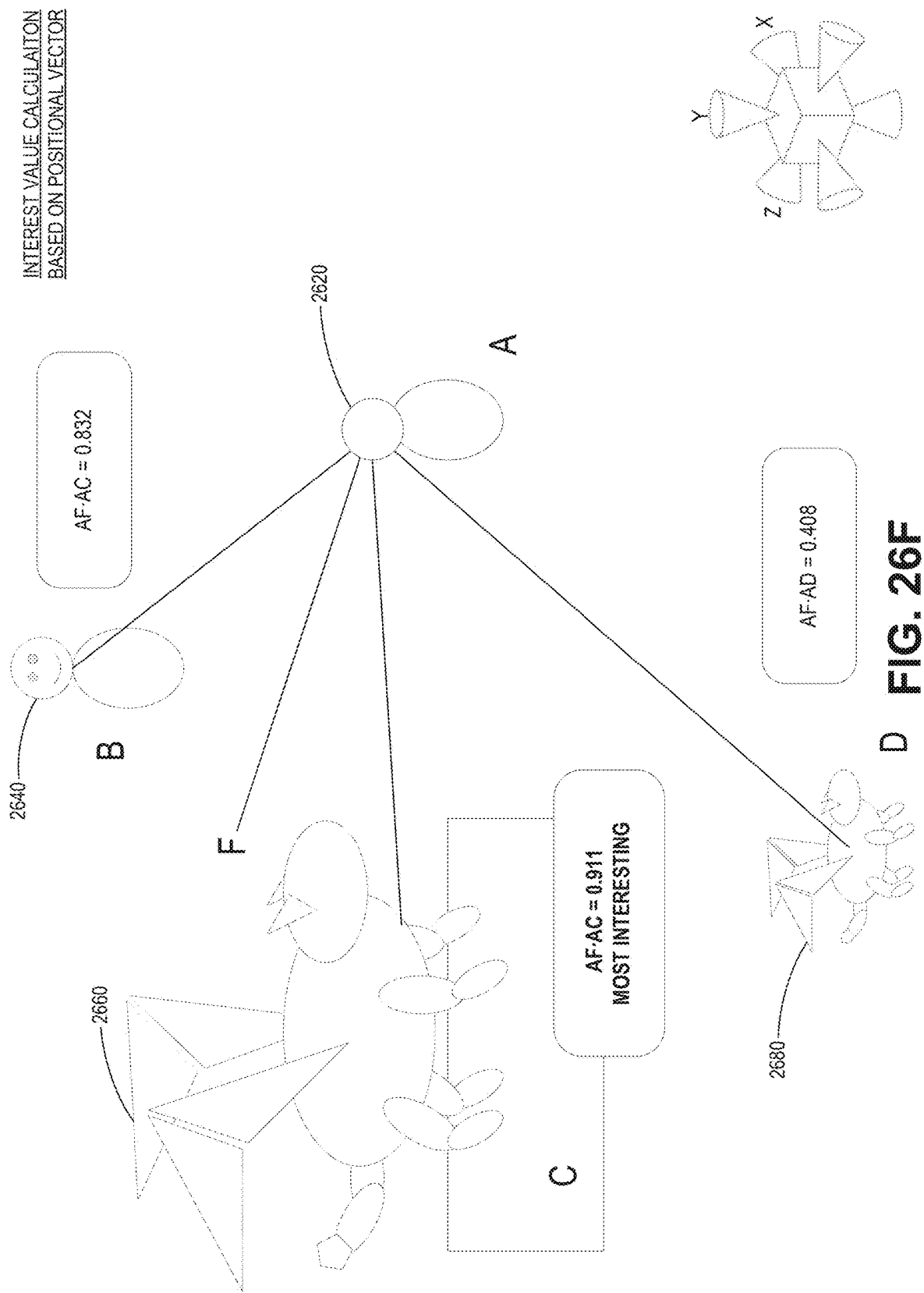


FIG. 26E



COMMUNICATE SELECTION OF AN OBJECT OF
INTEREST TO ANOTHER WEARABLE DEVICE

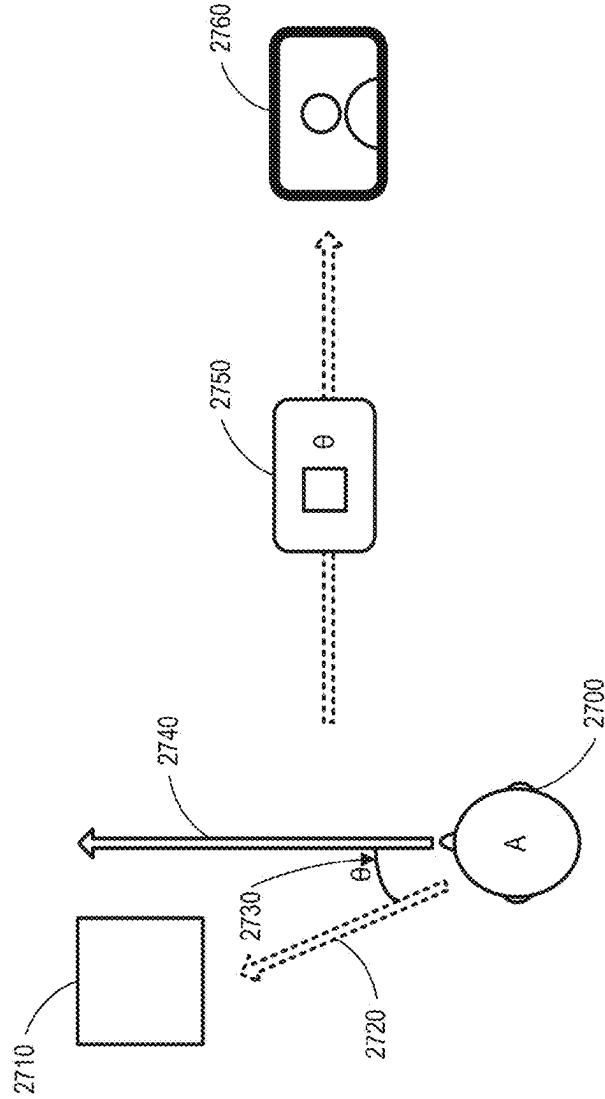


FIG. 27

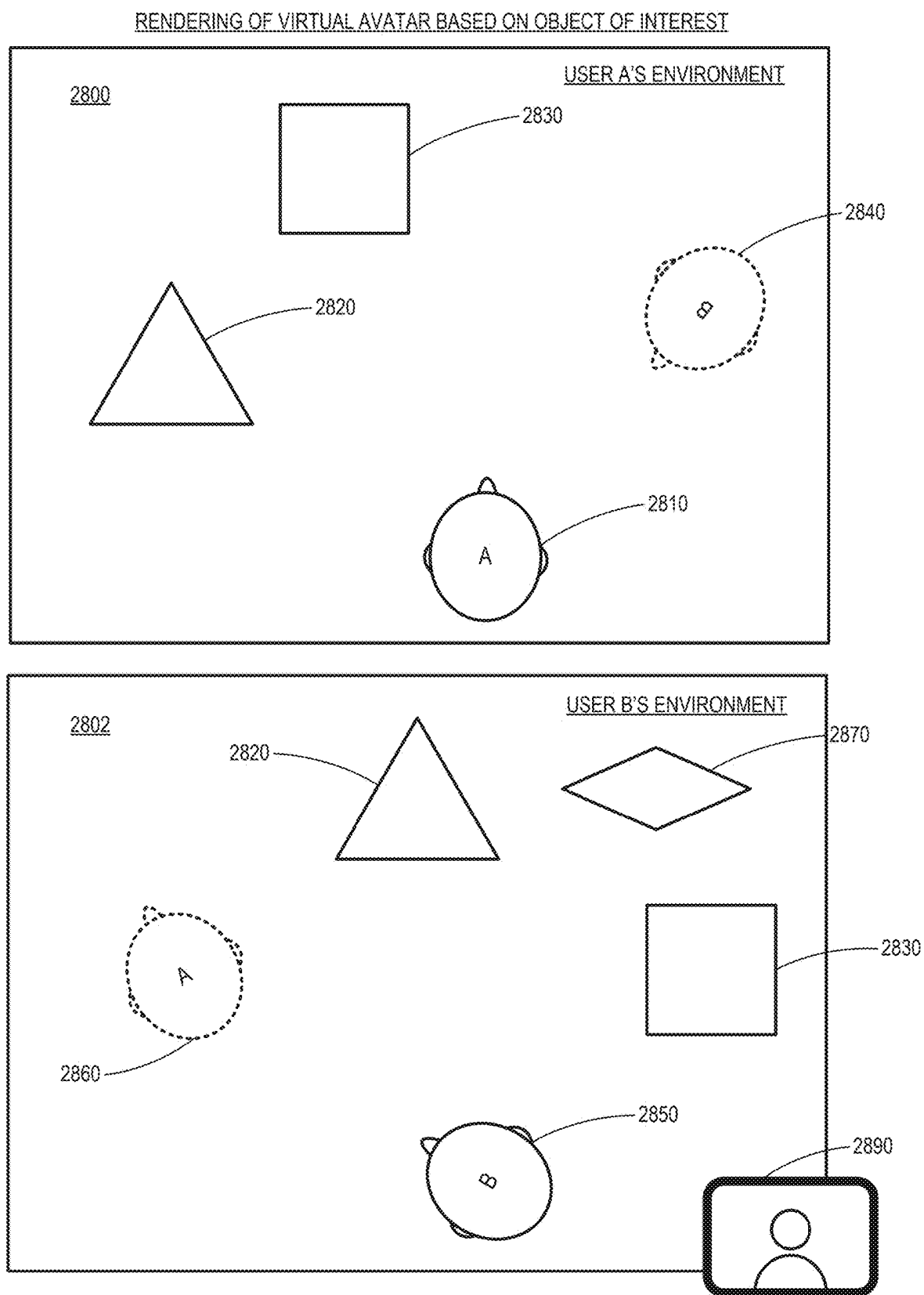


FIG. 28A

RENDERING OF VIRTUAL AVATAR BASED ON OBJECT OF INTEREST

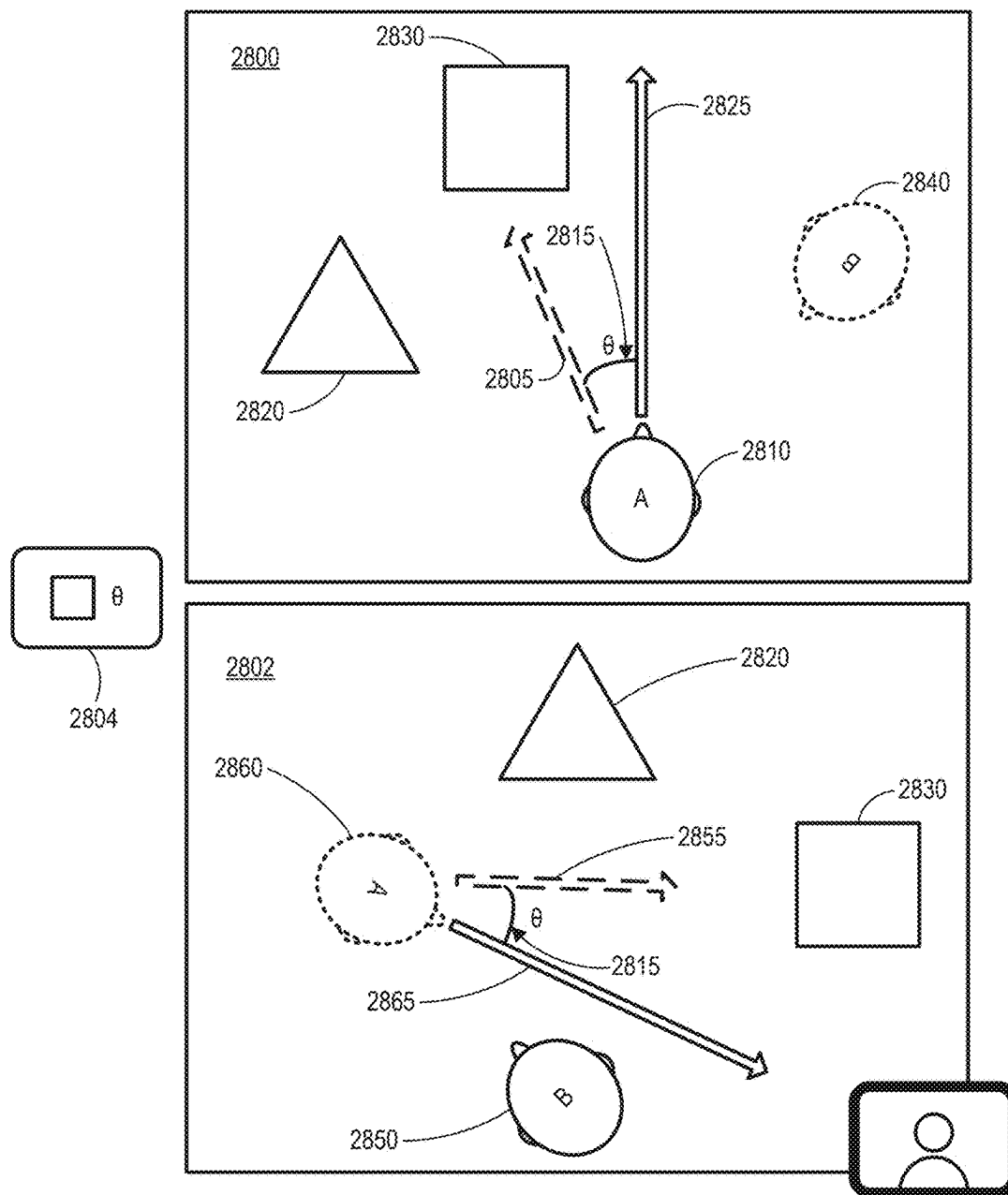
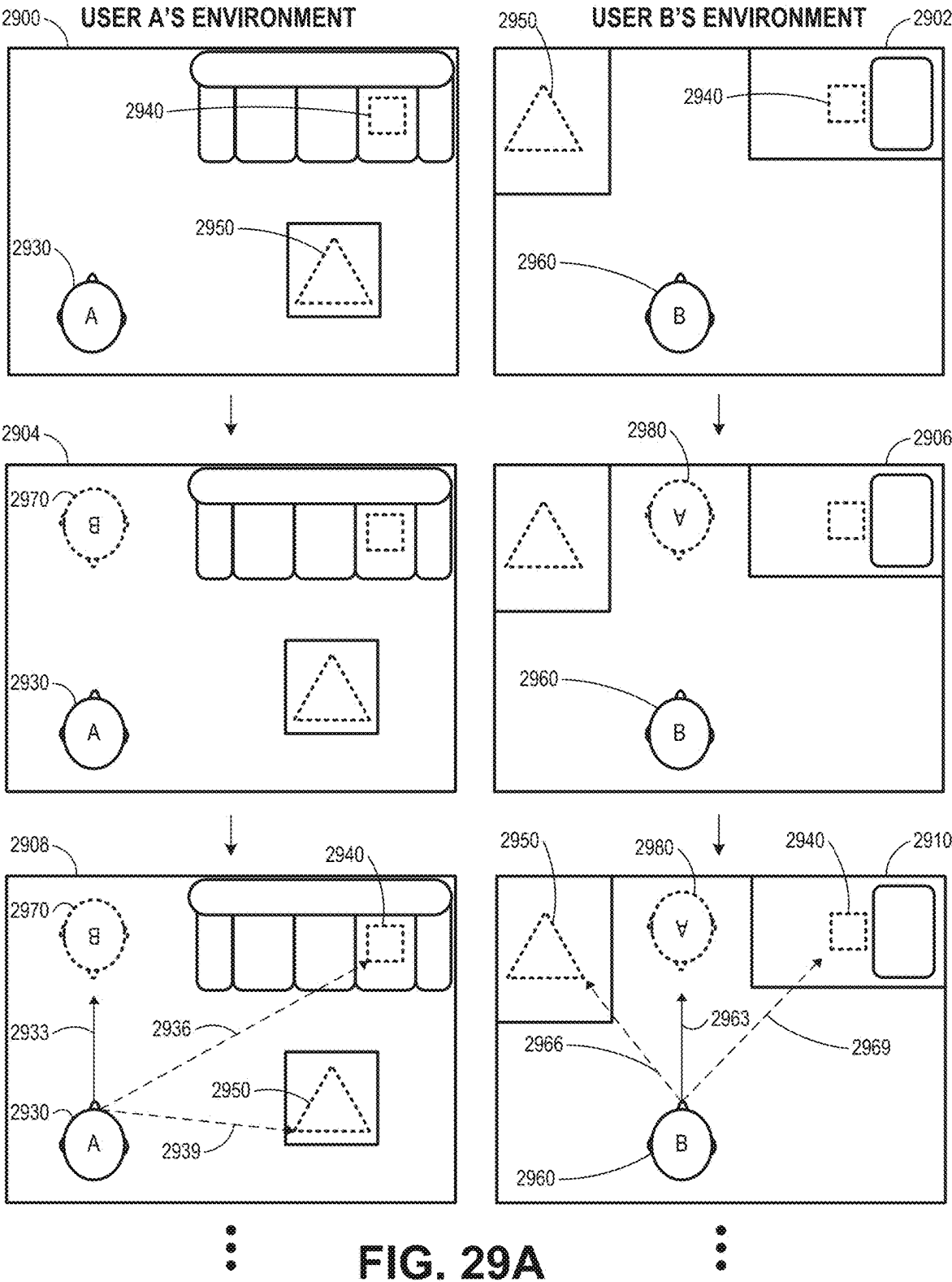


FIG. 28B

RECALCULATION OF OBJECT OF INTEREST



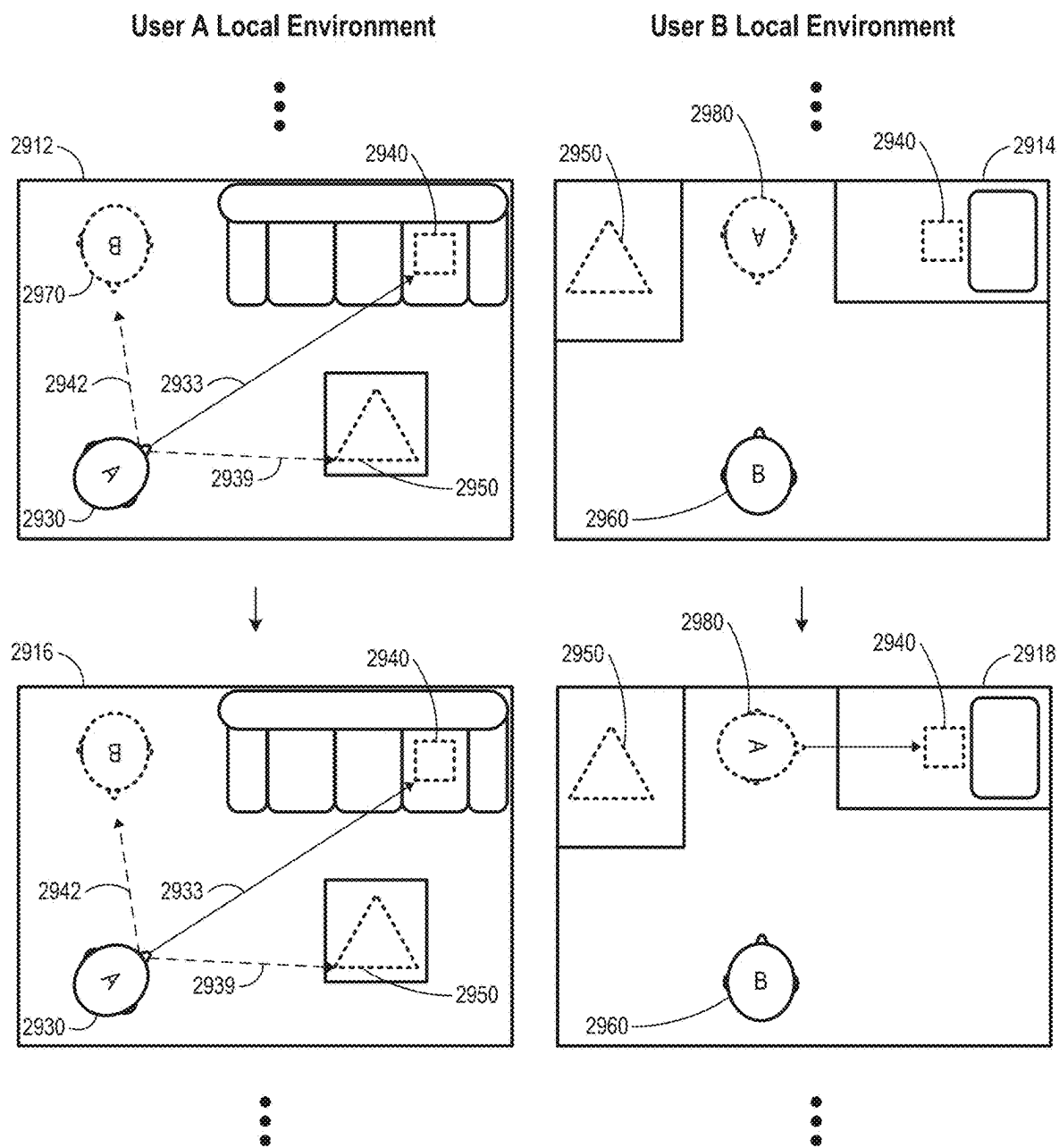


FIG. 29B

RECALCULATION OF OBJECT OF INTEREST

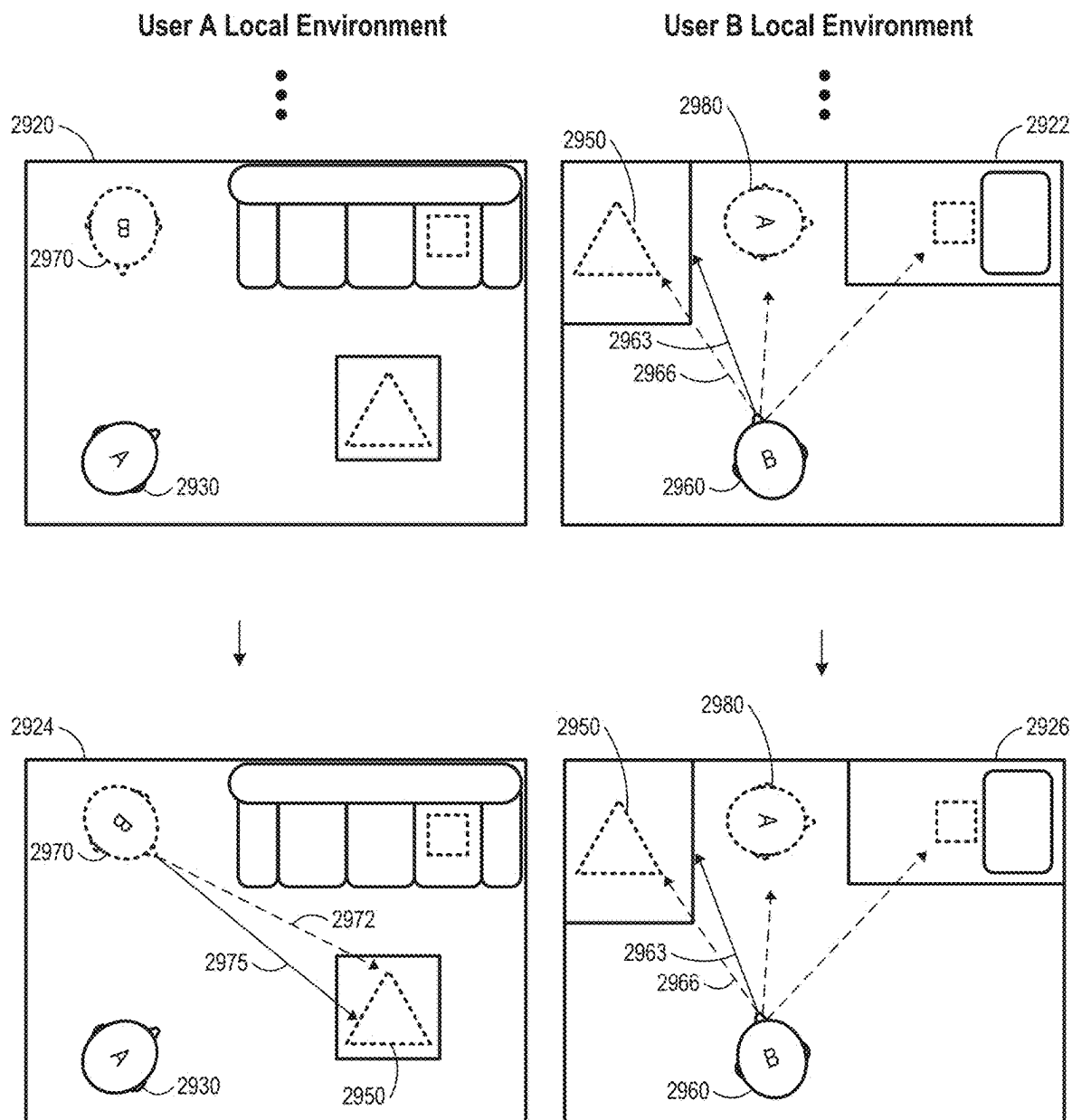


FIG. 29C

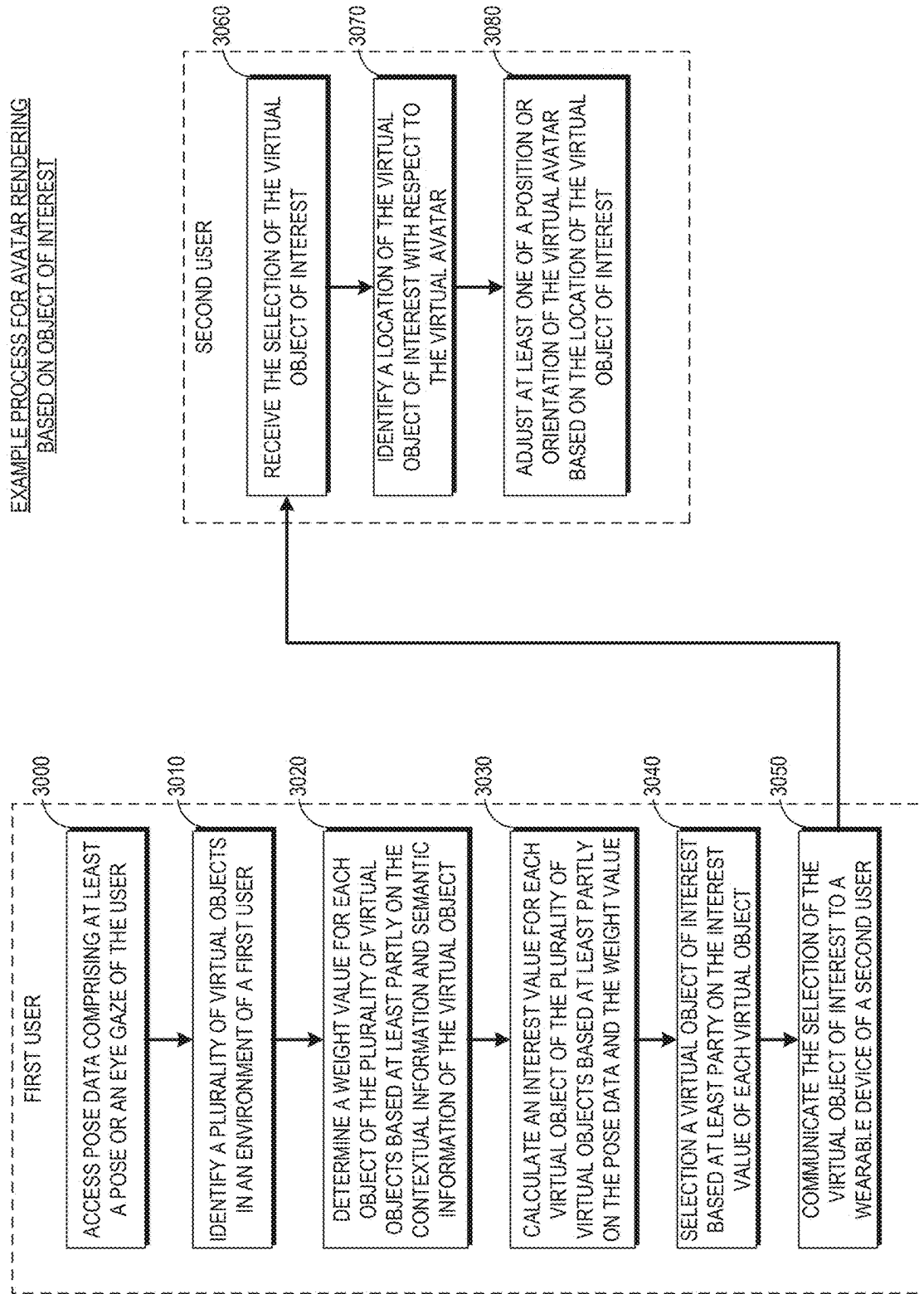


FIG. 30A

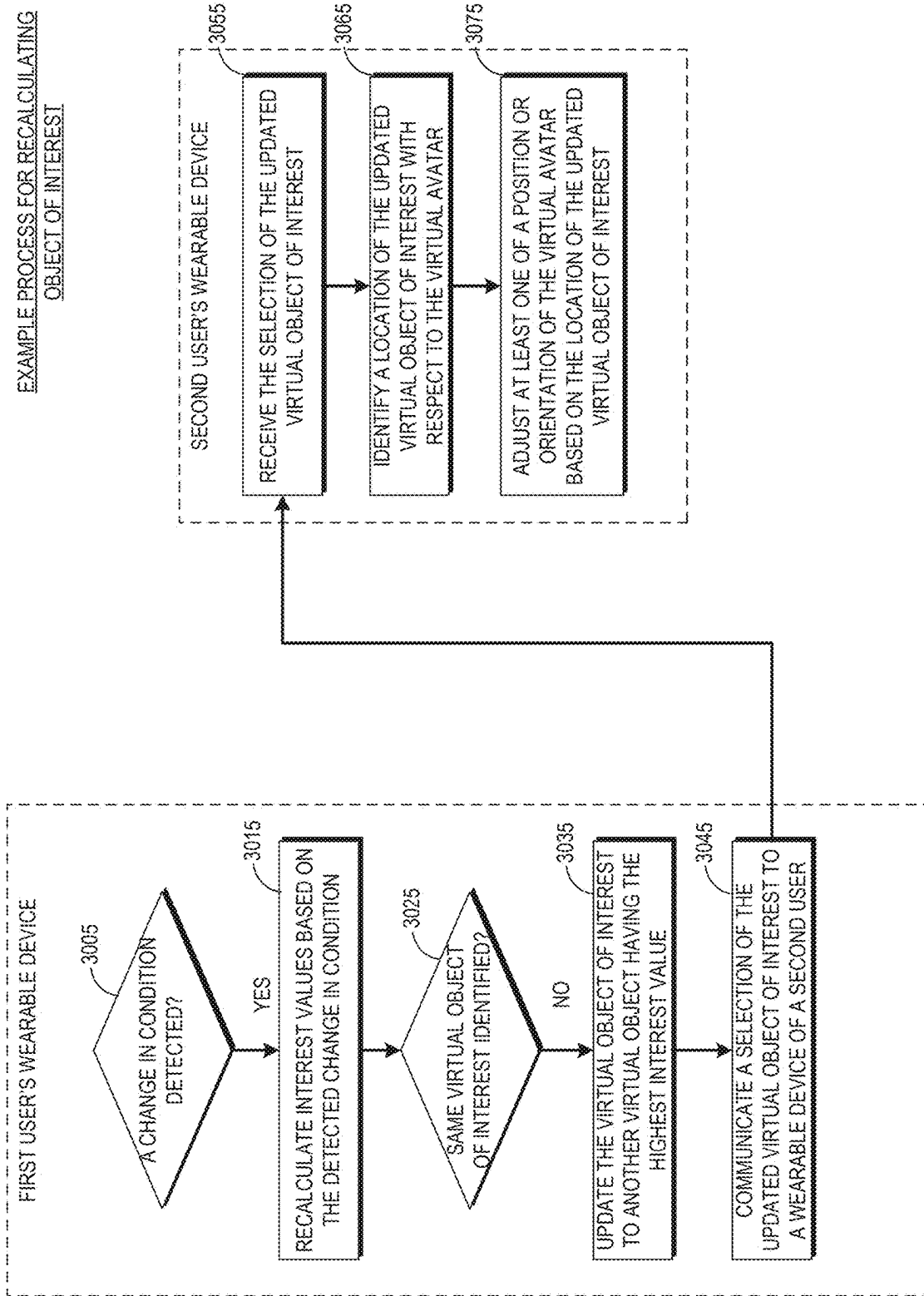


FIG. 30B

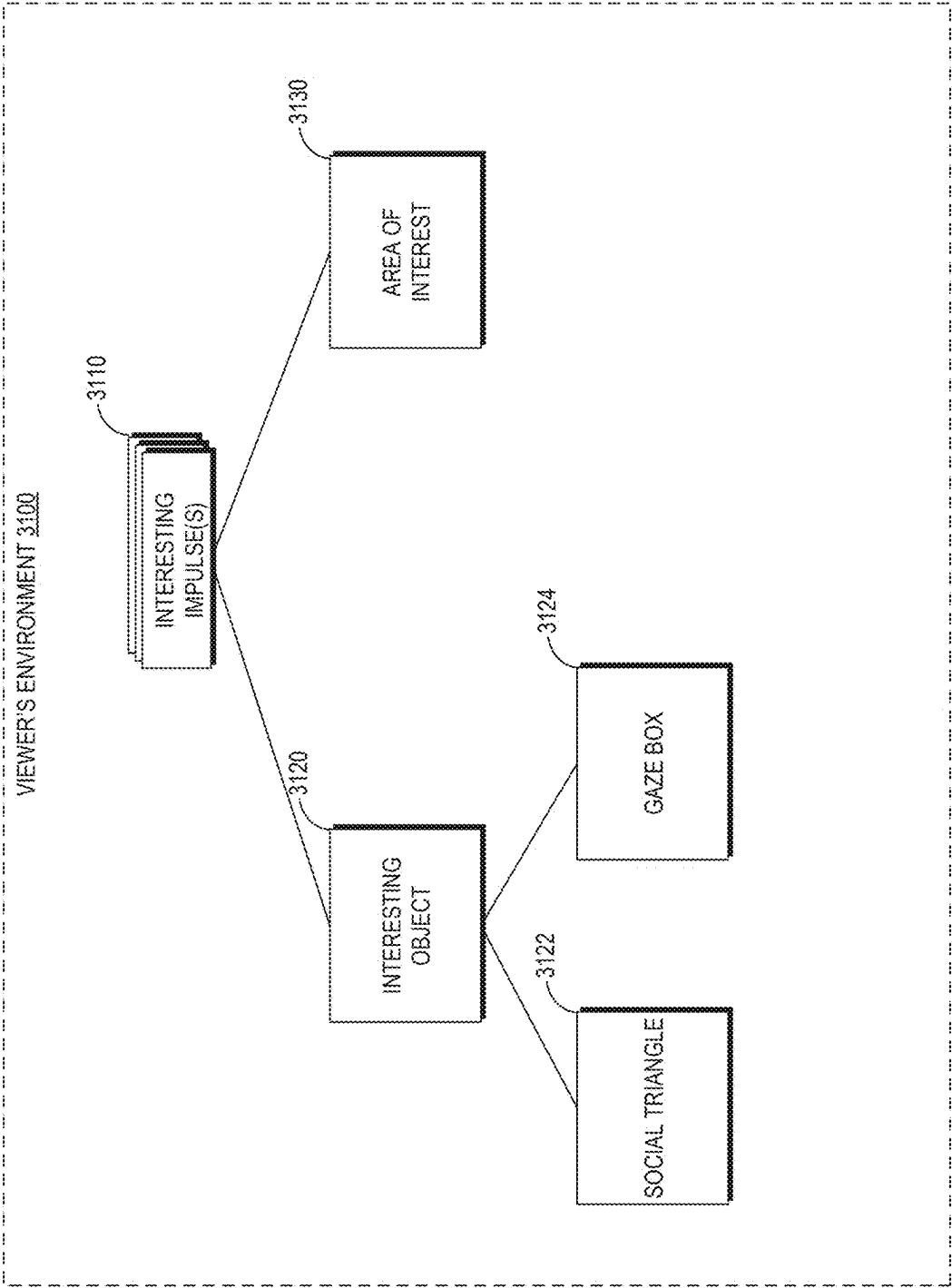


FIG. 31A

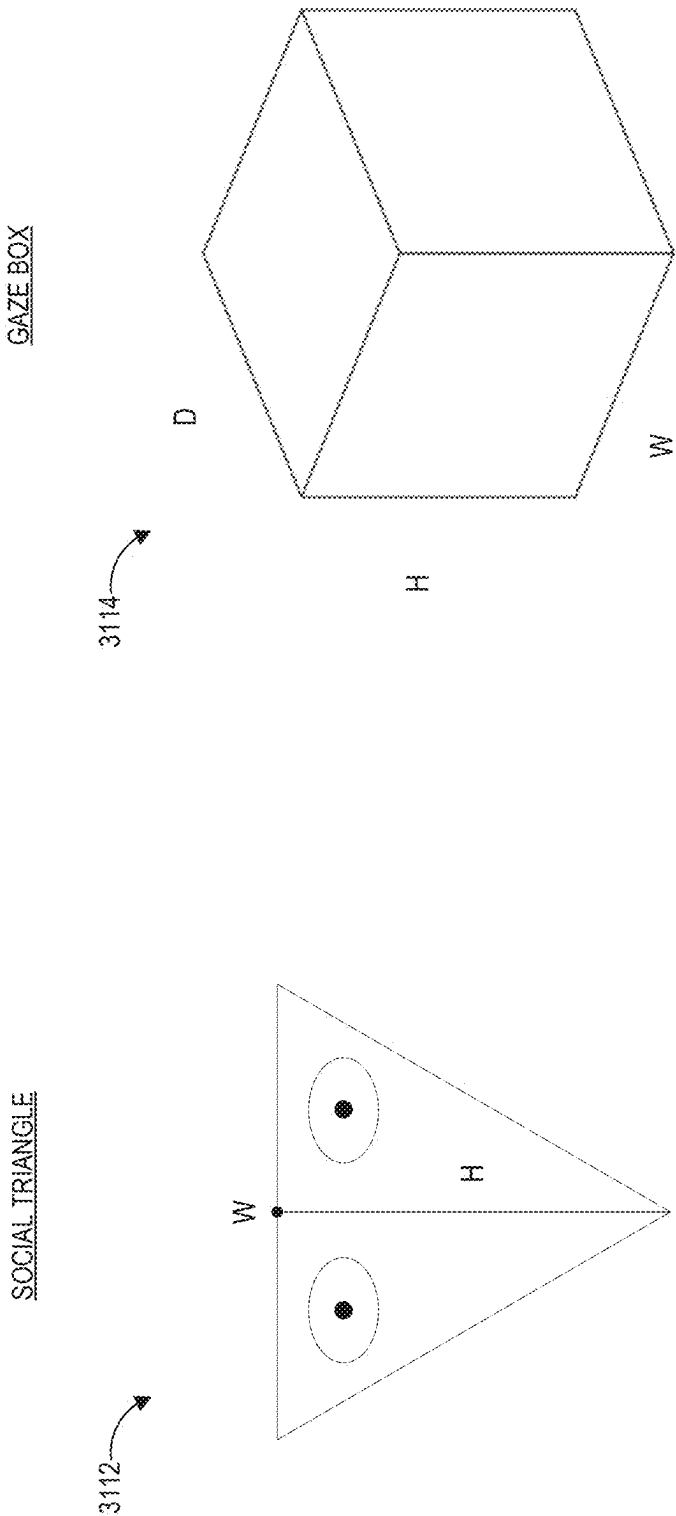
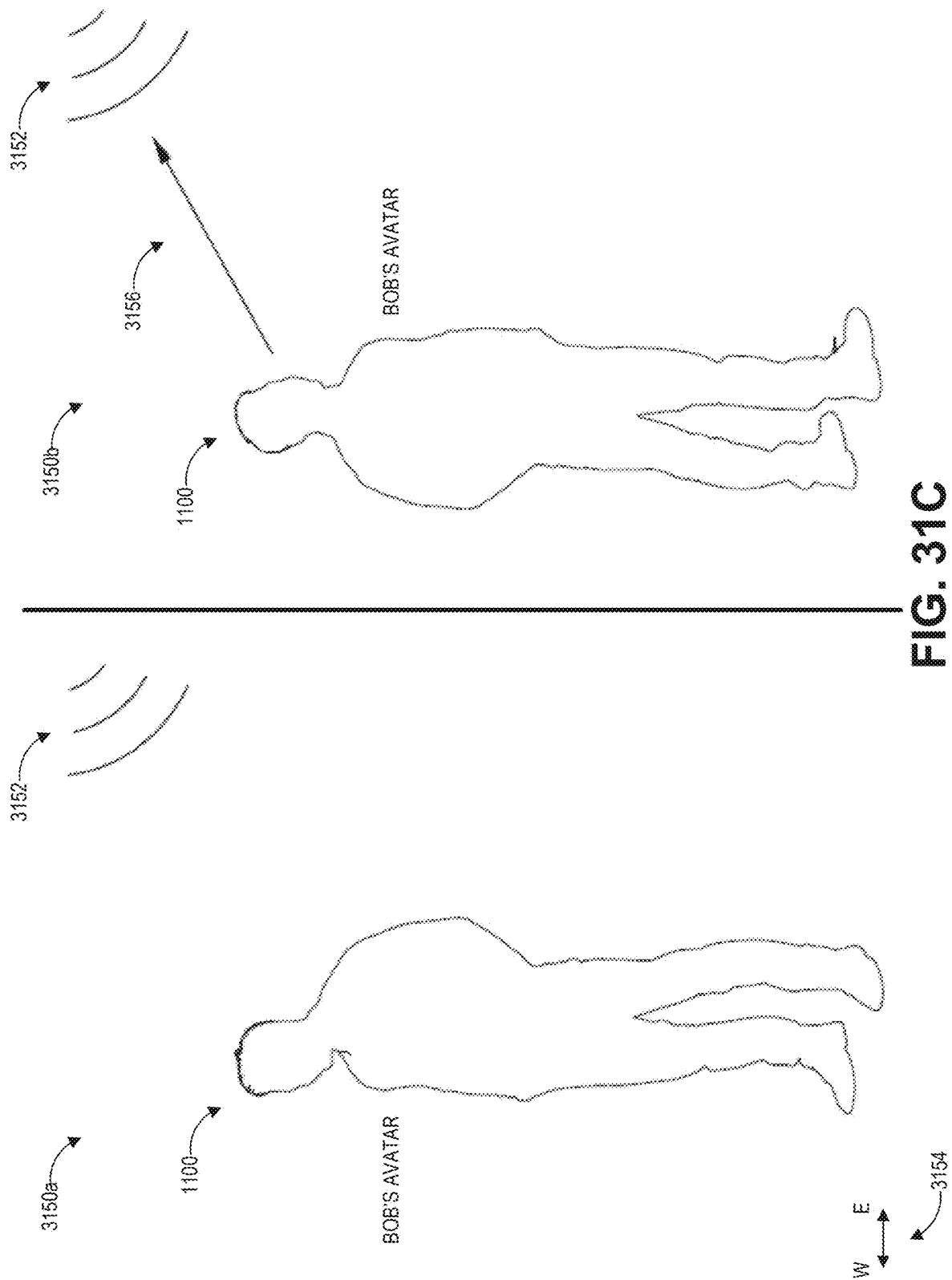


FIG. 31B



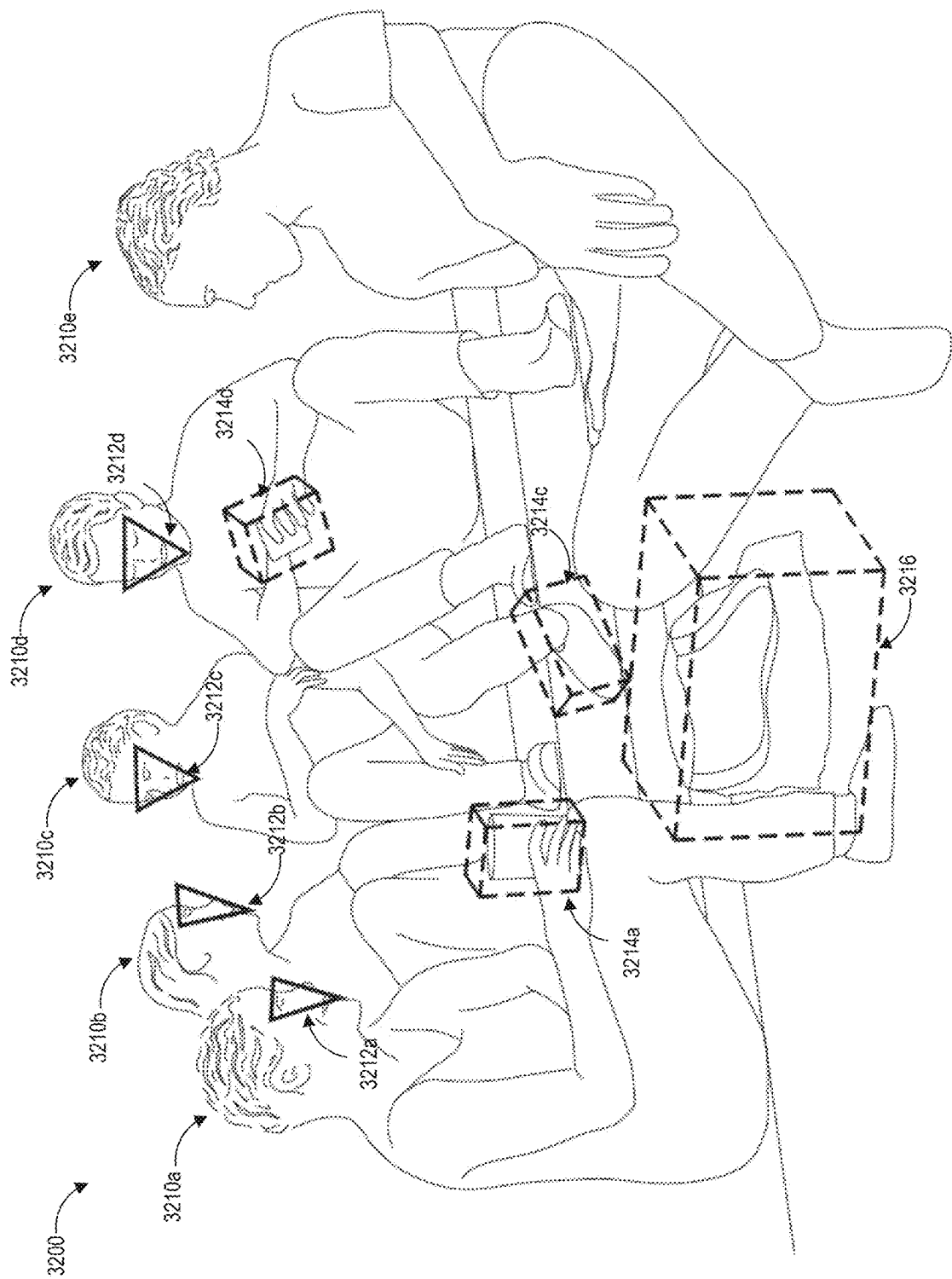


FIG. 32A

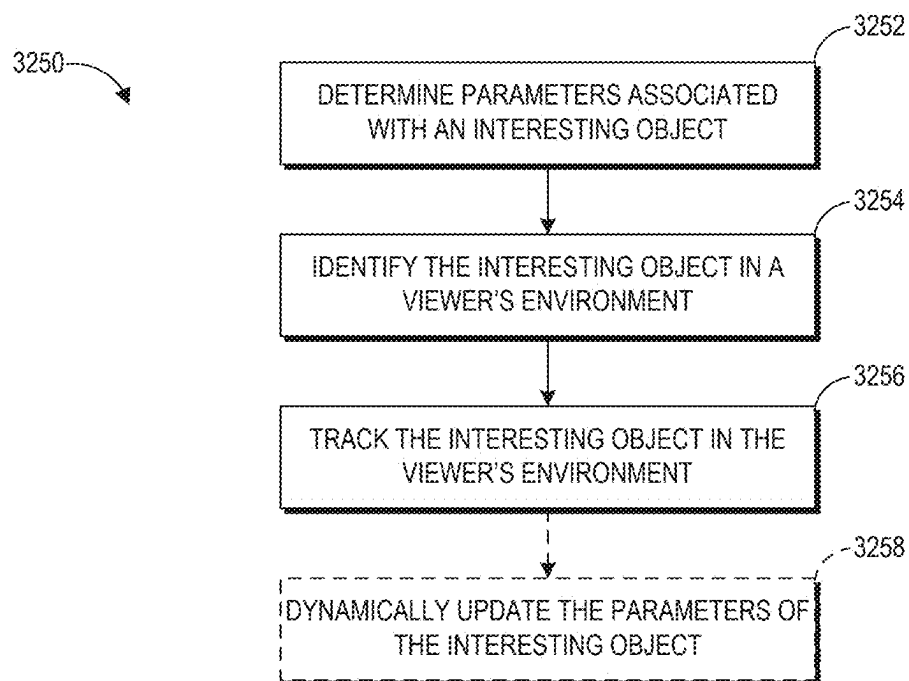


FIG. 32B

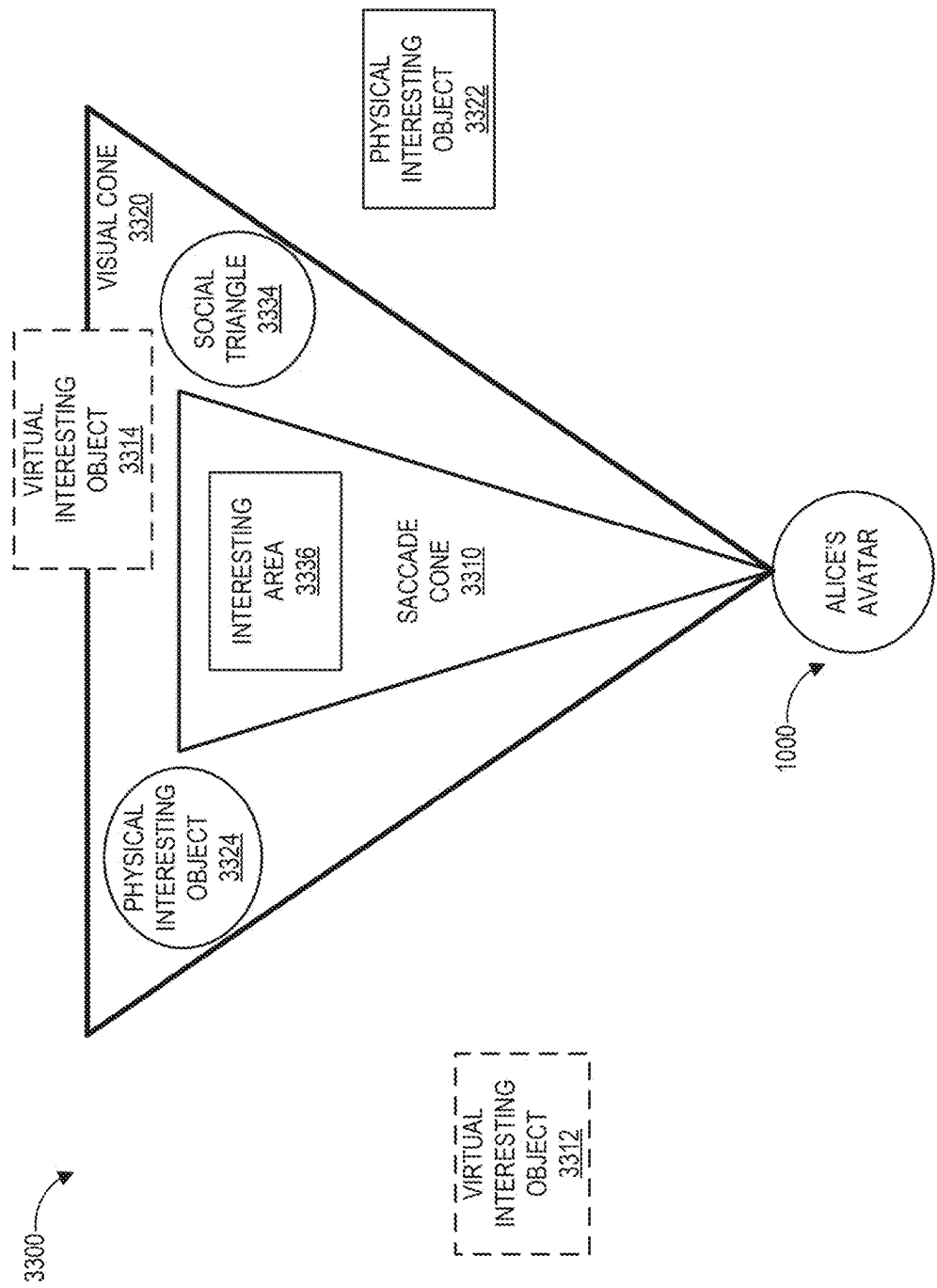


FIG. 33

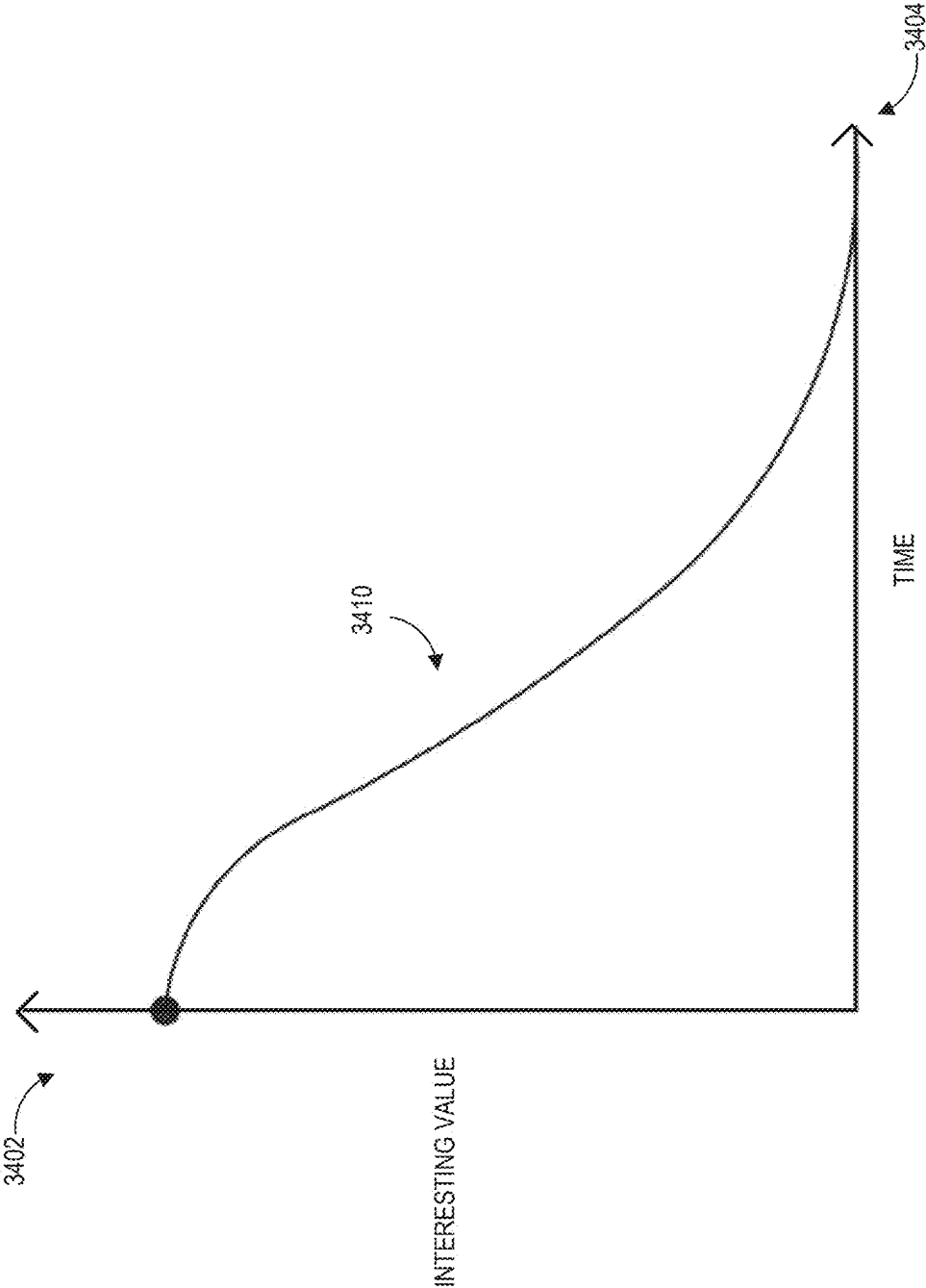


FIG. 34

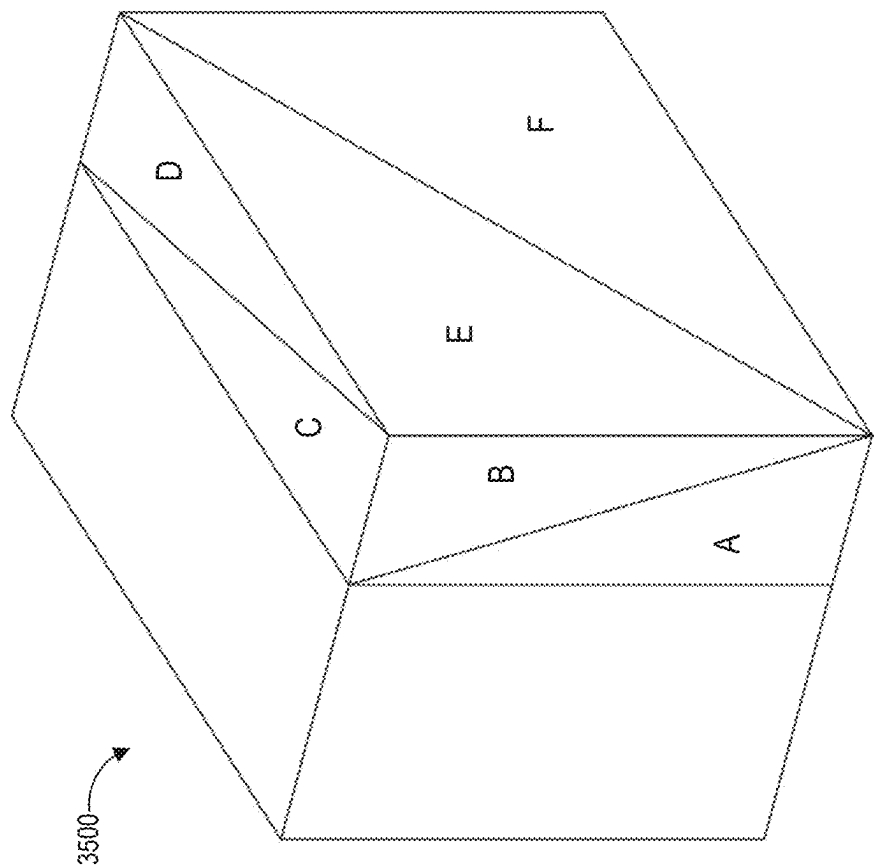


FIG. 35

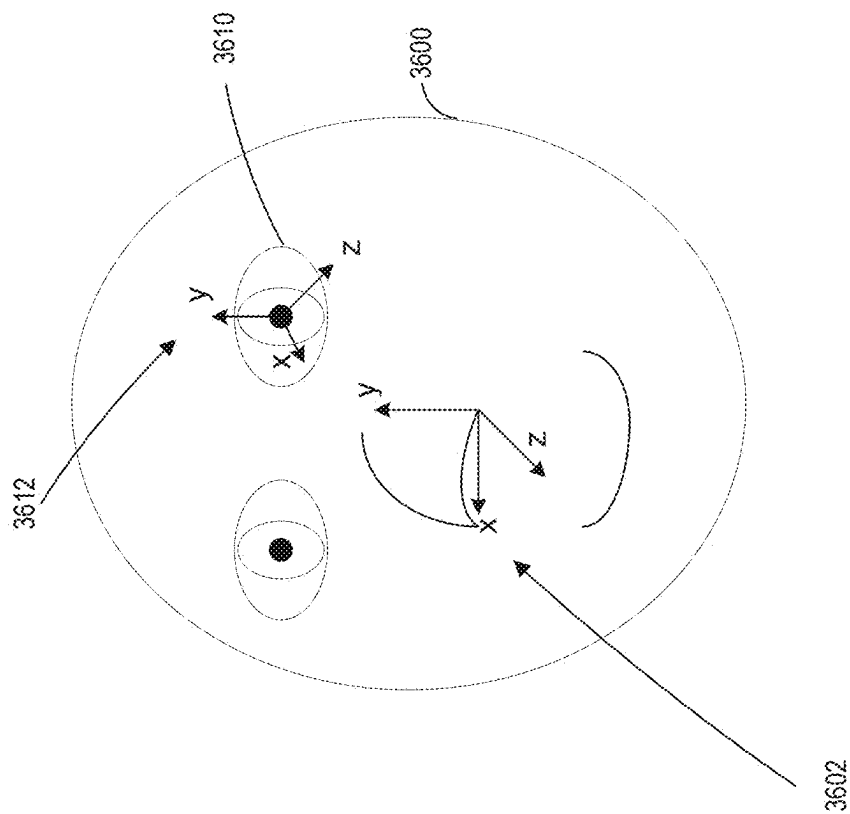


FIG. 36

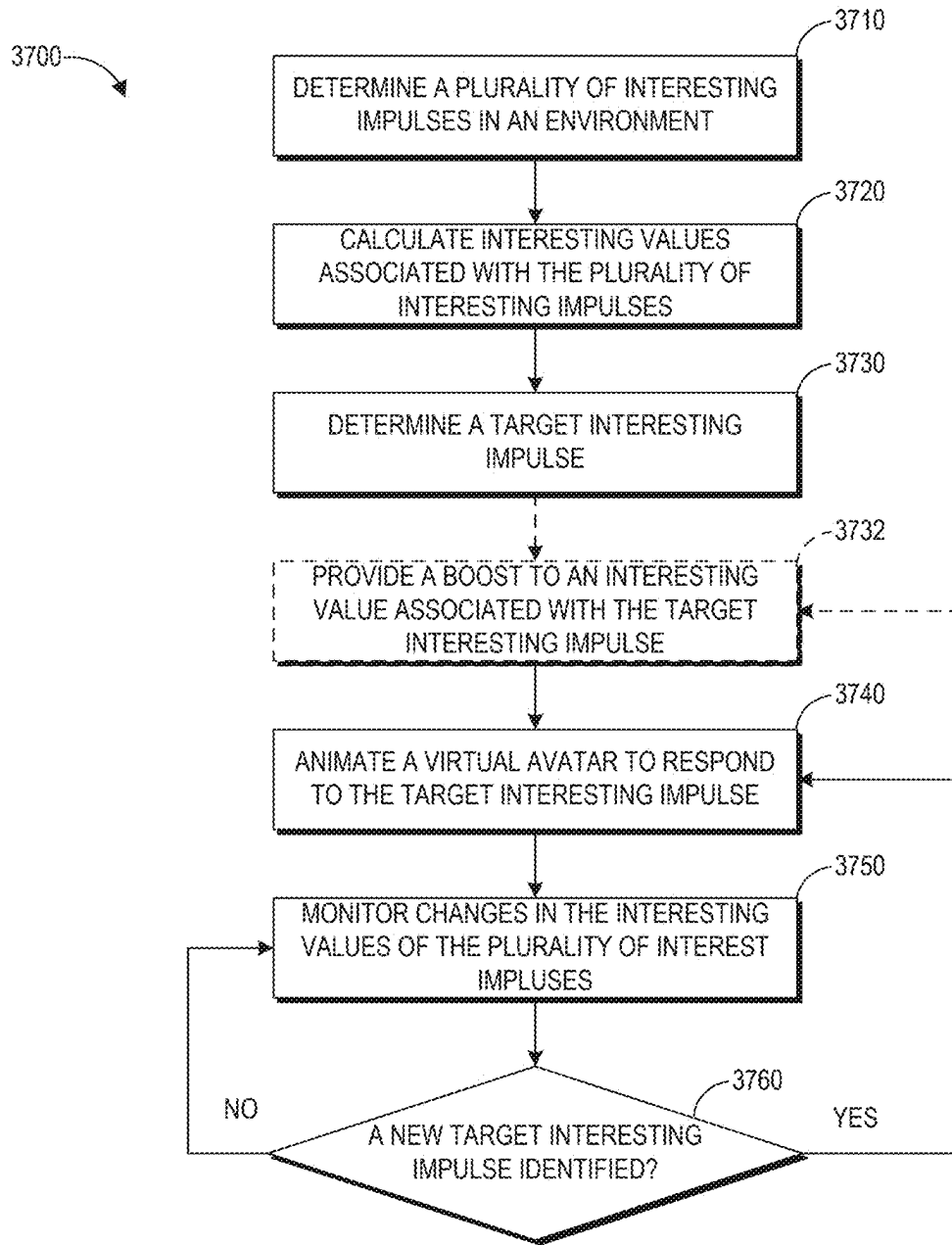


FIG. 37

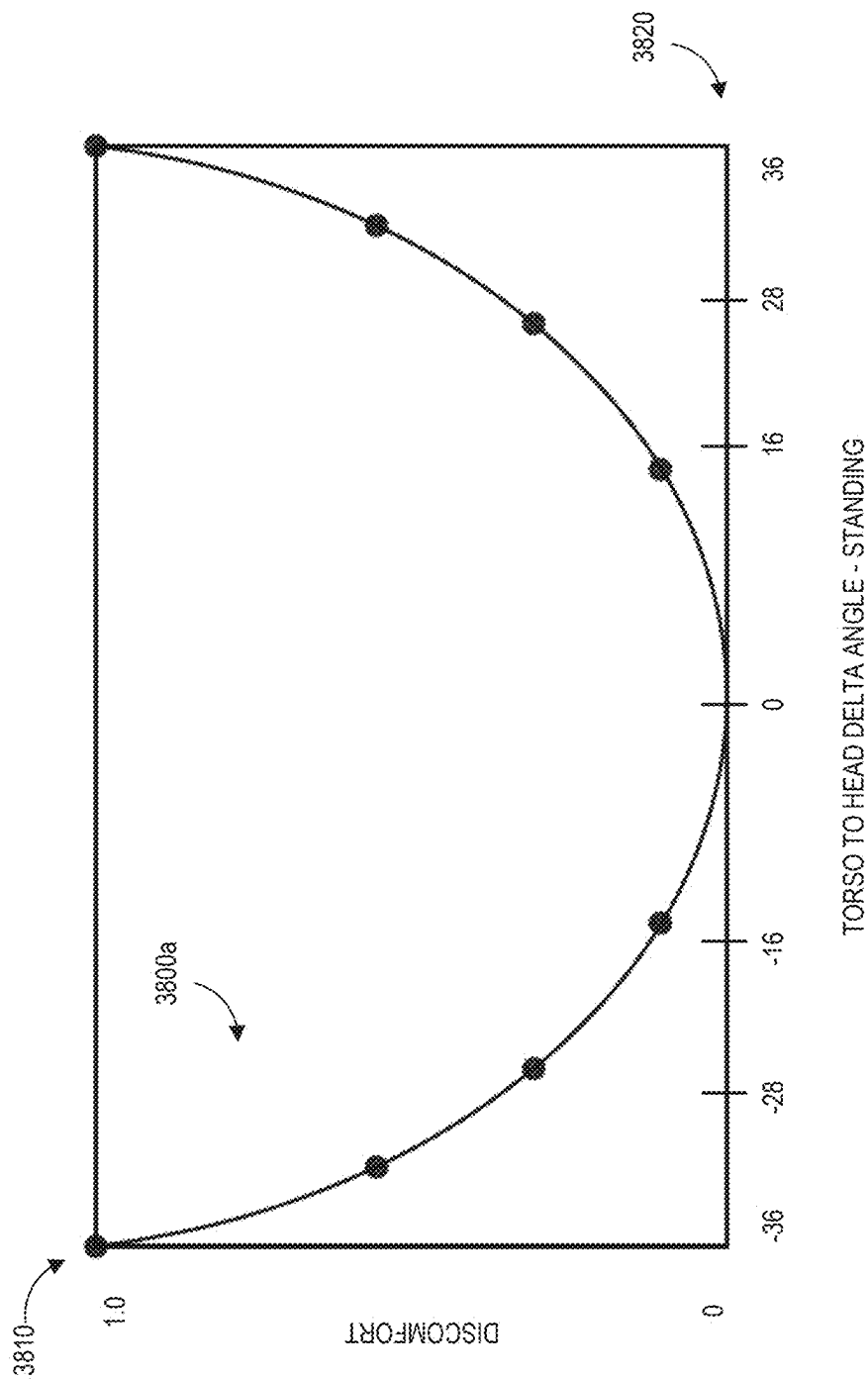


FIG. 38A

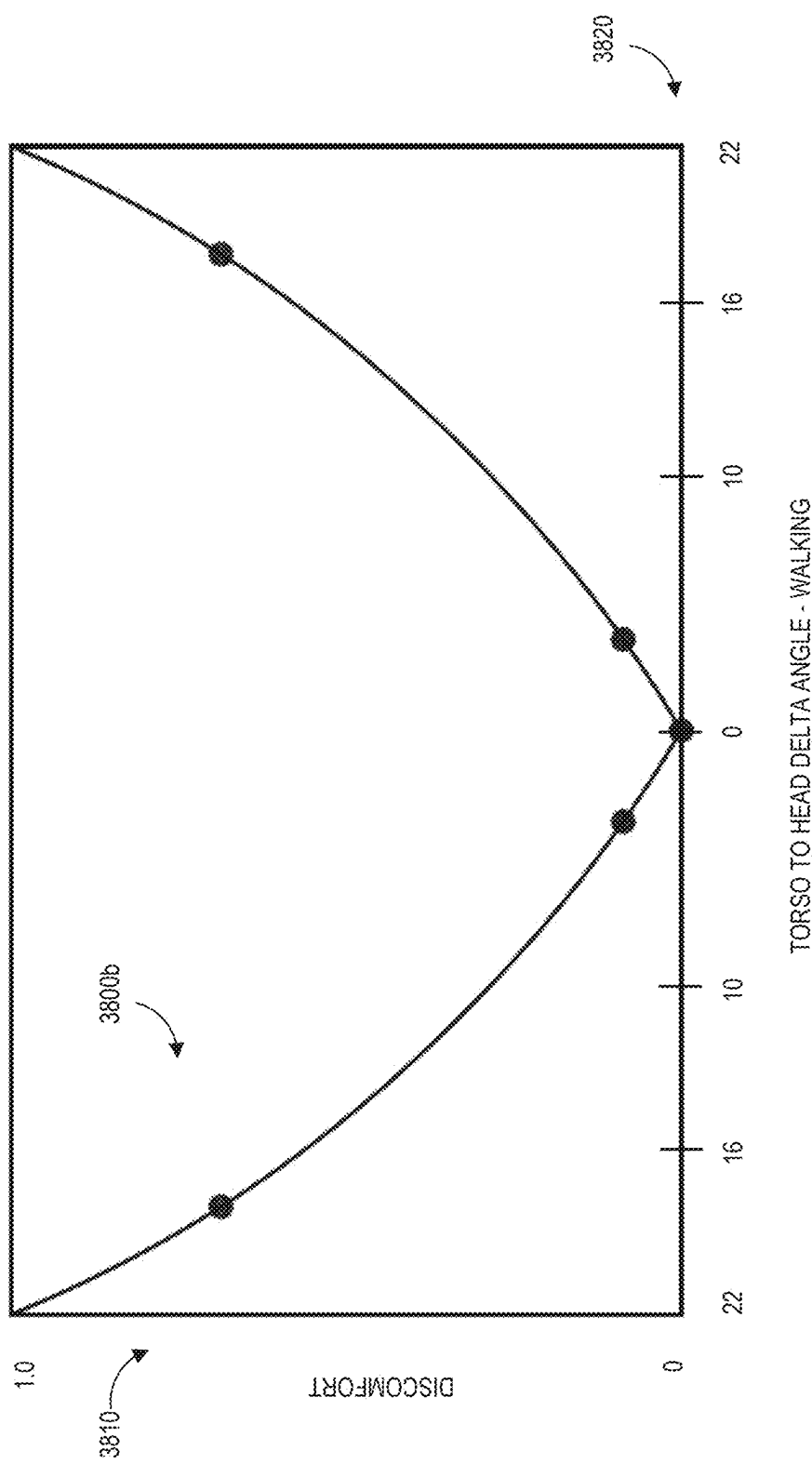


FIG. 38B

EXAMPLE PROCESS OF ANIMATING A VIRTUAL
AVATAR WITH A DISCOMFORT CURVE

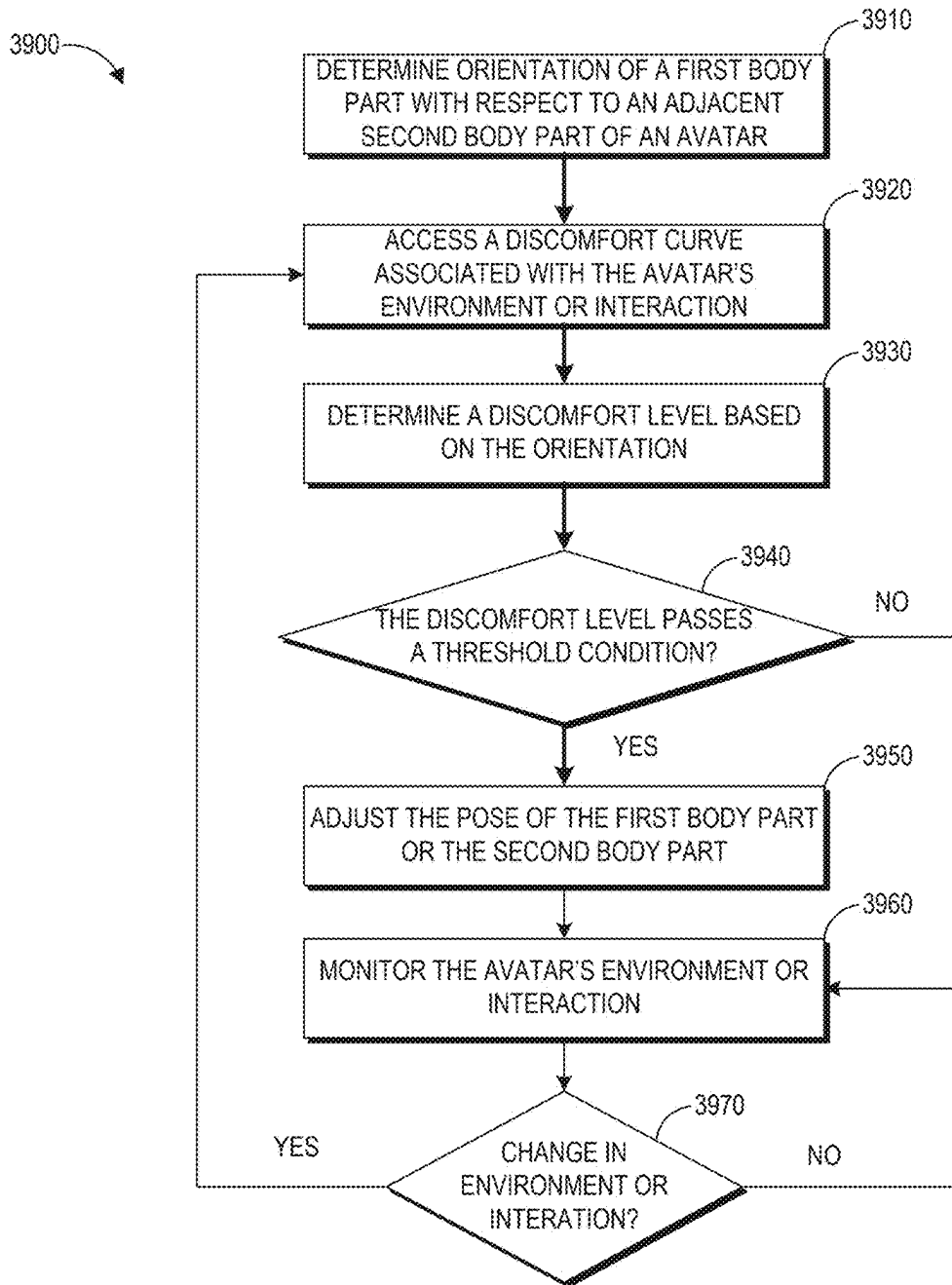


FIG. 39

CONTEXTUAL-BASED RENDERING OF VIRTUAL AVATARS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional application of U.S. patent application Ser. No. 18/320,904, filed May 19, 2023, which is a divisional application of U.S. patent application Ser. No. 17/833,508, filed Jun. 6, 2022, now U.S. Pat. No. 11,699,255, which is a divisional application of U.S. patent application Ser. No. 16/208,119, filed Dec. 3, 2018, now U.S. Pat. No. 11,380,039, which claims the benefit of priority to U.S. Provisional Patent Application No. 62/598,905, filed Dec. 14, 2017, which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present disclosure relates to virtual reality and augmented reality imaging and visualization systems and more particularly to dynamically adjusting and rendering virtual avatars based on contextual information.

BACKGROUND

[0003] Modern computing and display technologies have facilitated the development of systems for so called “virtual reality”, “augmented reality”, or “mixed reality” experiences, wherein digitally reproduced images or portions thereof are presented to a user in a manner wherein they seem to be, or may be perceived as, real. A virtual reality, or “VR”, scenario typically involves presentation of digital or virtual image information without transparency to other actual real-world visual input; an augmented reality, or “AR”, scenario typically involves presentation of digital or virtual image information as an augmentation to visualization of the actual world around the user; a mixed reality, or “MR”, related to merging real and virtual worlds to produce new environments where physical and virtual objects co-exist and interact in real time. As it turns out, the human visual perception system is very complex, and producing a VR, AR, or MR technology that facilitates a comfortable, natural-feeling, rich presentation of virtual image elements amongst other virtual or real-world imagery elements is challenging. Systems and methods disclosed herein address various challenges related to VR, AR and MR technology.

SUMMARY

[0004] Various examples of a mixed reality system for adjusting and rendering virtual avatars based on contextual information are disclosed.

[0005] Examples of systems and methods for rendering an avatar in a mixed reality environment are disclosed. The systems and methods may be configured to automatically scale an avatar or to render an avatar based on a determined intention of a user, an interesting impulse, environmental stimuli, or user saccade points. The disclosed systems and methods may apply discomfort curves when rendering an avatar. The disclosed systems and methods may provide a more realistic interaction between a human user and an avatar.

[0006] Details of one or more implementations of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become appar-

ent from the description, the drawings, and the claims. Neither this summary nor the following detailed description purports to define or limit the scope of the inventive subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 depicts an illustration of a mixed reality scenario with certain virtual reality objects, and certain physical objects viewed by a person.

[0008] FIG. 2 schematically illustrates an example of a wearable system.

[0009] FIG. 3 schematically illustrates example components of a wearable system.

[0010] FIG. 4 schematically illustrates an example of a waveguide stack of a wearable device for outputting image information to a user.

[0011] FIG. 5 is a process flow diagram of an example of a method for interacting with a virtual user interface.

[0012] FIG. 6A is a block diagram of another example of a wearable system which can comprise an avatar processing and rendering system.

[0013] FIG. 6B illustrates example components of an avatar processing and rendering system.

[0014] FIG. 7 is a block diagram of an example of a wearable system including various inputs into the wearable system.

[0015] FIG. 8 is a process flow diagram of an example of a method of rendering virtual content in relation to recognized objects.

[0016] FIG. 9A schematically illustrates an overall system view depicting multiple wearable systems interacting with each other.

[0017] FIG. 9B illustrates an example telepresence session.

[0018] FIG. 10 illustrates an example of an avatar as perceived by a user of a wearable system.

[0019] FIGS. 11A-11D illustrate example scenes of an avatar in various environments, where the avatar has an unnatural appearance or interaction.

[0020] FIGS. 12A and 12B illustrate two scenes of scaling an avatar, where the avatar is spawned on the same surface as the viewer.

[0021] FIGS. 13A and 13B illustrate an example of automatically scaling the avatar based on contextual information, wherein the avatar is not rendered on the same surface as the viewer.

[0022] FIG. 14 illustrates an example of dynamically scaling an avatar in response to a viewer's movement.

[0023] FIG. 15 illustrates another example of dynamically scaling an avatar in response to movement of the avatar or a user.

[0024] FIGS. 16A-1, 16A-2, and 16B illustrate examples of data flow diagrams for automatically scaling the avatar based on contextual factors.

[0025] FIGS. 17A-17B illustrate an example of identifying planar surfaces from a meshed representation of the environment.

[0026] FIGS. 18A and 18B illustrate example processes for automatically scaling an avatar based on contextual information.

[0027] FIG. 19A describes an example of a system for decomposing a user interaction.

[0028] FIG. 19B describes an example of a system for mapping a user's interaction to an avatar in a remote environment.

[0029] FIG. 20 illustrates an example process of intent based rendering of a virtual avatar.

[0030] FIG. 21A illustrates an example process for determining intent based on head pose tracking and eye gaze tracking.

[0031] FIG. 21B illustrates an example process for mapping an interaction with an environment to an avatar's interaction with another environment.

[0032] FIG. 21C illustrates an example of environment reconstruction and analysis subprocesses.

[0033] FIGS. 22 and 23 illustrate two user interaction experiences where a user's interaction with an environment is mapped to the user's avatar in another environment based on intent.

[0034] FIG. 24 shows a block diagram of an overview of intent-based virtual avatar rendering based on an object of interest.

[0035] FIGS. 25A-25D illustrate examples for determining an item of interest where the item of interest is shared between two users.

[0036] FIGS. 26A-26F illustrate examples of calculating interest values for each object in a virtual environment to determine an item of interest.

[0037] FIG. 27 shows an example of information that may be communicated from one wearable device to another wearable device during intent based rendering by the other wearable device.

[0038] FIGS. 28A and 28B illustrate examples of rendering a virtual avatar based on an item of interest.

[0039] FIGS. 29A, 29B, and 29C illustrate examples of recalculating an item of interest.

[0040] FIGS. 30A and 30B illustrate example processes of intent based avatar rendering based on an object of interest.

[0041] FIGS. 31A-31C illustrate examples of categories of interesting impulses that may occur or be present in a viewer's environment.

[0042] FIG. 32A illustrates an example of generating interesting impulses based on real world stimuli.

[0043] FIG. 32B illustrates an example process of configuring parameters associated with an interesting impulse and tracking the interesting impulse.

[0044] FIG. 33 illustrates an example of identifying a target interesting impulse.

[0045] FIG. 34 illustrates an example of interestingness decay based on time.

[0046] FIG. 35 illustrates an example of determining saccade points associated with an object which is selected as a target interesting impulse.

[0047] FIG. 36 illustrates an example of eye pose and face transform for animating an avatar based on saccade points.

[0048] FIG. 37 illustrates an example process of avatar rendering based on environmental stimuli.

[0049] FIGS. 38A and 38B illustrate examples of discomfort curves of the torso in relation to the head.

[0050] FIG. 39 illustrates an example process of applying a discomfort curve for rendering an avatar.

[0051] Throughout the drawings, reference numbers may be re-used to indicate correspondence between referenced elements. The drawings are provided to illustrate example embodiments described herein and are not intended to limit the scope of the disclosure.

DETAILED DESCRIPTION

Overview

[0052] A virtual avatar may be a virtual representation of a real or fictional person (or creature or personified object) in an AR/VR/MR environment. For example, during a telepresence session in which two AR/VR/MR users are interacting with each other, a viewer can perceive an avatar of another user in the viewer's environment and thereby create a tangible sense of the other user's presence in the viewer's environment. The avatar can also provide a way for users to interact with each other and do things together in a shared virtual environment. For example, a student attending an online class can perceive other students' or teachers' avatars in a virtual classroom and can interact with the avatars of the other students or the teacher.

[0053] When placing an avatar in a user's physical environment for an AR/VR/MR environment, a size of the avatar needs to be determined. When spawning (the initial rendering of the avatar) in a three-dimensional (3D) space, the avatar could, in practice, be any size (e.g., tiny, human-sized, or gigantic). The avatar could maintain a 1:1 size to its human counterpart, but this may not make sense in certain environments (due to, for example, lack of space, privacy concerns, etc.). An improperly sized avatar can create awkward social interactions or create user fatigue when interacting with an avatar. For example, if an avatar is too big or too small relative to a viewer, the viewer may need to position his head or body at an uncomfortable position in order to engage in an eye-to-eye conversation with the avatar. Further, an improperly sized avatar can convey the wrong social message such as an implied superiority (e.g., when the avatar is bigger than the user) or inferiority (e.g., when the avatar is smaller than the user) between the avatar's human counterpart and the viewer. Additional examples related to problems caused by an improperly sized avatar are further described with reference to FIG. 11A and examples of solutions to this problem are described with reference to FIGS. 12A-18B.

[0054] Advantageously, in some embodiments, the wearable system described herein can automatically determine an appropriate size for an avatar at spawning and can re-scale the avatar throughout some or all parts of the interaction with other users (or avatars) based on contextual information regarding the interaction. Some example contextual information can include the position of the user, the rendering location of the avatar in the environment of the other user, a relative height difference between the user and the avatar, presence of objects in the rendering environment (e.g., whether there are chairs for an avatar to sit on or whether movement of the avatar would cause the avatar to pass through solid objects such as tables), etc. The wearable system can automatically scale the avatar in a manner that increases or maximizes direct eye contact based on the contextual information, and therefore facilitates avatar-human communication. Details for scaling the avatar based on contextual information are further described in FIGS. 12A-18B.

[0055] An avatar can be animated based on its human counterpart, where a human's interaction is mapped to his avatar. A one-to-one mapping between a user and an avatar can be employed in an AR/VR/MR environment, such that the avatar's action is a direct mapping of a user's action. For example, if a user looks left, its avatar also looks left. If the

user stands up, its avatar stands up. If a user walks in a certain direction, its avatar walks in that direction. This one-to-one mapping may work in a VR environment because the participants of the VR environment (including the user's virtual avatar) are seeing the same shared virtual content. However, in an AR/MR environment, each user's physical environment and the way the other user's avatar appears within it might be different, because the two (or more) users may be in very different environments. For example, Bob might be in a living room in a house, and Alice might be in a room in an office building. Bob might see Alice's avatar across from Bob in Bob's environment (the living room), whereas Alice sees Bob's avatar located to the left of Alice in Alice's environment (the office room). As another example, since avatars are virtual objects and can be resizable, Bob can shrink the size of Alice's avatar and place it on a table in his living room, whereas Alice might be in a large office room and may choose to have Bob's avatar stand in the corner and be life-sized.

[0056] Such sizing may result in Bob looking down at Alice's avatar when Bob is talking to her avatar. If a one-to-one mapping is applied to Bob's avatar based on Bob's action, Bob's avatar rendered in Alice's office may look unusual to Alice, because Bob's avatar would be looking at the floor while talking to Alice (since the real Bob is looking down at Alice's avatar). On the other hand, it may be desirable that certain aspects of the user be preserved and mapped to the avatar in another user's environment. For example, a user nodding his or her head in agreement or shaking his or her head in disagreement can be conveyed to the other user by mapping such motions to the avatar. Additional examples describing problems in one-to-one mapping are further provided with reference to FIGS. 11B-11D. Examples of solutions to such problems are described with reference to FIGS. 19A-30B.

[0057] Advantageously, in some embodiments, the wearable system can analyze an interaction of a user and break the interaction into a world component and a local component. A world component can include the portion of the interaction that interacts with an environment. For example, a world motion may include walking from point A to point B, climbing up a ladder, sitting or standing, facing a certain direction, and interacting with an object (virtual or physical) in the environment. As further described herein, the world component can be described with respect to a world reference frame which is associated with an environment. A local component can include an action relative to a user (which can be described relative to a body-fixed reference frame). For example, if Alice is nodding her head or shaking her head, that motion has meaning based on the angle of her head with respect to her torso (or body). As another example, Alice can turn around 180 degrees and nod her head. These two motions can be considered as local because they are localized with respect to her torso and may not require interactions with the environment. As yet another example, waving a hand can be a local motion, because it can be defined with respect to the user's body. Some movements may have a local portion and a world portion. For example, a user may gesture by pointing a finger, which has local attributes relative to the user's body as well as world aspects if the user is pointing at an object in the user's environment. For example, if Alice is pointing to the avatar of Bob in Alice's environment, the intent determined from Alice's hand gesture is that Alice points to Bob. In Bob's environ-

ment, Alice might appear in a different relative location or orientation to Bob and if her hand gesture were rendered in one-to-one correspondence, Alice's avatar might not point at Bob and thus might not reflect Alice's intent. The wearable device of Bob can map Alice's intent to Bob's environment so that Alice's avatar is rendered as pointing at Bob.

[0058] The wearable system can extract intent of a user's interaction based on contextual information associated with the user's environment, the user's movements, the user's intentions, and so forth. The wearable system can accordingly map the world motion of the user's interaction to an avatar's action based on the avatar's environment and map the local action of the user's interaction directly to the avatar. The mapping of the world motion can include adjusting one or more characteristics of the avatar such as, e.g., the movement, position, orientation, size, facial expression, pose, eye gaze, etc., to be compatible with the physical environment in which the avatar is rendered (rather than simply mapping the characteristics in a direct one-to-one fashion).

[0059] For example, when Alice walks to a chair and sits down on the chair, the wearable system can automatically find a chair in Bob's environment (or another sit-able surface if there is no chair) by accessing information from the world map of Bob's environment and rendering Alice as sitting on the chair. As another example, the wearable system may determine that Alice intends to interact with an object of interest (e.g., a tree or a virtual book) in her environment. The wearable system can automatically reorient Alice's avatar to interact with the object of interest in Bob's environment, where the location of the object of interest may not be the same as that in Alice's environment. For example, if a direct one-to-one mapping of the virtual book would cause it to be rendered inside or underneath a table in Bob's environment, Bob's wearable system may instead render the virtual book as lying on top of the table, which will provide Bob with a more natural interaction with Alice and the virtual book.

[0060] While the wearable system may remap the world motions, the wearable system can preserve local motions, such as nodding, hand tilting or shakes (which can indicate confusion, agreement or refusal) of a user's interaction. For example, Alice can shake her head and walk toward a chair. This interaction includes a world motion, such as walking toward a chair, and a local motion, such as shaking her head. The wearable system can adjust Alice's avatar's direction of walking based on the location of a chair in Bob's environment, but in the meantime, render Alice's avatar as shaking her head. Further descriptions related to intent-based rendering of avatars are described with reference to FIGS. 19-31.

[0061] In certain implementations, the wearable system can also map a local component to the avatar based on intent. For example, if Alice gives a thumbs up sign, the wearable system can interpret this as an emblem gesture (e.g., a gesture that is consciously used and consciously understood and which is used as a substitute for words and is closer to sign language than everyday body language) and map a more expressive thumbs up animation to Alice's avatar in Bob's environment to convey the same intention. This can apply to other common symbolic or emblem gestures, such as, e.g., waving your hand with an open palm gesture, or giving an okay sign with the okay gesture.

[0062] In addition to or as an alternative to animating a virtual avatar based on interactions of its human counterpart, advantageously, in some embodiments, the wearable system can also animate an avatar based on the environment that the avatar is rendered in. The wearable system can render the virtual avatar such that the virtual avatar can appear to be able to make its decision based on interactions with the environment (e.g., the wearable system may display the avatar as sitting if there is a chair present in the environment but display the avatar as standing if no chair is present).

[0063] The wearable system can discover interesting impulses in the viewer's environment, such as an object of interest, an area of interest, a sound, or a component of an object of interest, and cause the avatar to automatically respond to the interesting impulse (such as, e.g., turning around to look at the interesting impulses). As one example, the wearable system of the viewer may detect a sudden loud noise, and the wearable system of the viewer can automatically reorient the avatar to look in the direction of where the loud noise came from.

[0064] The wearable system can animate the avatar based on the impulse regardless of whether the viewer is in a telepresence session with the avatar's human counterpart. In situations where the viewer is in a telepresence session, the avatar can respond to the interesting impulse in the viewer's environment even though such interesting impulse does not appear in the human counterpart's environment. For example, Alice and Bob may be in telepresence session. Alice's avatar may initially face Bob as Bob is talking. When Bob's wearable system detects a loud noise in Bob's environment, Alice's avatar may switch its attention from Bob to the direction of the loud noise (and thus may look away from Bob). Further descriptions related to animating an avatar based on a viewer's environment are described with reference to FIGS. 31A-37.

[0065] In certain implementations, the wearable system can learn the behaviors of one or more users (e.g., including the avatar's human counterpart) and drive the avatar's animation based on such learning even though the human counterpart user may or may not be present (either remotely or in the same environment). For example, the wearable system can learn (e.g., based on data from devices sensors) how the user interacts with others from the user's eye gaze direction and frequency of eye contact in relation to speaking voices or objects in the user's environment. The wearable system can accordingly drive the avatar's animation based on the learned behaviors of the user. As an example, if a user does not respond to country music (e.g., does not look at the sound source playing the country music), the avatar associated with the user may also be rendered such that the avatar does not respond to the country music.

[0066] Although the examples in this disclosure describe animating a human-shaped avatar, similar techniques can also be applied to animals, fictitious creatures, objects (e.g., the virtual book described above), etc. For example, Alice's system may detect a dog moving in her environment. Bob's wearable system can present movement of a virtual dog in Bob's environment. Bob's wearable system may display the virtual dog's movement in Bob's environment based on obstacles in Bob's environment (e.g., by having the virtual dog move on a trajectory that does not cause the virtual dog to pass through physical objects).

[0067] Accordingly, embodiments of the disclosed systems and methods may provide for a much more realistic

interaction between a user of the wearable system and avatars in the user's environment.

Examples of 3D Display of a Wearable System

[0068] A wearable system (also referred to herein as an augmented reality (AR) system) can be configured to present 2D or 3D virtual images to a user. The images may be still images, frames of a video, or a video, in combination or the like. At least a portion of the wearable system can be implemented on a wearable device that can present a VR, AR, or MR environment, alone or in combination, for user interaction. The wearable device can be used interchangeably as an AR device (ARD). Further, for the purpose of the present disclosure, the term "AR" is used interchangeably with the term "MR".

[0069] FIG. 1 depicts an illustration of a mixed reality scenario with certain virtual reality objects, and certain physical objects viewed by a person. In FIG. 1, an MR scene 100 is depicted wherein a user of an MR technology sees a real-world park-like setting 110 featuring people, trees, buildings in the background, and a concrete platform 120. In addition to these items, the user of the MR technology also perceives that he "sees" a robot statue 130 standing upon the real-world platform 120, and a cartoon-like avatar character 140 flying by which seems to be a personification of a bumble bee, even though these elements do not exist in the real world.

[0070] In order for the 3D display to produce a true sensation of depth, and more specifically, a simulated sensation of surface depth, it may be desirable for each point in the display's visual field to generate an accommodative response corresponding to its virtual depth. If the accommodative response to a display point does not correspond to the virtual depth of that point, as determined by the binocular depth cues of convergence and stereopsis, the human eye may experience an accommodation conflict, resulting in unstable imaging, harmful eye strain, headaches, and, in the absence of accommodation information, almost a complete lack of surface depth.

[0071] VR, AR, and MR experiences can be provided by display systems having displays in which images corresponding to a plurality of depth planes are provided to a viewer. The images may be different for each depth plane (e.g., provide slightly different presentations of a scene or object) and may be separately focused by the viewer's eyes, thereby helping to provide the user with depth cues based on the accommodation of the eye required to bring into focus different image features for the scene located on different depth plane or based on observing different image features on different depth planes being out of focus. As discussed elsewhere herein, such depth cues provide credible perceptions of depth.

[0072] FIG. 2 illustrates an example of wearable system 200 which can be configured to provide an AR/VR/MR scene. The wearable system 200 can also be referred to as the AR system 200. The wearable system 200 includes a display 220, and various mechanical and electronic modules and systems to support the functioning of display 220. The display 220 may be coupled to a frame 230, which is wearable by a user, wearer, or viewer 210. The display 220 can be positioned in front of the eyes of the user 210. The display 220 can present AR/VR/MR content to a user. The display 220 can comprise a head mounted display (HMD) that is worn on the head of the user.

[0073] In some embodiments, a speaker 240 is coupled to the frame 230 and positioned adjacent the ear canal of the user (in some embodiments, another speaker, not shown, is positioned adjacent the other ear canal of the user to provide for stereo/shapeable sound control). The display 220 can include an audio sensor (e.g., a microphone) 232 for detecting an audio stream from the environment and capture ambient sound. In some embodiments, one or more other audio sensors, not shown, are positioned to provide stereo sound reception. Stereo sound reception can be used to determine the location of a sound source. The wearable system 200 can perform voice or speech recognition on the audio stream.

[0074] The wearable system 200 can include an outward-facing imaging system 464 (shown in FIG. 4) which observes the world in the environment around the user. The wearable system 200 can also include an inward-facing imaging system 462 (shown in FIG. 4) which can track the eye movements of the user. The inward-facing imaging system may track either one eye's movements or both eyes' movements. The inward-facing imaging system 462 may be attached to the frame 230 and may be in electrical communication with the processing modules 260 or 270, which may process image information acquired by the inward-facing imaging system to determine, e.g., the pupil diameters or orientations of the eyes, eye movements, or eye pose of the user 210. The inward-facing imaging system 462 may include one or more cameras. For example, at least one camera may be used to image each eye. The images acquired by the cameras may be used to determine pupil size or eye pose for each eye separately, thereby allowing presentation of image information to each eye to be dynamically tailored to that eye.

[0075] As an example, the wearable system 200 can use the outward-facing imaging system 464 or the inward-facing imaging system 462 to acquire images of a pose of the user. The images may be still images, frames of a video, or a video.

[0076] The display 220 can be operatively coupled 250, such as by a wired lead or wireless connectivity, to a local data processing module 260 which may be mounted in a variety of configurations, such as fixedly attached to the frame 230, fixedly attached to a helmet or hat worn by the user, embedded in headphones, or otherwise removably attached to the user 210 (e.g., in a backpack-style configuration, in a belt-coupling style configuration).

[0077] The local processing and data module 260 may comprise a hardware processor, as well as digital memory, such as non-volatile memory (e.g., flash memory), both of which may be utilized to assist in the processing, caching, and storage of data. The data may include data a) captured from sensors (which may be, e.g., operatively coupled to the frame 230 or otherwise attached to the user 210), such as image capture devices (e.g., cameras in the inward-facing imaging system or the outward-facing imaging system), audio sensors (e.g., microphones), inertial measurement units (IMUs), accelerometers, compasses, global positioning system (GPS) units, radio devices, or gyroscopes; or b) acquired or processed using remote processing module 270 or remote data repository 280, possibly for passage to the display 220 after such processing or retrieval. The local processing and data module 260 may be operatively coupled by communication links 262 or 264, such as via wired or wireless communication links, to the remote processing

module 270 or remote data repository 280 such that these remote modules are available as resources to the local processing and data module 260. In addition, remote processing module 280 and remote data repository 280 may be operatively coupled to each other.

[0078] In some embodiments, the remote processing module 270 may comprise one or more processors configured to analyze and process data or image information. In some embodiments, the remote data repository 280 may comprise a digital data storage facility, which may be available through the internet or other networking configuration in a "cloud" resource configuration. In some embodiments, all data is stored and all computations are performed in the local processing and data module, allowing fully autonomous use from a remote module.

Example Components of a Wearable System

[0079] FIG. 3 schematically illustrates example components of a wearable system. FIG. 3 shows a wearable system 200 which can include a display 220 and a frame 230. A blown-up view 202 schematically illustrates various components of the wearable system 200. In certain implementations, one or more of the components illustrated in FIG. 3 can be part of the display 220. The various components alone or in combination can collect a variety of data (such as e.g., audio or visual data) associated with the user of the wearable system 200 or the user's environment. It should be appreciated that other embodiments may have additional or fewer components depending on the application for which the wearable system is used. Nevertheless, FIG. 3 provides a basic idea of some of the various components and types of data that may be collected, analyzed, and stored through the wearable system.

[0080] FIG. 3 shows an example wearable system 200 which can include the display 220. The display 220 can comprise a display lens 226 that may be mounted to a user's head or a housing or frame 230, which corresponds to the frame 230. The display lens 226 may comprise one or more transparent mirrors positioned by the housing 230 in front of the user's eyes 302, 304 and may be configured to bounce projected light 338 into the eyes 302, 304 and facilitate beam shaping, while also allowing for transmission of at least some light from the local environment. The wavefront of the projected light beam 338 may be bent or focused to coincide with a desired focal distance of the projected light. As illustrated, two wide-field-of-view machine vision cameras 316 (also referred to as world cameras) can be coupled to the housing 230 to image the environment around the user. These cameras 316 can be dual capture visible light/non-visible (e.g., infrared) light cameras. The cameras 316 may be part of the outward-facing imaging system 464 shown in FIG. 4. Image acquired by the world cameras 316 can be processed by the pose processor 336. For example, the pose processor 336 can implement one or more object recognizers 708 (e.g., shown in FIG. 7) to identify a pose of a user or another person in the user's environment or to identify a physical object in the user's environment.

[0081] With continued reference to FIG. 3, a pair of scanned-laser shaped-wavefront (e.g., for depth) light projector modules with display mirrors and optics configured to project light 338 into the eyes 302, 304 are shown. The depicted view also shows two miniature infrared cameras 324 paired with infrared light (such as light emitting diodes "LED"s), which are configured to be able to track the eyes

302, 304 of the user to support rendering and user input. The cameras **324** may be part of the inward-facing imaging system **462** shown in FIG. 4. The wearable system **200** can further feature a sensor assembly **339**, which may comprise X, Y, and Z axis accelerometer capability as well as a magnetic compass and X, Y, and Z axis gyro capability, preferably providing data at a relatively high frequency, such as 200 Hz. The sensor assembly **339** may be part of the IMU described with reference to FIG. 2A. The depicted system **200** can also comprise a head pose processor **336**, such as an ASIC (application specific integrated circuit), FPGA (field programmable gate array), or ARM processor (advanced reduced-instruction-set machine), which may be configured to calculate real or near-real time user head pose from wide field of view image information output from the capture devices **316**. The head pose processor **336** can be a hardware processor and can be implemented as part of the local processing and data module **260** shown in FIG. 2A.

[0082] The wearable system can also include one or more depth sensors **234**. The depth sensor **234** can be configured to measure the distance between an object in an environment to a wearable device. The depth sensor **234** may include a laser scanner (e.g., a LIDAR), an ultrasonic depth sensor, or a depth sensing camera. In certain implementations, where the cameras **316** have depth sensing ability, the cameras **316** may also be considered as depth sensors **234**.

[0083] Also shown is a processor **332** configured to execute digital or analog processing to derive pose from the gyro, compass, or accelerometer data from the sensor assembly **339**. The processor **332** may be part of the local processing and data module **260** shown in FIG. 2. The wearable system **200** as shown in FIG. 3 can also include a position system such as, e.g., a GPS **337** (global positioning system) to assist with pose and positioning analyses. In addition, the GPS may further provide remotely-based (e.g., cloud-based) information about the user's environment. This information may be used for recognizing objects or information in user's environment.

[0084] The wearable system may combine data acquired by the GPS **337** and a remote computing system (such as, e.g., the remote processing module **270**, another user's ARD, etc.), which can provide more information about the user's environment. As one example, the wearable system can determine the user's location based on GPS data and retrieve a world map (e.g., by communicating with a remote processing module **270**) including virtual objects associated with the user's location. As another example, the wearable system **200** can monitor the environment using the world cameras **316** (which may be part of the outward-facing imaging system **464** shown in FIG. 4). Based on the images acquired by the world cameras **316**, the wearable system **200** can detect objects in the environment (e.g., by using one or more object recognizers **708** shown in FIG. 7). The wearable system can further use data acquired by the GPS **337** to interpret the characters.

[0085] The wearable system **200** may also comprise a rendering engine **334** which can be configured to provide rendering information that is local to the user to facilitate operation of the scanners and imaging into the eyes of the user, for the user's view of the world. The rendering engine **334** may be implemented by a hardware processor (such as, e.g., a central processing unit or a graphics processing unit). In some embodiments, the rendering engine is part of the local processing and data module **260**. The rendering engine

334 can be communicatively coupled (e.g., via wired or wireless links) to other components of the wearable system **200**. For example, the rendering engine **334**, can be coupled to the eye cameras **324** via communication link **274**, and be coupled to a projecting subsystem **318** (which can project light into user's eyes **302, 304** via a scanned laser arrangement in a manner similar to a retinal scanning display) via the communication link **272**. The rendering engine **334** can also be in communication with other processing units such as, e.g., the sensor pose processor **332** and the image pose processor **336** via links **276** and **294** respectively.

[0086] The cameras **324** (e.g., mini infrared cameras) may be utilized to track the eye pose to support rendering and user input. Some example eye poses may include where the user is looking or at what depth he or she is focusing (which may be estimated with eye vergence). The GPS **337**, gyros, compass, and accelerometers **339** may be utilized to provide coarse or fast pose estimates. One or more of the cameras **316** can acquire images and pose, which in conjunction with data from an associated cloud computing resource, may be utilized to map the local environment and share user views with others.

[0087] The example components depicted in FIG. 3 are for illustration purposes only. Multiple sensors and other functional modules are shown together for ease of illustration and description. Some embodiments may include only one or a subset of these sensors or modules. Further, the locations of these components are not limited to the positions depicted in FIG. 3. Some components may be mounted to or housed within other components, such as a belt-mounted component, a hand-held component, or a helmet component. As one example, the image pose processor **336**, sensor pose processor **332**, and rendering engine **334** may be positioned in a belt pack and configured to communicate with other components of the wearable system via wireless communication, such as ultra-wideband, Wi-Fi, Bluetooth, etc., or via wired communication. The depicted housing **230** preferably is head-mountable and wearable by the user. However, some components of the wearable system **200** may be worn to other portions of the user's body. For example, the speaker **240** may be inserted into the ears of a user to provide sound to the user.

[0088] Regarding the projection of light **338** into the eyes **302, 304** of the user, in some embodiment, the cameras **324** may be utilized to measure where the centers of a user's eyes are geometrically verged to, which, in general, coincides with a position of focus, or "depth of focus", of the eyes. A 3-dimensional surface of all points the eyes verge to can be referred to as the "horopter". The focal distance may take on a finite number of depths, or may be infinitely varying. Light projected from the vergence distance appears to be focused to the subject eye **302, 304**, while light in front of or behind the vergence distance is blurred. Examples of wearable devices and other display systems of the present disclosure are also described in U.S. Patent Publication No. 2016/0270656, which is incorporated by reference herein in its entirety.

[0089] The human visual system is complicated and providing a realistic perception of depth is challenging. Viewers of an object may perceive the object as being three-dimensional due to a combination of vergence and accommodation. Vergence movements (e.g., rolling movements of the pupils toward or away from each other to converge the lines of sight of the eyes to fixate upon an object) of the two eyes

relative to each other are closely associated with focusing (or “accommodation”) of the lenses of the eyes. Under normal conditions, changing the focus of the lenses of the eyes, or accommodating the eyes, to change focus from one object to another object at a different distance will automatically cause a matching change in vergence to the same distance, under a relationship known as the “accommodation-vergence reflex.” Likewise, a change in vergence will trigger a matching change in accommodation, under normal conditions. Display systems that provide a better match between accommodation and vergence may form more realistic and comfortable simulations of three-dimensional imagery.

[0090] Further spatially coherent light with a beam diameter of less than about 0.7 millimeters can be correctly resolved by the human eye regardless of where the eye focuses. Thus, to create an illusion of proper focal depth, the eye vergence may be tracked with the cameras **324**, and the rendering engine **334** and projection subsystem **318** may be utilized to render all objects on or close to the horopter in focus, and all other objects at varying degrees of defocus (e.g., using intentionally-created blurring). Preferably, the system **220** renders to the user at a frame rate of about 60 frames per second or greater. As described above, preferably, the cameras **324** may be utilized for eye tracking, and software may be configured to pick up not only vergence geometry but also focus location cues to serve as user inputs. Preferably, such a display system is configured with brightness and contrast suitable for day or night use.

[0091] In some embodiments, the display system preferably has latency of less than about 20 milliseconds for visual object alignment, less than about 0.1 degree of angular alignment, and about 1 arc minute of resolution, which, without being limited by theory, is believed to be approximately the limit of the human eye. The display system **220** may be integrated with a localization system, which may involve GPS elements, optical tracking, compass, accelerometers, or other data sources, to assist with position and pose determination; localization information may be utilized to facilitate accurate rendering in the user’s view of the pertinent world (e.g., such information would facilitate the glasses to know where they are with respect to the real world).

[0092] In some embodiments, the wearable system **200** is configured to display one or more virtual images based on the accommodation of the user’s eyes. Unlike prior 3D display approaches that force the user to focus where the images are being projected, in some embodiments, the wearable system is configured to automatically vary the focus of projected virtual content to allow for a more comfortable viewing of one or more images presented to the user. For example, if the user’s eyes have a current focus of 1 m, the image may be projected to coincide with the user’s focus. If the user shifts focus to 3 m, the image is projected to coincide with the new focus. Thus, rather than forcing the user to a predetermined focus, the wearable system **200** of some embodiments allows the user’s eye to function in a more natural manner.

[0093] Such a wearable system **200** may eliminate or reduce the incidences of eye strain, headaches, and other physiological symptoms typically observed with respect to virtual reality devices. To achieve this, various embodiments of the wearable system **200** are configured to project virtual images at varying focal distances, through one or more

variable focus elements (VFEs). In one or more embodiments, 3D perception may be achieved through a multi-plane focus system that projects images at fixed focal planes away from the user. Other embodiments employ variable plane focus, wherein the focal plane is moved back and forth in the z-direction to coincide with the user’s present state of focus.

[0094] In both the multi-plane focus systems and variable plane focus systems, wearable system **200** may employ eye tracking to determine a vergence of the user’s eyes, determine the user’s current focus, and project the virtual image at the determined focus. In other embodiments, wearable system **200** comprises a light modulator that variably projects, through a fiber scanner, or other light generating source, light beams of varying focus in a raster pattern across the retina. Thus, the ability of the display of the wearable system **200** to project images at varying focal distances not only eases accommodation for the user to view objects in 3D, but may also be used to compensate for user ocular anomalies, as further described in U.S. Patent Publication No. 2016/0270656, which is incorporated by reference herein in its entirety. In some other embodiments, a spatial light modulator may project the images to the user through various optical components. For example, as described further below, the spatial light modulator may project the images onto one or more waveguides, which then transmit the images to the user.

Waveguide Stack Assembly

[0095] FIG. 4 illustrates an example of a waveguide stack for outputting image information to a user. A wearable system **400** includes a stack of waveguides, or stacked waveguide assembly **480** that may be utilized to provide three-dimensional perception to the eye/brain using a plurality of waveguides **432b**, **434b**, **436b**, **438b**, **4400b**. In some embodiments, the wearable system **400** may correspond to wearable system **200** of FIG. 2, with FIG. 4 schematically showing some parts of that wearable system **200** in greater detail. For example, in some embodiments, the waveguide assembly **480** may be integrated into the display **220** of FIG. 2.

[0096] With continued reference to FIG. 4, the waveguide assembly **480** may also include a plurality of features **458**, **456**, **454**, **452** between the waveguides. In some embodiments, the features **458**, **456**, **454**, **452** may be lenses. In other embodiments, the features **458**, **456**, **454**, **452** may not be lenses. Rather, they may simply be spacers (e.g., cladding layers or structures for forming air gaps).

[0097] The waveguides **432b**, **434b**, **436b**, **438b**, **4400b** or the plurality of lenses **458**, **456**, **454**, **452** may be configured to send image information to the eye with various levels of wavefront curvature or light ray divergence. Each waveguide level may be associated with a particular depth plane and may be configured to output image information corresponding to that depth plane. Image injection devices **420**, **422**, **424**, **426**, **428** may be utilized to inject image information into the waveguides **440b**, **438b**, **436b**, **434b**, **432b**, each of which may be configured to distribute incoming light across each respective waveguide, for output toward the eye **410**. Light exits an output surface of the image injection devices **420**, **422**, **424**, **426**, **428** and is injected into a corresponding input edge of the waveguides **440b**, **438b**, **436b**, **434b**, **432b**. In some embodiments, a single beam of light (e.g., a collimated beam) may be injected into each

waveguide to output an entire field of cloned collimated beams that are directed toward the eye **410** at particular angles (and amounts of divergence) corresponding to the depth plane associated with a particular waveguide.

[0098] In some embodiments, the image injection devices **420**, **422**, **424**, **426**, **428** are discrete displays that each produce image information for injection into a corresponding waveguide **440b**, **438b**, **436b**, **434b**, **432b**, respectively. In some other embodiments, the image injection devices **420**, **422**, **424**, **426**, **428** are the output ends of a single multiplexed display which may, e.g., pipe image information via one or more optical conduits (such as fiber optic cables) to each of the image injection devices **420**, **422**, **424**, **426**, **428**.

[0099] A controller **460** controls the operation of the stacked waveguide assembly **480** and the image injection devices **420**, **422**, **424**, **426**, **428**. The controller **460** includes programming (e.g., instructions in a non-transitory computer-readable medium) that regulates the timing and provision of image information to the waveguides **440b**, **438b**, **436b**, **434b**, **432b**. In some embodiments, the controller **460** may be a single integral device, or a distributed system connected by wired or wireless communication channels. The controller **460** may be part of the processing modules **260** or **270** (illustrated in FIG. 2) in some embodiments.

[0100] The waveguides **440b**, **438b**, **436b**, **434b**, **432b** may be configured to propagate light within each respective waveguide by total internal reflection (TIR). The waveguides **440b**, **438b**, **436b**, **434b**, **432b** may each be planar or have another shape (e.g., curved), with major top and bottom surfaces and edges extending between those major top and bottom surfaces. In the illustrated configuration, the waveguides **440b**, **438b**, **436b**, **434b**, **432b** may each include light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** that are configured to extract light out of a waveguide by redirecting the light, propagating within each respective waveguide, out of the waveguide to output image information to the eye **410**. Extracted light may also be referred to as outcoupled light, and light extracting optical elements may also be referred to as outcoupling optical elements. An extracted beam of light is outputted by the waveguide at locations at which the light propagating in the waveguide strikes a light redirecting element. The light extracting optical elements (**440a**, **438a**, **436a**, **434a**, **432a**) may, for example, be reflective or diffractive optical features. While illustrated disposed at the bottom major surfaces of the waveguides **440b**, **438b**, **436b**, **434b**, **432b** for ease of description and drawing clarity, in some embodiments, the light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** may be disposed at the top or bottom major surfaces, or may be disposed directly in the volume of the waveguides **440b**, **438b**, **436b**, **434b**, **432b**. In some embodiments, the light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** may be formed in a layer of material that is attached to a transparent substrate to form the waveguides **440b**, **438b**, **436b**, **434b**, **432b**. In some other embodiments, the waveguides **440b**, **438b**, **436b**, **434b**, **432b** may be a monolithic piece of material and the light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** may be formed on a surface or in the interior of that piece of material.

[0101] With continued reference to FIG. 4, as discussed herein, each waveguide **440b**, **438b**, **436b**, **434b**, **432b** is configured to output light to form an image corresponding to a particular depth plane. For example, the waveguide **432b**

nearest the eye may be configured to deliver collimated light, as injected into such waveguide **432b**, to the eye **410**. The collimated light may be representative of the optical infinity focal plane. The next waveguide up **434b** may be configured to send out collimated light which passes through the first lens **452** (e.g., a negative lens) before it can reach the eye **410**. First lens **452** may be configured to create a slight convex wavefront curvature so that the eye/brain interprets light coming from that next waveguide up **434b** as coming from a first focal plane closer inward toward the eye **410** from optical infinity. Similarly, the third up waveguide **436b** passes its output light through both the first lens **452** and second lens **454** before reaching the eye **410**. The combined optical power of the first and second lenses **452** and **454** may be configured to create another incremental amount of wavefront curvature, so that the eye/brain interprets light coming from the third waveguide **436b** as coming from a second focal plane that is even closer inward toward the person from optical infinity than was light from the next waveguide up **434b**.

[0102] The other waveguide layers (e.g., waveguides **438b**, **440b**) and lenses (e.g., lenses **456**, **458**) are similarly configured, with the highest waveguide **440b** in the stack sending its output through all of the lenses between it and the eye for an aggregate focal power representative of the closest focal plane to the person. To compensate for the stack of lenses **458**, **456**, **454**, **452** when viewing/interpreting light coming from the world **470** on the other side of the stacked waveguide assembly **480**, a compensating lens layer **430** may be disposed at the top of the stack to compensate for the aggregate power of the lens stack **458**, **456**, **454**, **452** below. Such a configuration provides as many perceived focal planes as there are available waveguide/lens pairings. Both the light extracting optical elements of the waveguides and the focusing aspects of the lenses may be static (e.g., not dynamic or electro-active). In some alternative embodiments, either or both may be dynamic using electro-active features.

[0103] With continued reference to FIG. 4, the light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** may be configured to both redirect light out of their respective waveguides and to output this light with the appropriate amount of divergence or collimation for a particular depth plane associated with the waveguide. As a result, waveguides having different associated depth planes may have different configurations of light extracting optical elements, which output light with a different amount of divergence depending on the associated depth plane. In some embodiments, as discussed herein, the light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** may be volumetric or surface features, which may be configured to output light at specific angles. For example, the light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** may be volume holograms, surface holograms, and/or diffraction gratings. Light extracting optical elements, such as diffraction gratings, are described in U.S. Patent Publication No. 2015/0178939, published Jun. 25, 2015, which is incorporated by reference herein in its entirety.

[0104] In some embodiments, the light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** are diffractive features that form a diffraction pattern, or “diffractive optical element” (also referred to herein as a “DOE”). Preferably, the DOE has a relatively low diffraction efficiency so that only a portion of the light of the beam is deflected away

toward the eye **410** with each intersection of the DOE, while the rest continues to move through a waveguide via total internal reflection. The light carrying the image information can thus be divided into a number of related exit beams that exit the waveguide at a multiplicity of locations and the result is a fairly uniform pattern of exit emission toward the eye **304** for this particular collimated beam bouncing around within a waveguide.

[0105] In some embodiments, one or more DOEs may be switchable between “on” state in which they actively diffract, and “off” state in which they do not significantly diffract. For instance, a switchable DOE may comprise a layer of polymer dispersed liquid crystal, in which microdroplets comprise a diffraction pattern in a host medium, and the refractive index of the microdroplets can be switched to substantially match the refractive index of the host material (in which case the pattern does not appreciably diffract incident light) or the microdroplet can be switched to an index that does not match that of the host medium (in which case the pattern actively diffracts incident light).

[0106] In some embodiments, the number and distribution of depth planes or depth of field may be varied dynamically based on the pupil sizes or orientations of the eyes of the viewer. Depth of field may change inversely with a viewer’s pupil size. As a result, as the sizes of the pupils of the viewer’s eyes decrease, the depth of field increases such that one plane that is not discernible because the location of that plane is beyond the depth of focus of the eye may become discernible and appear more in focus with reduction of pupil size and commensurate with the increase in depth of field. Likewise, the number of spaced apart depth planes used to present different images to the viewer may be decreased with the decreased pupil size. For example, a viewer may not be able to clearly perceive the details of both a first depth plane and a second depth plane at one pupil size without adjusting the accommodation of the eye away from one depth plane and to the other depth plane. These two depth planes may, however, be sufficiently in focus at the same time to the user at another pupil size without changing accommodation.

[0107] In some embodiments, the display system may vary the number of waveguides receiving image information based upon determinations of pupil size or orientation, or upon receiving electrical signals indicative of particular pupil size or orientation. For example, if the user’s eyes are unable to distinguish between two depth planes associated with two waveguides, then the controller **460** (which may be an embodiment of the local processing and data module **260**) can be configured or programmed to cease providing image information to one of these waveguides. Advantageously, this may reduce the processing burden on the system, thereby increasing the responsiveness of the system. In embodiments in which the DOEs for a waveguide are switchable between the on and off states, the DOEs may be switched to the off state when the waveguide does receive image information.

[0108] In some embodiments, it may be desirable to have an exit beam meet the condition of having a diameter that is less than the diameter of the eye of a viewer. However, meeting this condition may be challenging in view of the variability in size of the viewer’s pupils. In some embodiments, this condition is met over a wide range of pupil sizes by varying the size of the exit beam in response to determinations of the size of the viewer’s pupil. For example, as the pupil size decreases, the size of the exit beam may also

decrease. In some embodiments, the exit beam size may be varied using a variable aperture.

[0109] The wearable system **400** can include an outward-facing imaging system **464** (e.g., a digital camera) that images a portion of the world **470**. This portion of the world **470** may be referred to as the field of view (FOV) of a world camera and the imaging system **464** is sometimes referred to as an FOV camera. The FOV of the world camera may or may not be the same as the FOV of a viewer **210** which encompasses a portion of the world **470** the viewer **210** perceives at a given time. For example, in some situations, the FOV of the world camera may be larger than the viewer **210** of the viewer **210** of the wearable system **400**. The entire region available for viewing or imaging by a viewer may be referred to as the field of regard (FOR). The FOR may include $4\times$ steradians of solid angle surrounding the wearable system **400** because the wearer can move his body, head, or eyes to perceive substantially any direction in space. In other contexts, the wearer’s movements may be more constricted, and accordingly the wearer’s FOR may subtend a smaller solid angle. Images obtained from the outward-facing imaging system **464** can be used to track gestures made by the user (e.g., hand or finger gestures), detect objects in the world **470** in front of the user, and so forth.

[0110] The wearable system **400** can include an audio sensor **232**, e.g., a microphone, to capture ambient sound. As described above, in some embodiments, one or more other audio sensors can be positioned to provide stereo sound reception useful to the determination of location of a speech source. The audio sensor **232** can comprise a directional microphone, as another example, which can also provide such useful directional information as to where the audio source is located. The wearable system **400** can use information from both the outward-facing imaging system **464** and the audio sensor **230** in locating a source of speech, or to determine an active speaker at a particular moment in time, etc. For example, the wearable system **400** can use the voice recognition alone or in combination with a reflected image of the speaker (e.g., as seen in a mirror) to determine the identity of the speaker. As another example, the wearable system **400** can determine a position of the speaker in an environment based on sound acquired from directional microphones. The wearable system **400** can parse the sound coming from the speaker’s position with speech recognition algorithms to determine the content of the speech and use voice recognition techniques to determine the identity (e.g., name or other demographic information) of the speaker.

[0111] The wearable system **400** can also include an inward-facing imaging system **466** (e.g., a digital camera), which observes the movements of the user, such as the eye movements and the facial movements. The inward-facing imaging system **466** may be used to capture images of the eye **410** to determine the size and/or orientation of the pupil of the eye **304**. The inward-facing imaging system **466** can be used to obtain images for use in determining the direction the user is looking (e.g., eye pose) or for biometric identification of the user (e.g., via iris identification). In some embodiments, at least one camera may be utilized for each eye, to separately determine the pupil size or eye pose of each eye independently, thereby allowing the presentation of image information to each eye to be dynamically tailored to that eye. In some other embodiments, the pupil diameter or orientation of only a single eye **410** (e.g., using only a single

camera per pair of eyes) is determined and assumed to be similar for both eyes of the user. The images obtained by the inward-facing imaging system 466 may be analyzed to determine the user's eye pose or mood, which can be used by the wearable system 400 to decide which audio or visual content should be presented to the user. The wearable system 400 may also determine head pose (e.g., head position or head orientation) using a pose sensor, e.g., sensors such as IMUs, accelerometers, gyroscopes, etc.

[0112] The wearable system 400 can include a user input device 466 by which the user can input commands to the controller 460 to interact with the wearable system 400. For example, the user input device 466 can include a trackpad, a touchscreen, a joystick, a multiple degree-of-freedom (DOF) controller, a capacitive sensing device, a game controller, a keyboard, a mouse, a directional pad (D-pad), a wand, a haptic device, a totem (e.g., functioning as a virtual user input device), and so forth. A multi-DOF controller can sense user input in some or all possible translations (e.g., left/right, forward/backward, or up/down) or rotations (e.g., yaw, pitch, or roll) of the controller. A multi-DOF controller which supports the translation movements may be referred to as a 3DOF while a multi-DOF controller which supports the translations and rotations may be referred to as 6DOF. In some cases, the user may use a finger (e.g., a thumb) to press or swipe on a touch-sensitive input device to provide input to the wearable system 400 (e.g., to provide user input to a user interface provided by the wearable system 400). The user input device 466 may be held by the user's hand during the use of the wearable system 400. The user input device 466 can be in wired or wireless communication with the wearable system 400.

Other Components of the Wearable System

[0113] In many implementations, the wearable system may include other components in addition or in alternative to the components of the wearable system described above. The wearable system may, for example, include one or more haptic devices or components. The haptic devices or components may be operable to provide a tactile sensation to a user. For example, the haptic devices or components may provide a tactile sensation of pressure or texture when touching virtual content (e.g., virtual objects, virtual tools, other virtual constructs). The tactile sensation may replicate a feel of a physical object which a virtual object represents, or may replicate a feel of an imagined object or character (e.g., a dragon) which the virtual content represents. In some implementations, haptic devices or components may be worn by the user (e.g., a user wearable glove). In some implementations, haptic devices or components may be held by the user.

[0114] The wearable system may, for example, include one or more physical objects which are manipulable by the user to allow input or interaction with the wearable system. These physical objects may be referred to herein as totems. Some totems may take the form of inanimate objects, such as for example, a piece of metal or plastic, a wall, a surface of table. In certain implementations, the totems may not actually have any physical input structures (e.g., keys, triggers, joystick, trackball, rocker switch). Instead, the totem may simply provide a physical surface, and the wearable system may render a user interface so as to appear to a user to be on one or more surfaces of the totem. For example, the wearable system may render an image of a

computer keyboard and trackpad to appear to reside on one or more surfaces of a totem. For example, the wearable system may render a virtual computer keyboard and virtual trackpad to appear on a surface of a thin rectangular plate of aluminum, which can serve as a totem. The rectangular plate does not itself have any physical keys or trackpad or sensors. However, the wearable system may detect user manipulation or interaction or touches with the rectangular plate as selections or inputs made via the virtual keyboard or virtual trackpad. The user input device 466 (shown in FIG. 4) may be an embodiment of a totem, which may include a trackpad, a touchpad, a trigger, a joystick, a trackball, a rocker or virtual switch, a mouse, a keyboard, a multi-degree-of-freedom controller, or another physical input device. A user may use the totem, alone or in combination with poses, to interact with the wearable system or other users.

[0115] Examples of haptic devices and totems usable with the wearable devices, HMD, and display systems of the present disclosure are described in U.S. Patent Publication No. 2015/0016777, which is incorporated by reference herein in its entirety.

Example Processes of User Interactions with a Wearable System

[0116] FIG. 5 is a process flow diagram of an example of a method 500 for interacting with a virtual user interface. The method 500 may be performed by the wearable system described herein. Embodiments of the method 500 can be used by the wearable system to detect persons or documents in the FOV of the wearable system.

[0117] At block 510, the wearable system may identify a particular UI. The type of UI may be predetermined by the user. The wearable system may identify that a particular UI needs to be populated based on a user input (e.g., gesture, visual data, audio data, sensory data, direct command, etc.). The UI can be specific to a security scenario where the wearer of the system is observing users who present documents to the wearer (e.g., at a travel checkpoint). At block 520, the wearable system may generate data for the virtual UI. For example, data associated with the confines, general structure, shape of the UI etc., may be generated. In addition, the wearable system may determine map coordinates of the user's physical location so that the wearable system can display the UI in relation to the user's physical location. For example, if the UI is body centric, the wearable system may determine the coordinates of the user's physical stance, head pose, or eye pose such that a ring UI can be displayed around the user or a planar UI can be displayed on a wall or in front of the user. In the security context described herein, the UI may be displayed as if the UI were surrounding the traveler who is presenting documents to the wearer of the system, so that the wearer can readily view the UI while looking at the traveler and the traveler's documents. If the UI is hand centric, the map coordinates of the user's hands may be determined. These map points may be derived through data received through the FOV cameras, sensory input, or any other type of collected data.

[0118] At block 530, the wearable system may send the data to the display from the cloud or the data may be sent from a local database to the display components. At block 540, the UI is displayed to the user based on the sent data. For example, a light field display can project the virtual UI into one or both of the user's eyes. Once the virtual UI has been created, the wearable system may simply wait for a

command from the user to generate more virtual content on the virtual UI at block 550. For example, the UI may be a body centric ring around the user's body or the body of a person in the user's environment (e.g., a traveler). The wearable system may then wait for the command (a gesture, a head or eye movement, voice command, input from a user input device, etc.), and if it is recognized (block 560), virtual content associated with the command may be displayed to the user (block 570).

Examples of Avatar Rendering in Mixed Reality

[0119] A wearable system may employ various mapping related techniques in order to achieve high depth of field in the rendered light fields. In mapping out the virtual world, it is advantageous to know all the features and points in the real world to accurately portray virtual objects in relation to the real world. To this end, FOV images captured from users of the wearable system can be added to a world model by including new pictures that convey information about various points and features of the real world. For example, the wearable system can collect a set of map points (such as 2D points or 3D points) and find new map points to render a more accurate version of the world model. The world model of a first user can be communicated (e.g., over a network such as a cloud network) to a second user so that the second user can experience the world surrounding the first user.

[0120] FIG. 6A is a block diagram of another example of a wearable system which can comprise an avatar processing and rendering system 690 in a mixed reality environment. The wearable system 600 may be part of the wearable system 200 shown in FIG. 2. In this example, the wearable system 600 can comprise a map 620, which may include at least a portion of the data in the map database 710 (shown in FIG. 7). The map may partly reside locally on the wearable system, and may partly reside at networked storage locations accessible by wired or wireless network (e.g., in a cloud system). A pose process 610 may be executed on the wearable computing architecture (e.g., processing module 260 or controller 460) and utilize data from the map 620 to determine position and orientation of the wearable computing hardware or user. Pose data may be computed from data collected on the fly as the user is experiencing the system and operating in the world. The data may comprise images, data from sensors (such as inertial measurement units, which generally comprise accelerometer and gyroscope components) and surface information pertinent to objects in the real or virtual environment.

[0121] A sparse point representation may be the output of a simultaneous localization and mapping (e.g., SLAM or vSLAM, referring to a configuration wherein the input is images/visual only) process. The system can be configured to not only find out where in the world the various components are, but what the world is made of. Pose may be a building block that achieves many goals, including populating the map and using the data from the map.

[0122] In one embodiment, a sparse point position may not be completely adequate on its own, and further information may be needed to produce a multifocal AR, VR, or MR experience. Dense representations, generally referring to depth map information, may be utilized to fill this gap at least in part. Such information may be computed from a process referred to as Stereo 640, wherein depth information is determined using a technique such as triangulation or time-of-flight sensing. Image information and active pat-

terns (such as infrared patterns created using active projectors), images acquired from image cameras, or hand gestures/totem 650 may serve as input to the Stereo process 640. A significant amount of depth map information may be fused together, and some of this may be summarized with a surface representation. For example, mathematically definable surfaces may be efficient (e.g., relative to a large point cloud) and digestible inputs to other processing devices like game engines. Thus, the output of the stereo process (e.g., a depth map) 640 may be combined in the fusion process 630. Pose 610 may be an input to this fusion process 630 as well, and the output of fusion 630 becomes an input to populating the map process 620. Sub-surfaces may connect with each other, such as in topographical mapping, to form larger surfaces, and the map becomes a large hybrid of points and surfaces.

[0123] To resolve various aspects in a mixed reality process 660, various inputs may be utilized. For example, in the embodiment depicted in FIG. 6A, Game parameters may be inputs to determine that the user of the system is playing a monster battling game with one or more monsters at various locations, monsters dying or running away under various conditions (such as if the user shoots the monster), walls or other objects at various locations, and the like. The world map may include information regarding the location of the objects or semantic information of the objects (e.g., classifications such as whether the object is flat or round, horizontal or vertical, a table or a lamp, etc.) and the world map can be another valuable input to mixed reality. Pose relative to the world becomes an input as well and plays a key role to almost any interactive system.

[0124] Controls or inputs from the user are another input to the wearable system 600. As described herein, user inputs can include visual input, gestures, totems, audio input, sensory input, etc. In order to move around or play a game, for example, the user may need to instruct the wearable system 600 regarding what he or she wants to do. Beyond just moving oneself in space, there are various forms of user controls that may be utilized. In one embodiment, a totem (e.g. a user input device), or an object such as a toy gun may be held by the user and tracked by the system. The system preferably will be configured to know that the user is holding the item and understand what kind of interaction the user is having with the item (e.g., if the totem or object is a gun, the system may be configured to understand location and orientation, as well as whether the user is clicking a trigger or other sensed button or element which may be equipped with a sensor, such as an IMU, which may assist in determining what is going on, even when such activity is not within the field of view of any of the cameras.)

[0125] Hand gesture tracking or recognition may also provide input information. The wearable system 600 may be configured to track and interpret hand gestures for button presses, for gesturing left or right, stop, grab, hold, etc. For example, in one configuration, the user may want to flip through emails or a calendar in a non-gaming environment, or do a "fist bump" with another person or player. The wearable system 600 may be configured to leverage a minimum amount of hand gesture, which may or may not be dynamic. For example, the gestures may be simple static gestures like open hand for stop, thumbs up for ok, thumbs down for not ok; or a hand flip right, or left, or up/down for directional commands.

[0126] Eye tracking is another input (e.g., tracking where the user is looking to control the display technology to render at a specific depth or range). In one embodiment, vergence of the eyes may be determined using triangulation, and then using a vergence/accommodation model developed for that particular person, accommodation may be determined. Eye tracking can be performed by the eye camera(s) to determine eye gaze (e.g., direction or orientation of one or both eyes). Other techniques can be used for eye tracking such as, e.g., measurement of electrical potentials by electrodes placed near the eye(s) (e.g., electrooculography).

[0127] Speech tracking can be another input can be used alone or in combination with other inputs (e.g., totem tracking, eye tracking, gesture tracking, etc.). Speech tracking may include speech recognition, voice recognition, alone or in combination. The system 600 can include an audio sensor (e.g., a microphone) that receives an audio stream from the environment. The system 600 can incorporate voice recognition technology to determine who is speaking (e.g., whether the speech is from the wearer of the ARD or another person or voice (e.g., a recorded voice transmitted by a loudspeaker in the environment)) as well as speech recognition technology to determine what is being said. The local data & processing module 260 or the remote processing module 270 can process the audio data from the microphone (or audio data in another stream such as, e.g., a video stream being watched by the user) to identify content of the speech by applying various speech recognition algorithms, such as, e.g., hidden Markov models, dynamic time warping (DTW)-based speech recognitions, neural networks, deep learning algorithms such as deep feedforward and recurrent neural networks, end-to-end automatic speech recognitions, machine learning algorithms (described with reference to FIG. 7), or other algorithms that uses acoustic modeling or language modeling, etc.

[0128] The local data & processing module 260 or the remote processing module 270 can also apply voice recognition algorithms which can identify the identity of the speaker, such as whether the speaker is the user 210 of the wearable system 600 or another person with whom the user is conversing. Some example voice recognition algorithms can include frequency estimation, hidden Markov models, Gaussian mixture models, pattern matching algorithms, neural networks, matrix representation, Vector Quantization, speaker diarisation, decision trees, and dynamic time warping (DTW) technique. Voice recognition techniques can also include anti-speaker techniques, such as cohort models, and world models. Spectral features may be used in representing speaker characteristics. The local data & processing module or the remote data processing module 270 can use various machine learning algorithms described with reference to FIG. 7 to perform the voice recognition.

[0129] An implementation of a wearable system can use these user controls or inputs via a UI. UI elements (e.g., controls, popup windows, bubbles, data entry fields, etc.) can be used, for example, to dismiss a display of information, e.g., graphics or semantic information of an object.

[0130] With regard to the camera systems, the example wearable system 600 shown in FIG. 6A can include three pairs of cameras: a relative wide FOV or passive SLAM pair of cameras arranged to the sides of the user's face, a different pair of cameras oriented in front of the user to handle the stereo imaging process 640 and also to capture hand gestures and totem/object tracking in front of the user's face. The

FOV cameras and the pair of cameras for the stereo process 640 may be a part of the outward-facing imaging system 464 (shown in FIG. 4). The wearable system 600 can include eye tracking cameras (which may be a part of an inward-facing imaging system 462 shown in FIG. 4) oriented toward the eyes of the user in order to triangulate eye vectors and other information. The wearable system 600 may also comprise one or more textured light projectors (such as infrared (IR) projectors) to inject texture into a scene.

[0131] The wearable system 600 can comprise an avatar processing and rendering system 690. The avatar processing and rendering system 690 can be configured to generate, update, animate, and render an avatar based on contextual information. Some or all of the avatar processing and rendering system 690 can be implemented as part of the local processing and data module 260 or the remote processing module 262, 264 alone or in combination. In various embodiments, multiple avatar processing and rendering systems 690 (e.g., as implemented on different wearable devices) can be used for rendering the virtual avatar 670. For example, a first user's wearable device may be used to determine the first user's intent, while a second user's wearable device can determine an avatar's characteristics and render the avatar of the first user based on the intent received from the first user's wearable device. The first user's wearable device and the second user's wearable device (or other such wearable devices) can communicate via a network, for example, as will be described with reference to FIGS. 9A and 9B.

[0132] FIG. 6B illustrates an example avatar processing and rendering system 690. The example avatar processing and rendering system 690 can comprise a 3D model processing system 680, a contextual information analysis system 688, an avatar autoscaler 692, an intent mapping system 694, an anatomy adjustment system 698, a stimuli response system 696, alone or in combination. The system 690 is intended to illustrate functionalities for avatar processing and rendering and is not intended to be limiting. For example, in certain implementations, one or more of these systems may be part of another system. For example, portions of the contextual information analysis system 688 may be part of the avatar autoscaler 692, intent mapping system 694, stimuli response system 696, or anatomy adjustment system 698, individually or in combination.

[0133] The contextual information analysis system 688 can be configured to determine environment and object information based on one or more device sensors described with reference to FIGS. 2 and 3. For example, the contextual information analysis system 688 can analyze environments and objects (including physical or virtual objects) of a user's environment or an environment in which the user's avatar is rendered, using images acquired by the outward-facing imaging system 464 of the user or the viewer of the user's avatar. The contextual information analysis system 688 can analyze such images alone or in combination with a data acquired from location data or world maps (e.g., maps 620, 710, 910) to determine the location and layout of objects in the environments. The contextual information analysis system 688 can also access biological features of the user or human in general for animating the virtual avatar 670 realistically. For example, the contextual information analysis system 688 can generate a discomfort curve which can be applied to the avatar such that a portion of the user's avatar's body (e.g., the head) is not at an uncomfortable (or unreal-

istic) position with respect to the other portions of the user's body (e.g., the avatar's head is not turned 270 degrees). In certain implementations, one or more object recognizers 708 (shown in FIG. 7) may be implemented as part of the contextual information analysis system 688.

[0134] The avatar autoscaler 692, the intent mapping system 694, and the stimuli response system 696, and anatomy adjustment system 698 can be configured to determine the avatar's characteristics based on contextual information. Some example characteristics of the avatar can include the size, appearance, position, orientation, movement, pose, expression, etc. The avatar autoscaler 692 can be configured to automatically scale the avatar such that the user does not have to look at the avatar at an uncomfortable pose. For example, the avatar autoscaler 692 can increase or decrease the size of the avatar to bring the avatar to the user's eye level such that the user does not need to look down at the avatar or look up at the avatar respectively. The intent mapping system 694 can determine an intent of a user's interaction and map the intent to an avatar (rather than the exact user interaction) based on the environment that the avatar is rendered in. For example, an intent of a first user may be to communicate with a second user in a telepresence session (see, e.g., FIG. 9B). Typically, two people face each other when communicating. The intent mapping system 694 of the first user's wearable system can determine that such a face-to-face intent exists during the telepresence session and can cause the first user's wearable system to render the second user's avatar to be facing the first user. If the second user were to physically turn around, instead of rendering the second user's avatar in a turned position (which would cause the back of the second user's avatar to be rendered to the first user), the first user's intent mapping system 694 can continue to render the second avatar's face to the first user, which is the inferred intent of the telepresence session (e.g., face-to-face intent in this example).

[0135] The stimuli response system 696 can identify an object of interest in the environment and determine an avatar's response to the object of interest. For example, the stimuli response system 696 can identify a sound source in an avatar's environment and automatically turn the avatar to look at the sound source. The stimuli response system 696 can also determine a threshold termination condition. For example, the stimuli response system 696 can cause the avatar to go back to its original pose after the sound source disappears or after a period of time has elapsed.

[0136] The anatomy adjustment system 698 can be configured to adjust the user's pose based on biological features. For example, the anatomy adjustment system 698 can be configured to adjust relative positions between the user's head and the user's torso or between the user's upper body and lower body based on a discomfort curve (see, e.g., the description with reference to FIGS. 38A-39).

[0137] The 3D model processing system 680 can be configured to animate and cause the display 220 to render a virtual avatar 670. The 3D model processing system 680 can include a virtual character processing system 682 and a movement processing system 684. The virtual character processing system 682 can be configured to generate and update a 3D model of a user (for creating and animating the virtual avatar). The movement processing system 684 can be configured to animate the avatar, such as, e.g., by changing the avatar's pose, by moving the avatar around in a user's environment, or by animating the avatar's facial expres-

sions, etc. As will further be described with reference to FIG. 10, the virtual avatar can be animated using rigging techniques (e.g., skeletal system or blendshape animation techniques) where an avatar is represented in two parts: a surface representation (e.g., a deformable mesh) that is used to render the outward appearance of the virtual avatar and a hierarchical set of interconnected joints (e.g., a skeleton) for animating the mesh. In some implementations, the virtual character processing system 682 can be configured to edit or generate surface representations, while the movement processing system 684 can be used to animate the avatar by moving the avatar, deforming the mesh, etc.

Examples of Mapping a User's Environment

[0138] FIG. 7 is a block diagram of an example of an MR environment 700. The MR environment 700 may be configured to receive input (e.g., visual input 702 from the user's wearable system, stationary input 704 such as room cameras, sensory input 706 from various sensors, gestures, totems, eye tracking, user input from the user input device 466, etc.) from one or more user wearable systems (e.g., wearable system 200 or display system 220) or stationary room systems (e.g., room cameras, etc.). The wearable systems can use various sensors (e.g., accelerometers, gyroscopes, temperature sensors, movement sensors, depth sensors, GPS sensors, inward-facing imaging system, outward-facing imaging system, etc.) to determine the location and various other attributes of the environment of the user. This information may further be supplemented with information from stationary cameras in the room that may provide images or various cues from a different point of view. The image data acquired by the cameras (such as the room cameras and/or the cameras of the outward-facing imaging system) may be reduced to a set of mapping points.

[0139] One or more object recognizers 708 can crawl through the received data (e.g., the collection of points) and recognize or map points, tag images, attach semantic information to objects with the help of a map database 710. The map database 710 may comprise various points collected over time and their corresponding objects. The various devices and the map database can be connected to each other through a network (e.g., LAN, WAN, etc.) to access the cloud.

[0140] Based on this information and collection of points in the map database, the object recognizers 708a to 708n may recognize objects in an environment. For example, the object recognizers can recognize faces, persons, windows, walls, user input devices, televisions, documents (e.g., travel tickets, driver's license, passport as described in the security examples herein), other objects in the user's environment, etc. One or more object recognizers may be specialized for objects with certain characteristics. For example, the object recognizer 708a may be used to recognize faces, while another object recognizer may be used recognize documents.

[0141] The object recognitions may be performed using a variety of computer vision techniques. For example, the wearable system can analyze the images acquired by the outward-facing imaging system 464 (shown in FIG. 4) to perform scene reconstruction, event detection, video tracking, object recognition (e.g., persons or documents), object pose estimation, facial recognition (e.g., from a person in the environment or an image on a document), learning, indexing, motion estimation, or image analysis (e.g., identifying

indicia within documents such as photos, signatures, identification information, travel information, etc.), and so forth. One or more computer vision algorithms may be used to perform these tasks. Non-limiting examples of computer vision algorithms include: Scale-invariant feature transform (SIFT), speeded up robust features (SURF), oriented FAST and rotated BRIEF (ORB), binary robust invariant scalable keypoints (BRISK), fast retina keypoint (FREAK), Viola-Jones algorithm, Eigenfaces approach, Lucas-Kanade algorithm, Horn-Schunck algorithm, Mean-shift algorithm, visual simultaneous location and mapping (vSLAM) techniques, a sequential Bayesian estimator (e.g., Kalman filter, extended Kalman filter, etc.), bundle adjustment, Adaptive thresholding (and other thresholding techniques), Iterative Closest Point (ICP), Semi Global Matching (SGM), Semi Global Block Matching (SGBM), Feature Point Histograms, various machine learning algorithms (such as e.g., support vector machine, k-nearest neighbors algorithm, Naive Bayes, neural network (including convolutional or deep neural networks), or other supervised/unsupervised models, etc.), and so forth.

[0142] The object recognitions can additionally or alternatively be performed by a variety of machine learning algorithms. Once trained, the machine learning algorithm can be stored by the HMD. Some examples of machine learning algorithms can include supervised or non-supervised machine learning algorithms, including regression algorithms (such as, for example, Ordinary Least Squares Regression), instance-based algorithms (such as, for example, Learning Vector Quantization), decision tree algorithms (such as, for example, classification and regression trees), Bayesian algorithms (such as, for example, Naive Bayes), clustering algorithms (such as, for example, k-means clustering), association rule learning algorithms (such as, for example, a-priori algorithms), artificial neural network algorithms (such as, for example, Perceptron), deep learning algorithms (such as, for example, Deep Boltzmann Machine, or deep neural network), dimensionality reduction algorithms (such as, for example, Principal Component Analysis), ensemble algorithms (such as, for example, Stacked Generalization), and/or other machine learning algorithms. In some embodiments, individual models can be customized for individual data sets. For example, the wearable device can generate or store a base model. The base model may be used as a starting point to generate additional models specific to a data type (e.g., a particular user in the telepresence session), a data set (e.g., a set of additional images obtained of the user in the telepresence session), conditional situations, or other variations. In some embodiments, the wearable HMD can be configured to utilize a plurality of techniques to generate models for analysis of the aggregated data. Other techniques may include using pre-defined thresholds or data values.

[0143] Based on this information and collection of points in the map database, the object recognizers 708a to 708n may recognize objects and supplement objects with semantic information to give life to the objects. For example, if the object recognizer recognizes a set of points to be a door, the system may attach some semantic information (e.g., the door has a hinge and has a 90 degree movement about the hinge). If the object recognizer recognizes a set of points to be a mirror, the system may attach semantic information that the mirror has a reflective surface that can reflect images of objects in the room. The semantic information can include

affordances of the objects as described herein. For example, the semantic information may include a normal of the object. The system can assign a vector whose direction indicates the normal of the object. Over time the map database grows as the system (which may reside locally or may be accessible through a wireless network) accumulates more data from the world. Once the objects are recognized, the information may be transmitted to one or more wearable systems. For example, the MR environment 700 may include information about a scene happening in California. The environment 700 may be transmitted to one or more users in New York. Based on data received from an FOV camera and other inputs, the object recognizers and other software components can map the points collected from the various images, recognize objects etc., such that the scene may be accurately “passed over” to a second user, who may be in a different part of the world. The environment 700 may also use a topological map for localization purposes.

[0144] FIG. 8 is a process flow diagram of an example of a method 800 of rendering virtual content in relation to recognized objects. The method 800 describes how a virtual scene may be presented to a user of the wearable system. The user may be geographically remote from the scene. For example, the user may be in New York, but may want to view a scene that is presently going on in California, or may want to go on a walk with a friend who resides in California.

[0145] At block 810, the wearable system may receive input from the user and other users regarding the environment of the user. This may be achieved through various input devices, and knowledge already possessed in the map database. The user's FOV camera, sensors, GPS, eye tracking, etc., convey information to the system at block 810. The system may determine sparse points based on this information at block 820. The sparse points may be used in determining pose data (e.g., head pose, eye pose, body pose, or hand gestures) that can be used in displaying and understanding the orientation and position of various objects in the user's surroundings. The object recognizers 708a-708n may crawl through these collected points and recognize one or more objects using a map database at block 830. This information may then be conveyed to the user's individual wearable system at block 840, and the desired virtual scene may be accordingly displayed to the user at block 850. For example, the desired virtual scene (e.g., user in CA) may be displayed at the appropriate orientation, position, etc., in relation to the various objects and other surroundings of the user in New York.

Example Communications Among Multiple Wearable Systems

[0146] FIG. 9A schematically illustrates an overall system view depicting multiple user devices interacting with each other. The computing environment 900 includes user devices 930a, 930b, 930c. The user devices 930a, 930b, and 930c can communicate with each other through a network 990. The user devices 930a-930c can each include a network interface to communicate via the network 990 with a remote computing system 920 (which may also include a network interface 971). The network 990 may be a LAN, WAN, peer-to-peer network, radio, Bluetooth, or any other network. The computing environment 900 can also include one or more remote computing systems 920. The remote computing system 920 may include server computer systems that are clustered and located at different geographic locations.

The user devices **930a**, **930b**, and **930c** may communicate with the remote computing system **920** via the network **990**.

[0147] The remote computing system **920** may include a remote data repository **980** which can maintain information about a specific user's physical and/or virtual worlds. Data storage **980** can store information related to users, users' environment (e.g., world maps of the user's environment), or configurations of avatars of the users. The remote data repository may be an embodiment of the remote data repository **280** shown in FIG. 2. The remote computing system **920** may also include a remote processing module **970**. The remote processing module **970** may be an embodiment of the remote processing module **270** shown in FIG. 2. The remote processing module **970** may include one or more processors which can communicate with the user devices (**930a**, **930b**, **930c**) and the remote data repository **980**. The processors can process information obtained from user devices and other sources. In some implementations, at least a portion of the processing or storage can be provided by the local processing and data module **260** (as shown in FIG. 2). The remote computing system **920** may enable a given user to share information about the specific user's own physical and/or virtual worlds with another user.

[0148] The user device may be a wearable device (such as an HMD or an ARD), a computer, a mobile device, or any other devices alone or in combination. For example, the user devices **930b** and **930c** may be an embodiment of the wearable system **200** shown in FIG. 2 (or the wearable system **400** shown in FIG. 4) which can be configured to present AR/VR/MR content.

[0149] One or more of the user devices can be used with the user input device **466** shown in FIG. 4. A user device can obtain information about the user and the user's environment (e.g., using the outward-facing imaging system **464** shown in FIG. 4). The user device and/or remote computing system **1220** can construct, update, and build a collection of images, points and other information using the information obtained from the user devices. For example, the user device may process raw information acquired and send the processed information to the remote computing system **1220** for further processing. The user device may also send the raw information to the remote computing system **1220** for processing. The user device may receive the processed information from the remote computing system **1220** and provide final processing before projecting to the user. The user device may also process the information obtained and pass the processed information to other user devices. The user device may communicate with the remote data repository **1280** while processing acquired information. Multiple user devices and/or multiple server computer systems may participate in the construction and/or processing of acquired images.

[0150] The information on the physical worlds may be developed over time and may be based on the information collected by different user devices. Models of virtual worlds may also be developed over time and be based on the inputs of different users. Such information and models can sometimes be referred to herein as a world map or a world model. As described with reference to FIGS. 6 and 7, information acquired by the user devices may be used to construct a world map **910**. The world map **910** may include at least a portion of the map **620** described in FIG. 6A. Various object recognizers (e.g. **708a**, **708b**, **708c** . . . **708n**) may be used to recognize objects and tag images, as well as to attach

semantic information to the objects. These object recognizers are also described in FIG. 7.

[0151] The remote data repository **980** can be used to store data and to facilitate the construction of the world map **910**. The user device can constantly update information about the user's environment and receive information about the world map **910**. The world map **910** may be created by the user or by someone else. As discussed herein, user devices (e.g. **930a**, **930b**, **930c**) and remote computing system **920**, alone or in combination, may construct and/or update the world map **910**. For example, a user device may be in communication with the remote processing module **970** and the remote data repository **980**. The user device may acquire and/or process information about the user and the user's environment. The remote processing module **970** may be in communication with the remote data repository **980** and user devices (e.g. **930a**, **930b**, **930c**) to process information about the user and the user's environment. The remote computing system **920** can modify the information acquired by the user devices (e.g. **930a**, **930b**, **930c**), such as, e.g. selectively cropping a user's image, modifying the user's background, adding virtual objects to the user's environment, annotating a user's speech with auxiliary information, etc. The remote computing system **920** can send the processed information to the same and/or different user devices.

Examples of a Telepresence Session

[0152] FIG. 9B depicts an example where two users of respective wearable systems are conducting a telepresence session. Two users (named Alice **912** and Bob **914** in this example) are shown in this figure. The two users are wearing their respective wearable devices **902** and **904** which can include an HMD described with reference to FIG. 2 (e.g., the display device **220** of the system **200**) for representing a virtual avatar of the other user in the telepresence session. The two users can conduct a telepresence session using the wearable device. Note that the vertical line in FIG. 9B separating the two users is intended to illustrate that Alice and Bob may (but need not) be in two different locations while they communicate via telepresence (e.g., Alice may be inside her office in Atlanta while Bob is outdoors in Boston).

[0153] As described with reference to FIG. 9A, the wearable devices **902** and **904** may be in communication with each other or with other user devices and computer systems. For example, Alice's wearable device **902** may be in communication with Bob's wearable device **904**, e.g., via the network **990** (shown in FIG. 9A). The wearable devices **902** and **904** can track the users' environments and movements in the environments (e.g., via the respective outward-facing imaging system **464**, or one or more location sensors) and speech (e.g., via the respective audio sensor **232**). The wearable devices **902** and **904** can also track the users' eye movements or gaze based on data acquired by the inward-facing imaging system **462**. In some situations, the wearable device can also capture or track a user's facial expressions or other body movements (e.g., arm or leg movements) where a user is near a reflective surface and the outward-facing imaging system **464** can obtain reflected images of the user to observe the user's facial expressions or other body movements.

[0154] A wearable device can use information acquired of a first user and the environment to animate a virtual avatar that will be rendered by a second user's wearable device to create a tangible sense of presence of the first user in the

second user's environment. For example, the wearable devices **902** and **904**, the remote computing system **920**, alone or in combination, may process Alice's images or movements for presentation by Bob's wearable device **904** or may process Bob's images or movements for presentation by Alice's wearable device **902**. As further described herein, the avatars can be rendered based on contextual information such as, e.g., a user's intent, an environment of the user or an environment in which the avatar is rendered, or other biological features of a human.

[0155] Although the examples only refer to two users, the techniques described herein should not be limited to two users. Multiple users (e.g., two, three, four, five, six, or more) using wearables (or other telepresence devices) may participate in a telepresence session. A particular user's wearable device can present to that particular user the avatars of the other users during the telepresence session. Further, while the examples in this figure show users as standing in an environment, the users are not required to stand. Any of the users may stand, sit, kneel, lie down, walk or run, or be in any position or movement during a telepresence session. The user may also be in a physical environment other than described in examples herein. The users may be in separate environments or may be in the same environment while conducting the telepresence session. Not all users are required to wear their respective HMDs in the telepresence session. For example, Alice may use other image acquisition and display devices such as a webcam and computer screen while Bob wears the wearable device **904**.

Examples of a Virtual Avatar

[0156] FIG. **10** illustrates an example of an avatar as perceived by a user of a wearable system. The example avatar **1000** shown in FIG. **10** can be an avatar of Alice (shown in FIG. **9B**) standing behind a physical plant in a room. An avatar can include various characteristics, such as for example, size, appearance (e.g., skin color, complexion, hair style, clothes, facial features (e.g., wrinkle, mole, blemish, pimple, dimple, etc.)), position, orientation, movement, pose, expression, etc. These characteristics may be based on the user associated with the avatar (e.g., the avatar **1000** of Alice may have some or all characteristics of the actual person Alice). As further described herein, the avatar **1000** can be animated based on contextual information, which can include adjustments to one or more of the characteristics of the avatar **1000**. Although generally described herein as representing the physical appearance of the person (e.g., Alice), this is for illustration and not limitation. Alice's avatar could represent the appearance of another real or fictional human being besides Alice, a personified object, a creature, or any other real or fictitious representation. Further, the plant in FIG. **10** need not be physical, but could be a virtual representation of a plant that is presented to the user by the wearable system. Also, additional or different virtual content than shown in FIG. **10** could be presented to the user.

Example Control Systems for Animating an Avatar

[0157] As described with reference to FIG. **6B**, an avatar can be animated by the wearable system using rigging techniques. A goal of rigging is to provide pleasing, high-fidelity deformations of an avatar based upon simple, human-understandable controls. Generally, the most appealing deformations are based at least partly on real-world

samples (e.g., photogrammetric scans of real humans performing body movements, articulations, facial contortions, expressions, etc.) or art-directed development (which may be based on real-world sampling). Real-time control of avatars in a mixed reality environment can be provided by embodiments of the avatar processing and rendering system **690** described with reference to FIG. **6B**.

[0158] Rigging includes techniques for transferring information about deformation of the body of an avatar (e.g., facial contortions) onto a mesh. A mesh can be a collection of 3D points (e.g., vertices) along with a set of polygons that share these vertices. FIG. **10** shows an example of a mesh **1010** around an eye of the avatar **1000**. Animating a mesh includes deforming the mesh by moving some or all of the vertices to new positions in 3D space. These positions can be influenced by the position or orientation of the underlying bones of the rig (described below) or through user controls parameterized by time or other state information for animations such as facial expressions. The control system for these deformations of the mesh is often referred to as a rig. The example avatar processing and rendering system **690** of FIG. **6B** includes a 3D model processing system **680**, which can implement the rig.

[0159] Since moving each vertex independently to achieve a desired deformation may be quite time-consuming and effort-intensive, rigs typically provide common, desirable deformations as computerized commands that make it easier to control the mesh. For high-end visual effects productions such as movies, there may be sufficient production time for rigs to perform massive mathematical computations to achieve highly realistic animation effects. But for real-time applications (such as in mixed reality), deformation speed can be very advantageous and different rigging techniques may be used. Rigs often utilize deformations that rely on skeletal systems and/or blendshapes.

Example Skeletal Systems

[0160] Skeletal systems represent deformations as a collection of joints in a hierarchy. Joints (also called bones) primarily represent transformations in space including translation, rotation, and change in scale. Radius and length of the joint may be represented. The skeletal system is a hierarchy representing parent-child relationships among joints, e.g., the elbow joint is a child of the shoulder and the wrist is a child of the elbow joint. A child joint can transform relative to its parent's joint such that the child joint inherits the transformation of the parent. For example, moving the shoulder results in moving all the joints down to the tips of the fingers. Despite its name, a skeleton need not represent a real world skeleton but can describe the hierarchies used in the rig to control deformations of the mesh. For example, hair can be represented as a series of joints in a chain, skin motions due to an avatar's facial contortions (e.g., representing an avatar's expressions such as smiling, frowning, laughing, speaking, blinking, etc.) can be represented by a series of facial joints controlled by a facial rig, muscle deformation can be modeled by joints, and motion of clothing can be represented by a grid of joints.

[0161] Skeletal systems can include a low level (also referred to as low order in some situations) core skeleton that might resemble a biological skeleton of an avatar. This core skeleton may not map exactly to a real set of anatomically correct bones, but can resemble the real set of bones by having at least a sub-set of the bones in analogous orienta-

tions and locations. For example, a clavicle bone can be roughly parallel to the ground, roughly located between the neck and shoulder, but may not be the exact same length or position. Higher order joint structures representing muscles, clothing, hair, etc. can be layered on top of the low level skeleton. The rig may animate only the core skeleton, and the higher order joint structures can be driven algorithmically by rigging logic based upon the core skeleton's animation using, for example, skinning techniques (e.g. vertex weighting methods such as linear blend skinning (LBS)). Real-time rigging systems (such as the avatar processing and rendering system 690) may enforce limits on the number of joints that can be assigned to a given vertex (e.g., 8 or fewer) to provide for efficient, real-time processing by the 3D model processing system 680.

Blendshapes

[0162] Blendshapes include deformations of the mesh where some or all vertices are moved in 3D space by a desired amount based on a weight. Each vertex may have its own custom motion for a specific blendshape target, and moving the vertices simultaneously will generate the desired shape. Degrees of the blendshape can be applied by using blendshape weights. The rig may apply blendshapes in combination to achieve a desired deformation. For example, to produce a smile, the rig may apply blendshapes for lip corner pull, raising the upper lip, lowering the lower lip, moving the eyes, brows, nose, and dimples.

Example Rigging Techniques

[0163] A rig is often built in layers with lower, simpler layers driving higher order layers, which produce more realistic mesh deformations. The rig can implement both skeletal systems and blendshapes driven by rigging control logic. The control logic can include constraints among the joints (e.g., aim, orientation, and position constraints to provide specific movements or parent-child joint constraints); dynamics (e.g., for hair and clothing); pose-based deformations (PSDs, where the pose of the skeleton is used to drive a deformation based on distances from defined poses); machine learning techniques (e.g., those described with reference to FIG. 7) in which a desired higher level output (e.g., a facial expression) is learned from a set of lower level inputs (of the skeletal system or blendshapes); etc. Some machine learning techniques can utilize radial basis functions (RBFs).

[0164] In some embodiments, the 3D model processing system 680 animates an avatar in the mixed reality environment in real-time to be interactive (with users of the MR system) and to provide appropriate, contextual avatar behavior (e.g., intent-based behavior) in the user's environment. The system 680 may drive a layered avatar control system comprising a core skeletal hierarchy, which further drives a system of expressions, constraints, transforms (e.g., movement of vertices in 3D space such as translation, rotation, scaling, shear), etc. that control higher level deformations of the avatar (e.g., blendshapes, correctives) to produce a desired movement and expression of the avatar.

Example Problems of Realistically and Dynamically Rendering a Virtual Avatar in an Environment

[0165] FIGS. 11A-11D illustrate example scenes of an avatar in various environments, where the virtual avatar may

have an unnatural appearance or cause an unrealistic interaction. The avatar 1100 may be an avatar of Bob. As described with reference to FIG. 9B, the avatar 1100 may be animated based on Bob's characteristics including, e.g., intentions, poses, movements, expressions, or actions.

[0166] FIG. 11A illustrates an example scene 1102 where three users 1112, 1114, and 1116 are interacting with the avatar 1100 during a telepresence session. However, as shown in this example, Bob's avatar 1100 is relatively small compared to the three users 1112, 1114, and 1116, which may lead to awkward interactions, because humans often feel most comfortable communicating with each other while maintaining eye contact and approximate eye height with each other. Thus, due to the difference in sight lines between the avatar and the three users, the three users may need to pose themselves at uncomfortable positions in order to look at the avatar 1100, or maintain (or alter) social dynamics in a conversation. For example, the user 1112 is kneeling down in order to look at the avatar's eyes; the user 1114 is looking down at the avatar; and the user 1116 bends his body forward to engage in conversation with the avatar 1100. To reduce a user's physical strain caused by an improperly sized avatar, advantageously, in some implementations, the wearable system can automatically scale the avatar to increase or decrease the size of the avatar based on contextual information such as, e.g., the height level of the other user's eyes. Such adjustment can be implemented in a manner that increases or maximizes direct eye contact between the avatar and the others, and therefore facilitates avatar-human communication. For example, the avatar can be scaled such that the wearable device can render the avatar's head at a viewer's eye level, and thus the user may not have to experience physical strain while interacting with the avatar. Detailed descriptions and examples of dynamically scaling an avatar based on contextual information are further described with reference to FIGS. 12A-18B.

[0167] As described with reference to FIGS. 6B and 10, an avatar of a user can be animated based on characteristics of the user. However, a one-to-one mapping of the user's characteristics into an avatar's characteristics can be problematic because it can create unnatural user interactions or convey the wrong message or intent of the user to a viewer. FIGS. 11B-11D illustrate some example scenarios where a one-to-one mapping (which animates between a user and an avatar) can create problems.

[0168] FIG. 11B illustrates a scene where Bob is talking to Charlie during a telepresence session. The scene in this figure includes two environments 1120a and 1120b. The environment 1120a is where Bob resides. The environment 1120b is where Charlie 1118 resides and includes a physical table 1122 with Charlie sitting on a chair next to the table 1122. Charlie can perceive, e.g., via the display 220, Bob's avatar 1100. In the environment 1120a, Bob is facing west (as shown by the coordinate 1128). To animate Bob's avatar 1100, Bob's characteristics are mapped as one-to-one to Bob's avatar 1100 in FIG. 11B. This mapping, however, is problematic because it does not take into account Charlie's environment and it creates an unnatural or unpleasant user interaction experience with the avatar 1100. For example, Bob's avatar is taller than Charlie because Charlie is sitting on a chair, and Charlie may need to strain his neck to maintain communication with Bob's avatar 1100. As another example, Bob's avatar 1100 is facing to the west because Bob is facing to the west. However, Charlie is to the east of

Bob's avatar **1100**. Thus, Charlie perceives the back of Bob's avatar and cannot observe Bob's facial expressions as reflected by Bob's avatar **1100**. This orientation of Bob's avatar **1100** relative to Charlie may also convey an inaccurate social message (e.g., Bob does not want to engage with Charlie or Bob is angry at Charlie), even though Bob intends to be in a friendly conversation with Charlie.

[0169] FIG. 11C illustrates a scene where Bob's avatar **1100** is rendered without taking into account physical objects in Charlie's environment. This scene illustrates two environments **1130a** and **1130b**. Bob is located in the environment **1130a** and Charlie is in the environment **1130b**. As illustrated, Bob is sitting on a chair **1124** in the environment **1130a**. Due to one-to-one mapping of Bob's pose to Bob's avatar's pose that is illustrated in this example, Bob's avatar **1100** is also rendered with a sitting pose in Charlie's environment **1130b**. However, there is no chair in Charlie's environment. As a result, Bob's avatar **1100** is rendered as sitting in mid-air which can create an unnatural appearance of Bob's avatar **1100**.

[0170] FIG. 11D illustrates an example scene where one-to-one mapping causes unrealistic movement of a virtual avatar. The scene in FIG. 11D illustrates two environments **1140a** and **1140b**. Bob is moving eastbound in his environment **1140a**. To map Bob's movement **1142** to the environment **1140b** where Bob's avatar **1100** is rendered, Bob's avatar **1100** also moves eastbound (e.g., from position **1142a** to position **1142b**). However, the environment **1140b** has a table **1126**. By directly mapping Bob's movement **1142** to Bob's avatar's **1100** movement, Bob's avatar **1100** moves straight into the table and appears to be trapped in table **1126**, which creates an unnatural and unrealistic movement and appearance of Bob's avatar **1100**.

[0171] Advantageously, in some implementations, the wearable system **200** can be configured to render an avatar based on contextual information relating to the environment where the avatar is displayed or to convey the intent of a user (rather than a direct, one-to-one mapping), and thus may avoid unnatural or unrealistic appearances or interactions by an avatar. For example, the wearable system **200** can analyze the contextual information and Bob's action to determine the intent of Bob's action. The wearable system **200** can adjust the characteristics of Bob's avatar to reflect Bob's intent in view of Bob's action and contextual information about the environment in which Bob's avatar is to be rendered.

[0172] For example, with reference to FIG. 11B, rather than rendering the avatar **1100** facing westward, the wearable system **200** can turn the avatar around to face Charlie **1118** because Bob intends to converse with Charlie in a friendly manner, which normally occurs face-to-face. However, if Bob is angry at Charlie (e.g., as determined by the tone, content, volume of Bob's speech as detected by a microphone on Bob's system, or Bob's facial expression), the wearable system **200** can keep Bob's orientation such that Bob faces away from Charlie.

[0173] As another example, rather than rendering Bob's avatar **1100** sitting in mid-air (as shown in FIG. 11C), the wearable system **200** can automatically identify an object with a horizontal surface suitable for sitting (e.g., a bed or a sofa) in Charlie's environment and can render Bob's avatar **1100** as sitting on the identified surface (rather than in mid-air). If there is no place in Charlie's environment **1130b** that Bob's avatar **1100** can sit (e.g., all chairs have been occupied by either human or other avatars or there are no

sit-table surfaces), the wearable system may instead render Bob's avatar as standing or render a virtual chair for the virtual avatar to sit in.

[0174] As yet another example, with reference to FIG. 11D, rather than rendering Bob's avatar as walking into or through the table, the wearable system can detect the presence of the table **1126** as an obstacle on the route of Bob's avatar in the environment **1140b** (e.g., based on a world map **910** of the environment **1140b** or based on images acquired by the outward-facing imaging system **464** of a viewer's wearable device in the environment **1140b**). The wearable system **200** can accordingly reroute the avatar **1100** to circumvent the table **1126** or to stop prior to the table. Further details related to intent-based rendering are described with reference to FIGS. 19A-22.

Examples of Scaling a Virtual Avatar Based on Contextual Information

[0175] As described with reference to FIG. 11A, an improperly scaled avatar can result in physical strain for a viewer of the avatar and may increase the likelihood of an inappropriate social interaction between the avatar and the user. For example, improperly scaling an avatar may incur discomfort or pain (e.g., neck pain) for a user (e.g., because the user has to look up or look down at the avatar). Such improper scaling may also provide for an awkward social dynamic for a user. As an example, an improperly sized avatar (e.g., an avatar shorter than the viewer) may be rendered as looking at an improper or inappropriate region of the viewer's body. As another example, differing sight lines or eye levels between the user and the avatar may improperly imply social inferiority or superiority.

[0176] For example, in friendly conversations, the eyes of a user are typically directed toward a region called the social triangle of the other user's face. The social triangle is formed with a first side on a line between the user's eyes and a vertex at the user's mouth. Eye contact within the social triangle is considered friendly and neutral, whereas eye gaze directed outside the social triangle can convey a power imbalance (e.g., eye gaze directed above the social triangle, toward the other person's forehead), anger, or that the conversation is serious. Thus, an avatar rendered taller than the viewer may be tend to be viewed as looking at a region above the viewer's social triangle, which can create a psychological effect for the viewer that the avatar is superior to the viewer. Thus, incorrect-sizing of the avatar can lead to awkward or unpleasant encounters between a human and an avatar that were not intended between the actual human participants of the conversation.

[0177] In some wearable devices, a user can manually scale an avatar so that the size of the avatar is at a comfortable height. However, such manual control may take more time to complete and require the user to make refined adjustments to the avatar, which can cause muscle fatigue of a user and require more expert control from the user. Other wearable devices may use scaling methods that seek to maintain a 1:1 scale between the avatar and the user (e.g., an avatar is automatically scaled at the same height as the user). However, this technique can produce inappropriate sight lines if the avatar is standing on a surface higher than the surface on which the user is sitting or standing (e.g., where the avatar looks over the user's head).

[0178] Advantageously, in some embodiments, the wearable system **200** can automatically scale the virtual avatar

based on contextual information regarding the rendering position of the avatar in the environment and the position or eye-height of the user in the environment. The wearable system **200** can calculate the size of the virtual avatar based on contextual factors such as, e.g., the rendering location of the avatar, the user's position, the height of the user, the relative positions between the user and the avatar, the height of surface that the avatar will be rendered on, the height of the surface the user is standing or sitting on, alone or in combination. The wearable system **200** can make the initial rendering of the avatar (called spawning) such that the avatar is rendered with the appropriate height based at least in part on such contextual factors. The wearable system **200** can also dynamically scale the size of the virtual avatar in response to a change in the contextual information, such as, e.g., as the avatar or the user moves around in the environment.

[0179] For example, prior to or at the time of spawning an avatar, the wearable system can determine the user's head height (and therefore the eye height, since the eyes are typically about halfway between top and bottom of the head or about 4 to 6 inches below the top of the head) and compute a distance from the base surface of the avatar (e.g., the surface that the avatar will be spawned on) to the user's eye height. This distance can be used to scale the avatar so that its resulting head and sight lines are the same height as the user. The wearable system can identify environment surfaces (e.g., the surface the user is on or the surface the avatar will be spawned on) and adjust the avatar height based on these surfaces or the relative height difference between the user and avatar surfaces. For example, the wearable system can scan for the floor and measure the height of the head with respect to the floor plane. The wearable system can determine a head pose of the user (e.g., via data from IMUs) and compute environment surfaces relative to the user's head pose or a common coordinate system shared by both the environment and the head pose. Based on this information, the wearable system can calculate a size of the avatar and instruct the display **220** to display the avatar as superimposed on the environment.

[0180] In certain implementations, as the user moves (or the avatar moves) around in the environment, the wearable system can continuously track the user's head pose and environment surfaces and dynamically adjust the size of the avatar based on these contextual factors in a similar fashion as when the avatar is originally spawned. In some embodiments, these techniques for automatically scaling an avatar (either at spawning or in real-time as the avatar moves) can advantageously allow direct eye contact to be made while minimizing neck strain, facilitate user-avatar communication, and minimize the amount of manual adjustments a user needs to make when placing avatars in the user's local environment, thereby allowing both participants (e.g., avatar and its viewer) to communicate eye-to-eye, creating a comfortable two-way interaction.

[0181] In some implementations, the wearable system **200** can allow a user to turn-off (temporarily or permanently) automatic, dynamic re-scaling of the avatar. For example, if the user frequently stands up and sits down during a telepresence session, the user may not wish the avatar to correspondingly re-scale, which may lead to an uncomfortable interaction since humans do not dynamically change size during conversations. The wearable system can be configured to switch among different modes of avatar scaling

options. For example, the wearable system may provide three scaling options: (1) automatic adjustment based on contextual information, (2) manual control, and (3) 1:1 scaling (where the avatar is rendered as the same size as the viewer or its human counterpart). The wearable system can set the default to be automatically adjustable based on contextual information. The user can switch this default option to other options based on user inputs (such as, e.g., via the user input device **466**, poses, or hand gestures, etc.). In other implementations, the wearable system may smoothly interpolate between size changes so that the avatar is rendered as smoothly changing size over a short time period (e.g., a few to tens of seconds) rather than abruptly changing size.

Examples of Spawning a Scaled Avatar

[0182] The wearable system can automatically scale an avatar based on contextual information to allow eye-to-eye communication between the avatar and a viewer. The calculation of the avatar's height can be performed upon initial spawning of the avatar into the viewer's environment. The wearable system can identify a rendering location of the avatar at the spawning site. The rendering location of the avatar can be a horizontal support platform (or surface), such as, e.g., a ground, table, a chair's sitting surface, etc. In some situation, the support platform is not horizontal and may be inclined or vertical (if the user is laying down, for example).

[0183] The wearable system can calculate the height of the avatar based on the current head position of the user (regardless of whether the user is standing or sitting) and the location of the horizontal support platform at the spawning site for the avatar. The wearable system can compute the estimated height of eyes above this platform (which may be a distance perpendicular and vertical to the platform) for computing a scale factor for adjusting the avatar's size. The estimated height of the eyes above the platform can be based on a distance between the eyes and the platform. In some implementations, the wearable system can compute an eye level which may be a 1D, 2D, 3D, or other mathematical representations of a level where the eyes are looking straight ahead. The estimated avatar's height can be calculated based on the difference between the eye level and the level of the platform.

[0184] FIGS. **12A** and **12B** illustrate two scenes of scaling avatar, where the avatar is spawned on the same surface as the viewer. The scene **1200a** in FIG. **12A** shows an improperly scaled avatar while the scene **1200b** in FIG. **12B** shows a scaled avatar that maintains roughly the same eye height as the viewer. In these two figures, the example virtual avatar **1000** can be Alice's avatar while the user **914** may be Bob as identified in FIG. **9B**. Both Alice and Bob may wear the wearable device as described with reference to FIG. **2**. In these examples, Bob is standing on the ground (as represented by the ground plane **1214**) while Alice's avatar **912** will also be spawned on the ground in this example.

[0185] FIG. **12A** illustrates an example where Alice's avatar **1000** is too small such that the viewer (Bob) needs to look down when interacting with the Alice's avatar **1000**. The height of Alice's avatar **100** and Bob can be measured from a common ground position line **1210**, which may be part of the ground plane **1214**. The ground position line **1210** may connect a position of the user **914** and a position of the virtual avatar **1000** along the ground plane **1214**.

[0186] FIG. 12A also shows Bob's 914 eye level (as illustrated by the user eye line 1206) and the avatar's eye level (as illustrated by the avatar eye line 1228), which is below Bob's eye level 1206. The avatar eye line 1228 and user eye line 1206 are shown as parallel to the ground position line 1210 and intersecting an eye of the virtual avatar 1000 and the user 914, respectively, but other types of eye lines or representations illustrating a line of sight are also possible in various implementations. Each of the user eye line 1206 and avatar eye line 1228 may correspond to respective planes (not shown) that encompass the corresponding eye line and that are parallel to the ground plane 1214. One or both of the user eye line 1206 and the avatar eye line 1228 may be parallel to the ground plane 1214.

[0187] To determine the size of the avatar, the wearable system (such as, e.g., the avatar autoscaler 692 in the avatar processing and rendering system 690) can calculate a height 1202 of the viewer 914 and a height 1224 of the avatar 1000. The avatar's height and the viewer's height can be measured from the avatar and the user's respective eye lines vertically to the ground surface 1214 on which the avatar is rendered and on which the viewer stands. As illustrated in FIG. 12A, an avatar eye height 1224 may be determined between the avatar eye line 1228 and the ground position line 1210. Similarly, a user eye height 1202 may be determined between the user eye line 1206 and the ground position line 1210. The user eye height 1202 intersects the user's 914 eye as illustrated in FIG. 12A, however, in other implementations, the user (or avatar) height may be referenced to the top of the user's (or avatar's) head or some other convenient reference position.

[0188] In certain implementations, the system may be configured to determine a distance 1242 between the user 914 and the rendering position of the virtual avatar 1000. The distance 1242 may be used to display the virtual avatar 1000 at a more comfortable position or apparent depth for the user 914. For example, the wearable system may increase the size of the avatar if the avatar is relatively far away from the viewer so that the viewer may have a better view of the avatar.

[0189] In the example shown in FIG. 12A, the avatar 1000 is not properly sized because the user eye line 1206 is not collinearly aligned with an avatar eye line 1228, since the avatar eye line 1228 is lower than the user eye line 1206. This suggests that the avatar 1000 is too small, causing Bob to tilt his head downward to interact with Alice's avatar. Although this shows that the avatar is shorter than the viewer, the avatar size may also be improper if the avatar is taller than the viewer, which would cause Bob to tilt his head upward to interact with Alice's avatar.

[0190] FIG. 12B shows a virtual avatar 1000 whose size is properly rendered relative to Bob in the sense that their respective eye heights are comparable. In this example, the virtual avatar 1000 is scaled based on the viewer 914's eye height. Scaling the virtual avatar 1000 may include matching the avatar eye height 1224 and the user eye height 1202. Details related to algorithms on how to calculate the viewer's height and a target height of the avatar are described with reference to FIGS. 16A-18B.

[0191] In some situations, the user 914 or the avatar 1000 (or both) may be on a surface different from the ground plane 1214 (e.g., the user 914 or the avatar 1000 is sitting on a chair). FIGS. 13A and 13B illustrate an example of

automatically scaling the avatar based on contextual information, wherein the avatar is not rendered on the same surface as the viewer.

[0192] The scene 1300a shows that Bob is sitting on a chair 1322 in front of a table 1324. Bob can initiate a telepresence session with Alice which will cause Alice's avatar 1000 to be rendered on the table 1324. To spawn Alice's avatar 1000 at a comfortable height for Bob to interact with, the wearable system can calculate a height difference between the user's eye level and the platform which Alice's avatar 1000 will be rendered on. In this example, the height difference 1302 is measured from Bob's eye level 1306 to the table's 1324 surface. This height difference 1302 can then be applied to scale the avatar 1000. In this example, although the table 1324 has a height 1332 (as measured from the ground), this height 1332 is not required for calculating the size of the avatar because the avatar will be rendered on the tabletop not on the ground.

[0193] The scene 1300b in FIG. 13B shows a scene where Alice's avatar 1000 is scaled and rendered on the table 1324 top. In this example, the scale of Alice's avatar 1000 is based on the height 1302 (measured from the user's eye level to the table top). For example, Alice's avatar's eye height (as measured from the avatar's 1000 eyes to the table top) can equal to the height 1302.

Examples of Dynamically Resizing an Avatar

[0194] In some situations, an avatar or a viewer of the avatar can move around in an environment and one or both may move to positions where there is a relative height difference between the horizontal planes on which the user is positioned and on which the avatar is rendered. The wearable system 200 can dynamically resize the avatar based on the updated relative height information as the viewer or the avatar moves. For example, as either Alice's avatar moves (e.g., because Alice moves) or Bob moves, the height difference between the Alice's avatar and Bob may change. FIGS. 14 and 15 illustrate examples of dynamically scaling an avatar to maintain the horizontal eye contact.

[0195] FIG. 14 illustrates an example of dynamically scaling an avatar in response to a user's movement where the user is above the ground plane of the avatar. Bob may initially be at the position shown in FIG. 12B and engage in horizontal eye contact with Alice's avatar 1000. In the scene 1400 shown in FIG. 14, Bob is now standing on an object (e.g., a stool 1438) which is above the ground plane 1214 of the avatar. If Alice's avatar 1000 were not dynamically re-scaled, Bob would look down at the avatar 1000, because Bob has been elevated by the height 1422 of the stool 1438. The wearable system 200 can adjust the height of Alice's avatar to account for this additional height 1422 of Bob, by increasing the avatar's height (relative to the height 1202 shown in FIG. 12B) to a value represented by dashed arrow 1434.

[0196] FIG. 15 illustrates a further example of dynamically scaling an avatar in response to an avatar's further movement relative to FIG. 14 (e.g., Alice's movement). As shown in the scene 1500, Alice has moved up from the ground to a platform 1548 (which has a height 1558), while the Bob is still standing on the platform 1438 as described in FIG. 14. Because the ground planes for Alice's avatar and Bob are now closer together, the height 1524 of Alice's avatar is reduced (by the height 1558 of the platform 1548)

relative to the avatar's height **1434** shown in FIG. **14**, so that Bob and Alice's avatar maintain eye contact.

[0197] As further described below with reference to FIGS. **17A** and **17B**, the wearable system **200** can detect the platform **1548** as a new surface that the avatar is to be rendered as standing on. Thus, the wearable system can calculate an eye height of the avatar **1000** with respect to platform **1548** (rather than the ground **1214**) and the eye height of Bob with respect to the platform **1438** (rather than the ground plane **1214**). The wearable system can then calculate the avatar height based on Bob's height and the relative difference between the horizontal planes on which each is standing.

[0198] Formula (1) describes an example formula for calculating the avatar's target height based on the concepts illustrated with reference to FIGS. **12A-15**.

$$\text{Avatar}_H = \text{User}_H + \text{UserSupportPlane}_H - \text{AvatarSupportPlane}_H \quad (1)$$

[0199] where Avatar_H represents the target height of the avatar to be rendered by the user's wearable system, User_H represents the user's height as measured from the support plane of the user, $\text{UserSupportPlane}_H$ represents the height of the support plane on which the user is situated, and the $\text{AvatarSupportPlane}_H$ represents the height of the plane on which the avatar is to be rendered. Applying this formula to the examples in FIG. **15**, Alice's avatar's target height **1524**=Bob's height **1202**+Bob's platform height **1422**−Alice's avatar's platform height **1558**. The expression $\text{UserSupportPlane}_H - \text{AvatarSupportPlane}_H$ may be referred to as a height modifier, because this expression modifies the user's height to determine the avatar's rendered height.

[0200] Formula (2) describes another example formula for calculating avatar's target height.

$$\text{Avatar}_H = \text{UserToAvatarSupportPlane}_H \quad (2)$$

where Avatar_H represents the target height of the avatar, $\text{UserToAvatarSupportPlane}_H$ represents the user's height as measured from the support surface on which the avatar is standing. The wearable system can detect the support surface using ray casting and techniques described in FIGS. **16A-17B** and can determine the right hand side of Formula (2) using trigonometric techniques by measuring an angle between the user's eye height and the avatar's support surface and a distance from the user to the avatar support surface.

[0201] Although the examples in FIGS. **14** and **15** illustrate increasing an avatar's height, similar techniques can also be applied to decrease the height of the avatar. For example, the height modifier may be negative where the avatar **1000** is standing at a surface which is higher than the surface on which the user **914** is standing. Further, similar techniques can also be applied when the user or the avatar is in other poses, such as sitting. For example, while a user is standing, an avatar may walk to a chair and sit on a chair. The wearable system can automatically increase the avatar's size so that the horizontal eye contact may be maintained. As another example, the user may find a chair and sit down while the avatar is standing. The wearable system can detect

that the user's eye height has decreased (due to sitting down) and the wearable system can automatically reduce the size of the avatar to bring the avatar to the user's eye height.

Examples of Analyzing Contextual Factors for Automatically Scaling an Avatar

[0202] As described herein, the wearable system **200** can be configured to automatically identify contextual factors to calculate a target height for a virtual avatar for spawning the virtual avatar or for dynamically adjusting the size of the virtual avatar in real-time.

[0203] FIGS. **16A-1-16B** illustrate example data flow diagrams for automatically scaling the avatar based on contextual factors. Some example contextual factors can include the user's head position, a rendering location of the avatar, a user's body position (e.g., the user's foot position), heights of surfaces the user and the avatar are positioned on (or a relative height difference between them), etc. The example data flow diagram **1600** can be implemented by the wearable system **200** described herein, for example, by the avatar autoscaler **692** of the avatar processing and rendering system **690** of FIG. **6B**.

[0204] The wearable system can include one or more device sensors **1674**, such as those described with reference to FIGS. **2** and **3**. The data acquired from the device sensors **1674** can be used to determine the environment of the user (such as e.g., to identify objects in the user's environment or to detect surfaces in the user's environment) as well as to determine the user's position with respect to the environment.

[0205] For example, the IMUs can acquire user data such as, e.g., the user's head pose or body movements. The outward-facing imaging system **464** can acquire images of the user's environment. The data from the IMUs and the outward-facing imaging system **464** may be an input for determining head position. The wearable system can detect a position, orientation, or movement of the head with respect to a reference frame associated with the user's environment (also referred to as a world frame). The reference frame may be a set of map points based on which the wearable system can translate the movement of the user to an action or command. In some implementations, camera calibration **1688** may be performed for determining the head localization **1682** in the world frame. The camera calibration **1688** may result in a mapping of a user's head pose as determined from the IMUs (or other hardware sensors of a wearable device) to a head location in the world frame. As further described with reference to the avatar autoscaler **692** in FIG. **6B** (and the subprocess **1650** described with reference to FIG. **16B**), such head localization **1682** in the world frame can be fed into the avatar autoscaler **692** and can be utilized as an input for determining a user's head position **1604** for automatically scaling an avatar.

[0206] The device sensors can include one or more depth sensors **234** (e.g., LIDAR, time of flight sensors, or ultrasound sensors), or world cameras (which may be part of the outward-facing imaging system **464**) where the world cameras have depth sensing ability (e.g., an RGB-D camera or stereoscopic cameras). For example, a depth sensor can acquire depth data of objects in the environment, such as, for example, how far away the objects are from the user. The depth data can be used to create an environment point cloud **1678** which can comprise 3D mathematical representations of the user's environment (which may take into account

objects in the user's environment). This environment point cloud **1678** may be stored in (or accessed from) the map database **710** shown in FIG. 7.

[0207] As will be further described with reference to FIGS. 17A and 17B, the wearable system can identify major horizontal planes (such as, e.g., tabletops, grounds, walls, chair surfaces, platforms, etc.) based on the environment point cloud **1678**. The major horizontal planes can include environment surfaces on which the user or the avatar may be positioned.

[0208] The wearable system can convert the point cloud to a meshed environment, such as, e.g., a polygon (e.g., triangle) mesh, and extract major horizontal planes from the mesh. In certain implementations, the wearable system can estimate planes directly from the point cloud without converting the cloud of points to a mesh. As an example of estimating planes directly from the point cloud, the wearable system can determine one or more depth points based on images acquired by the outward-facing imaging system alone or in combination with the depth sensors. The depth points may be mapped by the system onto a world reference frame (for representing the user's environment). The depth points may correspond to one or more points in the environment of the user. The wearable system may be configured to extract one or more surfaces from the one or more depth points. The one or more surfaces extracted from the depth point(s) may include one or more triangles. Vertices of each of the one or more triangles may comprise neighboring depth points.

[0209] As shown in FIG. 16A-1, with depth camera calibration **1688** the wearable system can convert this point cloud **1678** into meshed environment in a world reference frame (which can be used for head localization in block **1682**) as shown in the block **1680**. Depth camera calibration can include information on how to relate the positions of the point cloud obtained from the depth camera to positions in the wearable's frame of reference or the environment's frame of reference. Depth camera calibration may be advantageous, because it can permit locating the points in the same reference frame as the environment and camera frames, so that the wearable system knows where those points are located in the working coordinate system.

[0210] The meshed environment may be a 3D meshed environment. The meshed environment may comprise one or more surface triangles. Each surface triangle may comprise vertices corresponding to adjacent depth points. The wearable system can be configured to construct a signed distance field function from the point cloud and use a triangulation algorithm, such as, e.g., the Marching Cubes algorithm to convert the point cloud into a surface representation of triangles, such as a polygon (e.g., triangle) mesh. In some embodiments, the surface representation can be determined directly from the point cloud rather than from the meshed environment.

[0211] At block **1684** the wearable system can approximate a planar environment in a world reference frame, which may include plane extractions from the mesh. Plane extractions can group the triangles into areas of similar orientation. Further processing can be done of these meshed areas (as identified from plane extractions) to extract pure planar regions representing flat areas in the environment.

[0212] At block **1686**, the wearable system can perform further processing to extract major horizontal planes from the environment. The wearable system may be configured to

determine major horizontal planes based on the orientation, size, or shape of the surfaces from the regions identified from block **1684**. For example, the wearable system can identify horizontal surfaces that are large enough to allow a user or an avatar to stand on as the major horizontal planes. In some implementations, the wearable system can identify a major horizontal plane by finding a first intersection point of a ray with a physical horizontal surface whose normal at the intersection point is closely aligned to the gravity vector (which can be determined by an IMU on the wearable system). Additional examples of identifying major horizontal planes in the scene from the meshed environment are further described with reference to FIGS. 17A-17C.

[0213] FIG. 16A-2 illustrates another example data flow diagram for extracting major horizontal planes from the scene. The process **1640** starts from acquiring data from the depth camera **1642**, IMU **1644**, and/or the camera **1646** (which may be eye cameras or world cameras, RGB, RGB-D, infrared, or grayscale cameras). The data acquired from the IMU **1644**, the camera **1646**, and/or the depth camera **1642** can be used to determine head pose **1648** of a user. If the depth camera **1642** inputs to the head pose determination **1648**, then a camera may not be needed (indicated as Alt 1 in FIG. 16A-2). As described with reference to block **1682**, the position of the head pose may be determined with reference to a world frame of the user's environment.

[0214] At block **1652**, the wearable system can perform world reconstruction based on images acquired by depth camera **1642**. Output from the camera **1646** can optionally be used for world reconstruction (indicated as Alt 2 in FIG. 16A-2). The world reconstruction can involve stitching all images acquired by the cameras **1642** and/or the camera **1646** together for the user's environment (e.g., the user's room), can utilize the head pose **1648** for real world location correspondence, and convert the images of the environment in a mesh representing the environment. As described with reference to FIG. 16A-1 the world reconstruction may be based on a 3D point cloud generated based on images acquired by the depth camera **1642**. The world reconstruction can be based on data from time "i" (indicated as @time i in FIG. 16A-2) and can be updated as further data is received by the system (e.g., at times i+1, i+2, etc.).

[0215] At block **1654**, the wearable system can perform plane extraction based on the mesh. The wearable system can use techniques described with reference to block **1684** in FIG. 16A-1 to perform such extraction. For example, the wearable system can determine planes based on neighboring triangles in the mesh.

[0216] At block **1656**, the wearable system can extract major horizontal planes from the scene. The wearable system can use techniques described with reference to block **1686** in FIG. 16A-1 to perform such extraction. The major horizontal planes may include planes on which an avatar will be rendered on or which a user is standing.

[0217] In an alternate embodiment, the wearable system can directly perform plane extraction from the depth camera **1642** and IMU **1644**. The depth camera provides point locations of objects and the IMU provides directionality from gravity. This technique can provide a plane with a vector indicating which way is up (e.g., the direction opposite to the local gravitational direction indicated by the IMU). The wearable system can then spawn a new avatar on the first horizontal (or first suitable horizontal) surface detected, without the need for world reconstruction **1652**.

[0218] Information on major horizontal planes and the head location in the world frame may be passed by either of the processes **1600**, **1640** to the avatar autoscaler **692** for automatically scaling an avatar. FIG. 16B illustrates an example subprocess **1650** which can be implemented by the avatar autoscaler **692** (shown in FIG. 6B) as part of the wearable system **200**.

[0219] Upon receiving the head location information, the avatar autoscaler can, at block **1604**, determine a head position of the user wearing the wearable device. This action may be part of the head-tracking (which can also be referred to as inside-out head tracking because the user is wearing, for example, a device with one or more cameras facing outward toward the environment).

[0220] Upon receiving the major horizontal planes information from the scene, at block **1616**, the avatar autoscaler can determine a horizontal ground plane which supports the user. The horizontal ground plane can be the first large horizontal ground plane that is below the head (e.g., in situations where the user is standing on the ground). As described with reference to FIGS. 16A-1 and 16A-2, this ground plane can be extracted from 3D environment point cloud (either directly or by converting the 3D environment point clouds to a mesh first).

[0221] Given the 3D head position (as determined from block **1604**), the wearable system at block **1608** can project head position to the first large horizontal plane underneath the head. This can be performed by casting a ray down from the head to identify the largest horizontal planar region that intersects that ray. This action can be applied to identify a floor level (referred to as variable “user_feet” which indicates where the feet of the user appear) associated with where the user stands. In the situations where the user is not standing on the ground, the first large horizontal plane may not be the same as the horizontal ground plane.

[0222] At block **1620**, the avatar autoscaler **692** can estimate the user’s eye height from the horizontal support plane. The avatar autoscaler **692** can then take the vertical distance between the head and user_feet to estimate the height of the user. With reference to formulae (1) and (2), $User_H = user_head - user_feet$, where user_head represents the position of the head as determined from the block **1604** (which may correspond to the user’s eye level in certain implementations).

[0223] At block **1612**, the wearable system can identify the avatar’s rendering location in the scene. As will further be described herein, the avatar’s rendering location can be determined by the avatar processing and rendering system **690**. For example, the avatar autoscaler **692** can receive or access the avatar’s rendering location based on outputs from the intent mapping system **694**.

[0224] With the avatar’s rendering location, at block **1642**, the avatar autoscaler **692** can compute how tall the avatar should be to ensure the avatar and the user have similar eye height levels.

[0225] At block **1626**, the avatar autoscaler **629** can calculate the avatar’s target height. Formula (3) below illustrates an example of calculating the avatar’s target height.

$$Avatar_H = User_H - AvatarBase_H \quad (3)$$

[0226] $Avatar_H = User_H - AvatarBase_H$ where $AvatarBase_H$ is the height of the base of the avatar’s feet with respect to $User_H$. In situations where the avatar’s feet are standing on a higher location than the user’s feet, the $AvatarBase_H$ is a positive number and the avatar’s height is less than the user’s height. But if the user is standing on a horizontal support plane which is higher than the avatar’s rendering location, the $AvatarBase_H$ may be a negative number, which increases the avatar’s height to maintain eye contact with the user. If the avatar and the user are standing on surfaces with equal height, then $AvatarBase_H$ equals to 0.

[0227] At block **1632**, the avatar can be scaled so that its height matches the target height. In certain implementations, to avoid deforming the avatar, scaling the avatar can include resizing the height, width, and length of the avatar proportionally when matching the height of the avatar to the target height.

[0228] In certain implementations, the avatar autoscaler **692** can be left running (e.g., as a background process or agent) when the wearable system is being used, so that if the user dynamically changes his or her eye height, a new $Avatar_H$ can be scaled dynamically to maintain the user eye height with that of the avatar. For example, if the user’s head height changes because of some action such as, e.g., stepping on a stool, the system can quickly compute the new $Avatar_H$ to maintain eye contact between viewer and the avatar. Similarly, if the avatar changes its standing location to a different surface, the avatar’s scale can adjust accordingly to maintain horizontal eye contact.

Examples of Identifying Surfaces in a Physical Environment

[0229] As described with reference to FIGS. 16A-16B, the wearable system can identify planar surfaces in the user’s environment. These planar surfaces may include the horizontal ground plane (which can correspond to the ground of the environment), a horizontal support surface on which the user stands, and a surface on which the avatar is rendered. FIGS. 17A-17B illustrate an example of identifying planar surfaces from a meshed environment.

[0230] FIGS. 17A-17B illustrate a mesh environment **1740** (FIG. 17A) that is converted to a surface environment **1750** (FIG. 17B) which can include a plurality of planar surfaces. The mesh environment **1740** can be reconstructed from a 3D point cloud as described with reference to FIG. 16A-1. The reconstructed mesh environment **1740** may include one or more surfaces, such as, e.g., vertical meshed surfaces **1742** or horizontal meshed surfaces **1744**. The meshed surfaces may represent portions of objects **1748** in the environment (e.g., a tabletop can be represented as a horizontal meshed surface). A user or an avatar may be positioned on or near one or more reconstructed mesh surfaces, and as the user or the avatar moves, the height of the surface on which the user or avatar is positioned may change.

[0231] The wearable system may be configured to convert the reconstructed mesh environment **1740** into a surface environment **1750** which includes a plurality of planar surfaces extracted from the mesh. FIG. 17B illustrates extracted surfaces of the reconstructed mesh environment **1740**. Some example extracted surfaces can include the vertical surface **1752**, the horizontal surface **1758**, and the ground **1756**. Planar extractions can include grouping triangles of the mesh environment **1740** into areas of similar

orientation, which tends to indicate the grouping of triangles is part of a common surface. For example, the triangles **1704a** and **1704b** shown in FIG. 17B can be grouped together as representing a flat, horizontal surface, and the triangles **1702a** and **1702b** can be grouped together as representing a flat, vertical surface. Further processing can be done of these mesh areas to extract pure planar regions representing flat areas in the environment. Based on this planar extraction technique, the wearable system can extract major horizontal planes in the scene, on which the user, avatar, or objects may be situated.

Example Processes of Automatically Scaling an Avatar

[0232] FIGS. 18A and 18B illustrate example process of automatically scaling an avatar. The processes **1800a** and **1800b** in FIGS. 18A and 18B respectively can be performed by the wearable system **200**. For example, one or more blocks in these two processes may be performed by the avatar autoscaler **692** described with reference to FIG. 6B. Further, the positions and heights described herein may be calculated with respect to a world reference frame.

[0233] The process **1800a** describes calculating a target height for the avatar and rendering an avatar based on the target height. This process **1800a** may be applied prior to when an avatar is spawned.

[0234] At block **1812**, the wearable system determines a head position of a user **1812**. The head position may be determined based on data acquired from device sensors such as IMUs or world cameras. The head position may be determined with respect to the user's environment (e.g., as represented by a world frame).

[0235] At block **1814**, the wearable system can determine a surface in the user's environment **1814**. The surface may be the ground, the surface the user is standing on, or the surface that the avatar will be or is rendered on. As described with reference to FIGS. 16A-17C, the surface can be extracted from a set of 3D points (e.g., 3D environment point clouds) or extracted from a mesh of the environment (which may be generated from the 3D environment point clouds).

[0236] At block **1816**, the wearable system can calculate a user's height based on the surface and the user's head position. For example, the wearable system can calculate a user's height by computing the difference in height between the user's head position and the surface. In certain implementations, the user's height may be measured as the difference between the user's eye level and the avatar's feet which are situated on the surface.

[0237] At block **1818**, the wearable system can calculate a target height of an avatar based at least partly on the user's height. The wearable system can use any one of the formulas (1)-(3) described herein to calculate the target height.

[0238] At block **1820**, the wearable system can automatically scale the avatar based on the target height. For example, the wearable system can change the size of the avatar (which may include changing height, width, or length, alone or in combination) such that the height of the avatar matches the target height.

[0239] At block **1822**, the wearable system can cause a display to render the avatar at the target height in a mixed reality environment. For example, the wearable can cause the display **220** to render the avatar as standing on the ground or a platform. The wearable system can be config-

ured to display the virtual avatar appearing with reference to a world frame. The world frame may be overlaid on real objects in and/or features of the environment of the user.

[0240] This process **1800b** illustrates another example of scaling avatar. This process **1800b** may be applied after the avatar is spawned.

[0241] At block **1832**, the wearable system can monitor a user's position and an avatar's position in an environment. For example, the wearable system can monitor the user or the avatar's movements during a telepresence session. The wearable system can monitor the user's actions based on data acquired from the device sensors. The wearable system can monitor the avatar's movements based on the movements of the user associated with the avatar alone or in combination with outputs of one or more components of the avatar processing and rendering system **690** (such as, e.g., whether an intent-based rendering is enabled).

[0242] At block **1834**, the wearable system can detect a change in a relative height between the user and the avatar. For example, the wearable system can detect that a user or the avatar has sit on a chair, stepped up to or step down from a platform, etc., based on the user or avatar's position or movements.

[0243] At block **1836**, the wearable system can perform blocks **1812-1818** in the process **1800a** in response to detecting the change of the relative height. Based on the target height (generated by block **1818**), the wearable system can dynamically adjust the size of the avatar based on the target height. For example, the wearable system can increase or decrease the size of the avatar based on whether the avatar's feet are at a lower or higher position (respectively) than the user's feet.

Examples of Intent Based Rendering of a Virtual Avatar

[0244] As described with reference to FIGS. 11B-11D, the one-to-one mapping of a user interaction (such as, e.g., a head or body pose, a gesture, movement, eye gaze, etc.) into an avatar action can be problematic because it may create awkward or unusual results that do not make sense in the environment where the avatar is rendered. Advantageously, in some embodiments, the wearable system **200** can determine which part of an interaction is a world component (e.g., movements or interactions with an object of interest) that may be different in a remote environment, and which part of the interaction is a local component which does not require interactions with the environment (such as, e.g., nodding yes or no). The wearable system **200** (such as, e.g., the avatar processing and rendering system **690** or the intent mapping system **694**) can decompose a user interaction into two parts: the world component and the local component. The world component can be rendered (for an avatar) in the other user's environment based on the user's intent such that the intent of the world component is preserved but the action of the avatar for carrying out the intent may be modified in the other user's environment (e.g., by walking on a different route, sitting on a different object, facing a different direction, etc.). The local component can be rendered as a backchannel communication such that the local motion is preserved.

[0245] As an example, Alice may be actively moving around in her environment, the wearable system may convey some of her translational motion to Bob's environment (in which Alice's avatar is rendered). The wearable system can

re-interpret Alice's movement in Alice's world frame to match the motion in Bob's world frame as suggested by the user's intent. For example, Alice may walk forward toward Bob's avatar in Alice's environment. Decomposing intent from Alice's and Bob's head poses can allow a wearable system to determine which direction is "forward" in each of Alice's and Bob's environments. As another example, if Alice walks to a chair and sits down, it will look unusual if there is no chair in Bob's environment and Alice's avatar is suddenly sitting in mid-air. The wearable system can be configured to focus on the intent of Alice's motion (sitting), identify a "sit-able" surface in Bob's environment (which may be a chair, sofa, etc.), move Alice's avatar to the sit-able surface, and render the avatar as sitting on the sit-able surface, even if the physical location, height of the sit-able surface in Bob's environment is different than the one Alice sits in. As another example, Alice may be looking down at Bob's avatar, while in the remote environment, Bob may be looking up at Alice's avatar.

[0246] In certain implementations, such remapping of intent can occur in real-time (e.g., when two users are conducting a telepresence session) the human counterpart of the avatar performs the interaction. In other situations, the remapping may not occur in real-time. For example, an avatar may serve as a messenger and delivers a message to a user. In this situation, the remapping of the avatar may not need to occur at the same time as the message is crafted or sent. Rather, the remapping of the avatar can occur when the avatar delivers the message (such as, e.g., when the user turns on the wearable device). The remapping may cause the avatar to look at the user (rather than a random location in the space) when delivering the message. By rendering the world motion based on the intent, the wearable system can advantageously reduce the likelihood of unnatural human-avatar interactions.

Examples of Decomposing a User Interaction

[0247] FIG. 19A describes an example of a system for decomposing a user interaction. In this example, Alice can be in an environment A 1902a. Alice can be a user of the wearable system 200, through which Alice can have a mixed reality experience with her environment A 1902a. Alice can perform various interactions 1912a which may be mapped to an avatar in a remote environment. An interaction 1912a of Alice can comprise a movement, pose (e.g., head pose, body pose or hand gesture, etc.), eye gaze, and so on, alone or in combination. In some situations, one or more of these interactions 1912a may also be referred to as an interaction event.

[0248] The wearable system can acquire data using one or more device sensors described with reference to FIGS. 2 and 3. Data acquired by the device sensors can be used to determine the user's interactions. Based on the data, the wearable system can determine the characteristics of the interaction 1912a, such as, e.g., the object of interest (also referred to as item of interest or IOI for short) which may be an object which the user is interacting with or is interested in, the type of the interaction (e.g., whether a user is walking, sitting, standing, etc.), and so on. Based on the characteristics of the interaction 1912a alone or in combination with the data acquired from the device sensors, the wearable system can determine whether the interaction 1912a comprises a world component and/or a local component.

[0249] As illustrated in FIG. 19A, the movement 1920a can be decomposed into a local motion 1922a and/or a world motion 1924a. The local motion 1922a can include a motion with respect to the fixed-body reference frame and may not interact with the environment A 1902a. The world motion 1924a can include a motion which involves an interaction with the environment A 1902a. Such world motion 1924a can be described with respect to the world frame. For example, Alice may be running in her environment. The arm movements may be considered as local motion 1922a while the leg movements (e.g., moving forward toward a direction in the environment) may be considered as world motion 1924a. As another example, Alice may be sitting on the chair with her legs crossed. The pose sitting with legs crossed is considered as a local component, while the interaction of sitting on the chair is considered as a world component. The wearable system of the viewer of Alice's avatar can render Alice's avatar sitting on a bed with legs crossed (if the remote environment does not have a chair).

[0250] The interactions 1912a can sometimes also involve interacting with an object of interest, such as, e.g., a physical object (e.g., a chair, table, a bed, a mirror, etc.) or a virtual object (e.g., another user's avatar, a virtual entertainment content, or other virtual applications) in the environment. Based on data acquired from the device sensors, the wearable system can identify an object of interest 1950a associated with the user Alice's interaction. For example, the object of interest 1950a can be identified from Alice's movement 1920a, pose 1930a, eye gaze 1940a, individually or in combination.

[0251] As further described herein, in certain implementations, the object of interest 1950a may include a component which attracts the user's attention. As described above, the eyes of a user are typically directed toward a region called the social triangle of the other user's face during friendly, neutral conversation. The social triangle is formed with a first side on a line between the user's eyes and a vertex at the user's mouth. The object of interest 1950a may include the social triangle. For example, Alice may look at the social triangle on Bob's face because humans tend to naturally looking at another's personal social triangle during a face-to-face conversation. The interactions with the object of interest can be considered as world components because they involve interacting with virtual or physical objects in the environment.

[0252] The wearable system can map the local component of an interaction 1912a to the virtual avatar 1970 using direct mapping 1962. For example, the wearable system can map the local motion 1922a, the pose 1930a, or the eye gaze 1940a into an action of the avatar 1970 using direct mapping 1962. As a result, the avatar's 1970 action can reflect the corresponding local component of the interaction 1912a performed by Alice (e.g., an avatar nods her head when Alice nods her head).

[0253] The wearable system can map the world component of an interaction 1912a using intent-based mapping 1964. As a result of the intent-based mapping 1964, the action of the avatar 1970 achieves the same purpose as the corresponding interaction 1912a performed by Alice, even though the action of the avatar 1970 may not be exactly the same as that of Alice. For example, the wearable system can map the world motion 1924a to the avatar 1970 based on environment features (such as, e.g., obstacles, layout of objects, etc.) in the environment in which the avatar 1970 is

rendered. As another example, the wearable system can move or reorient the virtual avatar **1970** in the remote environment such that the virtual avatar **1970** interacts with a similar object of interest **1950** in the remote environment. **[0254]** In certain implementations, by decomposing the user interaction into a world component and a local component, the techniques provided herein can provide an improved wearable device, by enabling faster processing, reducing storage requirements, and improving latency. For example, the technique does not require the entire animated avatar's geometry or the entire user's motion to be sent across to Bob's wearable device in the environment B **1902**. Alice's wearable device can send a subset of the sensor data that can be used to animate the avatar locally at Bob's wearable device, and thus Alice's device does not have to keep on sending geometry and animation updates for the entire avatar. Since the data sent by Alice's wearable device may be sparse, Bob's wearable device can update more frequently to get more responsive avatar animation.

Examples of Mapping a User Interaction to a Remote Environment

[0255] FIG. **19B** describes an example of a system for mapping a user's interaction to an avatar in a remote environment. The virtual avatar **1970** can be an avatar of Alice. FIG. **19B** shows that the virtual avatar has various characteristics such as, size **1972** (see examples described with reference to FIGS. **12A-18B**), movement **1920b**, pose, **1930b**, and gaze **1940b**. As described herein, virtual avatar **1970** can also include other characteristics not shown in this figure, such as e.g., facial expressions or looks. The avatar of Alice can be rendered in Bob's environment (shown as environment B **1902b**) by the wearable device **904**.

[0256] The virtual avatar **1970** can be rendered based on various contextual factors in the environment B **1902b**, such as, e.g., objects in the environment B **1902b** (in which the virtual avatar is rendered), an area of interest **1962**, contextual information **1964** of user Bob (such as, e.g., user Bob's position, orientation, interaction, etc.), and contextual information of the environment B **1966** (such as, e.g., layout of objects in the environment B **1966**). In addition to or in alternative to contextual factors in the environment B **1902b**, the virtual avatar **1970** can also be rendered based on Alice's interaction **1912a**. As described in FIG. **19A**, a world component of Alice's interaction **1912a** can be mapped to the virtual avatar **1970** based on intent while the local component of Alice's interaction **1912a** can be mapped to the virtual avatar **1970** via direct mapping **1962**.

[0257] As an example, Alice can interact with an object of interest **1950a** in Alice's environment. The wearable system can determine an object of interest **1950b** in the environment B **1902b** which corresponds to the object of interest **1950a**. For example, if Alice sits on a chair in the environment A **1902a**, the corresponding object of interest **1950b** may be a chair in the environment B **1902b** or another sit-able surface. As another example, Alice is having a face-to-face conversation with Bob's avatar in the environment A **1902a**. The wearable system can map Alice's interaction in environment A **1902a** to Bob's environment B **1902b** such that Bob is engaged in a face-to-face conversation of Alice's avatar (e.g., where Bob's wearable device can orient Alice's avatar toward Bob). Details related to rendering a virtual avatar based on an object of interest are further described with reference to FIGS. **24-30**. In certain implementations, the

intent-based mapping **1964** can also be applied to an area of interest **1962**. For example, Alice may look at a crowd of people in her environment **1902a**, although she does not look at a particular person in the crowd. Alice's interaction may be mapped to environment B **1902b**, where the virtual avatar of Alice **1970** may be orientated to look at a group of virtual avatar or a group of people in the environment B **1902b**. As will be further described herein, in certain implementations, the area of interest **1962** may also be determined based on environment B **1902b**, without intent-based mapping of Alice's interaction. For example, the wearable system can orientate the avatar **1970** to look in the direction of a sound source in the environment B **1902b**, even though a similar sound source is not present in the environment A **1902a**.

[0258] As another example, Alice's eye gaze **1940a** can be tracked to a fixation point in the environment A. However, because this fixation point is based on the Alice's local environment, when it is mapped directly to the environment B, it can become out of context. In some implementations, the wearable system can automatically identify an object or area of interest in the environment B such that that wearable system can compute a fixation point based on the object or area of interest, which may provide a more realistic eye gaze for the virtual avatar. Details related to rendering a virtual avatar based on objects or stimuli in a remote environment are further described herein in another section.

[0259] In addition to or in alternative to mapping the world component, the local component of an interaction **1912a** can also be mapped to the remote environment based on intent. For example, the user's interaction **1912a** may include waving one hand which can indicate that the user has negated a question. This intent of negation can be communicated to Bob's wearable device in the environment B **1902**. This intent can be mapped to a virtual avatar **1970** and Bob's wearable device can show the virtual avatar **1970** waving both hands. As another example, one type of user interaction may be mapped to a different type of user interaction to convey the same intent. For example, to map Alice's intent of negating a question (via waving her hand), Bob's wearable system can map this intent to a spoken sound "no", and place this sound to Bob via speaker **240** on Bob's wearable device.

[0260] Advantageously, in some embodiments, by remapping user's intent in an environment that the avatar is rendered, the wearable system can interact and communicate to adapt to a viewer's local physical surroundings. For example, Alice can be sitting at her desk talking to Bob's avatar **1100** on the table, while Bob is in a big empty room with Alice's avatar **1000** appearing to stand in front of him. Bob and Alice each has a very different environment, but can still successfully communicate. On the other hand, in a VR environment, both of Alice's avatar and Bob's avatar are transported to be at the same virtual space, and thus the avatars do not need to adapt to a user's local physical environment.

Example Processes of Intent Based Rendering of a Virtual Avatar

[0261] FIG. **20** illustrates an example process of intent based rendering of a virtual avatar. The example process **2000** can comprise two subprocesses **2010** and **2020**. The subprocess **2010** can be applied to decompose a user's interaction into a world component and a local component,

and determine intent associated the world component. The subprocess **2020** can be applied to render a virtual avatar in a remote environment based on the user's interaction and environment information of the remote environment. The subprocess **2010** can be performed by Alice's wearable device (referred to as wearable device A in FIGS. **20-21C**) and the subprocess **2020** can be performed by Bob's wearable device (referred to as wearable device B in FIGS. **20-21C**). One or more portions of the subprocess **2010** and **2020** can be performed by the intent mapping system **694** (in FIG. **6B**) as implemented on wearable device A and wearable device B, respectively.

[**0262**] With reference to the subprocess **2010**, at block **2012**, the wearable device A can identify a user interaction. The wearable device A can analyze data acquired from the device sensor to determine the user's interaction. Such example interactions are described with reference to interaction **1912a** shown in FIG. **19A**.

[**0263**] At block **2014**, the wearable device A can decompose the user interaction into a world component and a local component. As an example, pointing and gazing at an object of interest can be considered as a world component. As another example, moving around in Alice's environment or pose changes (e.g., sitting or lying down) can have a component relative to world space (e.g., the movement from location **1** to location **2**, or interact with a physical object for sitting or lying down). As yet another example, head position changing (e.g., nodding) includes data that is in the local space.

[**0264**] At block **2016**, the wearable device A can determine intent associated with the world component. The intent may include an object of interest that Alice is interacting with or is interested in interacting with. The wearable device A can determine what in the world space Alice is referencing. For example, the wearable device A can determine another avatar, shared digital content, or a neutral/ambiguous point (for eye gaze) as an object of interest. In some situations, there may be multiple possible things that Alice could be referencing. The wearable device A can leverage the contextual information associated with user Alice or the Alice's environment to determine the object of interest. As an example, Alice's hand gesture can point in the direction of a chair and the virtual content which Alice is sharing with other participants in a telepresence session. The wearable device A can determine that the virtual content is likely to be an object of interest because it is more pertinent to the conversation in the telepresence session.

[**0265**] Another example of contextual information can be the user's past or recent activity, for example, an object may have a higher likelihood to be an object of interest if it is the most-recently-used or referenced object. In certain implementations, multiple user inputs can be combined to determine the object of interest. For example, the wearable device can combine eye gaze direction and the user's hand gesture (e.g., pointing at a place in the user's environment) to determine the object of interest. FIGS. **24-30** describe additional examples of determining an object of interest using weight factors.

[**0266**] In some situations, the wearable device A may determine an intent without needing to find an object of interest in the wearable device A's environment. The wearable device A can make such determination of intent based on the nature of Alice's interaction. For example, if Alice is recording a message, even though the recipient (e.g., Bob's device) is not in Alice's space (and the wearable device A can't use a ray cast to identify an object of interest), the wearable device can automatically determine that the intent

(to implement on Alice's avatar) is to orient toward and look at the person receiving the message on playback. Thus, the wearable device A can automatically specify that the intent, in this example, is positioning eye gaze and orienting body pose toward the other live user in the avatar's environment when playing this recorded message.

[**0267**] At block **2018a**, the wearable device A can communicate the intent to the wearable device B and at block **2018b**, the wearable device B can communicate the information associated with the local component to wearable device B. Data related to the intent and data related to information associated with the local component may be transmitted separately so that the animation in local space is separated out from the world space animation or pose data. The information on the local component may be transmitted separately because such information (e.g., local animation data) is based at least on Alice's action in the local frame and sent over to the wearable device B (which can be mapped to avatar's local frame). World space information may need to be handled differently than the local component because it is in a different coordinate system (e.g., world space information is with reference to the world frame not the local frame of Alice or Alice's avatar). It can be possible to use the same data channel to send both, as long as each data or message type are differentiated (e.g., by providing a label for data associated with the world space and another label for data associated with the local component).

[**0268**] With reference to the subprocess **2020**, at block **2022**, the wearable device B can determine contextual information associated with the wearable device B. Such contextual information can be used to determine Alice's avatar's interaction with the wearable device B's environment. For example, the wearable device B can determine a corresponding object of interest in the wearable device B's environment. As an example, where the object of interest in wearable A's environment is a virtual object shared between the wearable device A and the wearable device B, the corresponding object of interest in the wearable device B's environment can be this shared virtual object. In some situations, the intent may include semantic intent, which can cause the wearable device B to find an object having similar classification or function or support similar user interaction in wearable device B's environment as the corresponding object of interest. For example, an object of interest in wearable device A's environment can include an office chair. The corresponding object of interest in wearable device B's environment may be an object with sit-able surface (such as a bed or a bench).

[**0269**] The wearable system can also determine where each world-space item is situated relative to Alice's avatar in wearable device B's environment. Examples of orienting the avatar based on layout of the environment are further described with reference to FIGS. **22-30B**.

[**0270**] At block **2026**, the wearable device B can determine the avatar's local action based on information associated with the local component received from the wearable device A. For example, the wearable device B can determine body pose including a position and an orientation for the avatar so that it can look or point at the appropriate world space content. In certain implementations, the avatar's actions in the local frame can be subject to collision checks and environment barriers in the world space. For example, a local motion may include raising an arm of the avatar. However, the avatar may be standing closely to a wall and the arm would appear to penetrate through the wall if the avatar raises the arm. The wearable device B may be configured not to animate this action or may be configured

to take an action with respect to the world frame (e.g., by first moving the avatar away from the wall (e.g., by taking a step back)) before rendering the animation of raising the arm.

[0271] At block 2028, the wearable device B can cause a display to render the virtual avatar based on the action with respect to the world space and the action for the local space as determined from blocks 2024 and 2026 respectively. The wearable system can map animation for the avatar's action in the local frame on top of the avatar's action in the world frame of the environment. For example, the wearable system can map a hand gesture (point at an object of interest) on top of a body orientation facing toward the object of interest (upon adjusting the orientation based on the intent).

[0272] In some implementations, the wearable device B can scan the environment (such as, e.g., the environment meshes and planes described with reference to FIGS. 17A and 17B) to obtain world space information in the environment B. With this information the wearable device B can determine Alice's avatar's position and movement with respect to environment landmarks (which may become objects of interest) in Alice's environment. The wearable system can treat actions, such as, e.g., locomotion, sitting and reaching out to objects as world component. In certain implementations, the wearable device B or A can have a set of preprogrammed actions that will automatically be modeled with respect to the world frame. The wearable device B can also determine a set of motions in the local space, such as, e.g., head pose changes relative to an initial starting pose (which may be set upon spawning the avatar). The wearable device B can compute relative locomotion with respect to a starting orientation and additively apply that motion to the initial orientation of the avatar so locomotion changes are relative to the spawning point.

[0273] Although in this example, blocks in the subprocess 2010 are described as being performed by the wearable device A while blocks in the subprocess 2020 are performed by the wearable device B, this is not a limitation on whether a block must be performed by the device shown in FIG. 20. For example, the wearable device A can capture Alice's interaction data and pass such data to a server or the wearable device B to extract the world component, the intent, and the local component.

Example Processes of Intent Based Rendering for Interactions with a Virtual Object Based on Head Pose and Eye Gaze Tracking

[0274] FIG. 21A illustrates an example process for determining intent based on head pose tracking and eye gaze tracking. The process 2130 shown in FIG. 21A can be applied to determine Alice's intent by mapping Alice's interaction in Alice's environment (also referred to as environment A in FIGS. 20-21C) to Alice's avatar rendered in Bob's environment (also referred to as environment B in FIGS. 20-21C). The process 2130 can be performed by Alice's wearable device 902, a remote computing system 920, individually or in combination. The process 2130 can also be implemented as part of the intent mapping system 694 shown in FIG. 6B.

[0275] The process 2130 starts off at 2132 with head pose tracking 2134a, which can track Alice's head movement data. Various techniques may be used for tracking head pose. For example, Alice's wearable device 902 can employ a combination of IMUs and cameras (e.g., cameras in the

outward-facing imaging system 464 or the inward-facing imaging system 462) to record rotational and translational motion of Alice. The movement data recorded by the wearable device 902 can be used to extract features in the environment A for estimating the head pose with respect to a world frame (shown as data block 2136a) associated with the environment A.

[0276] The wearable device 902 can also perform eye tracking 2134b to obtain a fixation point (point of focus) of the eyes in Alice's field of view in a head frame (as indicated in the data block 2136b, where the head frame is associated with a coordinate system local to Alice).

[0277] At block 2138, an eye gaze target point can be calculated based on a combination of the fixation point and the head pose. The result obtained from the computation in block 2138 can include the gaze fixation point with respect to the world frame and ray direction in the world frame (shown in the data block 2140a).

[0278] Based on the head pose and eye gaze, static virtual scene models in the world frame 2140b (which may describe static virtual objects in the environment A), and dynamic virtual objects (which may include avatars) in the scene 2140c (which may include virtual objects that are fixed at a given position or orientation in the environment A) can be used, at block 2142, to determine what virtual objects in the scene intersect with the gaze fixation point in the local space. For example, the wearable device 902 can perform a ray casting by casting a ray vector from the eyes to this fixation point. This ray vector can be used to determine what Alice is looking at in her field of view as perceived through the wearable device 902.

[0279] There may be three basic possibilities for things that Alice could be looking at. First, Alice could be looking at a physical object in her environment, such as, e.g., a chair or lamp. Second, Alice could be looking at a virtual object rendered in her AR/MR environment by the display 220 of her wearable device 902. Third, Alice could be looking at nothing in particular, such as, e.g., when lost in thought or thinking about something.

[0280] At block 2144, the wearable system can determine whether the gaze intersects with a virtual object, and if so, a different head pose for Alice's avatar may need to be computed if the objects are in a different relative position from Alice's avatar's perspective (as compared to Alice's perspective). If not, the process 2130 goes back to the start block 2132. In certain implementations, the virtual object as determined from block 2144 can be an object of interest.

[0281] At block 2146, the wearable system can extract semantic intent directives, such as interacting with a certain object. For example, the wearable system can determine an interaction with the object as intended by the user, such as, e.g., moving the object, staring at the object, modifying the object, talking to a virtual avatar of Bob, etc.

[0282] The intent of interacting with an object (or a person as determined from interacting with his avatar) and the head pose can be communicated to Bob's wearable device which can map Alice's head pose to Alice's avatar (as rendered by Bob's wearable device) based on the intent.

[0283] Algorithm (i) below describes an example pseudo-code implementation of the process 2130 in FIG. 21A.

Algorithm (i)

Given head pose H_W in a world coordinate frame W ,
 and eye fixation point F_H in the head frame H .
 Let P be the set of real physical objects in a user's immediate surroundings
 Let S be the set of 3-D static virtual objects rendered in the scene via the wearable display.
 Let D be the set of dynamic 3D virtual objects such as other avatars or moving objects.
 From H_W , and f_H ,
 Let $f_W = f_H H_W$
 represent the 3-D fixation point F_W with respect to the world frame W ,
 and H_W is a 4x4 transformation matrices representing a coordinate frame.
 Let e_H represent a reference point between the eyes of the head in the head frame H .
 Let $e_W = e_H H_W$ be the point e_H expressed in the world frame W .
 Let $g_W = f_W - e_W$ be a gaze direction ray pointing in the direction of the line of sight of the head looking towards the fixation point f_W and originating at e_W .
 The ray can be parameterized as $g_W(t) = e_W + t(f_W - e_W)$, t is in $[0, \infty]$,
 represents an infinite ray with $t=0$ corresponding to the point e_W and $t=1$ representing the fixation point f_W on this ray.
 For g_W , test intersection of this ray against P , S and D . Select the object O in the union of P , S , D , that intersects at the smallest value of t . This coincides with the closest object among P , S , D that intersects the ray $g_W(t)$.
 Let I_avatar be the set of intents for sending to a remote device for controlling the avatar.
 Let H_avatar be the current head pose for sending a remote device for controlling the avatar.
 If O is a member of S (static virtual objects), add the intent $lookat(S)$ to I_avatar .
 If O is a member of D (dynamic virtual objects), add the intent $lookat(D)$ to I_avatar .
 Set $H_avatar = H_W$.
 The output is the set of I_avatar and H_avatar

The output I_avatar and H_avatar can be communicated to Bob's wearable device for rendering Alice's avatar based on intent as shown in the block **2150**.

[0284] During an avatar control session (such as, e.g., during a telepresence session or during an interaction between a viewer and an avatar), the wearable device A can update Alice's avatar by sending the current head pose at regular, periodic, or irregular time intervals to the wearable device B. From Algorithm (i), the wearable device A can pass a set of intents (I_avatar), and the current head pose of Alice (H_avatar) to the wearable device B. To determine a baseline head pose ($H_baseline$) to be used in animating Alice's avatar by wearable device B, the wearable device perform the calculations in the paragraph immediately below this paragraph.

[0285] For the first time sample, a baseline head pose is defined as: $H_baseline = H_avatar$. If I_avatar is set, $H_baseline$ is computed to be the pose which the avatar's head points towards the object S for the intent ($lookat(S)$) in Bob's local space. Note that the location of S in Alice's local space may not be the same as the location of S in Bob's local space. However, by sending the intent $I_avatar = lookat(S)$, the wearable device B can compute head pose to maintain this intent of directing the head towards S in the avatar's remote environment. For every frame of image acquired by the wearable device A, the wearable device B can compute the final avatar's head pose, H_final by setting it initially to $H_baseline$ and adding on the relative pose ($H_relative$) between $H_baseline$ and H_avatar . The result is that the intent ($lookat(O)$) is preserved between the local user in his environment and the remote avatar looking at the same object or avatar O in the remote environment of the avatar.

[0286] Additional user experiences for intent based rendering for interacting with a virtual object of interest is further described with reference to FIGS. 24-30.

Example Processes of Intent Based Rendering for Interactions with an Environment

[0287] In addition to or as an alternative to mapping Alice's head pose and eye gaze based on intent, Alice's interaction with the environment can also be mapped to her avatar based on intent. The interaction with the environment may include interacting with a physical object of interest in the environment. For example, if an avatar of Alice appears in Bob's environment, rather than mimicking the exact motions of Alice, the wearable device of Bob can analyze Bob's environment and adjust Alice's avatar's behavior accordingly, such as, e.g., walking around furniture, sitting on empty chairs, or leaning on tables or shelves close to where Alice's avatar is standing in Bob's environment.

[0288] FIG. 21B illustrates an example process for mapping an interaction with an environment to an avatar's interaction with another environment. The process **2160** shown in FIG. 21B can be applied to determine Alice's intent for mapping Alice's interaction with Alice's environment to Alice's avatar rendered in Bob's environment. The process **2160** can also be implemented as part of the intent mapping system **694** shown in FIG. 6B. Although the process **2160** is described from the perspective of Bob's wearable device, the process **2160** can be performed by Alice's wearable device **902**, Bob's wearable device **904**, a remote computing system **920**, individually or in combination.

[0289] At block **2162a**, the wearable device B can retrieve participant sensor data **2160a**. For example, the participant sensor data **2160a** may include Alice's motion data acquired by device sensors of Alice's wearable device, Alice's intent as determined from Alice's motion data, the world component of Alice's interaction with her environment, the local component of Alice's interaction, etc., alone or in combination.

[0290] At block **2162b**, the wearable device B can determine a new relative motion and position of avatar in environment B in which Alice's avatar is rendered. The new relative motion and position can be determined based on the contextual information of the environment B (such as, e.g., the layout of objects in the environment B, objects in the environment B, etc.) or Alice's intent.

[0291] At block **2166**, the wearable device B can determine whether the result determined from block **2162b** will result in a collision with objects in the environment. The wearable device B uses one or more outputs from the environment reconstruction and analysis subprocesses **2164** shown in FIG. **21C** to perform such collision check.

[0292] If a collision is detected, the process **2160** can go back to the block **2162b** where a new relative motion and position of avatar is determined. For example, if an avatar's new position results in a collision or interference with an existing object in the environment, the wearable system can apply motion planning algorithms to determine a new collision-free position. One example of the motion planning algorithms is to back track steps from the new position towards the previous position until a collision no longer occurs. In some cases, a new direction can be calculated entirely to avoid a collision.

[0293] If the collision is not detected, at block **2168**, the wearable device can determine whether a mesh object is one having a semantic category that support user interaction. As described with reference to FIG. **7**, the object recognizers **708** can recognize the objects in an environment (where the object recognizers **708** may implement the environment reconstruction and analysis subprocesses **2164**) attach semantic information to the objects. The semantic information may include a category that an object is interactable by a person. For example, the object recognizer **708** can recognize what type of object a mesh is, such as e.g., a chair, table or wall or other piece of furniture.

[0294] If the object is not in an interactable category, the process **2160** ends at block **2178**. If the object is interactable, at block **2170**, the wearable system can determine interaction type based on semantic category, some example interactable categories can include: chair, wall, table, etc. Within each interactable category, the wearable device B can identify affordances as shown in blocks **2172a**, **2172b**, and **2172c**. The affordance of an object comprises a relationship between the object and the environment of the object which affords an opportunity for an action or use associated with the object. The affordances may be determined based on object's characteristics (which can be recorded as part of the semantic information). Some example object's characteristics can include the function, the orientation, the type, the location, the shape, or the size of the object. As examples, an affordance of a horizontal table is that objects can be set onto the table, and an affordance of a vertical wall is that objects may be hung from or projected onto the wall. The affordances may also be based on the environment in which the object is located (e.g., a table in an office can be used as a study table, while a table in a dining room is for dining purpose).

[0295] Once the category is identified, the wearable system can look for category-specific regions at blocks **2174a**, **2174b**, and **2174c**. In the case of a chair, this would be identifying the seat of a chair, the backrest, and perhaps armrests. These sub-parts of an object can imply various ways in an avatar can interact with it. For example, when an

avatar sits down, its rear end can rest on the seat of a chair and its back can be against the backrest of the arms resting on the armrests (see block **2176c**).

[0296] One method to position parts of an avatar's body on an object is to constrain a part of the avatar's body to a particular position and orientation is called inverse kinematics. Multiple body targets can be constrained and positioned at the same time using multiple inverse kinematics targets. [0297] FIG. **21C** illustrates an example of environment reconstruction and analysis subprocesses **2164**. The example subprocesses **2164** can be implemented as part of the object recognizer **708** shown in FIG. **7** or as part of the contextual information analysis system **688** shown in FIG. **6B**. The example subprocesses **2164** can include the subprocess **2180** for recognizing planes and the subprocess **2190** for identifying objects in the environment (e.g., by analyzing mesh segments).

[0298] With reference to subprocess **2180**, at block **2182**, the process can extract major planes in the environment. Major planes may be extracted using the techniques described with reference to FIGS. **16A** and **17A-17B**. For example, a wearable system can generate a 3D point cloud based on data acquired by the outward-facing imaging system **464** (which may have depth sensing ability). Planes can be extracted directly from the 3D point cloud or be extracted from meshes determined from the 3D point cloud. Although a 3D point cloud is used as an example here, planes can be extracted from other 3D models of the environment. For example, the planes may be extracted from the world map **910** which may or may not represent the environment as a 3D point cloud.

[0299] At block **2184**, the process can categorize planes into semantic categories, e.g., floor, wall, tabletop, etc. This block can be performed using techniques described with reference to FIGS. **17A** and **17B**. For example, planes may be identified from neighboring triangles of a mesh that is determined from image analysis of the environment.

[0300] With reference to the subprocess **2190**, at block **2192**, non-planar regions may be extracted from the environment mesh. At block **2194**, the nonplanar regions are separated out into mesh segments, where one or more mesh segments can correspond to a physical object in a physical environment. At block **2196b**, the process can compute bounded volumes for each segment to determine the boundary of the physical object. At block **2196a**, the mesh segments can be categorized into semantic categories which may correspond to what type an object is.

Example User Experiences for Intent Based Rendering for Interactions with an Environment

[0301] FIGS. **22** and **23** illustrate two user interaction experiences where a user's interaction with an environment is mapped to the user's avatar in another environment based on intent.

[0302] The scenario in FIG. **22** has four scenes **2202a**, **2202b**, **2204a**, and **2204b**. The scene **2202a** describes Alice's environment at time t_1 ; the scene **2202b** describes Bob's environment at time t_1 ; the scene **2204a** describes Alice's environment at time t_2 ; and the scene **2204b** describes Bob's environment at time t_2 . As shown in these two figures, Alice's environment is a living room environment, which can include a sofa **2222** while Bob's environment is a bedroom environment which can include a desk **2224** and a bed **2226**. In Alice's environment, Alice is facing

east while Bob's avatar **1100** is facing west (as shown by the coordinate system **2250**), such that Alice is facing Bob's avatar **1100**. In Bob's environment, Alice's avatar **1000** and Bob are also facing each other. However, Alice's avatar **1000** is facing south while Bob is facing north.

[0303] With reference to the scenes **2202a** and **2204a**, Alice is walking eastbound **2212** toward Bob **1100**. The wearable device of Alice can analyze Alice's motion and objects in Alice's environment (including Bob's avatar **1100**) to determine the intent of Alice's motion. In this example, the intent is to walk toward Bob (because Bob's avatar **1100** can serve as a virtual presence of Bob in Alice's environment). The wearable device of Alice can communicate this intent to Bob's wearable device for mapping Alice's motion to Bob's environment.

[0304] With reference to scenes **2202b** and **2204b**, because the orientation of Alice's avatar **1000** and Bob in the Bob's environment is different from Alice's orientation **912** and Bob's avatar's **1100** orientation in Alice's environment, to map Alice's intent to walk toward Bob, the wearable device of Bob renders Alice's avatar's motion as moving southbound **2214** toward Bob. In the scenario, if Alice's motion is mapped to Bob's environment, but the intent is not mapped Alice's avatar **1000** would walk eastbound into the bed **2226** which creates an unnatural user experience.

[0305] The scenario in FIG. **23** has four scenes **2202a**, **2202b**, **2302a**, and **2302b**. The scenes **2202a** and **2202b** are the same as those described in FIG. **22**. Scene **2302a** describes Alice's environment at time t_3 ; and the scene **2302b** describes Bob's environment at time t_3 .

[0306] With reference to the scenes **2202a** and **2302a**, Alice moves (as illustrated by the trajectory **2312a** to sit on the sofa **2222** in her environment. As illustrated in the scene **2302a**, Bob's avatar **1100** may respond to this movement of Alice (where Alice is considered as an object of interest for Bob's avatar). As a result, Bob's avatar **1100** can automatically reorient itself (as illustrated by the arrow **2214a**) such that Bob's avatar **1100** remains face-to-face contact with Alice. As described with reference to FIGS. **12A-18B**, in some implementations, Bob's avatar **1100** can also automatically be resized based on the height difference between Alice (after her sitting on the sofa **2222**) and Bob's avatar **1100** (as standing in the environment). For example, Bob's avatar **1100** can be dynamically scaled down to maintain eye-to-eye contact with Alice.

[0307] Alice's interaction with her environment can be mapped to Bob's environment based on the intent of her interaction. Bob's environment, however, does not have a sofa. The intent of Alice's interaction can be sit on a sit-able surface. The wearable device of Bob can automatically detect a sit-able surface (e.g., using the techniques described with reference to FIGS. **21B** and **21C**) and render Alice's avatar **1000** as sitting on the sit-able surface. As shown in the scene **2302b**, Bob can perceive, via his wearable device, that Alice's avatar moves along a trajectory **2312b** and sits on the bed **2226**. Alice's avatar **1000** does not sit on the table **2224**, however, because the table **2224**'s affordance does not include a sittable surface (e.g., a person won't sit on a table in a typical interaction).

Examples of Intent Based Rendering for Interactions with a Virtual Object of Interest

[0308] As described herein, each user in a mixed reality environment may have a different physical environments (or

may be in a different position of the same physical environment) but nevertheless shares at least a portion of the virtual experience (e.g., by interacting with the same objects in a virtual environment) with another user. However, because the users may be in completely different physical environments, the position and orientation of shared virtual objects and remote avatars may be unique to the particular user. The lack of a common environment layout makes it difficult to map a user's focus or communication cues in his own local environment to a remote avatar in a way that makes logical sense to other users in their own virtual environments. For example, if a user looked to the right to see a shared virtual object in his own local environment, having his remote avatar also look to the right may not make sense if the shared virtual object was actually to the avatar's left in the remote virtual environment.

[0309] As another example, in Alice's virtual experience displayed to her by her wearable system, Bob's avatar may be standing in front of Alice with a tree to his right and a boulder to his left. In Bob's virtual experience, Bob may be standing to the left of Alice's avatar, with both the boulder and the tree to her right. A naive approach of transmitting Bob's one-to-one head and hand movement may not make it clear to Alice what object Bob is looking at or interacting with. Simply forcing Bob's avatar's orientation to snap to a specific object may strip out subtle head and hand movements, which communicate Bob's intent and personality. Accordingly, as described with reference to FIGS. **19A-23**, the wearable systems described herein can remap Bob's interest point and movement from Bob's virtual environment to Alice's virtual environment, so if Bob is looking at the top of the tree in Bob's environment, Alice will see Bob's avatar looking at the top of the tree in her virtual environment, even if the tree is in a different relative location in Alice's virtual environment.

[0310] As described with reference to FIGS. **19A-23**, it may therefore be advantageous to map a user's interaction with objects in the environment onto his remote avatar using intent based rendering rather than using one-to-one mapping that directly maps the user's movements into an avatar's movements. In intent based rendering of a virtual avatar, the wearable system may map the remote virtual avatar's interaction based on the user's orientation in relation to an item of interest in the environment. The remote avatar may therefore turn or face towards the object the user is focused on rather than strictly mimicking the user's movements (as in a 1:1 mapping) which may cause the avatar to lose the interaction with the item of interest. For example, if a user turns to examine a virtual object to his right, the user's remote avatar may identify that same virtual object to the left of the remote avatar and turn to the left. The remote avatar therefore did not strictly mimic the user's movement but still captured the user's intent to face towards the virtual object. Once an object of interest has been identified, the wearable system of the user can manipulate an avatar to carry out the intent of the avatar.

[0311] FIG. **24** shows a block diagram of an overview of intent-based virtual avatar rendering based on an object of interest. Intent based rendering a virtual avatar can depend on a user's pose in a virtual environment. The block diagram may be implemented by the intent mapping system **684** described with reference to FIG. **6B**. In some embodiments, the user's pose may be determined by using a wearable system with an outward-facing system or an inward-facing

imaging system (e.g., for eye gaze) or IMUs, similar to the wearable system shown in FIG. 4. The user's pose data may be further divided into components body pose and eye gaze in blocks 2450 and 2460 respectively. In block 2470, the user's body pose and eye gaze data may be used to identify objects of interest the user may interact with in the virtual environment. A virtual avatar of the user may then be rendered to reflect the user's interaction event with the identified object of interest in block 2480.

[0312] Specific implementations of intent based rendering may rely on determining the user's focus point or item of interest. FIGS. 25A through 25D illustrate examples for determining an item of interest where the item of interest is shared between two users. FIG. 25A begins by illustrating how virtual objects and avatars may be placed in the virtual environments. User A 2510 and User B 2560 are placed in their own local virtual environments 2500 and 2550. User A 2510 may see a virtual triangle 2520, a virtual square 2530, and User 2560's remote avatar B 2540 in local virtual environment 2500. User B 2560 may also see virtual triangle 2520, virtual square 2530, and user A 2510's remote avatar A 2570 in local virtual environment 2550. However, the position and orientation of the virtual objects and remote avatars may be unique to each user. As an example, in FIG. 25A, the virtual triangle 2520 and virtual square 2530 appear to the left of user A 2510 in the local environment 2500. Meanwhile, the virtual square 2530 is to right of user B 2560 while the virtual triangle 2520 is in front of user B 2560 in local virtual environment 2550. Additionally, while remote avatar 2540 faces towards user 2510 in virtual environment 2500, user A's remote avatar 2570 faces away from user B 2560 in local virtual environment 2550.

[0313] FIG. 25B illustrates an example of how an item of interest can be identified once virtual objects or virtual avatars are placed throughout a virtual environment. Objects in the virtual environment can be marked as potential items of interest. In some embodiments, a vector may project from the user towards each potential item of interest. In FIG. 25B, a virtual triangle 2520, a virtual square 2530, and a remote avatar 2540 may all be identified as potential items of interest. The virtual triangle 2520, virtual square 2530, and remote avatar 2540 may all be represented by a corresponding vector 2502, 2504, and 2508 respectively. Each corresponding vector may then be compared to the user's current sight line vector 2506, which projects directly from user 2510 towards the direction the user 2510 faces. Although the corresponding vectors 2502, 2504, and 2508 in FIG. 25B extend from user 2510 towards each corresponding virtual object, different vectors or methods may be used in other embodiments. For example, the vector can point to an object's position in the virtual environment, which can be its local origin. Where that origin points in relation to a render model can vary depending on how the virtual object is defined in the render model. For some objects, the vector points towards the point closest to the user or the geometric center of the object. For a large object (e.g., one with a size greater than a threshold such as 3 ft., 6 ft., 8 ft., or more), it is possible to create sub-assemblies that are each tagged with their own local points of interest.

[0314] FIGS. 25C and 25D illustrate examples of determining an item of interest among a plurality of potential items of interest. An item of interest may be determined by calculating an interest value for all potential items of interest in the virtual scene. The interest value of an object may be

a number representing how strongly the user is focused on the object. In FIGS. 25C and 25D specifically, the interest value for a potential item of interest is calculated by computing a dot product between a vector representing the user's line of sight (V) and a vector representing the direction to the item of interest (referred to as vector B in FIG. 25C). The resulting scalar value can be multiplied by an interest weight W for the item of interest. Thus the interest value is a weighted dot product in these examples: interest value = $(V \cdot B) * W$. The dot product is useful, because the dot product is largest in positive magnitude when two vectors are parallel (zero angle between them), which indicates the two vectors are pointing in the same direction (the user is looking at the object). The dot product is zero when the two vectors are perpendicular to each other (ninety degree angle between them), which indicates the user is not looking toward the object. Thus, the dot product tends to select items of interest that are on or near the user's line of sight. For objects behind the user, the dot product is negative, therefore, the weighted dot product will tend to select items of interest that are in front of the user.

[0315] The interest weight W of an object may be a number representing how likely a user would want to focus on the object. A larger interest weight is indicative of greater interest in the object, and a smaller interest weight is indicative of less interest in the object. In FIG. 25C, the virtual triangle 2520, virtual square 2530, and remote avatar 2540 are potential items of interest. The interest value of virtual triangle 2520 may be calculated by computing the dot product between the user's sight line vector 2506 and the vector 2502. A similar calculation may be done to for virtual square 2530 and remote avatar 2540. Because the interest weight of the potential items of interest in FIG. 25C are equal ($W=1.0$ in this example), the potential item of interest with the highest interest value is the one closest to the user's sight line vector 2506. Therefore, in the particular example shown in FIG. 25C, virtual square 2530 is the item of interest, because the virtual square's interest value (0.90) is larger than the interest value for the virtual triangle (0.45) or the remote avatar (0.65). In certain embodiments, an object directly in the user's sight line vector may be the item of interest by default.

[0316] While the objects in the example shown in FIG. 25C have equal interest weights, other embodiments may have objects with different interest weights. FIG. 25D illustrates an example of determining an item of interest where the potential items of interest may have different interest weights. The scene in FIG. 25D is the same as the scene in FIG. 25C except that remote avatar 2540 has a higher interest weight ($W=1.5$) than virtual triangle 2520 and virtual square 2530 (which each have $W=1.0$). An object with a higher interest weight may become the item of interest even if it is further away from the sight line vector 2506 than other objects. Also, for objects that are along a common direction from the user (so that their dot products are all roughly equal), the object with the largest interest weight will be selected as the item of interest. In FIG. 25D, the remote avatar 2540 has a higher interest weight and overall higher interest value (0.975), so it becomes the item of interest even though virtual square 2530 is closer to sight line vector 2506, but with a lower interest value (0.90).

[0317] The interest weight of a potential item of interest may dynamically increase or decrease in response to events in the virtual scene. For example, the interest weight of the

remote avatar **2540** in FIG. **25D** may increase when remote avatar **2540** is speaking and may decrease when the remote avatar **2540** stops speaking to reflect that the user is likely more interested in an avatar when it is currently speaking. In the case where several users or avatars are speaking at once, they may each be weighted equally, and the item of interest can be the user or avatar that is nearest the user's view vector.

[0318] Objects which are being manipulated by a user may get more weight. For example, if the user is playing a game of virtual chess and her avatar opponent picks up the user's queen, then the queen can be weighted higher and become more interesting, since the queen is currently of interest in the chess game. Likewise, objects being pointed at, but not directly manipulated by the user or an avatar, may get a higher weight. In an example of a virtual birdwatching experience, a user and an avatar may be looking at a particular virtual bird, e.g., a Red-legged Kittiwake, which is the current item of interest. If the avatar points at another virtual bird in the virtual environment, e.g., a Whooping Motmot, then the interest weight of the Whooping Motmot can be increased sufficiently (e.g., compared to the interest value of the Red-legged Kittiwake) so that the item of interest changes from the Red-legged Kittiwake to the Motmot. FIGS. **26A** through **26F** show examples of calculating interest values for each object in a virtual environment to determine the item of interest, assuming each object has equal interest weights. Although the specific example illustrated by FIGS. **26A** through **26F** uses dot product multiplication of vectors and equal interest weights, other embodiments may use different methods (e.g., unequal interest weights for the different objects).

[0319] FIG. **26A** presents the initial scene of the example. In FIG. **26A**, a local user **2620**, a remote avatar **2640**, a large dragon **2660**, and a small dragon **2680** all occupy a unique position in three-dimensional space. The positions of the objects are represented by Cartesian coordinate values (x, y, z), although other coordinate systems can be used in other implementations. For example, the position of the local user is at (0, 2, 0), the position of the remote user B **2640** is at (4, 2, 6), and so forth.

[0320] FIG. **26B** replicates the scene presented in FIG. **26A**. However, FIG. **26B** also shows that each object from FIG. **26A** can be associated with a vector extending out from the user **2620** to each object. For example, the vector from the user to the object B **2640** is Vector AB and has Cartesian coordinate values (0.555, 0, 0.832). In this example, all of the vectors are normalized to have length 1.0 (e.g., they are unit vectors). A forward vector AF **2610** in FIG. **26B** represents the user's line of sight. In this example, the forward vector AF has coordinates (0, 0, 1).

[0321] FIGS. **26C** through **26E** illustrate an example of calculating the interest value of the three objects in the virtual scene. As is well known, the dot product of two vectors V1 and V2 having coordinate values, respectively, (V1x, V1y, V1z) and (V2x, V2y, V2z) is $V1 \cdot V2 = (V1x \cdot V2x) + (V1y \cdot V2y) + (V1z \cdot V2z)$. FIG. **26C** shows an example of calculating the interest value of the remote avatar **2640**. In FIG. **26C**, the interest value of remote avatar **2640** is the dot product of the forward vector AF **2610** and the vector AB **2630**, which points from the user **2620** to the remote avatar **2640**. Applying the dot product formula, the interest value of remote avatar **2640** relative to the user is $AF \cdot AB = (0 \cdot 0.555) + (0 \cdot 0) + (1 \cdot 0.832) = 0.832$. If an interest weight W not

equal to 1.0 were used for the remote avatar, this value of the dot product would be multiplied by W to arrive at the interest value.

[0322] FIGS. **26D** and **26E** shows examples of calculating the interest values for the large dragon **2660** and small dragon **2680** using the same method as illustrated in FIG. **26C**. As shown in FIG. **26D**, the interest value of the large dragon **2660** is the dot product of forward vector AF **2610** and vector AC **2650**, which equals 0.911. Similarly, as shown in FIG. **26E**, the interest value of the small dragon **2680** is the dot product of forward vector AF **2610** and the vector AD **2670**, which equals 0.408. FIG. **26F** shows an example of determining the item of interest by comparing the interest values calculated in FIGS. **26C**, **26D**, and **26E**. Because the large dragon **2660** has the highest interest value (0.911), it becomes the item of interest for the user **2620**. Note that the large dragon **2660** is the object closest to the user's line of sight AF and is accordingly selected as the item of interest in this example (where the interest weights are all equal). As objects move in the virtual environment, their coordinate values change, and the corresponding dot products between the user's forward vector and the vectors to the objects change. Thus, different objects in the virtual environment can become the item of interest over time. As described above, dynamically varying interest weights can be used for each object, so that the item of interest may be an object that is farther from the user's line of sight than another object.

[0323] In some embodiments, the interest weight W for an object can depend at least partly on the distance between the user and the object. For example, the interest weight may be larger if the distance to the object is smaller, thereby indicating that objects nearby the user will tend to be more interesting to the user. Interest values may also represent contextual factors in the environment. For example, in a virtual game, a dangerous game element (e.g., a large dragon) or a valuable item (e.g., a golden crown) may be assigned a higher interest value than a more neutral or passive game element (e.g., a rock or a non-threatening player), because the user will likely be more interested in fighting the dangerous game element or finding the valuable item than interacting with a neutral/passive game element.

[0324] FIG. **27** shows an example of information that may be communicated from one wearable device to another wearable device during intent based rendering by the other wearable device. In FIG. **27**, the virtual square **2710** is determined to be the item of interest, and the virtual square's corresponding vector **2720** and the user **2700**'s current sight line vector **2740** are offset by an angle θ **2730**. An identifier **2750** representing the virtual square and the offset angle (and optionally interest weights) may be communicated to a remote wearable system as an avatar's item of interest information. When the item of interest information is received by the remote system **2760**, the system **2760** may adjust the orientation of a remote avatar to face the item of interest in the remote virtual environment, and the system may rotate the remote avatar's head by the transmitted offset angle θ **2730**. While the offset angle in FIG. **27** represents an angle θ in a two-dimensional plane, the offset angle is not limited to a particular value or a two-dimensional plane. Other examples may involve multiple possible angles in three-dimensional space, for example, Euler angles such as roll, pitch, and yaw. The communication of the interest information from a first wearable system to another wear-

able system can occur over the network 990 described with reference to FIG. 9A. The communication of the interest information can occur continuously and the determination of the item of interest can be performed in real time so that avatar behavior appears natural and seamless. Thus, even though the virtual environments of different users can be very different, the avatars displayed to the users by their respective wearable systems appear natural and each user can readily determine which object is being interacted with, what the attention of the avatars is, and what the intended movements of the avatars are.

[0325] FIGS. 28A and 28B illustrate examples of rendering a virtual avatar based on an object of interest. FIG. 28A illustrates an example of an initial setup of two virtual environments. In user A's virtual environment 2800, user A 2810 may view a virtual triangle 2820 and a virtual square 2830 to his left and a remote virtual avatar 2840 (of user B) to his right. The remote avatar 2840 may initially face towards the user 2810. In user B's virtual environment 2802, user B 2850 may view the remote avatar 2860 (of user A) directly ahead and the virtual triangle 2820, the virtual diamond 2870, and the virtual square 2830 to his right. The remote avatar 2860 may initially face away from the user 2850.

[0326] In FIG. 28B, interest information 2804 relating to user 2810's item of interest and offset angle 1815 may be communicated to the wearable system of user B 2850 for rendering. In the example presented in FIG. 28B, the virtual square 2830 is the item of interest and is offset from sight line vector 2825 by offset angle 2815. When the interest information 2804 is communicated by user A's wearable system and processed by user B's wearable system, the remote avatar 2860 may be oriented to face the item of interest's position, and the head of remote avatar 2860 may be rotated by the transmitted offset angle 2815 so that the head's orientation with respect to the item of interest (square 2830) remains the same between environments 2800 and 2802 (e.g., where the user's head faces the item of interest in the environment 2800, the virtual avatar of the user can also face the item of interest rather than looking away from the item of interest).

[0327] FIGS. 29A, 29B, and 29C illustrate examples of recalculating an item of interest. For illustration purposes, FIGS. 29A, 29B, and 29C are split into two columns, one representing user 2930's virtual environment and the other representing user 2960's virtual environment during a teleconference. The two columns comprise a sequence of scenes representing an order of actions occurring in the virtual environments throughout this example. Although the following example relies on identifying an item of interest and offset angle, other embodiments may use other methods to recalculate an item of interest.

[0328] The example begins with scenes 2900 and 2902. User 2930 and user 2960 are each in their own local virtual environments. In these figures, objects with dashed outlines are shared virtual content between the two user's environments. Virtual triangle 2950 and virtual square 2940 are shared virtual objects that can be found in both local environments. In scenes 2904 and 2906, the teleconference begins, and remote avatars 2970 and 2980 are spawned in locations that make logical sense in the particular local environment. In scenes 2904 and 2906, virtual avatars 2970 and 2980 avoid spawning in areas where objects already

exist in the environment and by default spawn facing towards users 2930 and 2960 respectively.

[0329] In scenes 2908 and 2910, potential items of interest may be identified in the virtual environments, and an item of interest may be determined for each user. In certain embodiments, a remote avatar may be set as the most interesting item of interest by default, as shown in scenes 2908 and 2910.

[0330] Next, in scene 2912 of FIG. 29B, the user 2930 turns to face the virtual square 2940. The wearable system of the user 2930 may recalculate the item of interest with the highest interest value to the user 2930. For illustrative purposes, the virtual square 2940 is determined to be the new item of interest with the highest interest value to user A in this particular example. Scene 2914 shows user 2960's local environment while user 2930 turns towards virtual square 2940 but before user A's interest information is transmitted to the wearable system of user B. Because information regarding user 2930's item of interest has not yet been received in scene 2914, the remote avatar 2980 does not yet change position. However, upon receiving the transmitted information, user B's wearable system may render the remote avatar 2980 to move to match user 2930's intent. Because the virtual square 2940 was identified as user 2930's item of interest in scene 2912, the remote avatar 2980 may turn towards virtual square 2940 to match user 2930's intent in scene 2918.

[0331] In FIG. 29C, after the remote avatar 2980 moves to match user 2930's intent, user 2930 stops moving in scene 2920 and user 2960 turns away from remote avatar 2980 in scene 2922. The wearable system may recalculate the item of interest with the highest interest value to user 2960. While user 2960 in scene 2922 does not look directly at the virtual triangle 2950, the virtual triangle 2950 is determined to be user 2960's item of interest in scene 2922 for illustrative purposes. User B's wearable system communicates interest information regarding user 2960's item of interest and offset angle to user A's wearable system, which renders the remote avatar 2970 to move to match user 2960's intent. Specifically in scene 2924, the remote avatar 2970 may turn towards the virtual triangle 2950 and then rotate by the transmitted offset angle to match user 2960's intent from scene 2922, namely that user B is interested in the virtual triangle 2950.

[0332] Additional examples of rendering avatars are now described. In one example, an avatar spawns in an open spot, facing the highest item of interest for an initial scene, with a default preference to face the other avatar. If there is only one other avatar, this would mean facing the other avatar. If there are multiple avatars and one is speaking (e.g., having a higher interest weight), an avatar is spawned facing the speaker. If there are multiple avatars and they all have (or a majority has) the same item of interest, the wearable system can have the shared item of interest can be the newly spawned avatar's item of interest. If there are multiple avatars and there is not a common item of interest, the wearable system can have the newly spawned avatar's item of interest be the closest user.

[0333] As another example, when a user moves in their local environment, the user's avatar moves as well in the other user's virtual environments. The movement can be governed by the following rules. A head pose vector is drawn. A vector is drawn from the user's head to each potential item of interest. An interest value can be calculated

for each potential item of interest (e.g., using a weighted dot product). The avatar is moved to the same angle offset from the highest item of interest as the real user.

Example Processes for Rendering Avatars

[0334] FIG. 30A is a flow diagram of an example process for avatar rendering based on an object of interest. The process may be performed by the intent mapping system 694 described with reference to FIG. 6B. Similar to the wearable system described in FIG. 4 and the process described in FIG. 24, a wearable system may access a user's pose data in block 3000 using outward-facing, inward-facing imaging systems (e.g., for eye gaze), or an IMU, wherein a user's pose data may comprise body pose or eye gaze components. The wearable system may then identify a plurality of virtual objects in the user's environment in block 3010. After identifying a plurality of virtual objects in the user's environment, the wearable system may determine and assign a weight value for each virtual object in block 3020. The weight value for a virtual object may be based at least partly on contextual information and semantic information of the virtual object. For example, if a virtual avatar is speaking to the user, the virtual avatar may be assigned a higher weight value.

[0335] After determining a weight value for each virtual item, the wearable system may continue to block 3030 by calculating an interest value for each virtual object. The interest value for each virtual object may be calculated based at least partly on the user's pose data and the virtual object's weight value, for example, as a dot product weighted by an interest weight. After calculating an interest value for each virtual object, the process continues in block 3040 by selecting a virtual object based at least partly on the interest value of the virtual objects. For example, the process in certain embodiments may select the virtual object with the highest interest value among all virtual objects. After selecting a virtual object of interest, the selection may be communicated to a wearable device of a second user in block 3050 and received by the wearable device of the second user in block 3060.

[0336] Although this particular example process focuses on communicating the selection of the virtual object, information communicated to the wearable device of a second user is not limited to the identity of the selected virtual object. In certain implementations, the communicated information may include the identity of the selected object, an offset angle, and perhaps more. After receiving the communicated information in block 3060, the wearable system of the second user may identify the location of the selected virtual object of interest with respect to the virtual avatar of the first user in block 3070. Once the virtual object of interest has been identified with respect to the virtual avatar of the first user, the wearable system of the second user may adjust the position or orientation of the virtual avatar of the first user in block 3080 based on the location of the virtual object of interest.

[0337] FIG. 30B is a process flow diagram of an example process for avatar rendering based on an object of interest when there is a change in condition. The process may be performed by the intent mapping system 694 described with reference to FIG. 6B. If a change in condition is detected by a wearable system of a first user in block 3005, then the wearable system may recalculate interest values for each virtual object based on the detected change in condition in

block 3015. For the sake of example, the change in condition in certain embodiments may include changes in the user's pose, movement of a virtual item of interest, or other possibilities.

[0338] After recalculating interest values, the process may continue to block 3025 by determining the virtual object of interest with the highest interest value. In certain embodiments, the identified virtual object and the virtual object having the highest interest value before the change in condition occurred may differ. In such examples, the wearable system of the first user may update the virtual object of interest to the identified virtual object in block 3035. Once the wearable system of the first user has been identified and selected, the wearable system may communicate the selection to a wearable device of a second user in block 3045. In some embodiments, the communicated information may include an identifier of the selected virtual object of interest and an offset angle. However, additional or different information may be communicated in other examples.

[0339] When the wearable system of the second user receives the communicated selection in block 3055, the wearable system in block 3065 may identify the location of the updated virtual object of interest in relation to the position of the first user's virtual avatar. Once the wearable system has identified the location of the updated virtual object of interest, the wearable system in 3075 may adjust the position or orientation of the virtual avatar. For example, the virtual avatar in certain embodiments may be adjusted to face directly towards the location of the updated virtual object of interest.

Examples of Avatar Rendering Based on Environmental Stimuli

[0340] Systems and methods disclosed herein can render an avatar so that the avatar appears more real and lifelike to a user of a mixed reality system. Objects in the environment of a user can be categorized as real or virtual, and the wearable system can be configured such that the avatar interacts and makes decisions in that mixed reality environment. The avatar can be rendered so that the avatar gives an appearance of agency, presence, or naturalness to a user. The agency may be an attribute of the virtual avatar. When the agency is enabled for the virtual avatar, the virtual avatar can appear to act of its own accord (e.g., the avatar may appear to make its own decisions about what it finds interesting). The wearable system can create the appearance of agency for the virtual avatar (e.g., by images or animations of the virtual avatar). For example, to provide the appearance of agency, the wearable system can cause the avatar to automatically respond to an event or a stimulus in the viewer's environment, and to produce a particular effect or result as if the avatar were human. The agency can be determined by one or more contextual factors, such as, e.g., the environment of the virtual avatar, interactions of the viewer in the environment, and objects in the environment. In certain implementations, the contextual factors can also include characteristics of the avatar, such as, e.g., the avatar's beliefs, desires, and intentions. By rendering the avatar as more natural and lifelike, the user of the wearable system will be less likely to experience uncanny, eerie, or unpleasant feelings when interacting with the avatar. For example, the avatar can be rendered so as to reduce the likelihood of entering the so-called uncanny valley, which represents a dip

in human emotional response to an avatar that is almost, but not quite, human in its interactions.

[0341] In addition to or as an alternative to animating a virtual avatar based on virtual objects that are shared between two users, the wearable system can also animate the avatar based on the environment that the avatar is rendered in. The wearable system can make a virtual avatar appear lifelike by enabling a virtual avatar to make natural, human-like decisions to react to the mixed reality environment in which it is rendered in, which can give the virtual avatar an appearance of agency and presence. For example, the wearable system (e.g., via the stimuli response system 696) can automatically identify certain categories of environmental objects or environmental stimuli (referred to herein as interesting impulses) to which the avatar might react to, and automatically adjust the avatar's interaction with the viewer's environment based on the interesting impulses in the viewer's environment. Interesting impulses may be visual or audio stimuli (e.g., a movement, an appearance, or a sound) in the environment which can attract an avatar's attention. As further described with reference to FIGS. 31A-37 interesting impulses may be associated with an object of interest, an area of interest, a sound, a component of an object of interest, or other contextual factors of the viewer environment. The interesting impulses may be associated with a virtual or physical object, as well as a virtual or physical environment in which the virtual avatar is rendered.

[0342] The wearable system can execute the avatar's decision to interact with the interesting impulses in a manner to maintain the naturalness of the virtual avatar. For example, when there is a sound of an explosion in the viewer's environment, the wearable system can render the virtual avatar as running away from the source of the explosion (as if the virtual avatar is human). As another example, the virtual avatar may shift the eye gaze vector (which can include a gaze point and a direction of gaze) of the avatar within the social triangle of a user or person's face while talking to that person, which encourages more natural and non-threatening human-like interactions between the person and the avatar. As yet another example, the virtual avatar may lose interest after looking at an object for an extended period of time. Thus, the virtual avatar may identify another interesting impulse and moves its attention from the previous object to the other interesting impulse, which again is more typical of natural human-like behavior.

Examples of Interesting Impulses in an Environment

[0343] The wearable system 200 (e.g., the stimuli response system 696) can model a viewer's environment and categorize objects (e.g., real or virtual objects) to identify interesting impulses. An interesting impulse can have values for representing its inherent interestingness, which may decay or grow based on contextual factors, such as time elapsed, changes in objects or the environment (which may include changes in the object or portion of the environment identified as interesting impulse), or interactions of the human counterpart associated with the avatar. The interestingness value may change continuously or suddenly in response to a triggering event.

[0344] As an example where the interestingness value changes suddenly due to a triggering event, assuming an object of interest is a book, the interestingness value would increase in response to a page change. As an example where

the interestingness value changes continuously, an interesting impulse may be represented as a speaker's hand, and as the hand changes shape (e.g., to emphasize a point the speaker is making), the interestingness value may increase continuously. The changes to the interestingness value can include a change in speed or acceleration of the adjustment of the interestingness value. For example, in response to a triggering event on the object of interest, the acceleration or the speed associated with the growth of the interestingness value can suddenly increase. In some implementations, the speed or acceleration of adjustments to the interestingness value may remain constant. For example, the interestingness value can decay at a constant speed with the passage of time. FIGS. 31A-31C illustrates examples of categorization of the types of interesting impulses in a viewer's environment. In FIG. 31A, the viewer's environment can include a variety of interesting impulse(s) 3110. An interesting impulse 3110 may be an audio or a visual stimulus. For example, an interesting impulse 3110 may be a viewer's movement or a noise in the environment 3100.

[0345] As illustrated in FIG. 31A, the interesting impulse 3110 can include an interesting object 3120 or an area of interest 3130. The interesting object may be an example of the object of interest described with reference to FIGS. 19A-30B. In certain implementations, the interesting object can be a portion of the object of interest toward which the avatar looks. As will be further described below, the interesting object can be described as a polygonal structure for holding saccade points (which can be associated with the points of interest during saccadic eye movements, which are quick, simultaneous movements of both eyes between two or more phases of fixation in the same direction). An example interesting object 3120 can be a social triangle 3122, which can describe a region of a face that a person (or an avatar) focuses on during a conversation. FIG. 31B described below illustrates an example social triangle.

[0346] Another example interesting object 3120 may be a gaze box 3114 (also shown in FIG. 31B) which can be associated with the region of an object or another person that a person (or an avatar) focuses on (e.g., gazes at) during an interaction with the environment or the object of interest.

[0347] FIG. 31B illustrates an example social triangle 3112. The social triangle 3112 shown in FIG. 31B schematically illustrates a portion of a human face that people tend to focus on in a conversation. In the example in FIG. 31B, the social triangle 3112 is a triangle that covers the eyes from slightly over the eyebrow down to the mouth of a person.

[0348] The social triangle 3112 can have a variety of parameters such as, e.g., size, shape, boundary, area, etc. A boundary of the social triangle can constrain the saccade points during a saccadic eye movement of a second person (or an avatar) such that a saccade point at a given time does not land outside of the boundary of the social triangle of the first person. Thus, the saccade points of the eyes of the avatar looking at the person having the social triangle 3112 tend to be clustered within the social triangle 3112 and tend not to fall outside of the social triangle. The saccade points may be used by the wearable system to determine the avatar's eye gaze vector at a given time. As described herein, the saccadic movements may involve randomly placed saccade points which may cause the avatar's eye gaze vector to shift randomly within the social triangle 3112. The eye gaze vector can also move along a trajectory within the social triangle 3112 when animating the avatar's eye gaze. For

example, the eye gaze vector may first land on a point of the nose, then move to a point on the left eye, and further move to a point on the right eye, etc.

[0349] In certain implementations, possible saccade points can be kept in a data structure such as, e.g., a query-able database constructed from presence of virtual or physical objects in the user's environment. The possible saccade points may comprise the sum of visible spaces of interesting objects (e.g., saccade points within social triangles or gaze boxes). The query-able database may be part of the map database **710** shown in FIG. 7. The query-able database can also store other information, such as, e.g., information related to interesting objects (e.g., social triangles or gaze boxes), interesting impulses, etc., such that the wearable system can use the information to determine the eye gaze of the virtual avatar. The wearable system can automatically select (e.g., either randomly or following a defined sequence) saccade points of an interesting object by querying the query-able database. In certain implementations, the frequency of selections of saccade points can be associated with the interestingness value of the target interesting object. For example, a higher interestingness value may cause the frequency of selections to increase, and a lower interestingness value may cause the frequency of selections to decrease.

[0350] To randomly select the saccade points, the wearable system can make use of a variety of probabilistic models. For example, the wearable system can select another saccade point from a normal distribution around the current saccade point (e.g., a saccade point of which an avatar's eye gaze vector is currently placed) within an interesting object. As another example, the wearable system can use a distribution function specified in a texture channel and applied to the interesting object as a texture map. For example, where the virtual avatar is looking at a painting, the painting may have some parts that are more interesting to the human eyes than other parts. A texture map can be created to reflect that some parts are more interesting than others on an object. The texture map may have attributes similar to a heat map where brighter portions (e.g., portions with a higher interestingness value) of the texture map represent more interesting portions and darker portions represent less interesting portions (e.g., portions with a lower interestingness value). These maps may be defined by programmers or users, e.g., by manually inputting or indicating a region of interest. These maps can be created from eye tracking systems such as, e.g., based on the data acquired by cameras in the inward-facing imaging system **466**. For example, the wearable system can track the duration and positions of the viewer's eye gaze vector at a location of an object. The wearable system can identify the more interesting region as the region that has a higher density of points associated with eye gaze vectors or when the eye gaze vectors stays at the region (or points) for a longer duration of time, and vice versa. Alternatively, a user's personality, temperament, profile, or other attribute could at least partially define the texture map. For example, if a user is characterized as liking dogs and there is a dog in the painting, that portion of the painting would increase its interestingness value.

[0351] One or more of the parameters of the social triangle may be configurable. The parameters of the social triangle may be configured by a viewer of the avatar or may be preprogrammed into the wearable system. For example, in various embodiments, the boundary of the social triangle

may be configured by the wearable system to include less or more area than the social triangle **3112** shown in FIG. **31B**. As another, example, a viewer of the wearable system can specify a size of the social triangle, and the wearable system can determine a region of the face that meets the specified size as the social triangle (such as, e.g., by placing the triangle on the face such that the eyes, nose, or mouth fit within it). In certain implementations, the parameters of the social triangle may change based on the orientation or position of the face. For example, the social triangle **3212a** (shown in FIG. **32A**) has a smaller area than the social triangle **3212d** (also shown in FIG. **32A**) because the avatar (not shown in FIG. **32A**) can perceive the side of the person's **3210a** face (which has fewer facial features) but perceive the front of the person's **3212d** face (which has more facial features). Further, although the word "social triangle" is used in the disclosure (because this term is the conventional usage), the shape of the region represented by the social triangle need not be strictly triangular and can be any type of polygon (e.g., quadrilateral, pentagon, hexagon, etc.), convex planar shape (e.g., circle, ellipse, or oval), or a 3D shape.

[0352] In some situations, a portion of the face may be occluded, such that a person or an avatar may not be able to directly observe that portion of the face. For example, where a viewer is wearing the wearable device described herein, the head-mounted display may occlude the viewer's eyes and a portion of the viewer's nose. As a result, the saccadic points associated with the viewer's social triangle may land on a portion of the surface of the head-mounted display rather than on the viewer's eye region.

[0353] FIG. **31B** also illustrates an example of a gaze box **3114** which can include a 3D space which can capture the saccade points while a person is looking at an object. In this example, the gaze box **3114** is represented by a 6-sided rectangular cuboid gaze surface associated with the saccade points. It has a width (represented by "W"), a height (represented by "H"), and a depth (represented by "D"). In various implementations, the gaze box **3114** can have other 2D or 3D shapes, other than the cuboid illustrated in FIG. **31B** (e.g., the box may be polyhedral). The gaze box **3114** can also include similar parameters as the social triangle. As described herein with reference to social triangles, the parameters of the gaze box **3114** can also be configurable by a viewer or automatically by the wearable system. For example, the boundary of the gaze box or the size of the gaze box may be configured based on the types of objects. For example, as shown in FIG. **32A**, the gaze box **3216** associated with the backpack is larger than the gaze box **3214a** associated with a cup.

[0354] As will further be described with reference to FIGS. **31A-37**, the wearable system can simulate the avatar's interaction with a person or an object based on the interesting object. For example, the wearable system can animate the avatar's eye motion with the saccadic eye movements within the social triangle or gaze box. Thus, rather than focusing the avatar's eye at a particular point on the person or the object, the wearable system can simulate rapid eye movements from one point of interest to another within the interesting object and render the avatar's eye movements similar to human eye movements. Such saccadic movement of the avatar's eyes may not be readily apparent, but since it simulates actual human eye movements, rendering avatar saccadic eye movement can lead to feeling of

naturalness when interacting with the avatar, which also moves the interaction out from the uncanny valley.

[0355] As described with reference to FIG. 31A, the interesting impulse can also include an area of interest. The area of interest can be a general direction that the avatar is looking at in the viewer's space. It may be represented as a 2D area. However, it may not have a spatial dimensionality, but rather can be the directionality of the interesting impulse from the avatar's point of view. This type of interesting impulse can be useful for representing events in a general area or direction. For example, if a flash of light is detected (e.g., by the viewer's wearable device), the viewer's wearable device can represent the general direction of the light as the area of interest. Advantageously, in some embodiments, this would allow a virtual avatar to look in the direction of the light without committing to an exact position or set of potential saccade points.

[0356] Another example of an area of interest can be the general direction of a sound or a noise. FIG. 31C illustrates an example of avatar rendering based on a sound in a viewer's environment. FIG. 31C illustrates two mixed reality (MR) scenes 3150a and 3150b. The two MR scenes may be associated with Alice's environment, where Alice is a human (not shown) who can perceive Bob's avatar 1100 via Alice's wearable device 902. Alice's wearable device 902 may initially render Bob's avatar 1100 as facing west (as shown by the coordinate system 3154) and looking at its feet in the scene 3150a. Alice's wearable device 902 can detect a loud noise 3152 (e.g., via data acquired by the audio sensor 232) in Alice's environment. Without needing to pinpoint the origin of the noise 3152, Alice's wearable device 902 can determine that the noise came from the general direction 3156. The wearable device 902 can accordingly change the avatar's orientation and head pose to react to the noise. As shown in scene 3150b, the wearable device 902 can change the avatar's 1100 orientation from facing west to facing east, and change the head pose or the eye gaze direction of the avatar 1100 to look in the direction 3156, toward the noise 3152.

[0357] Because areas of interest may not have a fixed position in the viewer's environment, interesting areas are not required to be in the virtual avatar's visual cone (e.g., when a cone or ray casting is performed from the virtual avatar's perspective) to be eligible for the avatar to respond. Accordingly, as shown in FIG. 31C, the avatar can respond to a sound source coming from behind the avatar. The avatar is not limited to responding to sound sources that are in its field of view.

[0358] In addition to or as an alternative to the general direction of a sound or light, an area of interest can also be associated with a memory of previous stimuli such that the avatar may periodically check a region in the environment to determine if the previous stimuli are present. For example, in FIG. 31C, once the sound 3152 fades, the avatar 1100 may go back to its previous pose and orientation as shown in the scene 3150a. Time decay of interesting impulses is described further below. However, the wearable system may render the avatar 1100 as periodically changing its pose or orientation back to the one shown in the scene 3150b to look in the direction of the sound source, even though there may not be a sound in the direction 3152 at that particular time. As another example, the avatar 1100 may initially be looking at an item of interest (such as, e.g., a virtual dog) in the environment. The item of interest may leave the environ-

ment (e.g., because the virtual dog has moved to another room). However, the avatar 1100 may occasionally look at the location where the item of interest last appeared and to check to see if the item of interest reappears in the environment. Accordingly, once an item of interest or an interesting impulse has been identified in the environment, after a time delay to represent the decline of interest in the item or impulse, the avatar may be rendered as if it were periodically (or from time to time) checking back on the item or the impulse.

[0359] A virtual avatar can also respond to contextual factors, other than the interesting object 3112 or the area of interest 3114. A virtual avatar's behavior may change based on the characteristics of the environment the avatar is in. For example, where a virtual avatar is rendered in a conference room environment, the viewer's system may reduce the likelihood or frequencies that the virtual avatar checks on the area of interest associated with past stimuli (e.g., frequent checking may be inappropriate in a work or business environment). However, when the avatar is in a home environment, the virtual avatar may check on the area of interest associated with the past stimuli more frequently. As another example, a virtual avatar may be configured not to respond to certain types of stimuli based on the environment that the virtual avatar is in. Continuing with the same example above, if the virtual avatar is in a conference environment, the viewer's wearable system may be configured such that the virtual avatar is not responding to a ring tone from the wearable device or from another computing device, which indicates the arrival of an electronic message. Accordingly, the frequency for checking may be environmentally-dependent and may be in a range from, e.g., every few to tens of seconds to every few minutes, up to a few times an hour.

Examples of Generating Interesting Impulses in an Environment

[0360] A viewer's wearable system can detect the presence of interesting object 3112 or determine an area of interest 3114 at run time while the avatar is rendered by the wearable system. The wearable system can also generate interesting impulses at run time based on the presence of interesting object 3112 or area of interest 3114. The generated interesting impulse may cause the avatar to respond to the interesting impulse, e.g., by changing its pose, orientation, eye gaze direction, movement, by speaking, ceasing to speak, and so forth.

[0361] The interesting impulse can be tied to virtual or physical stimuli in the viewer's environment. Virtual stimuli can be explicitly tied to objects generated from a content engine that renders virtual content in the environment. As an example, an interesting impulse can be generated in response to a viewer flipping a page of a virtual book. As another example, an interesting impulse can be based on a facial expression (e.g., a smile) or a movement of another virtual avatar rendered in the same environment.

[0362] Real world stimuli can be generated based on data acquired by device sensors (such as, e.g., those shown in FIG. 3). The wearable system 200 (e.g., the stimuli response system 696) can analyze data acquired from the device sensors and process the data via detection and classification algorithms (such as those described with reference object recognizers 708 in FIG. 7) to determine the type of events (e.g., the presence of a certain object, a sound, or a light).

The wearable system **200** can perform such event detection using the local processing and data module **260**, alone or in combination with the remote processing module **270** (or the remote computing system **920**). The results of the detection and classification algorithms can then be processed by the wearable system **200** to create interesting impulses. In certain implementations, the interesting impulses may be stored by the wearable system **200** and can be part of the virtual avatar's knowledge base.

[0363] FIG. 32A illustrates an example of generating interesting impulses based on real world stimuli. In this figure, a viewer can perceive, via the wearable system, a group of people (persons **3210a-3210e**) in the scene **3200**. Examples of interesting impulses from this scene **3200** are the social triangles **3212a-3212d** associated with the respective persons **3210a-3210d**. The wearable system (e.g., the stimuli response system) can detect, e.g., based on one or more object recognizers **708**, the presence of the persons **3210a-3210e** in the environment. The one or more object recognizers **708** can employ various face detection algorithms or skeletal inference algorithms for detecting the presence of the persons' **3210a-3210d** faces in the environment. Once detected, the social triangles **3212a-3212d** may be inserted into the virtual avatar's knowledge base so that the wearable system can efficiently access this data. Advantageously, in some embodiments, the wearable system, by rendering the avatar as sharing attention with other humans (and the viewer) in the mixed reality environment, enhances the presence of the virtual avatar and improves the interactive experience between the viewer and the virtual avatar.

[0364] Although not shown in FIG. 32A, the viewer's social triangle can also be an interesting impulse, e.g., the avatar may want to interact directly with the viewer (e.g., the wearer of the wearable display device). The wearable system can obtain the position and orientation of the viewer's social triangle based on the viewer's head position (e.g., based on data acquired from the outward-facing imaging system **464**, the IMUs, etc.). The wearable system can calculate the head pose with respect to a world frame of the user's environment. The wearable system can also track and update the position and orientation of the viewer's social triangle as the viewer moves around in the environment. The viewer's social triangle can also be inserted into the virtual avatar's knowledge base for interactions by the virtual avatar.

[0365] In certain implementations, the interestingness of social triangles can be modulated on detected changes in facial expressions. The wearable system can modify the interestingness value associated with a social triangle of a person based on the facial expressions of the person. For example, as the expression on a real world human's face changes from a smile to a frown, the interestingness value of the associated social triangle may rise (due to this change in facial expressions).

[0366] The wearable system can also identify new social triangles based on audio data (e.g., data acquired by the audio sensor **232**). This can allow the virtual avatar to look at a speaking individual (who might have a larger interestingness value than non-speakers in the environment), which would increase the presence of the virtual character. For example, the wearable system can capture speech via the audio sensor **232** and detect the position of the speaker in the viewer's environment. Based on the position of the speaker, the wearable system can detect a previously undetected social triangle and create a new social triangle associated

with the speaker. For example, in FIG. 32, the wearable system did not detect a social triangle for the person **3210e** (e.g., the person **3210e** may have entered the scene before the wearable system updates its world map). However, the wearable system may capture speech by the person **3210e**. This speech data may cause the wearable system to re-analyze the region of the image associated with the person **3210e** and may accordingly identify a new social triangle (which is associated with the person **3210e**) in the scene **3200**. The wearable system can update its world map to reflect the presence of the person **3210e** and the social triangle associated with this person. Further, the wearable system may increase the interestingness value associated with the person **3210e** (or his or her social triangle), because humans tend to be interested in new people who enter an environment and tend to look in their direction.

[0367] The wearable system can also increase the interestingness value based on the audio data. For example, the wearable system can increase the interestingness value of a social triangle if the wearable system detects that the person associated with the social triangle is speaking. Alternatively, the wearable system can decrease the interestingness value of a social triangle if the wearable system detects that the person associated with the social triangle is not speaking or has not spoken for a period of time. Advantageously, in some embodiments, by increasing (or decreasing) the interestingness value of the social triangle based on the audio data, the wearable system can advantageously allow a virtual avatar to look at the speaking individual and avoid the interestingness value decay which may cause the virtual avatar to divert its attention to another object with a higher interestingness value.

[0368] FIG. 32A also shows a plurality of gaze boxes **3214a**, **3214c**, **3214d**, and **3216**. The gaze boxes can be generated for physical objects (such as a backpack and a cup, corresponding to the gaze boxes **3216** and **3214a** respectively) or a portion of the physical object (such as the foot or the hands, corresponding to the gaze boxes **3214c** and **3214d** respectively). The wearable system can identify the physical objects (or a portion thereof) using the object recognizer **708**. For example, the object recognizer **708** can include an image classifier which can provide an object's 3D position and boundaries, which can be transformed into gaze boxes. The gaze boxes can also be inserted into the virtual avatar's knowledge base which can later be used to determine the virtual avatar's attention.

[0369] The interestingness of the generated gaze boxes can be modulated using the object type (which can be determined by the object recognizer **708**, e.g., as semantic information associated with the object) and the personality (or disposition) of the virtual character. For example, a gaze box may be associated with a soda can and the virtual avatar has a thirsty trait. The interestingness value associated with the generated gaze box of the soda can may be increased. However, if the avatar is not thirsty now because the human counterpart just drank water or because the avatar just drank water (as animated in its virtual environment), the interestingness value associated with the soda can's gaze box may decrease. As another example, if an object of interest is a dog and the virtual avatar has a fear of dogs, the interestingness value associated with the dog (or the gaze box of the dog) can increase. In some implementations, as the dog moves closer to the virtual avatar, the amount of increase in the interestingness value may be faster to represent increased

fear. The high interestingness value may be cause the virtual avatar to perform a behavior that reflects a natural human interaction with the dog, e.g., to look at the dog or to move away from the dog. Continuing with this example, as the distance between the avatar and the dog increases, the interestingness value associated with the dog may decrease, and when the interestingness value drops below a threshold (which may represent that the dog is no longer a threat to the avatar), the avatar may be rendered so as to stop moving away from the dog. Thus, the increase and decrease of interestingness values associated with objects in the environment permit the wearable system to render avatar behavior that is natural and realistic.

[0370] The interestingness values of interesting objects (e.g., gaze boxes or social triangles) can also be modulated based on the attention of the viewer. For example, if the viewer is looking at a gaze box or a social triangle, the interestingness values associated with the gaze box or social triangle may also increase for virtual avatars in the viewer's environment, thereby increasing the likelihood that the avatar will be rendered to also look at the gaze box or social triangle.

[0371] FIG. 32B illustrates an example process of configuring parameters associated with an interesting impulse and tracking the interesting impulse. The example process 3250 can be performed by a viewer's wearable system which can be configured to render a virtual avatar in the viewer's environment.

[0372] At block 3252, the wearable system can determine parameters associated with an interesting object, such as, e.g., the boundary or shape of a social triangle or a gaze box. The wearable system can access the parameters pre-programmed into the system. The wearable system can also dynamically determine the parameters based on the object associated with the social triangle or gaze box. For example, the size or the shape of the social triangle for a face may vary depending of the size of the face or characteristics of the facial features (e.g., an elongated face may have a longer but narrower social triangle while a round face may have a wider social triangle).

[0373] At block 3254, the wearable system can identify the interesting object in a viewer's environment. In certain implementations, this block 3254 can be performed before the block 3252. The wearable system can make use of various facial recognition techniques described with reference to the object recognizer 708 in FIG. 7 to identify facial features in the viewer's environment and can fit a social triangle to a person's face in the environment based on the identified facial features (e.g., the social triangle may include at least portions of person's eyes, nose and face). The wearable system can also use object recognizer 708 to identify physical objects in the environment for determining the location of the gaze box.

[0374] Where the social triangle is associated with an avatar, the wearable system can determine the region for the social triangle by accessing the face data of the avatar. For example, the wearable system can access face parameters associated with the 3D model of the avatar to delineate a region of the avatar face based on the parameters of the social triangle. Where the social triangle is on a viewer, the wearable system can access a face model of the viewer and determine the social triangle based on the face model. Such a face model may be generated while the viewer is putting on a wearable device (e.g., by using the outward-facing

imaging system 464 to acquire images of the viewer's face). The face model may also be acquired by scanning the viewer's face with a camera in the viewer's environment.

[0375] At block 3256, the wearable system can track the interesting object in the viewer's environment. For example, the wearable system can track the movement of the person or the object with which the interesting object is associated. The wearable system can also track the head pose of the viewer, another person, or an avatar which may cause a change in the parameters of the interesting object (e.g., change in the location or size).

[0376] At block 3258, the wearable system can dynamically update the parameters of the interesting object. For example, as the viewer moves around, the wearable system may track the movement of the social triangle to be in accordance with the viewer's movement. As another example, as a person tilts his head, the size of the social triangle or the position of the social triangle can also change based on the head pose.

Examples of Identifying a Target Interesting Impulse

[0377] The wearable system can periodically scan through a knowledge base of interesting impulses and can select a most interesting impulse as the target interesting impulse. Once a target interesting impulse is selected, the wearable system can render the avatar as interacting with the target interesting impulse, such as, e.g., by orienting the avatar as if the avatar were focusing its attention on the target interesting impulse.

[0378] FIG. 33 illustrates an example of identifying a target interesting impulse. The environment 3300 in FIG. 33 may be a mixed reality environment in which Alice's Avatar 1000 is rendered. This mixed reality environment can be rendered by Bob's wearable device 904. The environment 3300 can include physical interesting objects 3324 and 3322 which may be part of a physical object in the environment. For example, the physical interesting objects 3324 and 3322 may be associated with gaze boxes. The environment 3300 can also include virtual interesting objects 3312 and 3314. The virtual interesting objects 3312 and 3314 may be virtual objects which may be shared by Alice's avatar 1000 and Bob in the mixed reality environment 3300. The environment 3334 can also include a social triangle 3334 which may be the social triangle of Bob (who may be a viewer of Alice's avatar 1000). The environment 3300 can also include an interesting area 3336. The physical interesting objects 3324, 3322, the virtual interesting objects 3312, 3314, the social triangle 3334, and the interesting area 3336 (e.g., the location, direction, or boundary of the interesting area) can be part of the virtual avatar's 1000 knowledge base.

[0379] In this example, Bob's wearable device 904 can render Alice's avatar to look at the interesting area 3336 by default (as indicated by the saccade cone 3310 indicating a region of eye movements of the avatar 1000). The interesting objects in the environment 3300, however, can be analyzed against a visual cone 3320 which may be part of a cone cast performed for the avatar 1000. In some situations, the saccade cone or the visual cone can also be referred to as the saccade frustum or the visual frustum, respectively.

[0380] When determining a target interesting impulse, Bob's wearable device 904 can perform the cone casting based on the virtual avatar's 1000 head pose and eye gaze. For example, the parameters of the cone can be generated

based on the avatar's **1000** current eye direction, horizontal angle or vertical angle of the virtual avatar's **1000** head, head speed, or eye speed. For example, during the cone casting, a virtual cone may be cast from the virtual avatar **1000** into the mixed reality environment (as shown by the visual cone **3320**). As the virtual avatar moves its head or eye gaze direction, the direction or movement (e.g., the movement speed) may be adjusted according to the avatar's head or eye movements and direction. In certain implementations, the horizontal angle or vertical angle of the cone are modulated by the virtual avatar's personality and disposition, while the head speed and current eye direction can be determined from the animation of the character. For example, if the avatar **1000** is in a thinking mode, the avatar **1000** may be looking down at a floor of the environment **3300**. As another example, if the avatar has an active personality, the virtual avatar may move its head around frequently.

[0381] In the example shown in FIG. 33, the visual cone **3320** can capture the physical interesting object **3324**, the interesting area **3336**, the social triangle **3334**, and the virtual interesting object **3314**. The wearable system can calculate the interestingness value (e.g., described with reference to FIGS. 30A and 30B) associated with these interesting objects and interesting area to determine a target interesting impulse. For example, the wearable system can select an interesting impulse as a target interesting impulse when that interesting impulse has the highest interestingness value.

[0382] In certain implementations, the wearable system maintains a list of interesting impulses for objects and areas within the avatar's field of view (which may be determined and represented by the visual cone **3320**). If an interesting impulse is not in the avatar's field of view, the interesting impulse may be culled from the list. For example, when the virtual avatar looks to its right, the physical interesting object **3324** may become outside of the visual cone **3320** while the physical interesting object **3322** may move inside of the visual cone **3320**. As a result, information of the physical interesting object **3324** is removed from the list of interesting impulses while information associated with the physical interesting object **3322** (which had been outside the avatar's field of view) may be added to the list of interesting impulses. The physical interesting object **3324** may remain in the avatar's knowledge base even though the physical interesting object **3324** is no longer in the virtual avatar **1000**'s field of view. Additionally or alternatively, as an object passes outside the avatar's field of view, the interestingness value associated with the object may be decreased to reflect the decreased likelihood that the avatar will interact with the object. Conversely, as an object passes into the avatar's field of view, the interestingness value associated with the object may be increased to reflect the increased likelihood that the avatar will interact with the object.

[0383] The list of interesting impulses may be sorted based on an interestingness value of the objects (real or virtual) in the environment **3300** or in the visual cone **3320**. The interestingness value can be calculated based on an inherent interestingness of an object minus an interestingness decay. The inherent interestingness of the object may be based on contextual factors, such as, e.g., the environment information, the interaction or triggering event associated with an interesting impulse, the object's characteristics (e.g., a moving object may have a higher interestingness value

than a static object), the avatar's (or its human counterpart's) personality, the characteristics or interactions of the viewer, etc. For example, an interesting object may have a boost in its interestingness value, if the avatar **1000** or a viewer is interacting with the interesting object. As another example, if the viewer is speaking, the social triangle **3334** associated with the viewer may have an increased interestingness value. In certain implementations, rather than increasing or decreasing the inherent interestingness, one or more of these contextual factors can also be used to adjust the interestingness decay as described herein.

[0384] The interestingness decay can be associated with a rate of decay or growth associated with the interestingness value. The interestingness decay can be based on the time or triggering events. For example, flipping a page of a virtual book can cause a decrease in the interestingness decay (which amounts to an increase in the interestingness value) or slow down the interestingness decay associated with the virtual book. As another example, a sound of explosion in the environment would cause a sudden increase (in addition to the inherent interestingness) to the interestingness value associated with the area having the explosion.

[0385] FIG. 34 illustrates an example of interestingness decay based on time. In this figure, an interestingness curve **3410** is plotted with respect to the axis **3402** (representing the amount of interest in an object) and the axis **3404** (representing the passage of time). The interestingness decay rate at a given time can be the tangent of the curve **3410** at the given time (e.g., a slope of the curve). This interestingness curve **3410** may be configured or customized for different objects (or areas). For example, an interestingness curve for a social triangle associated with the viewer may be different from an interestingness curve for an object, e.g., the decay for the object may be faster than the decay for the social triangle reflecting the fact that humans (and virtual avatars) are more likely to remain interested in other humans (or other avatars) than in non-human objects.

[0386] As shown in FIG. 34, an object's interestingness value tends to shrink over time. As a result, the object will less likely attract the virtual avatar's attention as time passes. In some embodiments, the curve **3410** may be configured such that the interestingness value will not fall below 0.

[0387] Although the example interestingness curve **3410** decays over time, in some implementations, the interestingness curve **3410** can also grow over time. For example, if an interesting impulse is the target interesting impulse, the interestingness may grow over time. For example, a person may enter the avatar's visual cone **3320**, causing an increase in interestingness value associated with the person. The person may then start to speak, which increases the interestingness value. The person may gesture while speaking, further increasing the interestingness value. Once the person stops speaking and gesturing, the interestingness value may start to decrease. Thus, the interestingness value for an object, human, or avatar, can increase or decrease over time or due to the occurrence of events (e.g., starting to speak, smiling, etc.). The interestingness curve **3410** can be generated and adjusted based on various mathematical functions, such exponential, polynomial, linear, power, sinusoidal, etc., alone or in combination. For example, a portion of the interestingness curve can be level (e.g., with the slope being 0) for a period of time. This portion of the interestingness curve may be preceded or followed by another portion of the curve with a non-zero slope (e.g., a positive

slope indicating the interestingness increases with time, or a negative slope indicating the interestingness decreases with time).

[0388] In certain implementations, the target interesting impulse, once determined, can be assigned an increment to the interestingness value to add hysteresis, which can advantageously reduce or prevent flip flopping between two interesting impulses (e.g., when the two interesting impulses have similar interestingness values) and can cause the virtual avatar to focus on the target interesting impulse.

Examples of Avatar Animation Based on a Target Interesting Impulse

[0389] Once an interesting impulse is selected as the target interesting impulse, the characteristics of the interesting impulse (e.g., the type of the interesting impulse, the type of reactions associated with the interesting impulse, etc.) can determine an avatar's interactions from overt behaviors such as potential animations and dialog lines to more subtle behaviors such as emotional response, eye attention, and saccade motion. For example, when the target interesting impulse is a sound, the viewer's wearable system can present a dialog line near the avatar stating "what happened?" and/or animate the avatar to look toward the direction of the sound. As another example, if the target interesting impulse is a social triangle of another's face, the wearable system can direct the avatar's attention to the social triangle by rendering the avatar's eye movements as saccadic motions within the social triangle.

[0390] The saccadic motion can be regulated through a saccade rate which can control how often an avatar's eye gaze is switched from one saccade point to another. To determine when and where an avatar's saccadic eye movement is switched, a sample saccade point can be selected from a randomized window of saccade timing, and once that time has expired, a new saccade point can be chosen. The saccade rate can be modulated by characteristics of the virtual avatar, such as, e.g., the disposition and emotion of the virtual avatar. For example, being excited or angry can increase the saccade rate, while being bored or lethargic can decrease the saccade rate. The saccade rate for an avatar can be representative of the saccade rate for humans. For example, a saccade can last from about 20 ms to about 200 ms, can have angular speeds from about 10 degrees per second to several hundred degrees per second, and so forth.

[0391] FIG. 35 illustrates an example of determining saccade points associated with an object which is selected as a target interesting impulse. This figure shows an interesting object 3500 which is determined to be a target interesting impulse. In this example, the interesting object 3500 can also be referred to as the most interesting object.

[0392] The most interesting object can include polygonal structures for holding saccade points. The most interesting object can be analyzed to identify a list of polygons on which a saccade point can lie. This list of polygons can be determined based on the position and orientation of the virtual avatar and the object. In the example in FIG. 35, the most interesting object 3500 includes 6 polygonal structures (surface polygons A-F, shown as triangles in FIG. 35) for holding the saccade points. In this example, these polygonal structures are on the surfaces nearest the user who is viewing the object 3500, because humans tend to look at the near, perceivable surfaces of an object. However, if the user walks to the other side of the interesting object 3500, the list of

polygons associated with the virtual avatar's saccade point can be different because one or more of the surface polygons may no longer be perceivable.

[0393] This list of polygons can be clipped to a saccade frustum (see e.g., saccade cone 3310 shown in FIG. 33). The saccade frustum can be a subset of the space of the visual frustum (see e.g., visual cone 3320 shown in FIG. 33) in the mixed reality environment. The saccade frustum can have reduced horizontal or vertical angles (e.g., as compared to the visual frustum 3320) and can be centered on the avatar's eye gaze direction. In certain implementations, the parameters associated with the saccade frustum (e.g., the location of the saccade frustum at a given time) can be modulated in a similar manner as the saccade rate modulation described herein.

[0394] Once clipped and triangulated as shown by polygons A-F in FIG. 35, the polygons A-F can be projected into two dimensions (from a 3D view shown in FIG. 35). A polygon among the polygons A-F can be randomly selected using the 2D area of the polygon to weight the random selection. For example, the probability of selecting the polygon A can be determined in accordance with the formula (4) below:

$$A_p = \frac{\text{Area } A}{\text{Area } A + \text{Area } B + \text{Area } C + \text{Area } D + \text{Area } E + \text{Area } F} \quad (4)$$

where Area A, B, C, D, E, F represents the areas of the polygons A, B, C, D, E, F (as projected into 2 dimensions), respectively. In certain implementations, each area can be associated with a weight which can allow a region of an interesting object to be configured to become more or less interesting as compared to other regions on the interesting object.

[0395] Once a polygon has been selected, a point (such as, e.g., a point with reference to a world frame) on the surface of that polygon can be randomly selected and returned as the saccade point. As described herein, the wearable system can animate the saccadic movements of the avatar's eyes by moving the avatar's eye gaze from a saccade point to another in accordance with a saccade rate (e.g., from about 10 degrees/sec to a few hundred degrees/sec). Angular separation between saccade points can be in the range from 0.1 degrees to 30 degrees or so.

[0396] The wearable system can use a selected saccade point to rapidly adjust the eye pose of the virtual avatar (e.g., by adjusting the orientation of the eyes) to focus on the point. The eye movements may also be based on the interocular convergence calculations which may be handled by the rig that controls the animation of the virtual avatar.

[0397] FIG. 36 illustrates an example of eye pose and face transform for animating an avatar based on saccade points. FIG. 36 shows an eye 3610 and a face 3600. The eye 3610 can be associated with an eye coordinate 3612 and the face 3600 can be associated with a face coordinate 3602.

[0398] The wearable system can record a resultant eye pose (e.g., as determined based on the selected saccade point) with respect to the face in a proprioception system. The proprioception system can maintain any kind of knowledge of the avatar's body, such as, e.g., knowing when the avatar's arm is raised, without needing for a user to look at the avatar. The proprioception system can also hold the formulas for rotations with respect to different body parts,

such as e.g., a relative rotation between the head and torso or between the head and eyes, which may be part of the discomfort curves described with reference to FIGS. 38A-39. The proprioception system may reference an avatar's local frame for maintaining the relative positions of the avatar's body parts. The proprioception system can be implemented as a component of the avatar processing and rendering system 690, e.g., as part of the anatomy adjustment system 698.

[0399] With reference back to FIG. 36, the resultant eye pose with respect to the face can be calculated as an eye to face delta. The eye to face delta can be broken down into two angles: a horizontal angle and a vertical angle, where the horizontal angle can be an angle with respect to the x-axis and z-axis shown in the coordinates 3612 and 3602, and the vertical angle can be an angle with respect to the x-axis and y-axis shown in the coordinates 3612 and 3602. The horizontal angle in the face coordinate 3602 can be used to drive an animation of the avatar's head turning from left to right (or right to left), while the vertical angle in the face coordinate 3602 can be used to drive an animation of the head pitching from looking down to looking up (or from looking up to looking down). The horizontal and vertical angles in the eye coordinate 3610 can be used to drive an animation of the eye 3610 from looking left or right or rolling up or down.

[0400] As further described with reference to FIGS. 38A-39, the animation of eye pose and head pose can be determined based at least partly on a discomfort curve which can reduce or minimize a value representative of the biological discomfort (as if the avatar were human) due to the relative positions between head and eye. For example, the avatar may turn the head slightly if the position of an eye is too close to the edge of the eye socket.

Examples Processes of Avatar Rendering Based on Environmental Stimuli

[0401] FIG. 37 illustrates an example process of avatar rendering based on environmental stimuli. The example process 3700 can be performed by a wearable system 200 of a viewer of a virtual avatar, for example, by the avatar processing and rendering system 690 described with reference to FIG. 6B.

[0402] At block 3710, the wearable system can determine a plurality of interesting impulses in an environment. The wearable system can make such determination based on an avatar's head pose and eye gaze direction. For example, the wearable system can determine objects, regions, sound, light, etc., in a viewer's environment and determine whether one or more of these are interesting objects or areas of interest.

[0403] The interesting impulses can be associated with respective interestingness values. At block 3720, the wearable system can calculate (or adjust) the interestingness values associated with the plurality of interesting impulses. For example, a triggering event associated with an interesting object or area of interest may boost or decrease the corresponding interestingness value. The emotion and disposition of an avatar can also cause the wearable system to find an interesting impulse to have a higher interestingness value than another interesting impulse. As yet another example, the characteristics (or interactions with) of the objects or the viewer may also cause some interesting impulses to have higher interestingness values than other

interesting impulses. For example, while a viewer is speaking, his social triangle may have a higher interestingness value which can cause the avatar to look at the viewer's social triangle. The wearable system can animate saccadic eye movements within the social triangle when the avatar is looking at the viewer's social triangle. For example, the avatar's saccadic movements can be random or quasi-random within the social triangle or may follow a trajectory (e.g., from left eye towards right eye towards nose and then back to left eye). The jumps between successive avatar saccadic eye movements may be constrained. For example, if an avatar's gaze vector is directed toward a person's chin, the next permitted eye gaze vector may be directed toward a position in the social triangle that is a fraction (less than one) of the size of the social triangle. A saccadic jump in a direction that is generally parallel with a previous jump may utilize larger displacements, whereas a saccadic jump in a direction that is generally perpendicular to a previous jump may utilize a smaller displacement. In some embodiments, the saccadic frequency, the angular velocity of the saccadic eye movement as a function of the angular amplitude of the jump, and other saccadic parameters can be modeled on the behavior of human saccadic eye movements.

[0404] At block 3730, the wearable system can determine a target interesting impulse. The wearable system can select an interesting impulse among a plurality of interesting impulses in the environment as the target interesting impulse. The wearable system can make the selection based on the interesting values associated with the respective interesting impulses. For example, the wearable system may associate the target interesting impulse with an interesting impulse having the highest interestingness value.

[0405] At optional block 3732, the wearable system can provide a boost to an interestingness value associated with the target interesting impulse. This can, in some embodiments, occur in order to prevent the avatar from switching its interactions between two interesting impulses having similar interestingness values, which may cause visually disruptive experiences for the viewer.

[0406] At block 3740, the wearable system can animate a virtual avatar to respond to the target interesting impulse. For example, the wearable system can change the avatar's body pose or eye gaze to look at or interact with the target interesting impulse. The wearable system can also move the avatar to be closer to or farther away from the target interesting impulse.

[0407] At block 3750, the wearable system can monitor changes in the interestingness values of the plurality of interesting impulses. The interestingness value may decay over time. The interestingness value may also increase in response to a change in the environment, object, or interactions of the avatar or the viewer. For example, the avatar may initially look at the viewer during a telepresence session between the viewer and the avatar's human counterpart. The viewer may start playing music in his environment. As a result, the interestingness value associated with the source of the music may increase.

[0408] At block 3760, the wearable system can determine whether there is a new target interesting impulse. If not, the process 3700 goes back to the block 3750 where the wearable system continues to monitor changes in the interestingness values. If there is a change in the target interesting impulse, the process can go back to the block 3740 where the virtual avatar can respond to the new target interesting

impulse. In certain implementations, rather than going back to the block 3740, the process 3700 can also go back to the optional block 3732.

Examples of Discomfort Curves

[0409] To provide a realistic and lifelike animation of a virtual avatar, the wearable system can animate the pose of the avatar to be lifelike. For example, the wearable system can turn the avatar's head and its body to look at an object of interest behind the avatar, to avoid the unrealistic situation where the avatar merely turns its head 180 degrees (without turning its body) to look at the object of interest. As further described below, a discomfort curve can be used to generate a value indicative of the discomfort a human would experience if one part of his or her body were contorted into a particular body pose relative to an adjacent body part. For example, a human can turn his or her head in a range of about 60 to 90 degrees without substantial discomfort. If the human were to try to turn his or her head by, for example, 120 degrees, such movement would lead to severe discomfort (if it were biomechanically possible at all). Instead of attempting just a head turn that is too large, humans naturally turn their bodies in conjunction with their heads, to look at objects requiring a large angular change in eye gaze direction. Accordingly, to mimic this natural human movement, if a discomfort level (e.g., measured from a discomfort curve) for a potential avatar movement passes a threshold (e.g., a pure head turn is too large), the avatar may move an additional avatar body part (e.g., the avatar's torso) in conjunction with the avatar's head to avoid unnatural-looking body movements (e.g., those that would cause discomfort in a real human, such as a head turning 180 degrees).

[0410] Bone feathering, non-linear fractions, or exponential proportioning are examples of techniques that can be applied to mimic the avatar's pose or movement to the natural movement of a real person. However, in these techniques, the head's rotation relative to the ground typically determines the simultaneous motion of the rest of the body, because the rest of the body moves with a fraction of the head's movement. For example, if an avatar's head needs to turn 45 degrees, then the head of the avatar can turn 45 degrees relative to the ground, and the body can turn a portion of the 45 degrees. Because turning is accomplished using a mathematical formula in these techniques, the simulated movement of an avatar may not necessarily reflect nuanced, non-simultaneous movement of a real human body. In addition, in these techniques, the entire body's movement is simultaneous because it is a simple calculation from a single number (e.g., the 45 degree turn in the above example).

[0411] To provide an improved technique and provide a more realistic animation of an avatar, the wearable system (e.g., the anatomy adjustment system 698) can adjust an avatar's animation of two adjacent body parts in accordance with a discomfort curve. A discomfort curve can represent boundaries for animations between two adjacent body parts, such as, e.g., a relative angle between two adjacent body parts (e.g., an avatar head relative to an avatar torso). By way of an example, there may be one discomfort curve governing the relative position between a virtual avatar's eyes and head, and a separate discomfort curve governing the relative positions between the head and the torso, and so forth. In this example, the discomfort curve governing the

head in relation to the torso may restrict the head's rotation to a maximum of 90 degrees with respect to the torso. By establishing a 90 degree boundary, the discomfort curve can reduce or prevent unrealistic movement such as the avatar's head rotating 135 degrees to the right.

[0412] Advantageously, in some embodiments, use of the discomfort curve can realistically mimic non-simultaneous movement between the two adjacent body parts, leading to natural movements of the avatar and avoiding movements that represent the uncanny valley. Rather than a simultaneous movement between of the two body parts, a discomfort due to the pose of a first portion of the body can cause a subsequent movement of a second portion of the body (e.g., a discomfort in an eye pose can cause a change in the head pose or a discomfort in a head pose can cause a change in torso pose). As a result, there may be a natural time delay built in to take into account the separate calculations for a first body part relative to the a second body part, resulting in non-simultaneous movement of the first and second body parts of the avatar. Because different types of discomforts (e.g., leaning too far forward v. leaning too far back) can result in different movements, a separate calculation may be performed by the wearable for a body part relative to one or more adjacent body parts.

[0413] FIGS. 38A and 38B illustrate examples of discomfort curves of the torso in relation to the head. The discomfort curve 3800a in FIG. 38A illustrates a discomfort curve when a virtual avatar is standing, while the discomfort curve 3800b in FIG. 38B illustrates a discomfort curve when the virtual avatar is walking.

[0414] The x-axis 3820 of the graphs shown in FIGS. 38A and 38B represents a torso to head delta angle. This delta angle can represent the relative angle between the head and the torso of the avatar. The y-axis 3810 of the graphs shows levels of discomfort ranging from 0 to 1, where discomfort can represent the amount of energy needed for an avatar to maintain a position. Other values of discomfort can be used, e.g., from 0 to 100, discrete levels (e.g., a graded scale from A to F, where A represents little or no discomfort and F represents extreme discomfort), and so forth. In FIGS. 38A and 38B, a discomfort level of 0 means that no energy is required to maintain the particular position (e.g., there would be no discomfort or physical stress to maintain the position), while higher discomfort levels require more energy to maintain (e.g., there would be more discomfort or more physical stress would be experienced to maintain the position). A discomfort level of 0 may reflect the natural pose of the body in a resting or neutral position. By way of an example, a human hand at rest is slightly curled, which can correspond to a discomfort level of 0.

[0415] The discomfort curves 3800a and 3800b of FIG. 38A and FIG. 38B respectively map discomfort levels associated with a delta angle of the torso in relation to the head. As the delta angle increases or decreases from 0, the discomfort level rises accordingly. The discomfort curves can feed into a spring-based animation system, where the magnitude of the discomfort can determine the movement of the adjacent body parts as if there were a spring joining them. For example, when the magnitude of the discomfort reaches a threshold level due to a certain torso to head delta angle, the wearable system can automatically move the torso or head to reduce the angle between the torso and the head (and thus to reduce the discomfort). Further, a high discomfort

fort level may cause a larger or faster movement of the torso or head for reducing the level of discomfort.

[0416] As an example of the time delay between movements of adjacent body parts, if the goal of the rendering system is to move the avatar so that its eyes point 100 degrees to the right, the rendering system may start to move just the avatar's head toward the right. The torso to head delta angle will start to increase and the discomfort level (from the curves **3800a**, **3800b**) will also start to increase. Once the discomfort level passes a threshold (say, e.g., 0.5), the rendering system will start to rotate the avatar's torso to the right as well, which will reduce the increase in the torso to head delta angle and maintain the discomfort level below the threshold. The viewer sees the avatar's head initially begin to turn to the right, followed by the avatar's torso rotating toward the right, until the avatar is rendered as looking at the target. Thus, in this example, the rendering system generates movement of the avatar that is natural and human-like. As another example, the avatar may initially turn its head to look at an interesting impulse. However, the interestingness of the interesting impulse may increase (e.g., due to changes of the interesting impulse or interactions with the viewer). As time passes, the discomfort level associated with the avatar's head pose (e.g., turning the head to look at the interesting impulse) may increase which may cause the avatar to turn its body to continue looking at the interesting impulse while reducing discomfort level. For relative body movements that might lead to extreme discomfort, the discomfort threshold can be set to a lower value so that movement of a first body part will relatively rapidly lead to movement of a second, adjacent body part. If the relative body motions are readily performed by humans without discomfort, the discomfort threshold can be set to a higher value since movement of just one body part may be sufficient to accomplish the motion. Individual discomfort thresholds can be set for each of the discomfort curves used by the rendering system and these discomfort curves (or thresholds) can be adjustable, modifiable, or dynamically changeable as further described below.

[0417] The discomfort curves can be interchanged or modified based on the virtual avatar's current mood or mode. As shown in FIGS. **38A** and **38B**, the discomfort curves can be different based on the virtual avatar's current interactions. The discomfort level in FIG. **38A** has a minimum of 0 and rises slowly as the delta angle increases or decreases further away from 0 degrees. The discomfort level in FIG. **38A** reaches the maximum value of 1 when the delta angle of the torso in relation to the head reaches 36 degrees while avatar is standing. However, as shown in FIG. **38B**, the discomfort level rises more quickly when the virtual avatar is walking. In the example presented in FIG. **38B**, the discomfort level reaches the maximum level of 1 when the delta angle of the torso in relation to the head reaches 22 degrees. In certain implementation, the discomfort curve of the torso in relation to the head therefore changes as the virtual avatar changes modes from standing to walking. For example, a virtual avatar may initially be standing, and thus the anatomy adjustment system **698** (shown in FIG. **6**) can apply the discomfort curve **3800a** to adjust the animation of the virtual avatar. However, the virtual avatar may start walking. Based on this change in the virtual avatar's behavior, the anatomy adjustment system **698** can switch to the discomfort curve **3800b** for animating the virtual avatar's head-torso rotation.

[0418] As another example, the tangents to the discomfort curve (e.g., slopes) may be larger where the virtual avatar is agitated than when the virtual avatar is calm. As a result, the virtual avatar may reach a threshold discomfort level faster when agitated than when the virtual avatar is calm.

[0419] A discomfort curve may be represented by a mathematical parameterization by the anatomy adjustment system **698** (e.g., a mathematical formula, spline, lookup table, etc.). The discomfort curve (or discomfort thresholds) may include parameters that can be adjusted to reflect avatar behavior (e.g., standing v. walking), mood (e.g., agitated v. calm), etc. so that the anatomy adjustment system **698** can readily determine whether discomfort exists in an avatar movement and adjust (or not adjust) adjacent avatar body parts accordingly.

[0420] As an example application of animating an avatar's body parts based on discomfort curves, the wearable system can utilize one vertical angle adjustment animation (e.g., animating a tilting movement of a head) and three horizontal angle adjustment animations. The three horizontal angle adjustment animations can be chained together such that a change in one part of the body can cause the change in another part of the body which can in turn cause a change in the third part of the body. This chain reaction can be based on the discomfort levels between two or more body parts (such that if a change in one part of the body increases the discomfort level over a threshold, the neighboring part can move accordingly to reduce the discomfort level, and further increase of the discomfort level (e.g., over another threshold) can result in movement of the third body part).

[0421] As an example, in a horizontal adjustment angle chain, the eye movements can drive the head's left-right rotation animation which can in turn drive the torso's left-right animation which can in turn drive the hip's left-right animation rotation. Although the examples in this paragraph describe one vertical angle adjustment animation and three horizontal angle adjustment animations, any number of other horizontal (or vertical) animations could be inserted into this chain (and any number of associated discomfort curves can be used). The techniques described herein are not limited to these numbers of horizontal or vertical angle adjustment animations. For example, some embodiments may include only one horizontal and one vertical angle change to animate a simple head turn while other embodiments may include a horizontal rotation and vertical shift to represent a head turn and an additional horizontal rotation to represent the movement of the torso.

[0422] In certain embodiments, behavioral responses can also be inserted at points along the curve. For example, if the discomfort gets too high, a dialog from the avatar can be triggered. For example, if the avatar movement would lead to a sufficiently large discomfort level, the avatar may be rendered to say "ouch!" As another example, an avatar behavior to choose another target interesting impulse can also be triggered if the discomfort gets too high. For example, rather than moving to look at a target object that requires a large head and body rotation, the avatar may move part of the way, and then gesture to the target object with a hand. Such combined motion can keep the discomfort level below appropriate thresholds and can advantageously mimic real human behavior.

Example Processes of Avatar Animation with Discomfort Curves

[0423] FIG. 39 illustrates an example process of animating a virtual avatar with a discomfort curve. The example process 3900 can be performed by a wearable system 200 (such as a wearable system of a viewer of an avatar) or the anatomy adjustment system 698 of the wearable system 200.

[0424] The process begins by determining an orientation of a first body part with respect to an adjacent second body part of an avatar at block 3910. In certain implementations, to determine the relative position of the two body parts of an avatar, the wearable system can communicate with a proprioception system which can track rotations of various parts of the body relative to each other, such as e.g., the rotation of the head relative to the torso. In certain implementations, the orientation of a first body part with respect to an adjacent second body part may be deconstructed into a horizontal angle component and a vertical angle component. For example, a head turn may be deconstructed into a horizontal angle of 30 degrees towards the right and a vertical angle of 15 degrees upwards in relation to the torso.

[0425] At block 3920, the wearable system can access a discomfort curve associated with the avatar's environment or interaction. For example, a discomfort curve can be automatically selected depending on the virtual avatar's mood or mode as described with reference to FIGS. 38A and 38B. The discomfort curve can describe the angles between the first body part and the second body part, and the discomfort levels for the corresponding angles. The discomfort curve can automatically be generated using deep learning or regression analysis of data based on motion tracking or manually created based on human intuition or artistic style (e.g., by an animator).

[0426] At block 3930, the wearable system can determine a discomfort level based on the orientation. For example, the wearable system can input an angle between the first body part and the second body part into a discomfort curve to calculate the discomfort level associated with the angle.

[0427] At block 3940, the wearable system can determine whether the discomfort level passes a threshold condition. The threshold condition may cause the avatar to change its pose. If the discomfort level does not pass the threshold condition, the process 3900 moves to block 3960, where the wearable system monitors the avatar's environment or interaction. If the discomfort level passes the threshold condition, the process 3900 moves to block 3950.

[0428] At block 3950, the wearable system can adjust the pose of the first body part or the second body part. As described with reference to FIGS. 38A and 38B, the wearable system can employ a spring-based avatar animation system such that a greater deviation from the threshold condition may cause the avatar to change its pose faster or in a greater degree. In certain implementations, the wearable system can use the horizontal and vertical angle components to drive an animation of the first body part in relation to the second body part according to the discomfort curve. For example, the horizontal angle may correspond to a discomfort curve while the vertical angle may correspond to another discomfort curve. The final pose or movement of the avatar may be a combination of movements in both the horizontal and vertical of the first or the second body part.

[0429] The animation of a body part may be bounded by limits proscribed by the discomfort curve to produce a more natural and lifelike animation. For example, the discomfort

curve governing the animation of the torso in relation to the head may prohibit the head from horizontally rotating more than 60 degrees in relation to the torso.

[0430] At block 3960, the wearable system monitors the avatar's environment or interaction. For example, the wearable system can monitor changes in the interesting impulse or viewer's pose. The wearable system can also monitor the response of avatar to the environment, such as, e.g., a change in the avatar's interaction or mood which may be due to a change in the environment or the viewer's interaction.

[0431] At block 3970, if the wearable system doesn't detect a change in the environment or interaction, the process 3900 loops back to block 3960. Otherwise, the process 3900 moves to the block 3920 where the wearable system can determine a discomfort curve appropriate for the avatar's mode or mood. For example, a new discomfort curve may be applied to govern the animation of an avatar's head and torso as the virtual avatar transitions from sitting in a chair to running across a room. While the discomfort curve for sitting may permit the head to rotate 45 degrees to the left and right in relation to the torso, the new discomfort curve may restrict the head from rotating more than 20 degrees in relation to the torso to reflect the fact that a real human tends to look straight ahead while running.

[0432] The example process shown in FIG. 39 may be applied for movements involving multiple body parts (e.g., leg movement can drive hip movement which can drive torso movement, etc.).

Additional Aspects

[0433] In a 1st aspect, a wearable system for automatically scaling a virtual avatar, the wearable system comprising: a device sensor configured to image an environment of a user; a pose sensor configured to measure a pose of the user; a display configured to display virtual content in a mixed reality environment; and a hardware processor programmed to: receive pose data from the pose sensor; calculate a head position of a user with reference to a world frame based at least partly on the pose data, wherein the world frame includes a reference frame associated with the user's environment; determine a major horizontal plane with reference to the world frame in the user's environment; calculate a height of the user based on the major horizontal plane and the user's head position; calculate a target height of an avatar based at least partly on the user's height; automatically scale the avatar based on the target height; and cause the display to automatically render the avatar at the target height.

[0434] In a 2nd aspect, the wearable system of aspect 1, wherein to calculate the user's height, the hardware processor is programmed to: perform a ray casting between the head position and the major horizontal plane; and calculate the user's height based on the vertical height difference between the head position and the major horizontal plane.

[0435] In a 3rd aspect, the wearable system of aspect 2, wherein the horizontal major plane is associated with a plane on which an avatar is rendered, and the target height of the avatar is the vertical height difference between the head position and the major horizontal plane.

[0436] In a 4th aspect, the wearable system of aspect 2 or 3, wherein the horizontal major plane is associated with a plane on which the user is standing, and the target height of the avatar is calculated based on the formula below: $Avatar_H = User_H + UserSupportPlane_H - AvatarSupportPlane_H$ where $Avatar_H$ is the target height of the avatar, $User_H$ is

user's height, and $\text{UserSupportPlane}_H$ represents a height of the major horizontal plane on which the user is situated, and the $\text{AvatarSupportPlane}_H$ represents a height of a plane on which the avatar is to be rendered.

[0437] In a 5th aspect, the wearable system of aspect 4, wherein the height of the major horizontal plane and the height of the plane on which the avatar is to be rendered are measured with reference to a ground plane which is below the major horizontal plane and the height of the plane on which the avatar is to be rendered.

[0438] In a 6th aspect, the wearable system of any one of aspects 1-5, wherein to determine the major horizontal plane, the hardware processor is programmed to: generate a three-dimensional (3D) point cloud based on environment image data acquired by the depth sensing camera; apply a mesh algorithm to extract surfaces from the mesh cloud; convert the surfaces into a 3D polygonal mesh describing the user's environment; and estimate the major horizontal plane based on the 3D polygonal mesh.

[0439] In a 7th aspect, the wearable system of aspect 6, wherein to estimate the major plane, the hardware process or is programmed to: detect groups of connected polygons with similar orientation.

[0440] In an 8th aspect, the wearable system of any one of aspects 1-7, wherein to determine the major horizontal plane, the hardware processor is programmed to extract a plane in a horizontal orientation directly from a 3D point cloud generated based on environment images acquired by the device sensor.

[0441] In a 9th aspect, the wearable system of any one of aspects 1-8, wherein the hardware processor is further programmed to: detect a change in a relative height between the user and the avatar; and in response to the change, dynamically scale the avatar based at least partly on an updated target height.

[0442] In any of aspects 1 to 9, the device sensor can comprise a depth-sensing camera or the pose sensor can comprise an inertial measurement unit.

[0443] In a 10th aspect, a method for automatically scaling a virtual avatar, the method comprising: receiving pose data of a user; calculating a head position of the user based at least partly on the pose data; determining a major horizontal plane in the user's environment; calculating a user's height based on the major horizontal plane and the user's head position; calculating a target height of an avatar based at least partly on the user's height; and determining a scale for adjusting a size of the avatar based on the target height.

[0444] In an 11th aspect, the method of aspect 10, further comprising: causing a display to automatically render the virtual avatar at the target height in a mixed reality environment.

[0445] In a 12th aspect, the method of aspect 10 or 11 wherein calculating the user's height comprises performing a ray casting from the user's head position and calculating the user's height based on the vertical height difference between the head position and the major horizontal plane.

[0446] In a 13th aspect, the method of aspect 12, wherein the major horizontal plane is associated with a plane on which an avatar is rendered, and the target height of the avatar is the vertical height difference between the head position and the major horizontal plane.

[0447] In a 14th aspect, the method of aspect 12 or 13, wherein the major horizontal plane is associated with a plane on which the user is standing, and the target height of the

avatar is calculated based on the formula below: $\text{Avatar}_H = \text{User}_H + \text{UserSupportPlane}_H - \text{AvatarSupportPlane}_H$ where Avatar_H is the target height of the avatar, User_H is user's height, and $\text{UserSupportPlane}_H$ represents a height of the major horizontal plane on which the user is situated, and the $\text{AvatarSupportPlane}_H$ represents a height of a plane on which the avatar is to be rendered.

[0448] In a 15th aspect, the method of aspect 14, wherein the height of the major horizontal plane and the height of the plane on which the avatar is to be rendered are measured with reference to a ground plane which is below the major horizontal plane and the height of the plane on which the avatar is to be rendered.

[0449] In a 16th aspect, the method of any one of aspects 10-15, wherein determining the major horizontal plane comprises: generating a three-dimensional (3D) point cloud based on environment image data acquired by the depth sensing camera; applying a mesh algorithm to extract surfaces from the mesh cloud; converting the surfaces into a 3D polygonal mesh describing the user's environment; and estimating the major horizontal plane based on the 3D polygonal mesh.

[0450] In a 17th aspect, the method of aspect 16, wherein estimating the major plane comprises detecting groups of connected polygons with similar orientation.

[0451] In an 18th aspect, the method of any one of aspects 10-17, wherein determining the major horizontal plane comprises extracting a plane in a horizontal orientation directly from a 3D model of the environment generated based on environment images.

[0452] In a 19th aspect, the method of any one of aspects 10-18, further comprising: detecting a change in a relative height between the user and the avatar; and in response to the change, dynamically resizing the avatar based at least partly on an updated scale.

[0453] In a 20th aspect, the method of any one of aspects 10-19, wherein the head position and the major horizontal plane are determined in reference to a world frame.

[0454] In a 21st aspect, a wearable device for determining an intent of an user interaction, the wearable device comprising: a device sensor configured to acquire a user's interaction data with the user's environment; and a hardware processor programmed to: identify a user interaction based on the interaction data acquired from the device sensor; decompose the user interaction into at least one of: a world component or a local component, wherein the world component comprises an action with reference to a world frame of the user's environment and a local component comprises an action with reference to a local frame of the user's body; determine an intent associated with the user's interaction based on contextual information; and communicate at least one of: the intent, information associated with the world component, or information associated with the local component to another wearable device to cause the other wearable device to map the user interaction to the user's avatar rendered by the other wearable device based on the intent.

[0455] In a 22nd aspect, the wearable device of aspect 21, wherein the user interaction comprises at least one of an interaction with the user's environment or an object of interest in the user's environment.

[0456] In a 23rd aspect, the wearable device of aspect 22, wherein the object of interest comprises a physical object in the environment or a virtual object.

[0457] In a 24th aspect, the wearable device of aspect 23, wherein the object of interest is a virtual avatar of another user.

[0458] In a 25th aspect, the wearable device of any one of aspects 21-24, wherein the user interaction is performed via at least one of a head pose, eye gaze, or body pose, or a movement in the user's environment.

[0459] In a 26th aspect, the wearable device of any one of aspects 21-25, wherein the device sensor comprises at least one of: an inertial measurement unit (IMU), an outward-facing camera, or an eye tracking camera.

[0460] In a 27th aspect, the wearable device of any one of aspects 21-26, wherein the contextual information comprises at least one of layout of objects in the user's environment, location of the objects in the environments, the user's position, or the user's orientation.

[0461] In a 28th aspect, the wearable device of any one of aspects 21-27, wherein the intent is associated with the world component of the user interaction.

[0462] In a 29th aspect, the wearable device of any one of aspects 21-28, wherein the information associated with the local component comprises an animation of the local component of the user interaction.

[0463] In a 30th aspect, a method for rendering a virtual avatar for a remote user interaction, the method comprising: determining contextual information associated with a first user's environment in which a second user's avatar is rendered; determining a second user's intent associated with a user interaction of the second user that occurred in the second user's environment; determining a world action of the second user's avatar with respect to the first user's world frame based on contextual information and the intent of the second user; animating a local action of the second user's avatar with respect to a local frame of the second user's avatar based on a local component of the user interaction of the second user that occurred in the second user's environment; and causing a display to render the avatar comprising the world action and the local action in a mixed reality environment.

[0464] In a 31st aspect, the method of aspect 30, wherein determining the second user's intent comprises at least one of: receiving intent data from the second user's wearable device; or determining the second user's intent from the interaction data of the second user's user interaction.

[0465] In a 32nd aspect, the method of aspect 30 or 31, wherein the second user's intent comprises an intent to interact with an object of interest.

[0466] In a 33rd aspect, the method of any one of aspects 30-32, wherein the second user's intent comprises an intent to communicate or interact with the first user.

[0467] In a 34th aspect, the method of any one of aspects 30-33, wherein the local action is further animated based on the second user's intent associated with the local action.

[0468] In a 35th aspect, the method of any one of aspects 30-34, wherein animating the local action comprises performing a collision detection of the avatar as a result of the local action and objects in the first user's environment.

[0469] In a 36th aspect, the method of any one of aspects 30-35, wherein the world action comprises at least a movement of the second user's avatar in the first user's environment or an interaction by the second user's avatar with an object in the first user's environment.

[0470] In a 37th aspect, a system of rendering a virtual avatar based on a first user's intent in a first environment, the

system comprising a hardware processor programmed to: extract environmental information of the first environment from a plurality of sensors; identify a plurality of potential objects of interest based at least partly on received environmental information; determine an object of interest based at least partly on received environmental information, wherein the determination comprises: extending a vector from the user towards each potential object of interest and a sight line vector extending from the user outwards directly in front of the first user; assigning an interest weight value to each potential object of interest, wherein the interest weight of a potential object of interest dynamically changes to reflect changes in the environmental information; and calculating an overall interest value for each potential object of interest, wherein the interest value for a potential object of interest is based at least partly on the interest weight value for the potential object of interest; communicate object of interest information to a remote computing device, wherein object of interest information comprises at least an identifier representing the object of interest; receive communicated object of interest information from a remote computing device; and adjust at least one of a position or orientation of the virtual avatar based at least partly on the received object of interest information from the remote computing device; and a non-transitory computer-readable medium configured to store instructions to execute a process of rendering a virtual avatar based at least partly on received object of interest information.

[0471] In a 38th aspect, the system of aspect 37, wherein environmental information comprises a head pose or eye pose of a user or the presence of a plurality of virtual objects.

[0472] In a 39th aspect, the system of aspect 37 or 38, wherein the plurality of sensors comprises an outward-facing imaging system which observes the world in an environment around the user, and an inward-facing imaging system which can track the movement of the user's eyes.

[0473] In a 40th aspect, the system of any one of aspects 37-39, wherein the determination of an object of interest further comprises calculating the dot product between a vector associated with a potential object of interest and the user's sight line vector.

[0474] In a 41st aspect, the system of any one of aspects 37-40, wherein the determination of an object of interest occurs whenever a change in condition in the environment around the user is detected, wherein the change in condition comprises: a change in the user's pose, a change in the user's position in the environment, a change in position of a potential object of interest.

[0475] In a 42nd aspect, the system of any one of aspects 37-41, wherein the determination of an object of interest is based at least partly on the head pose of the first user and the user's semantic interest.

[0476] In a 43rd aspect, the system of aspect 42, wherein the determination of an object of interest further comprises performing ray casting from the user towards the direction of the user's sight line vector.

[0477] In a 44th aspect, the system of any one of aspects 37-43, wherein the object of interest information further comprises an offset angle between a vector associated with a potential object of interest and a user's sight line vector.

[0478] In a 45th aspect, the system of any one of aspects 37-44, wherein the adjustment of at least one of a position or orientation of the virtual avatar comprises rotating at least a portion of the virtual avatar to face the location of the

object of interest by an offset angle between a vector associated with a potential object of interest and the user's sight line vector.

[0479] In a 46th aspect, a method of rendering a virtual avatar based on a first user's intent in a first environment, the method comprising: accessing pose data of a first user, wherein pose data comprises at least an eye pose or head pose; identifying a plurality of potential objects of interest in an environment of the first user based at least partly on the pose data of the first user; determining an object of interest to the first user from the plurality of potential objects of interest in the environment of the first user based at least partly on the pose data of the first user; communicating object of interest information to a second user, wherein object of interest information comprises at least an identifier representing the object of interest; receiving object of interest information from the first user; identifying a location of the object of interest in a second environment with respect to a virtual avatar in the second environment; and adjusting at least one of a position or orientation of the virtual avatar in a second environment based at least partly on the location of the object of interest in the second environment with respect to the virtual avatar in the second environment.

[0480] In a 47th aspect, the method of aspect 46, wherein accessing pose data of a first user is done by an outward-facing imaging system which observes the world in an environment around the first user, and an inward-facing imaging system which tracks the movement of the first user's eyes.

[0481] In a 48th aspect, the method of aspect 46 or 47, wherein determining an object of interest comprises: extending a vector from the user towards each potential object of interest and a sight line vector extending from the user outwards directly in front of the user; assigning an interest weight value to each potential object of interest, wherein the interest weight of a potential object of interest dynamically changes to reflect changes in the environment; and calculating an overall interest value for each potential object of interest, wherein the interest value for a potential object of interest is based at least partly on the interest weight value for the potential object of interest.

[0482] In a 49th aspect, the method of any one of aspects 46-48, wherein determining an object of interest further comprises calculating the dot product between a vector associated with a potential object of interest and the user's sight line vector.

[0483] In a 50th aspect, the method of any one of aspects 46-49, wherein determining an object of interest occurs whenever a change in condition in the environment around the user is detected, wherein the change in condition comprises: a change in the user's pose, a change in the user's position in the environment, a change in position of a potential object of interest.

[0484] In a 51st aspect, the method of any one of aspects 46-50, wherein determining an object of interest is based at least partly on the head pose of the first user and the user's semantic interest.

[0485] In a 52nd aspect, the method of aspect 51, wherein determining an object of interest further comprises performing ray casting from the user towards the direction of the user's sight line vector.

[0486] In a 53rd aspect, the method of any one of aspects 46-52, wherein the object of interest information further

comprises an offset angle between a vector associated with a potential object of interest and the user's sight line vector.

[0487] In a 54th aspect, the method of any one of aspects 46-53, wherein adjusting at least one of a position or orientation of the virtual avatar comprises rotating the virtual avatar to face the location of the object of interest by an offset angle between a vector associated with a potential object of interest and the user's sight line vector.

[0488] In a 55th aspect, a wearable system for adjusting an avatar's interaction based on contextual information, the wearable system comprising: a mixed reality display for rendering a virtual avatar in an environment of a viewer; a hardware processor programmed to: determine a plurality of interesting impulses in the environment; calculate interestingness values associated with the plurality of interesting impulses; determine a target interesting impulse based at least partly on the interestingness values associated with the plurality of interesting impulses; and cause the mixed reality display to render an animation of the avatar to respond to the target interesting impulse.

[0489] In a 56th aspect, the wearable system of aspect 55, wherein the plurality of interesting impulses comprises one or more of an interesting object or an area of interest.

[0490] In a 57th aspect, the wearable system of aspect 56, wherein the interesting object comprises a social triangle or a gaze box.

[0491] In a 58th aspect, the wearable system of any one of aspects 55-57, at least one of the interestingness values associated with the plurality of interesting impulses is calculated based on an inherent interestingness associated with at least one of the plurality of interesting impulses and one or more adjustment factors comprising at least one of: a time duration, a triggering event in the environment, or an interaction of the viewer.

[0492] In a 59th aspect, the wearable system of any one of aspects 55-58, wherein the hardware processor is further programmed to monitor changes in the interestingness values of the plurality of interesting impulses based at least partly on updates to values associated with the adjustment factors and update the target interesting impulse to another interesting impulse in response to a determination that a change to the interestingness value of the interesting impulse passes a threshold condition.

[0493] In a 60th aspect, the wearable system of any one of aspects 55-59, wherein to determine the target interesting impulse, the hardware processor is programmed to: select an interesting impulse from the plurality of interesting impulse as the target interesting impulse where the interesting impulse has the highest interestingness value.

[0494] In a 61st aspect, the wearable system of any one of aspects 55-60, wherein the hardware processor is further programmed to provide a boost to an interestingness value associated with an interesting impulse upon a determination that the interesting impulse is the target interesting impulse.

[0495] In a 62nd aspect, the wearable system of any one of aspects 55-61, wherein the target interesting impulse is an interesting object, to render the animation of the avatar to respond to the target interesting impulse, the hardware processor is programmed to: animate saccadic eye movements for the avatar based at least partly on saccade points associated with the interesting object.

[0496] In a 63rd aspect, the wearable system of aspect 62, wherein to animate saccadic eye movement at a given time, the hardware processor is programmed to: determine a

plurality of polygons on the interesting object for holding the saccade points; calculate a probability associated with a saccade point landing on each polygon of the plurality of polygons; select a polygon based on the probability associated with the polygon; randomly select a point on the surface of the polygon as the saccade point; adjust the avatar's eye pose to focus on the saccade point.

[0497] In a 64th aspect, the wearable system of aspect 63, wherein the probability calculated based on respective area associated with each polygon of the plurality of polygons.

[0498] In a 65th aspect, a method of animating an avatar with saccadic eye movements, the method comprising: identifying an interesting object in an environment of an avatar; determining a plurality of polygons on the interesting object for holding saccade points; calculating a probability associated with a saccade point landing on each polygon of the plurality of polygons; selecting a polygon based on the probability associated with the polygon; randomly selecting a point on the surface of the polygon as the saccade point; and adjusting the avatar's eye pose to focus on the saccade point at a given time.

[0499] In a 66th aspect, the method of aspect 65, wherein the interesting object comprises a social triangle or a gaze box.

[0500] In a 67th aspect, the method of aspect 65 or 66, wherein the interesting object is determined by calculating interestingness values associated with objects in the avatar's field of view.

[0501] In a 68th aspect, the method of aspect 67, wherein an interestingness value decays with a passage of time.

[0502] In a 69th aspect, the method of any one of aspects 65-68, further comprising: recalculating a saccade time after a randomized window of saccade timing has expired.

[0503] In a 70th aspect, the method of any one of aspects 65-69, wherein the plurality of polygons is determined based on the avatar's saccade frustum.

[0504] In a 71st aspect, a method for identifying a social triangle for animating an avatar's eye movements, comprising: accessing parameters of a social triangle associated with a face, wherein the social triangle includes a region of the face which is used for animating an avatar's eye movements when the avatar looks at the face; identifying a first feature, and a second feature associated with a face based at least partly on the parameters, wherein at least one of the first feature or the second feature comprises a plurality of sub-features; determining a first vertex, a second vertex, and a third vertex based at least partly on the first and second features; determining a bounded area formed by the first, second, and third vertices; and assigning the bounded area as the social triangle.

[0505] In a 72nd aspect, the method of aspect 71, wherein the avatar's eye movements are limited to points within the bounded area.

[0506] In a 73rd aspect, the method of aspect 71 or 72, wherein the first feature comprises a first sub-feature of a portion of a first eye and a second sub-feature of a portion of a second eye.

[0507] In a 74th aspect, the method of any one of aspects 71-73, wherein the second feature comprises at least one of: a portion of the chin or a portion of the mouth.

[0508] In a 75th aspect, the method of any one of aspects 71-74, wherein the first feature comprises at least a portion of a head-mounted display for rendering the avatar, the first

vertex comprises a first end of the head-mounted display, and the second vertex comprises a second end of the head-mounted display.

[0509] In a 76th aspect, the method of any one of aspects 71-75, further comprising calculating a gravity vector and wherein the second vertex is extracted by positioning the second vertex at a location based on the orthogonal distance of the first vertex from the gravity vector, wherein the gravity vector intersects the third vertex.

[0510] In a 77th aspect, a system for driving animation of a virtual avatar using a discomfort curve, the system comprising: a non-transitory computer-readable medium configured to store: an orientation of a first body part of a virtual avatar with respect to an adjacent second body part of the virtual avatar; and at least one discomfort curve governing motion of the first body part with respect to the adjacent second body part of the virtual avatar; and a hardware processor programmed to: determine an orientation of the first body part of the virtual avatar with respect to the adjacent second body part of the virtual avatar; access the at least one discomfort curve governing the motion of the first body part with respect to the adjacent second body part; and drive an animation of the first body part with respect to the adjacent second body part of the virtual avatar at least in part according to the at least one discomfort curve. In some embodiments of the 77th aspect, the hardware processor can execute a virtual agent to drive the animation. In some embodiments of the 77th aspect, to drive the animation, the hardware processor is programmed to reduce or minimize a value returned from the discomfort curve. In some embodiments of the 77th aspect, in to drive the animation, the hardware processor is programmed to keep a value returned from the discomfort curve below a threshold value.

[0511] In a 78th aspect, the system of aspect 77, wherein the hardware processor is programmed to deconstruct the orientation of the first body part with respect to the adjacent second body part into a horizontal angle component or a vertical angle component; and drive the animation with respect to the adjacent second body part based on at least one of the horizontal angle component or the vertical angle component.

[0512] In a 79th aspect, the system of aspect 77 or 78, wherein to drive the animation, the hardware processor is further programmed to turn the first body part or the second body part such that the first body part is at the orientation with respect to the second body part; determine a discomfort level after a duration of time based at least partly on the discomfort curve; and in response to a determination that the discomfort level exceeds a threshold, turn either the first body part or the second body part such that the first body part is at another orientation with respect to the second body part which has a reduced discomfort level.

[0513] In an 80th aspect, the system of any one of aspects 77-79, wherein the animation of the virtual avatar is driven in a way as to reduce or minimize a value returned from the discomfort curve.

[0514] In an 81st aspect, the system of any one of aspects 77-80, further comprising a proprioception system configured to track the orientation of the first body part with respect to the adjacent second body part.

[0515] In an 82nd aspect, the system of any one of aspects 77-81, wherein a discomfort curve describes a discomfort level with respect to an angle between the first body part and the second body part.

[0516] In an 83rd aspect, the system of any one of aspects 77-82, wherein the discomfort curve is automatically generated using a deep learning or regression analysis of pose data.

[0517] In an 84th aspect, the system of any one of aspects 77-83, wherein the animation of the first body part with respect to the adjacent second body part is prohibited if movement exceeds a discomfort threshold on the discomfort curve.

[0518] In an 85th aspect, the system of any one of aspects 77-84, wherein the hardware processor is further programmed to detect a change in a mood or mode of the virtual avatar based on changes in an environment in which the virtual avatar is rendered or an interaction of the virtual avatar.

[0519] In an 86th aspect, the system of aspect 85, wherein the discomfort curve is modified based on the virtual avatar's current mood or mode.

[0520] In an 87th aspect, the system of aspect 85 or 86, wherein the detection of the change in the virtual avatar's mode or mood causes an access of a new discomfort curve governing the movement of the first body part with respect to the adjacent second body part.

[0521] In an 88th aspect, the system of any one of aspects 77-87, wherein the change in the virtual avatar's mode comprises a movement of the virtual avatar in an environment.

[0522] In an 89th aspect, the system of any one of aspects 77-88, wherein the hardware processor is further programmed to: determine that a value of the at least one discomfort curve passes a threshold; and in response to the determination, cause performance of an additional avatar behavior.

[0523] In a 90th aspect, the system of aspect 89, wherein the additional avatar behavior comprises making a sound or making a gesture.

[0524] In a 91st aspect, the system of any one of aspects 77-90, wherein to drive the animation of the first body part with respect to the adjacent second body part, the hardware processor is programmed to: move the first body part until a first discomfort threshold of the at least one discomfort curve is reached; and move the second body part to reduce the value of the at least one discomfort curve to be below the first discomfort threshold.

[0525] In a 92nd aspect, the system of aspect 91, wherein the hardware processor is further programmed to: move a third body part of the virtual avatar to reduce the value of the at least one discomfort curve to be below a second discomfort threshold.

[0526] In an 93rd aspect, a method of driving animation of a virtual avatar using a discomfort curve, the method comprising: determining an orientation of a first body part of a virtual avatar with respect to a second body part of the virtual avatar; accessing at least one discomfort curve governing the motion of the first body part with respect to the second body part; driving an animation of the first body part with respect to the second body part at least in part according to the at least one discomfort curve. The second body part of the virtual avatar may be adjacent to the first body part of the virtual avatar.

[0527] In a 94th aspect, the method of aspect 93, further comprising: deconstructing the orientation of the first body part with respect to the adjacent second body part into a horizontal angle component or a vertical angle component;

and driving the animation with respect to the adjacent second body part at least in part based on at least one of the horizontal angle component or the vertical angle component.

[0528] In a 95th aspect, the method of aspect 93 or 94, wherein animating the virtual avatar is driven in a way as to reduce or minimize a value returned from the discomfort curve.

[0529] In a 96th aspect, the method of any one of aspects 93-95, wherein the orientation is determined based at least partly on data in a proprioception system.

[0530] In a 97th aspect, the method of aspect 96, wherein the proprioception system further stores discomfort curves and formulas for rotations between body parts.

[0531] In a 98th aspect, the method of any one of aspects 93-97, wherein the discomfort curve describes a discomfort level with respect to an angle of a body part relative to an adjacent body part.

[0532] In a 99th aspect, the method of any one of aspects 93-98, wherein the discomfort curve describes a level of effort for an avatar to maintain the first body part or the second body part in a pose.

[0533] In a 100th aspect, the method of any one of aspects 93-99, wherein the discomfort curve is generated automatically using deep learning or regression analysis of pose data.

[0534] In a 101st aspect, the method of any one of aspects 93-100, wherein animating the first body part with respect to the adjacent second body part is prohibited if the movement exceeds a discomfort threshold on the discomfort curve.

[0535] In a 102nd aspect, the method of any one of aspects 93-101, wherein the discomfort curve is modified based on the virtual avatar's current mood or mode.

[0536] In a 103rd aspect, the method of any one of aspects 93-102, wherein driving the animation of the first body part with respect to the adjacent second body comprises chaining motion of the second body part to motion of the first body part based at least in part on a value of the at least one discomfort curve.

[0537] In a 104th aspect, the method of aspect 103, wherein motion of the second body part does not occur until the value of the at least one discomfort curve passes a threshold.

Other Considerations

[0538] Each of the processes, methods, and algorithms described herein and/or depicted in the attached figures may be embodied in, and fully or partially automated by, code modules executed by one or more physical computing systems, hardware computer processors, application-specific circuitry, and/or electronic hardware configured to execute specific and particular computer instructions. For example, computing systems can include general purpose computers (e.g., servers) programmed with specific computer instructions or special purpose computers, special purpose circuitry, and so forth. A code module may be compiled and linked into an executable program, installed in a dynamic link library, or may be written in an interpreted programming language. In some implementations, particular operations and methods may be performed by circuitry that is specific to a given function.

[0539] Further, certain implementations of the functionality of the present disclosure are sufficiently mathematically, computationally, or technically complex that application-specific hardware or one or more physical computing devices (utilizing appropriate specialized executable instruc-

tions) may be necessary to perform the functionality, for example, due to the volume or complexity of the calculations involved or to provide results substantially in real-time. For example, animations or video may include many frames, with each frame having millions of pixels, and specifically programmed computer hardware is necessary to process the video data to provide a desired image processing task or application in a commercially reasonable amount of time.

[0540] Code modules or any type of data may be stored on any type of non-transitory computer-readable medium, such as physical computer storage including hard drives, solid state memory, random access memory (RAM), read only memory (ROM), optical disc, volatile or non-volatile storage, combinations of the same and/or the like. The methods and modules (or data) may also be transmitted as generated data signals (e.g., as part of a carrier wave or other analog or digital propagated signal) on a variety of computer-readable transmission mediums, including wireless-based and wired/cable-based mediums, and may take a variety of forms (e.g., as part of a single or multiplexed analog signal, or as multiple discrete digital packets or frames). The results of the disclosed processes or process steps or actions may be stored, persistently or otherwise, in any type of non-transitory, tangible computer storage or may be communicated via a computer-readable transmission medium.

[0541] Any processes, blocks, states, steps, or functionalities in flow diagrams described herein and/or depicted in the attached figures should be understood as potentially representing code modules, segments, or portions of code which include one or more executable instructions for implementing specific functions (e.g., logical or arithmetical) or steps in the process. The various processes, blocks, states, steps, or functionalities can be combined, rearranged, added to, deleted from, modified, or otherwise changed from the illustrative examples provided herein. In some embodiments, additional or different computing systems or code modules may perform some or all of the functionalities described herein. The methods and processes described herein are also not limited to any particular sequence, and the blocks, steps, or states relating thereto can be performed in other sequences that are appropriate, for example, in serial, in parallel, or in some other manner. Tasks or events may be added to or removed from the disclosed example embodiments. Moreover, the separation of various system components in the implementations described herein is for illustrative purposes and should not be understood as requiring such separation in all implementations. It should be understood that the described program components, methods, and systems can generally be integrated together in a single computer product or packaged into multiple computer products. Many implementation variations are possible.

[0542] The processes, methods, and systems may be implemented in a network (or distributed) computing environment. Network environments include enterprise-wide computer networks, intranets, local area networks (LAN), wide area networks (WAN), personal area networks (PAN), cloud computing networks, crowd-sourced computing networks, the Internet, and the World Wide Web. The network may be a wired or a wireless network or any other type of communication network.

[0543] The systems and methods of the disclosure each have several innovative aspects, no single one of which is solely responsible or required for the desirable attributes

disclosed herein. The various features and processes described above may be used independently of one another, or may be combined in various ways. All possible combinations and subcombinations are intended to fall within the scope of this disclosure. Various modifications to the implementations described in this disclosure may be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of this disclosure. Thus, the claims are not intended to be limited to the implementations shown herein, but are to be accorded the widest scope consistent with this disclosure, the principles and the novel features disclosed herein.

[0544] Certain features that are described in this specification in the context of separate implementations also can be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation also can be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination. No single feature or group of features is necessary or indispensable to each and every embodiment.

[0545] Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. In addition, the articles “a,” “an,” and “the” as used in this application and the appended claims are to be construed to mean “one or more” or “at least one” unless specified otherwise.

[0546] As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: A, B, or C” is intended to cover: A, B, C, A and B, A and C, B and C, and A, B, and C. Conjunctive language such as the phrase “at least one of X, Y and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be at least one of X, Y or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y and at least one of Z to each be present.

[0547] Similarly, while operations may be depicted in the drawings in a particular order, it is to be recognized that such

operations need not be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Further, the drawings may schematically depict one more example processes in the form of a flowchart. However, other operations that are not depicted can be incorporated in the example methods and processes that are schematically illustrated. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the illustrated operations. Additionally, the operations may be rearranged or reordered in other implementations. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products. Additionally, other implementations are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results.

What is claimed is:

1. A system of rendering a virtual avatar based on a first user's intent in a first environment, the system comprising a hardware processor programmed to:
 - extract environmental information of the first environment from a plurality of sensors;
 - identify a plurality of potential objects of interest based at least partly on received environmental information;
 - determine an object of interest based at least partly on received environmental information, wherein the object of interest is determined by:
 - extending a vector from the first user towards each potential object of interest and a sight line vector extending from the first user outwards directly in front of the first user;
 - assigning an interest weight value to each potential object of interest, wherein the interest weight value of a potential object of interest dynamically changes to reflect changes in the environmental information; and
 - calculating an overall interest value for each potential object of interest, wherein the overall interest value for a potential object of interest is based at least partly on the interest weight value for the potential object of interest;
 - communicate object of interest information to a remote computing device, wherein object of interest information comprises at least an identifier representing the object of interest;
 - receive communicated object of interest information from a remote computing device; and
 - adjust at least one of a position or orientation of the virtual avatar based at least partly on the object of interest information received from the remote computing device; and
- a non-transitory computer-readable medium configured to store instructions to execute a process of rendering a virtual avatar based at least partly on received object of interest information.
2. The system of claim 1, wherein environmental information comprises a head pose or eye pose of a user or a presence of a plurality of virtual objects.
3. The system of claim 1, wherein the plurality of sensors comprises an outward-facing imaging system which observes the world in an environment around the first user, and an inward-facing imaging system which can track movement of the first user's eyes.
4. The system of claim 1, wherein the object of interest is further determined by calculating a dot product between a vector associated with a potential object of interest and the first user's sight line vector.
5. The system of claim 1, wherein the object of interest is determined whenever a change in condition in an environment around the first user is detected, wherein the change in condition comprises: a change in the first user's pose, a change in the first user's position in the environment, a change in position of a potential object of interest.
6. The system of claim 1, wherein the object of interest is determined based at least partly on a head pose of the first user and the first user's semantic interest.
7. The system of claim 1, wherein the object of interest is further determined by performing ray casting from the first user towards a direction of the first user's sight line vector.
8. The system of claim 1, wherein the object of interest information further comprises an offset angle between a vector associated with a potential object of interest and a user's sight line vector.
9. The system of claim 1, wherein the at least one of the position or orientation of the virtual avatar is adjusted by rotating at least a portion of the virtual avatar to face a location of the object of interest by an offset angle between a vector associated with a potential object of interest and the first user's sight line vector.
10. A method of rendering a virtual avatar based on a first user's intent in a first environment, the method comprising:
 - accessing pose data of a first user, wherein pose data comprises at least an eye pose or head pose;
 - identifying a plurality of potential objects of interest in an environment of the first user based at least partly on the pose data of the first user;
 - determining an object of interest to the first user from the plurality of potential objects of interest in the environment of the first user based at least partly on the pose data of the first user;
 - communicating object of interest information to a second user, wherein object of interest information comprises at least an identifier representing the object of interest;
 - receiving object of interest information from the first user;
 - identifying a location of the object of interest in a second environment with respect to a virtual avatar in the second environment; and
 - adjusting at least one of a position or orientation of the virtual avatar in a second environment based at least partly on the location of the object of interest in the second environment with respect to the virtual avatar in the second environment.
11. The method of claim 10, wherein the accessing pose data of the first user is done by an outward-facing imaging system which observes the world in an environment around the first user, and an inward-facing imaging system which tracks movement of the first user's eyes.
12. The method of claim 10, wherein the determining the object of interest comprises:

extending a vector from the first user towards each potential object of interest and a sight line vector extending from the first user outwards directly in front of the first user;

assigning an interest weight value to each potential object of interest, wherein the interest weight value of a potential object of interest dynamically changes to reflect changes in the environment; and

calculating an overall interest value for each potential object of interest, wherein the overall interest value for a potential object of interest is based at least partly on the interest weight value for the potential object of interest.

13. The method of claim **10**, wherein the determining the object of interest further comprises calculating a dot product between a vector associated with a potential object of interest and the first user's sight line vector.

14. The method of claim **10**, wherein the determining the object of interest occurs whenever a change in condition in the environment around the first user is detected, wherein the change in condition comprises: a change in the first user's

pose, a change in the first user's position in the environment, a change in position of a potential object of interest.

15. The method of claim **10**, wherein the determining the object of interest is based at least partly on the head pose of the first user and the first user's semantic interest.

16. The method of claim **10**, wherein the determining the object of interest further comprises performing ray casting from the first user towards a direction of the first user's sight line vector.

17. The method of claim **10**, wherein the object of interest information further comprises an offset angle between a vector associated with a potential object of interest and the first user's sight line vector.

18. The method of claim **10**, wherein the adjusting the at least one of the position or orientation of the virtual avatar comprises rotating the virtual avatar to face the location of the object of interest by an offset angle between a vector associated with a potential object of interest and the first user's sight line vector.

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